GERMAN-MALAYSIAN INSTITUTE
NETWORK SECURITY

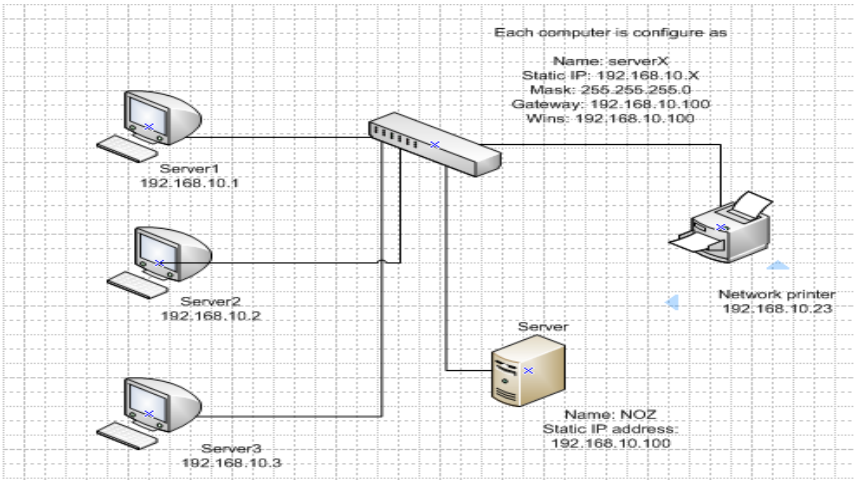
WINDOWS SERVER 2003

CIT 0553
OPERATING SYSTEM ADMINISTRATION 1

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Notes For Instructors

This section explains how the lab network of computers is arranged, the IP addresses and computer names used, and other details. Instructors should use this information to alter the examples in the labs and notes to match their own network configurations.



All computers use static IP addresses and are on the same subnet. All workstations that students use have the following configuration

Computer : Pentium D 2.8GHz, 512MB RAM, 80GB Disk

Partition 1 : C Drive, formatted as NTFS, 20GB, installed with Windows XP Pro SP2.
Partition 2 : D Drive, formatted as FAT32, 20GB, free space.
Used to install the Windows Server 2003 Operating System.
Partition 3 : E Drive, formatted as FAT32, 20GB, free space.
Partition 4 : F Drive, formatted as FAT32, 20GB, free space.

Students for other courses also use the computers. The use of the D: drive is reserved for the use of installing Windows 2003 Server. The computers dual-boot . the default operating system is Windows XP.

Source for installation files

Server NOZ has the share S2003 where the Windows 2003 files from the Server CD had been copied (specifically the I386 subdirectory). The share permissions have been set to

EVERYONE with READ access. The user account "**mstudent**" (password "**student**") provided by server NO is used to access the resource.

Computer Configuration

Each student group (students work in pairs) is allocated two computers, one acting as a server the other as a workstation. Windows Server 2003 is installed on one computer (drive D :) and Windows XP Professional is installed on the other computer (drive C :).

Generally, the instructor builds a list of computer names, domain names and IP addresses for the students to use. We use the real computer names and IP addresses (static, we do not use DHCP) for the lab exercises (this simplifies things).

Dual Boot

After installation, the computers will dual boot. Students are instructed in the lab exercise to modify the boot.ini file on each computer so that the computers will boot Windows XP (on Drive C :) so that other students can still use the computers in the normal way.

Removing Student Files

At the end of the course, the D: drive is reformatted and the C:\boot.ini file is modifies to remove references to the task of removing the student files and installation software.

Multiple Installations

We often have a large number of groups. This means that we allocate more than one installation of the software per computer. For example group 1can installed Windows Server 2003 on D:\w2001, and group 2 can install the same software in the folder E:\w2002.

As the groups are scheduled at different times of the day, this is not a problem. Some of the practical exercises use this same naming principle, for example, when creating shared folders. In this case, student group 1 would use D:\temp1 whilst student group 2 would use E:\temp2.

This concept is explained to each student group and they are instructed to write down their folder names in their booklets before beginning the install process.

Lab 1

Installing
Windows Server 2003

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Installing Windows Server 2003

In this section you will install a Windows Server 2003 from CD. This will comprise a Windows Server 2003 and a client computer running Windows XP Professional.

Active Directory

When you install Windows Server 2003 as the computer in charge of your network, you will also install active directory. This is like a large data store that holds information about users, policies computers and other information.

In a Windows 2003 network, all servers keep their own copy of the Active Directory, and when changes are made, the changes are sent to all the other servers (that are configured as domain controllers) in the network.

You will enable Active Directory when you install the Windows 2003 Server, thus creating a network for your organization. This active directory will be used in subsequent exercises.

File System Types

Windows Server 2003 supports the following file-systems. Which file system you choose depend upon a number of factors.

FAT[File Allocation Table]	Supported by MSDOS and Windows 9x. No file-level security. Partition sizes less than 2GB.
FAT32	Supported by MSDOS and Windows 9x. No file-level security. Partition sizes larger than 2GB but maximum 32GB.
NTFS [NT File System]	Windows NT, XP, 2000 and 2003 support this file system. Choose this file system if you want file level security [permissions at the file level]. If you intend to support Macintosh computers, or migrate Netware servers to Windows Server 2003, you must choose NTFS. The other features that NTFS supports are disk compression, disk quotas and file encryption.

FAT file systems can be converted to NTFS either during installation or afterwards. It is not possible to convert an NTFS partition to FAT (you have to remove the partition, recreate it, and then format it as a FAT partition)

Advantages Disadvantages

FAT Allows MS-DOS access to your files. Widely used. Files remain unprotected. Automatic disk restoration not possible. Allows access to files via an intruder using MS-DOS. Filenames restricted to 8.3 formats.

NTFS File level security possible. Automatic disk restoration possible. Supports long filenames Disk quotas. File compression and encryption. Only supported by Windows 2003, Windows 2000 or Windows NT. If the boot partition is NTFS, you cannot access any files if booting under MS-DOS.

You will use an NTFS file system because we are interested in security and will assign permission rights to folders and files.

Comparison Of Supported File Systems

Feature	**FAT**	**FAT32**	**NTFS**
Granular security			√
Compatibility	Can read FAT32; cannot read NTFS	Can read FAT; cannot read NTFS	Can read both FAT and FAT32
Support for Recovery Console in emergencies			√
Supports for becoming a domain controller			√
Can be converted	To FAT32 or NTFS	To NTFS	No conversion supported

Licensing Modes

The licensing mode refers to the way in which client licenses are managed. The two modes are:-

Per Seat License

A separated client license is required for each client computer on the network. This is preferred for large networks where client computers may connect to more than one server, and thus is easier to manage.

Per Server License

Each server is configured to supports a specific number of client computers, each of which is full licensed. On smaller networks where client computers may only access a single server, this option allows you to specify the maximum number of simultaneous connections that the server can support.

In the lab exercise, you will specify the "per server license" mode.

Different ways to install Windows Server 2003

Windows Server 2003 can be installed using one of TWO ways.

Network Install

The Windows files are placed on a network share. The computer must be booted with either a compatible operating system that has network access, or a client network boot disk that can access the network share.

CD-ROM Install

The Windows files reside in a CD-ROM drive that is supported by and recognized by Windows [check Hardware Compatibility List]. The computer must be booted with either a compatible operating system that recognizes the CD-ROM, or a boot disk that includes a CD-ROM driver.

During the install process, a folder Win_nt. ~bt is created on the target computer, and theWindows 2003 installation files are copied into that folder.

Network installation of Windows Server 2003

The windows files are copied from the distribution system installed on the hard disk, the installer boots the computer (from a system boot disk that has a network client) and maps a

network drive to the share point that holds the Windows files. The program WINNT.EXE is then executed.

If the computer is already running Windows (such as Windows 9x or NT workstation), it can be upgraded to Windows 2003 using the WINNT32.EXE program. The installer boots the computer system using the installing operating system, logs on, then accesses the shared network folder where the install files are located, and runs WINNT32.EXE to either install or upgrade to Windows 2003.

Installation Summary

Use WINNT.EXE if you are upgrading from MS-DOS or there is no operating system installed or the hard disk of the computer.

Use WINNT.EXE if you are upgrading from NT Server.

To create a Windows network, a Windows 2003 server is installed first. This server will hold information about the network, such as usernames and passwords, the names of the computers that belong to the network and other information. Windows 2003 stores this information in the Active Directory.

Using NTFS provides file-level security. This lets you secure files, folders, printers and other resources.

LAB 1. INSTALL WINDOWS SERVER 2003

This section requires you to work in groups (a group comprises two students) using two computers. One computer will act as a server for your network. The other computer will act as a client computer.

It is important that you perform the steps in the suggested order. Some steps are very dependent upon the earlier steps having been successfully completed.

Preliminary Setup

Before you start install Windows Server 2003, you have to make sure your PC have the following configuration:

Computer: Pentium D 2.8GHz, 512MB RAM, 80GB Disk

Partition 1: C Drive, formatted as NTFS, 20GB, installed with Windows XP Professional SP2.
(User name = **Administrator**, password = **student**.)

Partition 2: D Drive, formatted as FAT32, 20GB, free space. Used to install the Windows Server 2003 software.

Partition 3: E Drive, formatted as FAT32, 20GB, free space.

Partition 4: F Drive, formatted as FAT32, 20GB, free space.

PREPARING TO INSTALL WINDOWS SERVER 2003

Work in pairs, and install the Server first. You will need two computers, preferably side by side. You will be performing a network install.

The computer will need a name. This will identify the computer on the network from other computers, so must be a unique name. Please use the computer name assigned by your instructor. Your network will also require TCP/IP network details. Obtain these from your instructor.

The organization will also need a domain name. In a later exercise, you will install additional services (such as Domain Name Services) that rely on this name, please use the domain name assigned by your instructor.

Please enter the following information before continuing.

Name of This Computer	servergmi
Name of Organization	GMi
Role of This Computer	Server
Network Share for server Files	\\noz\s2003
Local Directory to store Server Files	D:\w2003x[where x represents 1, 2, 3 or 4] Ask your instructor.
Name of Installer	Administrator
Domain Name	3**CL**server**SN**.net (e.g. netsec.net)
TCP/IP Address	192.168.10.**SN**
TCP/IP Subnet Mask	255.255.255.0
TCP/IP Gateway	192.168.10.100

Note: **SN** = Station Number; **CL** = your class (A or B or C)

The computers you are using have Windows XP Professional SP2 installed on Drive C: and there is space allocated on Drive D: for your installation.

After installing Windows 2003, the computer will have two operating systems installed on it. Windows XP Professional SP2 will be on Drive C: and Windows Server 2003 will be on drive D:

Beware: You are going to install Windows Server 2003 onto a computer that has an operating system installed. The disk is already partitioned. Be very careful you follow the instructions, and if unsure, ASK your lab instructor.

YOU WILL INSTALL WINDOWS SERVER 2003 INTO DRIVE D: (D: = for class 3C; E: = for class 3B, F: = for class 3A)

Step 1: Install Windows Server 2003

1. **Boot** your PC using Windows 2003 CD.
2. Press **ENTER.** (To install Windows Server 2003)

TEXT MODE SETUP PHASE

During this step, the installation process displays the partition information and allows you to select the partition on which to install Windows 2003. The files are then copied to the partition specified. Once this is finished, the computer reboots.

3. **Windows Server 2003 Welcome Screen**

The welcome to Windows Setup screen will appear. Press Enter to continue.

Press ENTER to setup Windows Server 2003 on this computer.

4. **Windows Server 2003 License Agreement**

Read the terms of the license agreement.

If you accept (which, of course, you have to do to continue installation), press **F8** to continue.

5. **Windows Server 2003 Setup**

You are presented with options to Setup, repair or quit.

Press ENTER to setup Windows Server 2003 on this computer.

6. **Windows Server 2003 Setup: Partition List**

A list of available partitions is displayed.

Highlight the D: partition (move the highlight bar using the cursor down arrow key so the highlight bar is over Drive D :)

(D: = for class 3C; E: = for class 3B, F: = for class 3A)

Press ENTER to continue.

7. **Partition Options**

Windows 2003 setup displays a list of options, such as formatting the partition or leaving the file system intact. Highlight the option.

Select **"Format the partition using the NTFS file system"**

Then press ENTER to continue.

Wait until format finish. After finish, your PC will be automatically restarted.

Press "**ESC**" or wait 20 second and your PC will boot from hard disk

8. **Windows Server 2003 Setup: Copying Files**

The files are copied from the temporary installation folder to the D: partition folder.

Once the files have been copied, the computer will restart.

GUI MODE SETUP PHASE

In this phase, the installation process continues. You are prompted using setup wizards that allow you to select components and configurations.

9. **Installing Devices**

Windows 2003 will automatically detect hardware devices. This process can take a few minutes so please be patient.

10. **Regional and Language Options**

This allows you to specify your country and your keyboard layout. By default, text input language and method is: US Keyboard layout.

If you are not sure, ask your instructor.

Click Next to continue.

11. **Personalize your software**

Enter "**Student**" as the Name, "**GMi**" as the organization.

Click Next to continue.

12. **Product Key**

Enter "**Product Key**" as given by your instructor.

Click next to continue.

13. **Licensing Mode**

Select **Per Server** as the licensing mode and set the number of client connections to **10.**

Click next to continue.

14. **Computer Name and Administrative password**

Enter the computer name given to you earlier by your instructor. Set the administrative password to "**abc123!@#**" and reconfirm password to "**abc123!@#**".

Click next to continue.

15. **Date and Time Settings**

Select the time zone appropriate for your location.

(Kuala Lumpur, Singapore).

Click next to continue.

16. **Network Settings**

Windows 2003 will now install the network components.

Select Custom Settings

Click next to continue.

17. **Networking Components**

A list of components is presented

√	Client for Microsoft Networks
	Network Load Balancing
√	File and Printer Sharing for Microsoft Networks
√	Internet Protocol (TCP/IP)

Click next to continue.

18. **Workgroup or Computer Domain**

Select No, this computer is not on a network or is on a network without a domain.

Make this computer a member of the following workgroup.

SERVER

Click next to continue.

19. **Performing Final Tasks**

Please wait whilst Windows 2003 installs the start menu items, registers the components, save the settings and removes the temporary installation files.

20. **Completing the windows 2000 Setup wizard**

When finished, the computer will restart automatically.

Once the computer has restarted, logon as "**Administrator**". The next step will be to modify the boot-up sequence so that the computer reverts back to using Windows XP. This will allow other students to use these computers normally.

21. **First time Login**

When you first time login, the "**Manage Your Server**" dialog box will appear.
Mark the box as below to stop this dialog box displayed at logon.

☐ √ Don't display this page at logon

Close this page to continue.
Congratulation! You have finish install the Windows Server 2003.

Summary

In this section you installed Windows Server 2003 on a computer. In the following exercise you will install Active Directory and other services, creating a small network for you to administer.

Lab 2

Modifying The Startup Display List

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Lab 2: Modifying the Start Display List

You have successfully installed Windows Server 2003 on the computer and it is now running. You should currently be logged in as Administrator. Close any dialog boxes that are active.

After installing Windows Server 2003, the computer now has two operating systems on it. The default operating system that the computer starts has been changed from Windows XP to Windows Server 2003.

In this exercise, you will alter the startup settings of the computer so that it reverts back to Windows XP instead of Windows Server 2003. You will also modify the startup display list to include an adequate description of your installation, so that you can readily identify it from other installations on the same computer (the computer may end up with two or more Windows Server 2003 installations due to the size of student groups, so you need to know which your installation is).

The file **boot.ini** resides on the boot partition [drive c:] and allows the computer to display a list of installed operating systems at boot time.

Accessing Boot.ini

1. Click on My Computer
2. Click on Local Disk (C :)

You will note that the file **boot.ini** does not appear in the window. This is because the folder view for the current window does not display hidden files, so you will now need to change the folder view.

Changing the folder view

3. Click on the **Tools** option on the menu bar and select **Folder Options**
4. Select the **View** tab
5. Select the option **Show hidden files and folders**

6. Uncheck the box **Hide protected operating system files**
7. Click **YES** to the warning dialog box that appears concerning the hiding of protected operating system files.
8. Click **Apply**
9. Click **OK**

You will now have an icon that represents the file **boot.ini** in the window.

Editing boot.ini

10. Double click boot.ini

This will load boot.ini into a text editor (notepad) so you can edit it. The file will look similar to

[boot loader]
timeout=30
default=multi(0)disk(0)rdisk(0)partition(2)\WINDOWS

[operating systems]
multi(0)disk(0)rdisk(0)partition(2)\WINDOWS="Windows Server 2003, Enterprise" /fastdetect
multi(0)disk(0)rdisk(0)partition(1)\WINDOWS="Microsoft Windows XP Professional" noexecute=optin/fastdetect

11. Change the line

*multi(0)disk(0)rdisk(0)partition(2)\WINDOWS="**Windows Server 2003, Enterprise**" /fastdetect*
to
*multi(0)disk(0)rdisk(0)partition(2)\WINDOWS="**Your domain name here**" /fastdetect*

(replace "**Your domain name here**" with the domain name given to you by your instructor [e.g. netsec]).
Example:
*multi(0)disk(0)rdisk(0)partition(2)\WINDOWS="**netsec**" /fastdetect*

12. Save the file and exit notepad.

SPECIFYING BOOT OPTIONS

Currently, the computer will boot Windows Server 2003. This is not what it used to be prior to the installation you just performed.

In this section you will set the default boot operating system back to Windows XP. You could have done this when you were modifying **boot.ini** earlier (as that is where it is specified).

1. Open **System** (click Start, Control Panel, System)
2. Click the **Advance tab**
3. Click **Setting** on **Startup and Recovery** section
4. Select **"Microsoft Windows XP Professional" /noexecute=optin/fastdetect** as the startup operating system
5. Click **OK**
6. Click **OK** to close System Properties

RESTART THE COMPUTER

Lab 3

Installing and Configuring DNS

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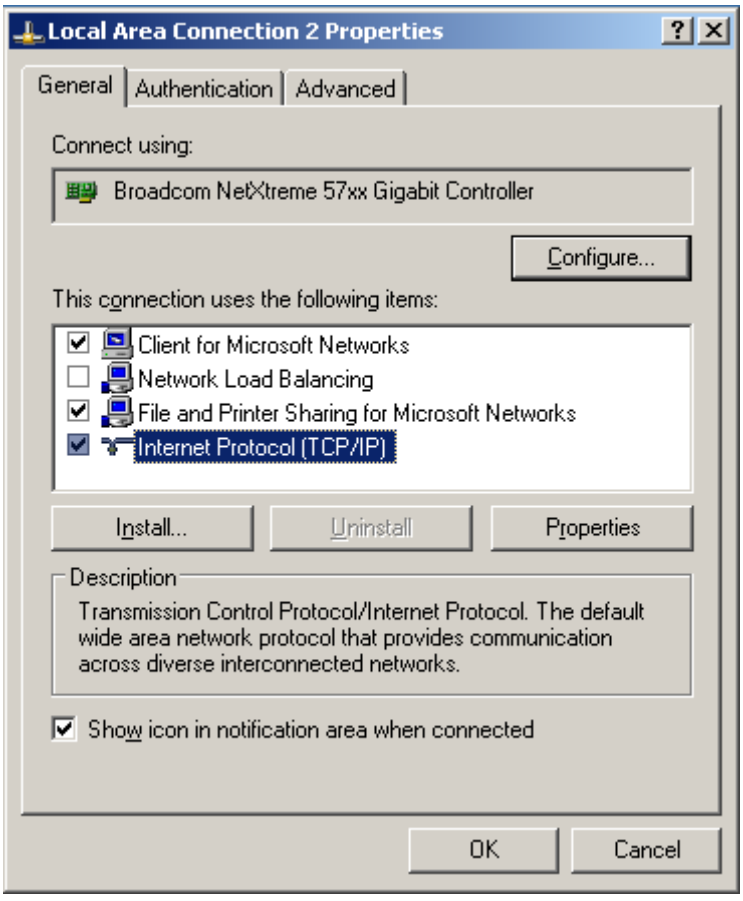
Prepared by :
Noor Azhar Bin Amran

LAB 3: DOMAIN NAME SERVICES

In this exercise you will implement a domain name server for your network. This server provides a means of resolving Internet domain host names to TCP/IP addresses.

The first step is required to ensure that you are using a static IP address and that the DNS setting on the computer have been correctly configured. Make sure your have hook up your IP to the network before you start.

1. Log on to Windows Server 2003 as "**Administrator"**
2. Close the dialog box "**Manage Your Server**"
3. Right-click My Network Places, select Properties,
4. Double-click **Local Area Connection** to display the "**Local Area Connection Properties"** dialog box.

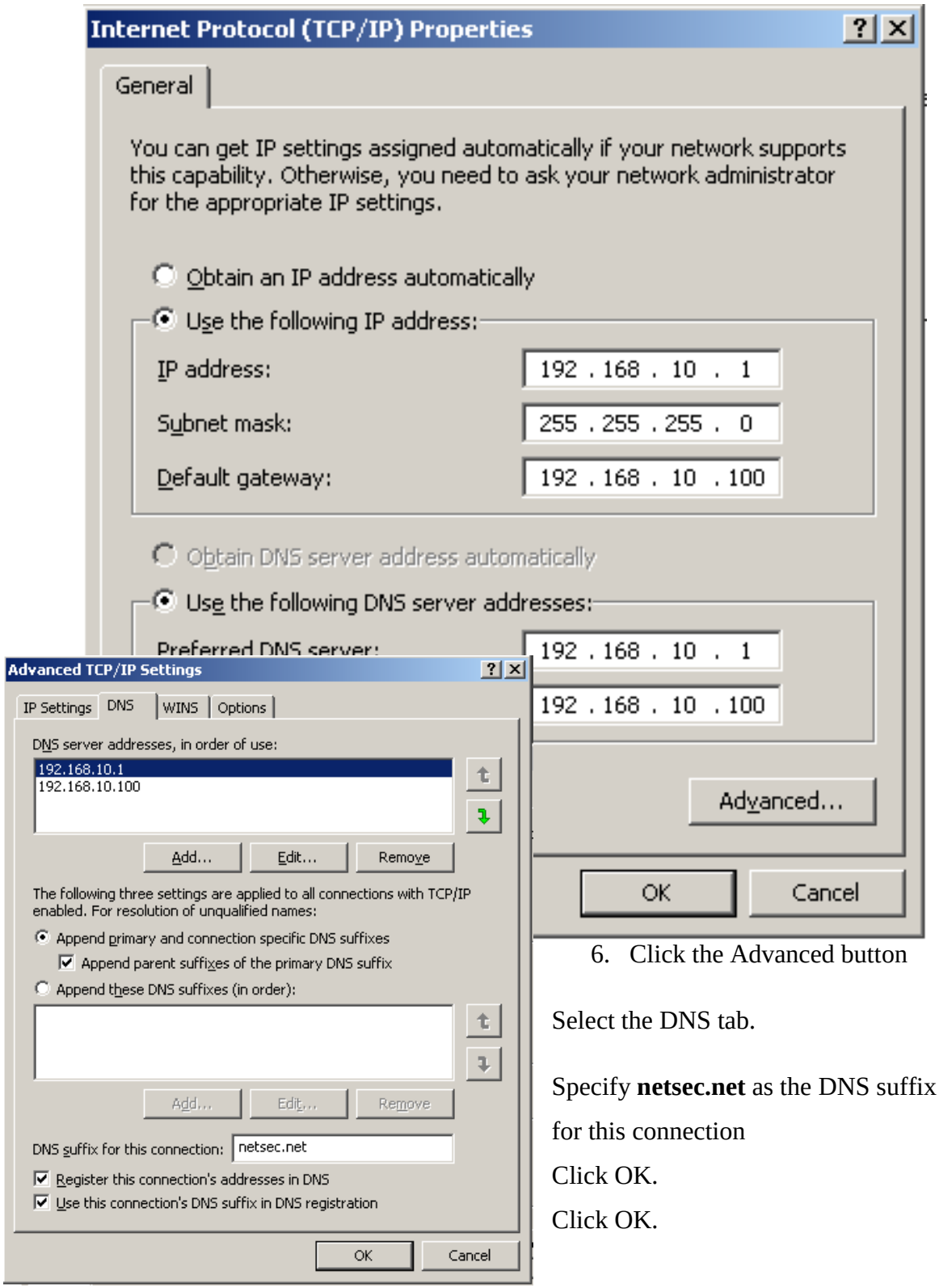


Click Internet Protocol (TCP/IP)

Click the "Properties" button

5. Ensure that the computer has a static IP address and that the DNS server address has been set to the same IP address.

This server is going to act also as a DNS server, thus needs to send requests to itself.



6. Click the Advanced button

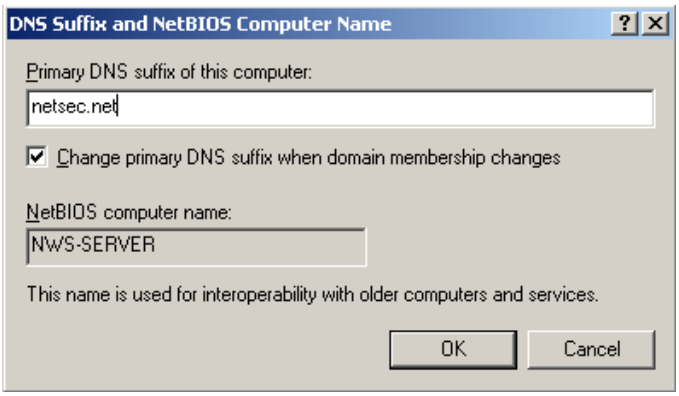
Select the DNS tab.

Specify **netsec.net** as the DNS suffix for this connection

Click OK.

Click OK.

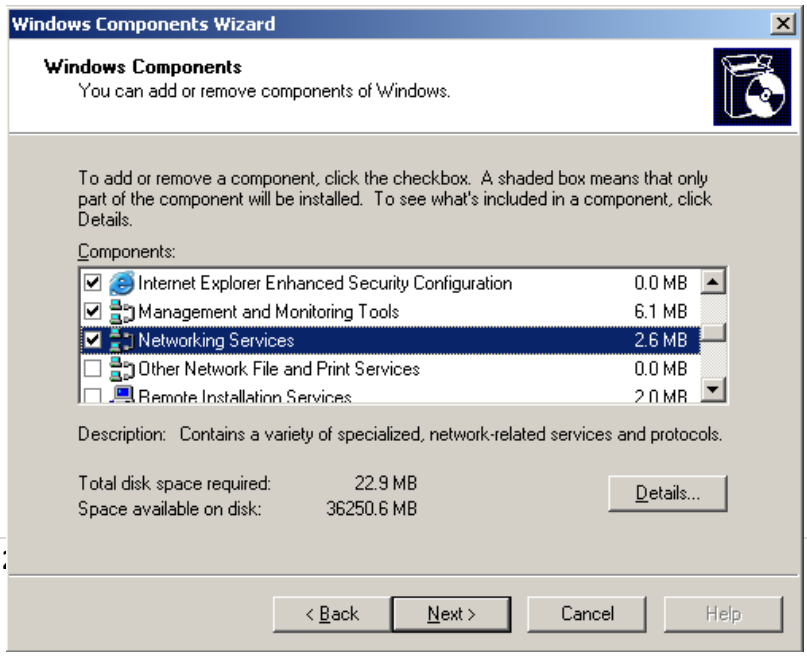
7. Close the remaining dialog boxes.
8. Right-click **My Computer** and select **Properties.**
9. Click the **Computer Name** tab then click **Change.**
10. Click **More.**
11. Specify **netsec.net** as the Primary DNS Suffix for This Computer.



12. Click **OK**.
13. Click **OK** to close the remaining dialog boxes, and then click **YES** to restart the computer.

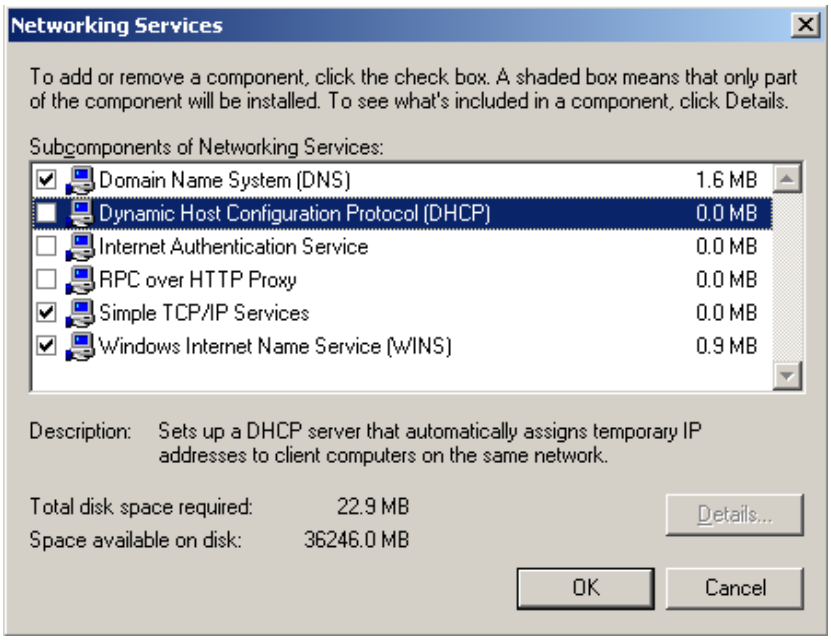
Now that you have verified that the computer is correctly configured to use DNS and has a static IP address, it is now time to proceed and install the DNS service on this computer.

14. Open Control Pane and click "Add Remove Programs", then click "Add/Remove Windows Components".



Scroll down the list and double-click the "**Networking Services"** option.

15. Scroll down the list and check the option "**Domain Name System (DNS)"**



After checking the option, click OK.

16. Click NEXT to install the DNS service.
17. Once the service has been installed, the Windows Components Wizard will appear.

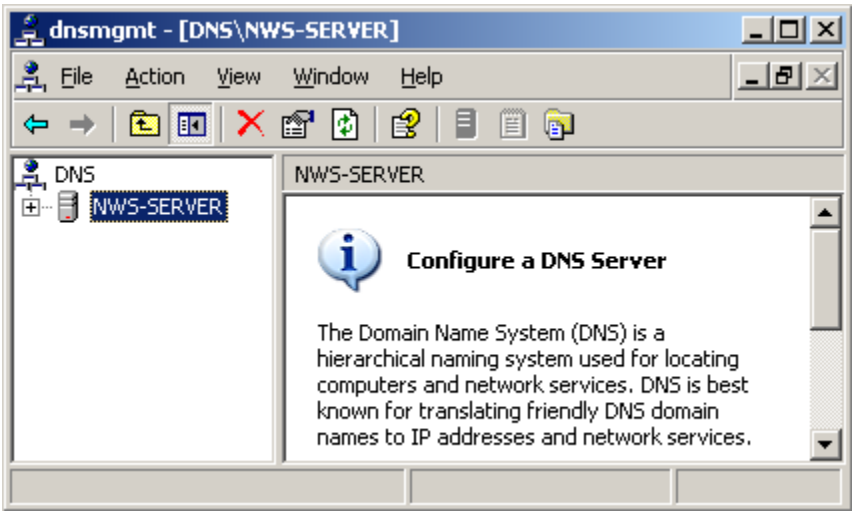


Click "Finish to close the wizard.

18. Close any remaining dialog boxes.

Now that the DNS service is installed, it is time to configure it. When the service is run for the first time, it asks to configure the root name server. If you connected to the internet, DO NOT configure a root name server (you would only do this if you were creating an Intranet and your company had no Internet connection).

19. Click Start, All Programs, Administrative Tools, DNS
20. The DNS management tool is now displayed. Your computer should appear as an icon in the left pane window.

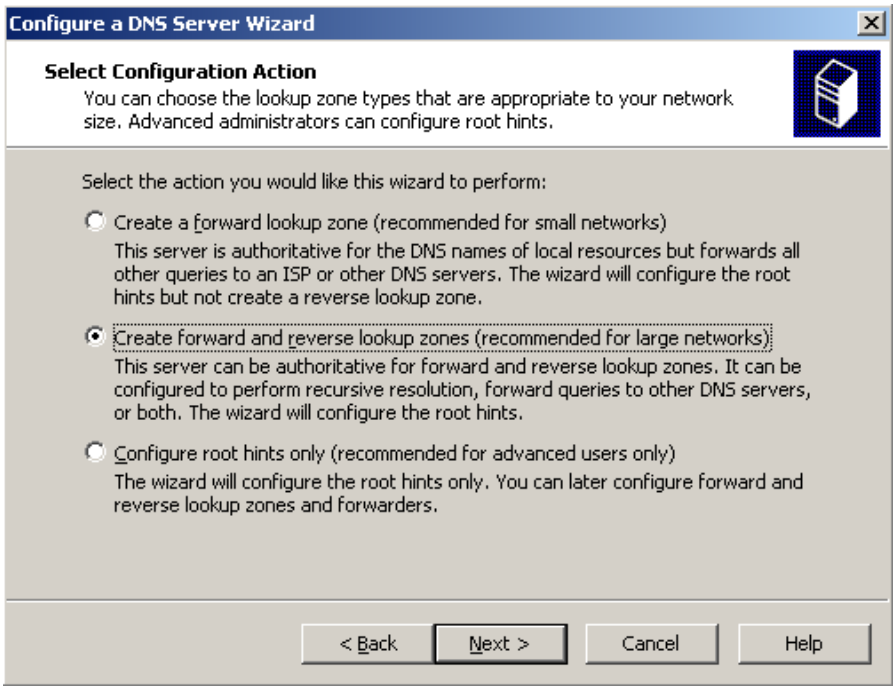


Click on the computer icon to display the message as shown in this example.

A DNS server uses zone files (forward and reverse) to perform lookup requests. In each zone file records are used to map computer names to IP addresses. In the following section you will create a forward and reverse zone for your network, and create records that map computer names to IP addresses.

21. Click **Action** then **Configure a DNS server.** The welcome to the **Configure a DNS Server Wizard** appears

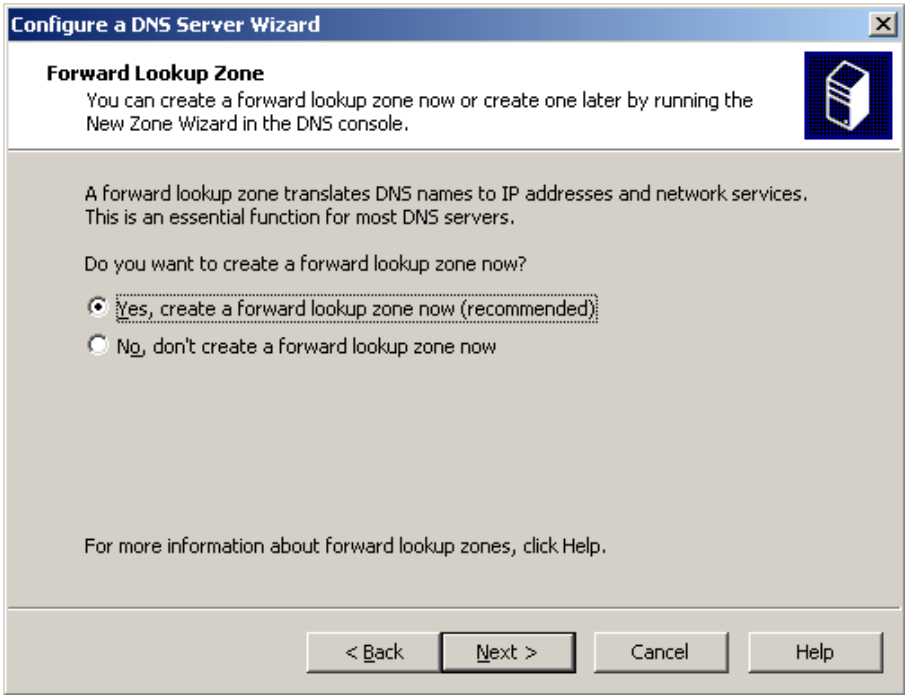
Click Next.



Specify the option "**Create a forward and reverse lookup zones".**

Click NEXT.

22. It is now time to create a forward lookup zone that will resolve computer names to IP addresses.

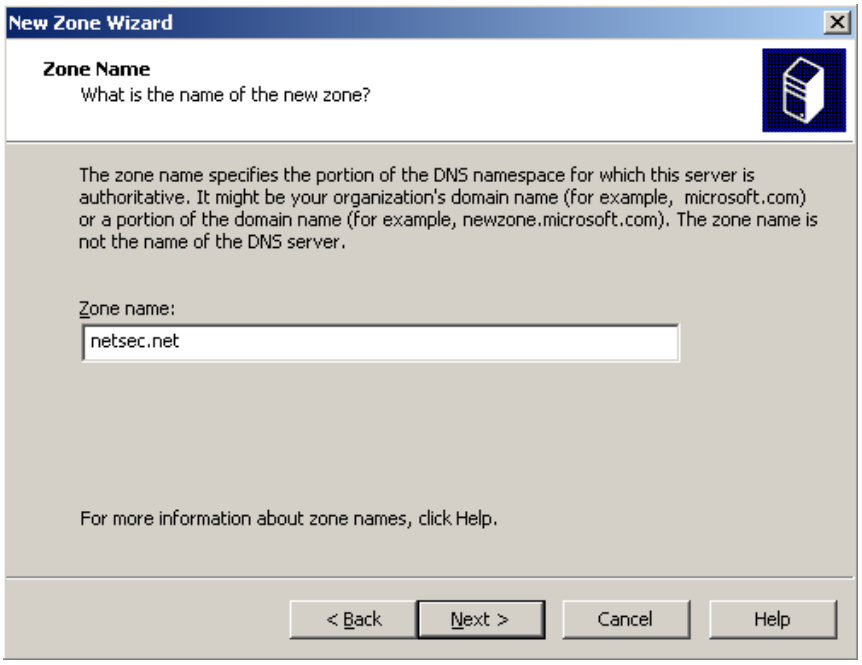


Select to option "**Yes, create a forward lookup zone now".**

Click Next.

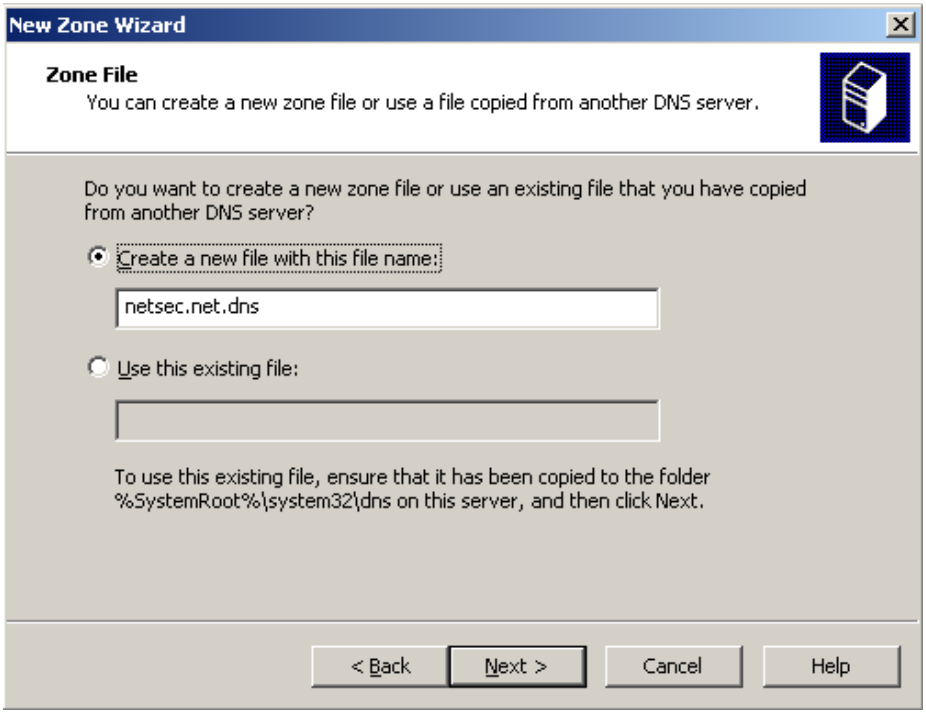
23. Select "Primary Zone". Click NEXT.

24. The New Zone Wizard dialog box requests the name for the zone. Enter the name that has been assigned to your domain as given to you by your instructor (this example uses netsec.net).



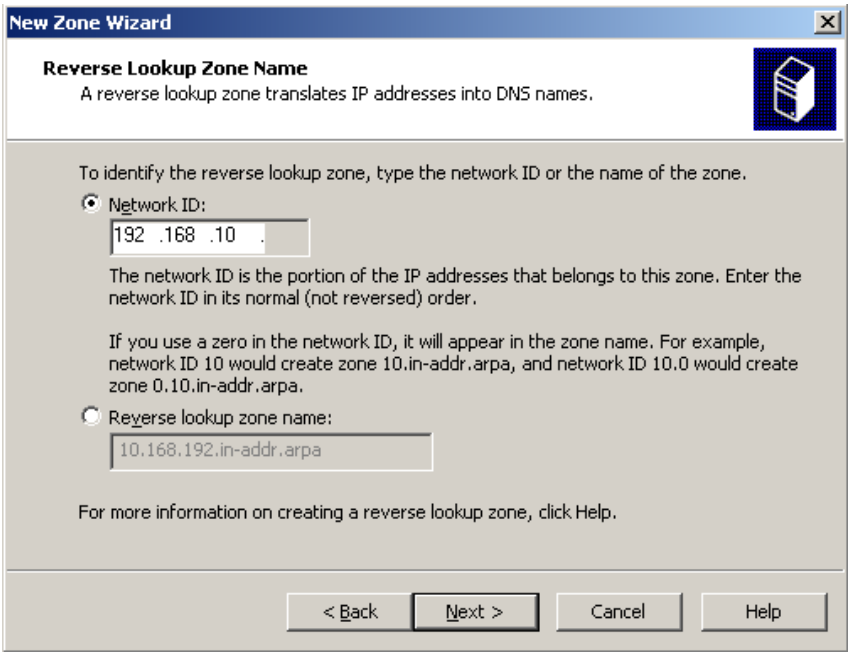
Once you have entered the correct name for the zone file, click NEXT.

25. The dialog box now displays the name that will be used to the new zone file.



Leave the filename as suggested then click NEXT.

26. Select the option “Do not allow dynamic updates”. Click NEXT.
27. Select the option “Yes, creates a reserve lookup zone now”. Click NEXT.
28. Select the zone type as “Primary Zone”. Click NEXT.
29. A reverse zone maps IP addresses to computer names, so it has to know what range of IP addresses it will be responsible for.



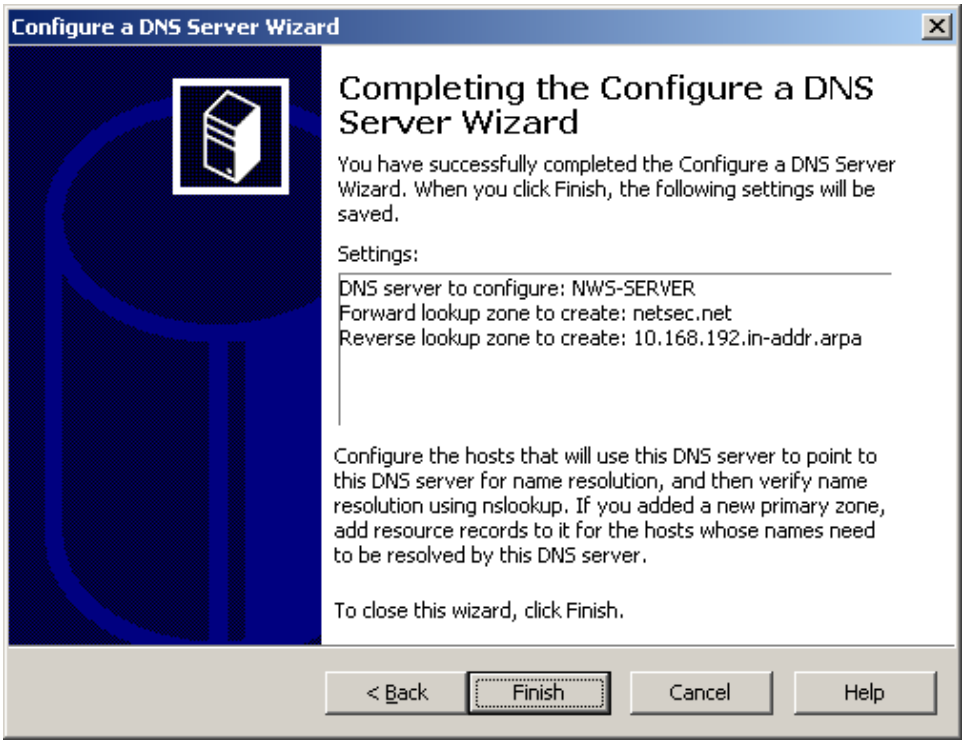
Enter the first 3 octets of the IP address that has been allocated to your network domain by the instructor.

After entering the network ID, click NEXT.

30. The wizard will display the name of the reverse zone file that it will create.
31. What is the name of the reverse file?

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32. Click NEXT.
33. Select the option “Do not allow dynamic updates”. Click NEXT.
34. Select the option “No, it should not forward queries”. Click NEXT.
35. Click FINISH so the wizard will create the reverse lookup zone.



Click FINISH to create the primary zone and the zone file.

You now successfully created a primary zone (netsec.net) and forward zone lookup file. Double-click on the computer icon to expand the tree and display the forward lookup zone.

36. What is listed under “Forward Lookup Zones”?

_____

_____

37. What are the entries listed under netsec.net?

_____

_____

38. What is listed under “Reverse Lookup Zones”?

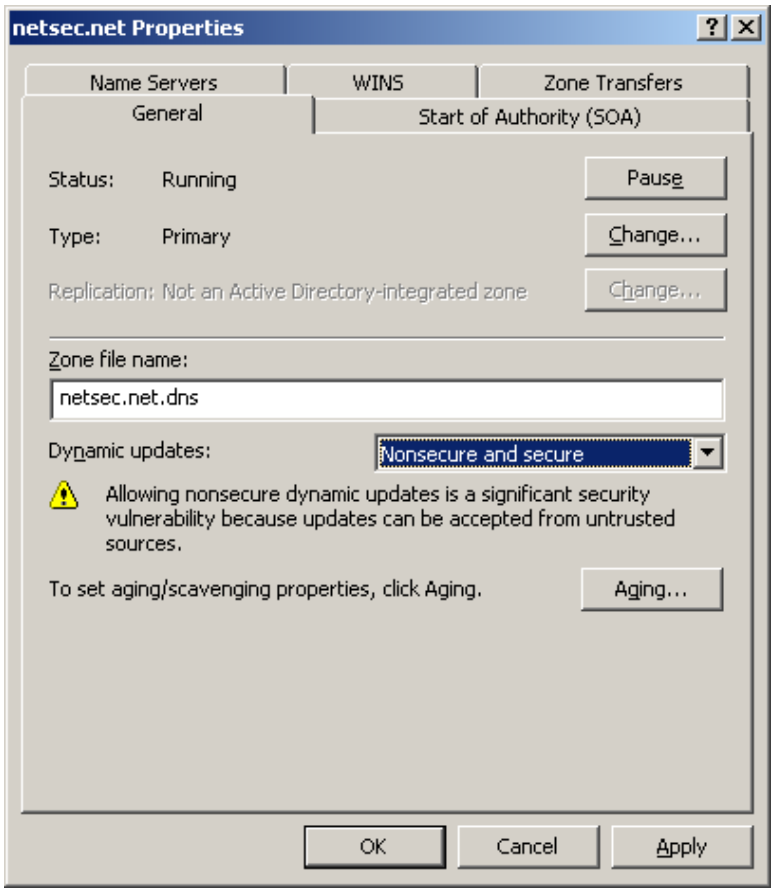
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_____

Dynamic DNS Updates

The Microsoft DNS supports dynamic updating of information, such as computer names and their IP addresses. In networks where computers are assigned IP addresses dynamically (using DHCP), this information needs to be integrated into DNS.

1. Expand the Forward Lookup Zones to reveal the domain (netsec.net)
2. Right-click the domain name and select properties.



The option "**Dynamic updates"** supports the ability to update DNS information dynamically.

To enable dynamic updates select "**Nonsecure and secure".**

Click APPLY.

Click OK.

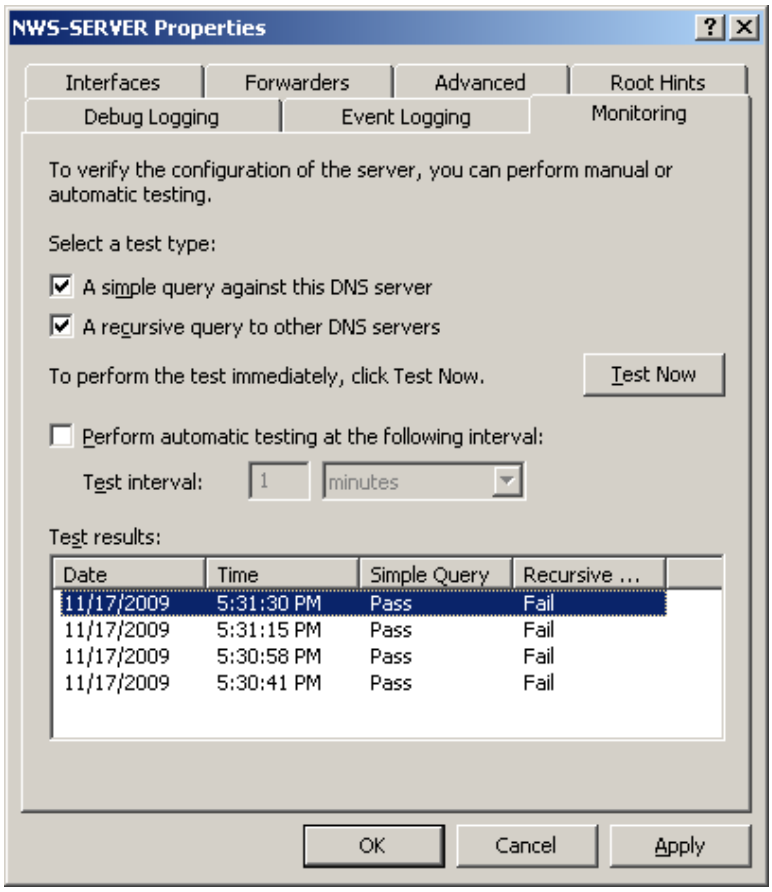
Repeat the procedure to enable dynamic updates for the **reverse zone.**

3. Close the DNS manage.

Testing the DNS

In this section you verify that the DNS is installed, running, and correctly configured.

1. In the DNS manager window, right-click the computer icon and select properties.



Click the Monitoring tab.

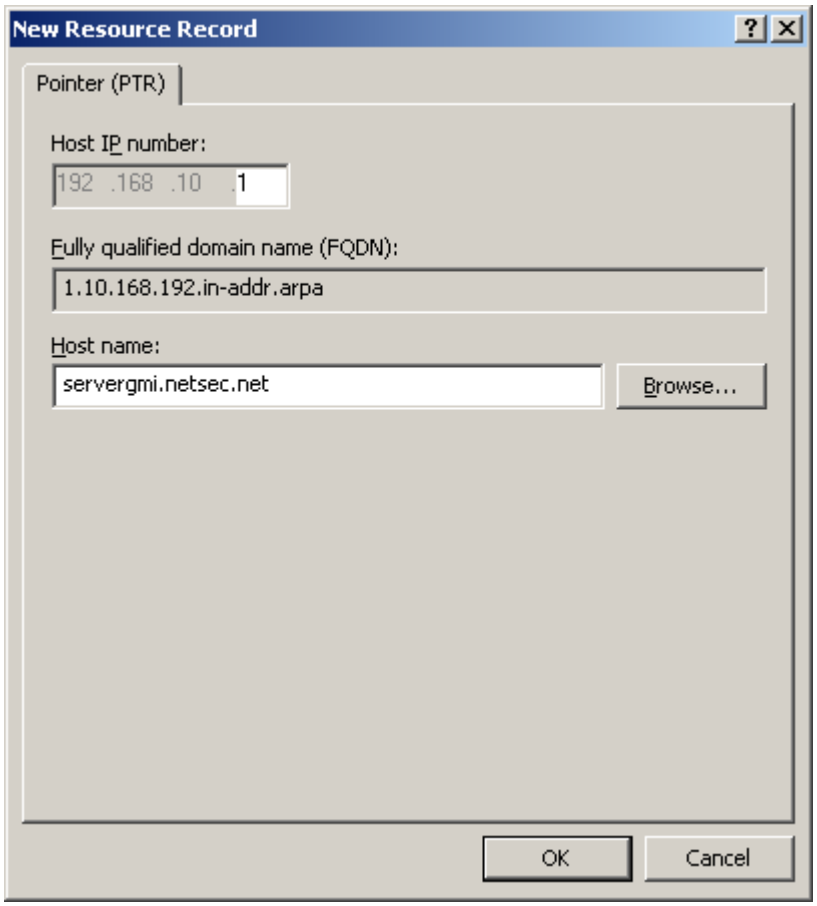
2. Enable both tests and click Test now
3. What was the result?

Do not proceed till the test results indicate a PASS.

Adding a PTR (pointer) record

In this exercise you will add a reverse lookup record (PTR) for a host computer. A reverse lookup record resolves an IP address back to a host name.

1. In the DNS manager window, double-click the computer icon and expand the "Reverse Lookup Zone" field.
2. Expand the subnet field.
3. Right-click the subnet field and select "New Pointer"



Enter the IP address and the server name, followed by .netsecx.net (be replace x with a number representing your domain).

Click OK.

Click "Refresh" button.

Using NSLOOKUP to query DNS

In this exercise you will use a client tool to check the operation of the DNS server. You will queue both a forward and reverse lookup.

1. Close the DNS manager window.
2. Click START, then RUN, and enter **nslookup** (then press ENTER).
3. A command prompt DOS window will appear with the program nslookup running in it.
4. The default sever name and IP address of the DNS server will be shown.

   What is the default server name?

   _____

   What is the IP address listed?

   _____

5. To perform a forward lookup (resolve a computer name to an IP address) enter the name of the computer (e.g. netsec.net) and press ENTER.

   What is the fully qualified domain name returned?

   _____

   What is the IP address given?

   _____

6. To perform a reverse lookup (resolve an IP address to a computer name), enter the IP address given in step 5.

   What is the fully qualified domain name returned?

   _____

Summary

The DNS server is a database that manages computer names, and their IP address. Zone file are used to store this information, within a zone, a forward lookup resolves computer names to IP addresses. A reverse zone resolves IP addresses to computer names.

A client tool such as NSLOOKUP can be used to test the operation of a DNS server.

Lab 4

Installing Active Directory

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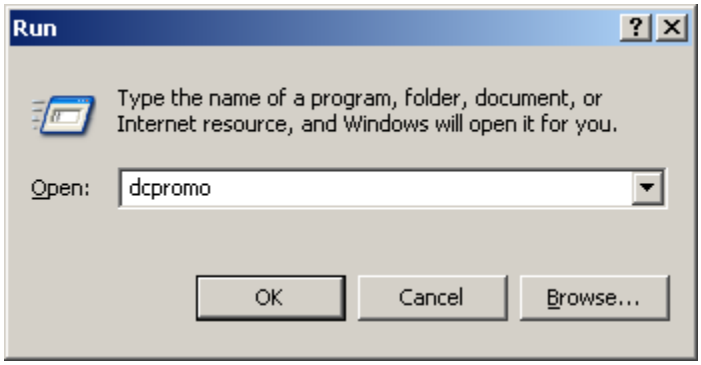
Prepared by :
Noor Azhar Bin Amran

LAB 4: INSTALLING ACTIVE DIRECTORY

In this exercise you will install active directory services (ADS) and change to native mode (when the server acts purely with ADS is installed, you will be able to take advantage of many of the new features of Windows 2003 in managing users, computers and sites.

Estimated time to complete: 20 minutes

1. Logon to the server as Administrator
2. Click Start, Run and type in **dcpromo**, then ENTER

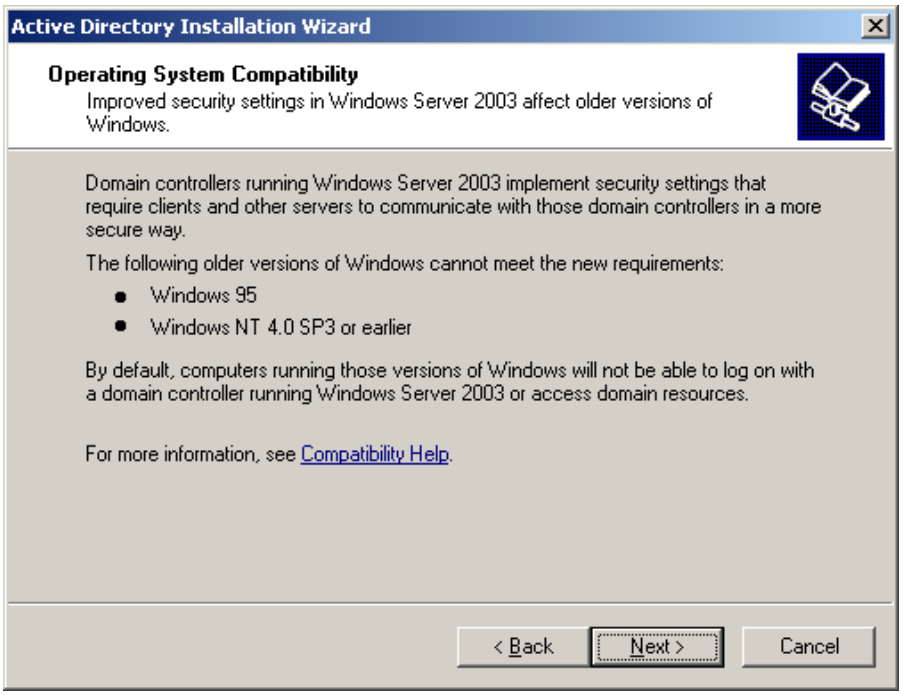


3. The ADS Wizard appears.



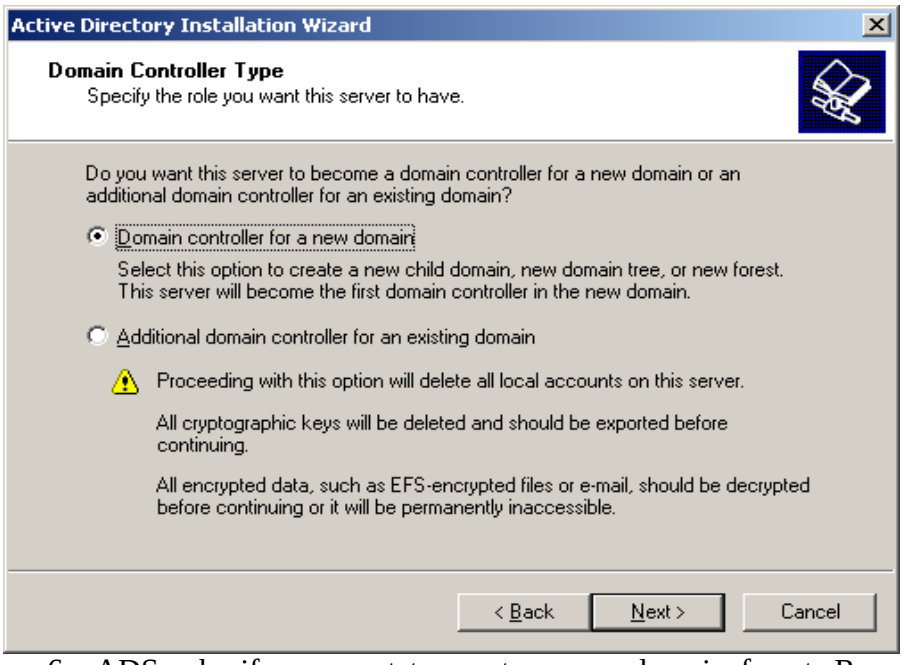
Click **Next.**

4. The ADS wizard alert about improvement security in Windows Server 2003 affects alder versions of Windows.



Click **Next.**

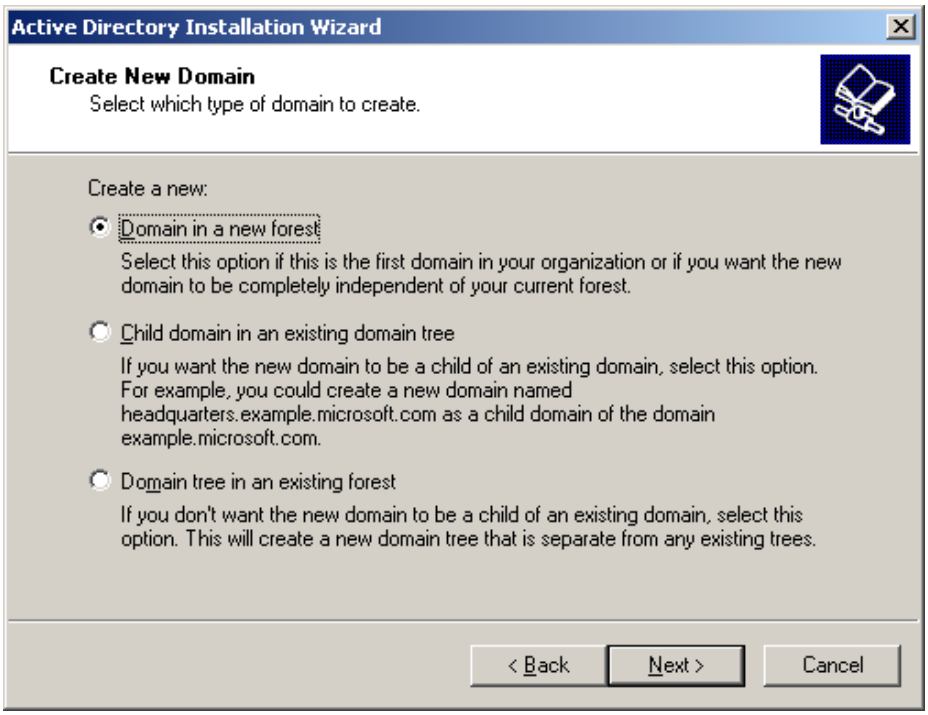
5. The ADS wizard asks for the Domain controller type



Select "Domain controller for a new domain".

Click **Next.**

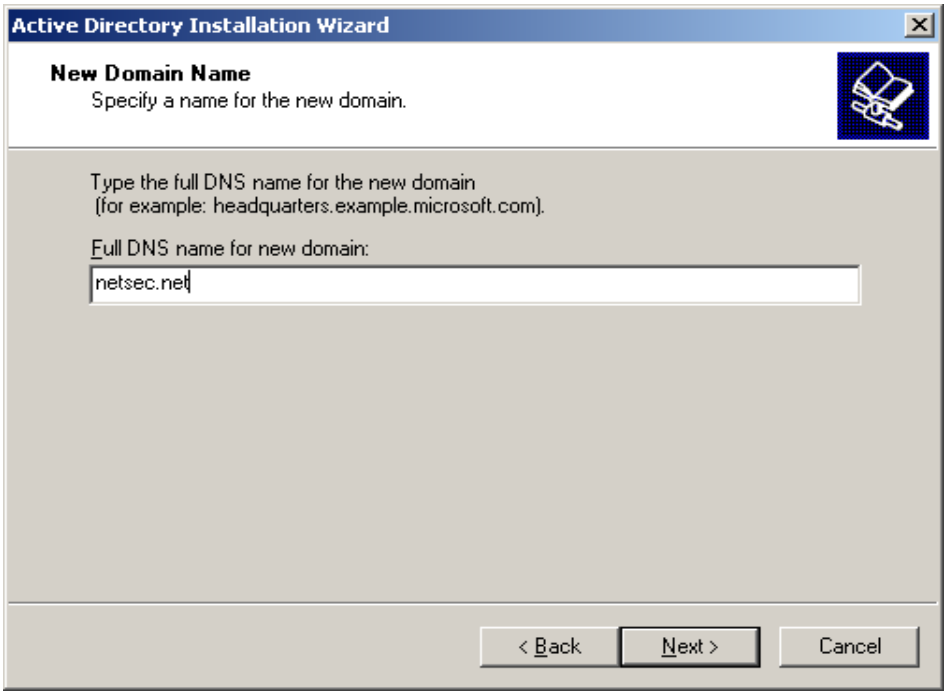
6. ADS asks if you want to create a new domain forest. Remember that ADS is not currently installed on this domain, so it is necessary to create a domain forest.



Select "**Create a new domain forest".**

Click **Next.**

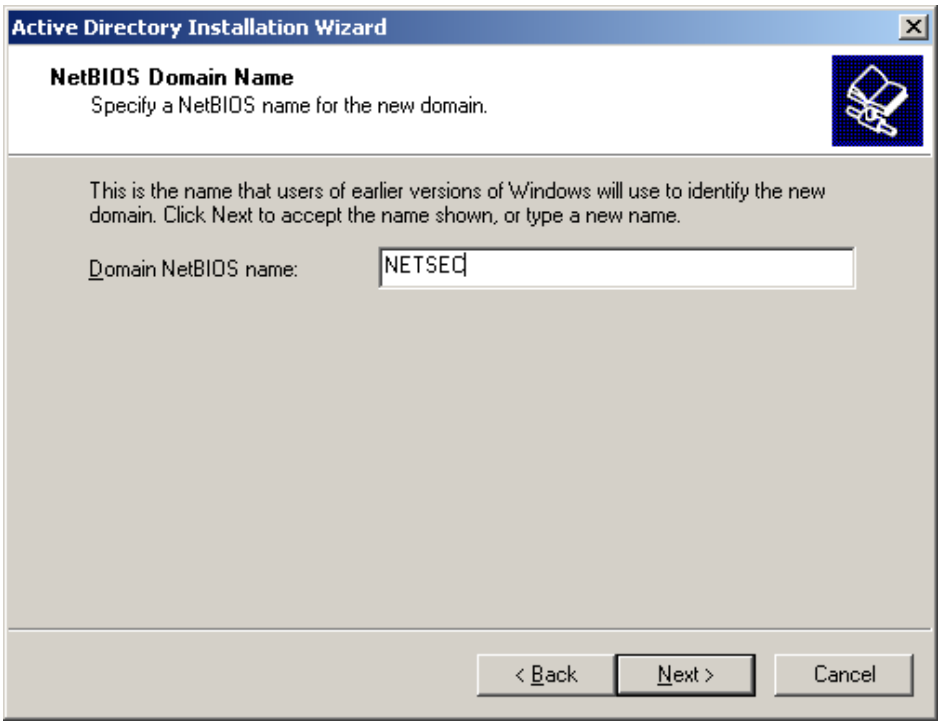
7. ADS prompts for the name to be given to this domain.



Enter the full name of your domain as supplied by your instructor, e.g., domain1.net

Click **Next.**

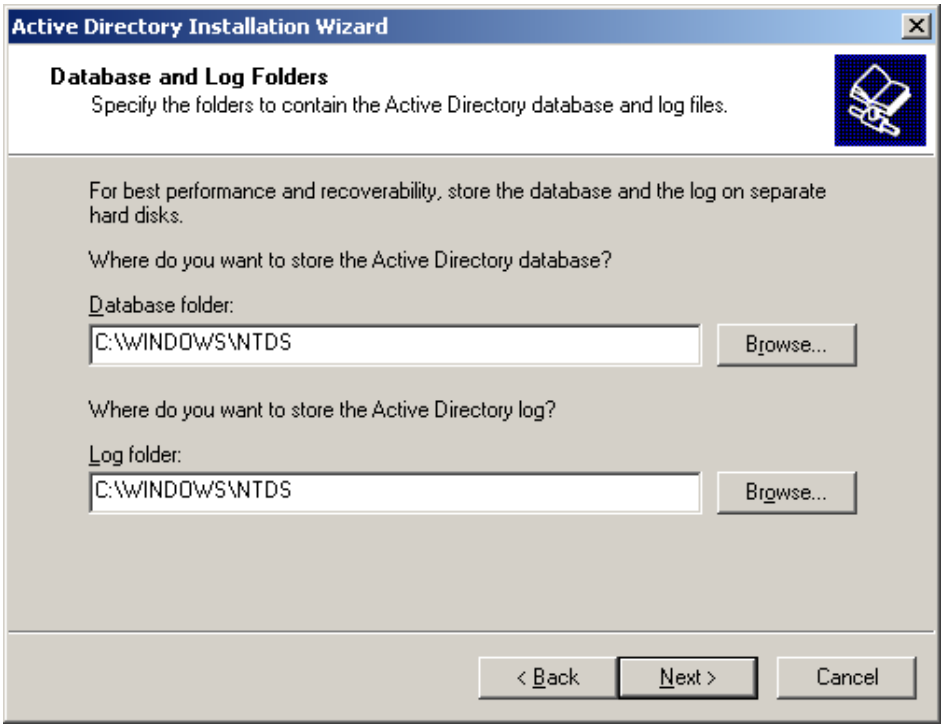
8. ADS prompts for the NetBIOS name.



ADS should full in the field correctly.

Click **Next.**

9. ADS prompts for the location of the database and log files.

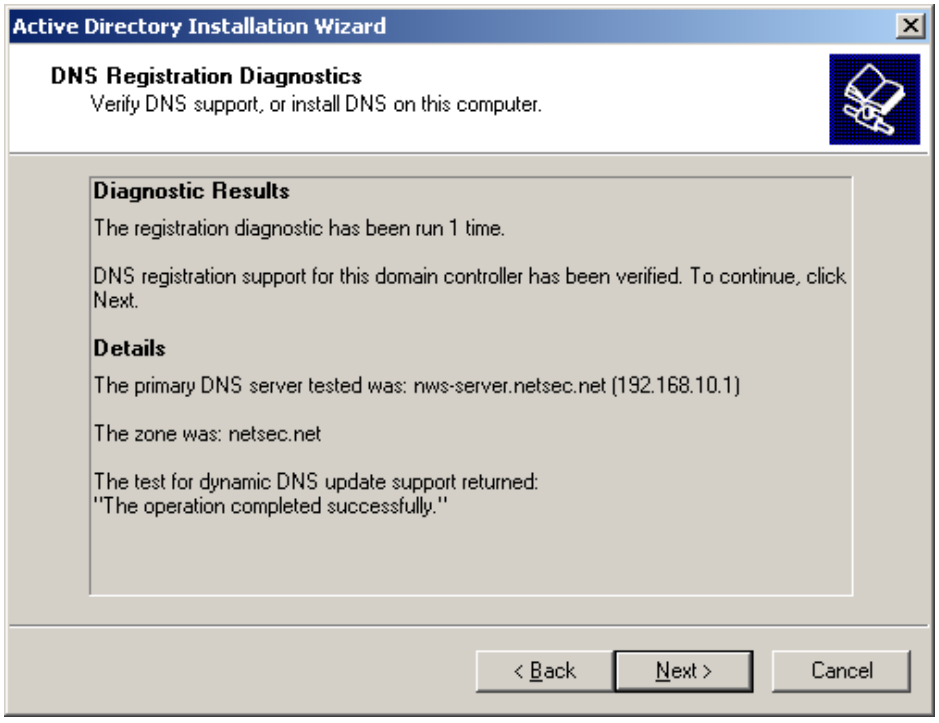


Click **Next.**

10. ADS prompts for the location of the SYSVOL.

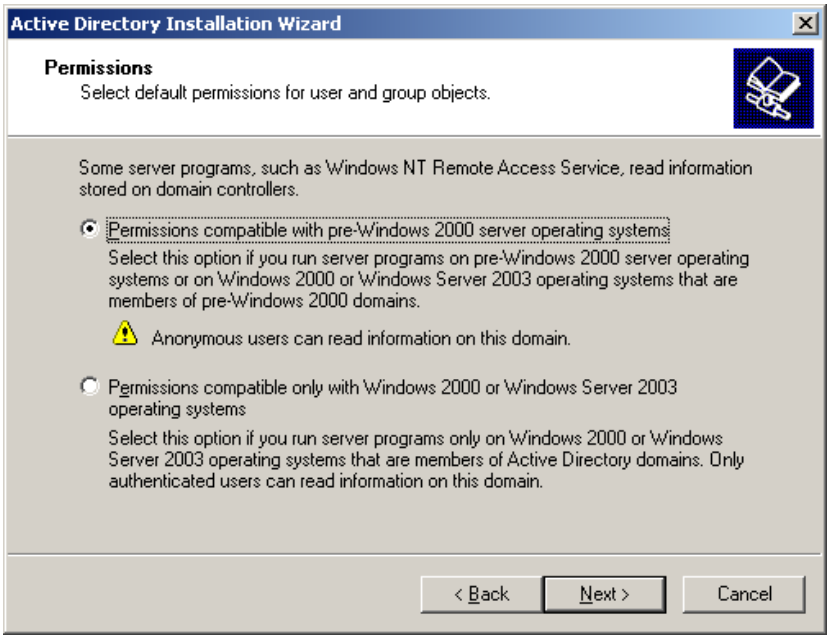
Click **Next.**

11. ADS displays a DNS Registration Diagnostics dialog box.



Click **Next.**

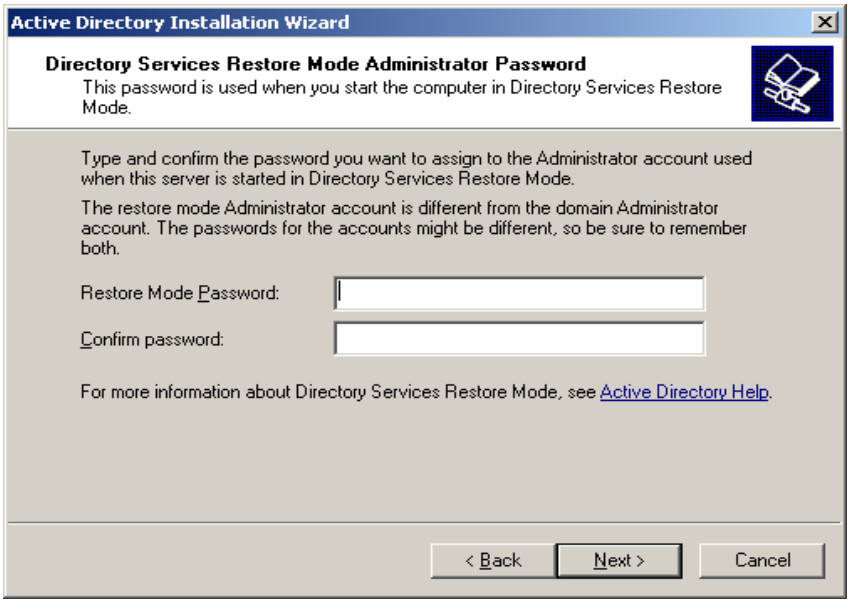
12. ADS displays a Permissions dialog box. This allows you to support existing Windows NT 4 servers in your network, or a network that only supports Windows 2000 Servers. If you had a network that has a Windows NT4 Remote Access Server, you would select the compatible option.



Select "**Permissions compatible with pre-Windows 2000 Servers operating systems".**

Click **Next.**

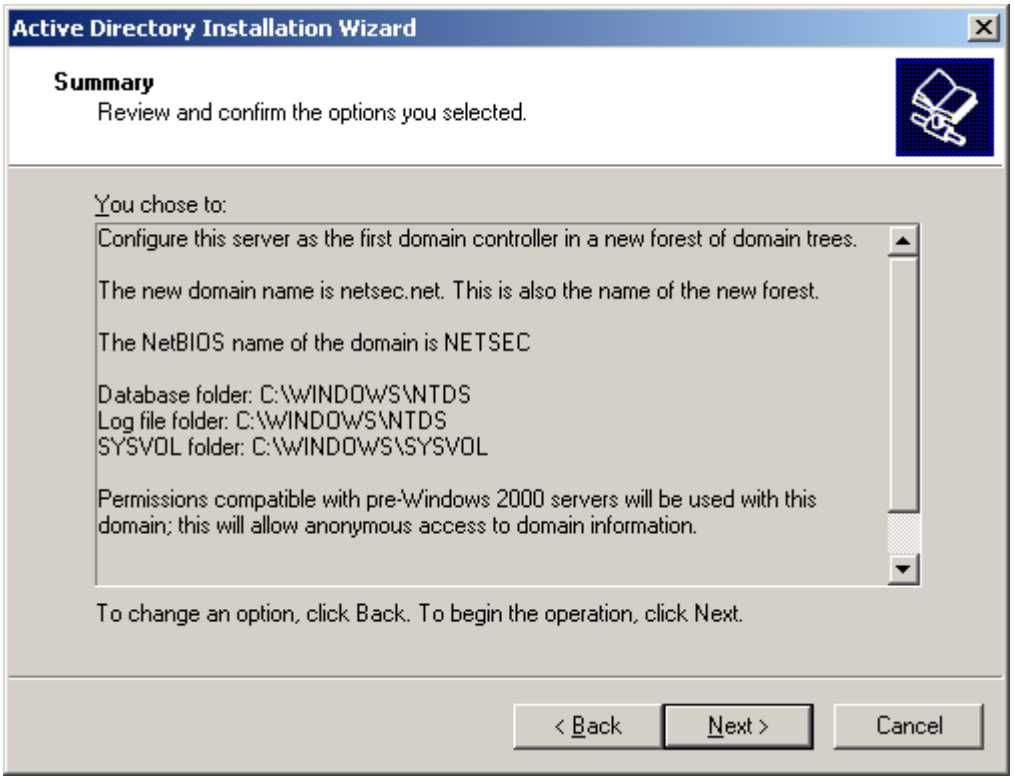
13. ADS displays the restore mode password dialog box. This allows you to specify a password if ADS is to be restored.



Specify "**abc123!@#**" as the password Enter "**abc123!@#** in the "Confirm password box.

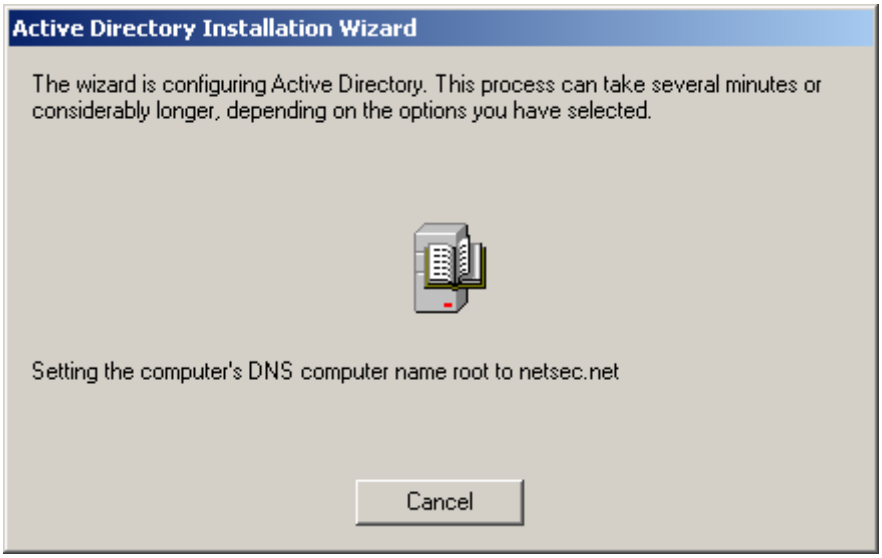
Click **Next.**

14. ADS displays a summary of your selections.



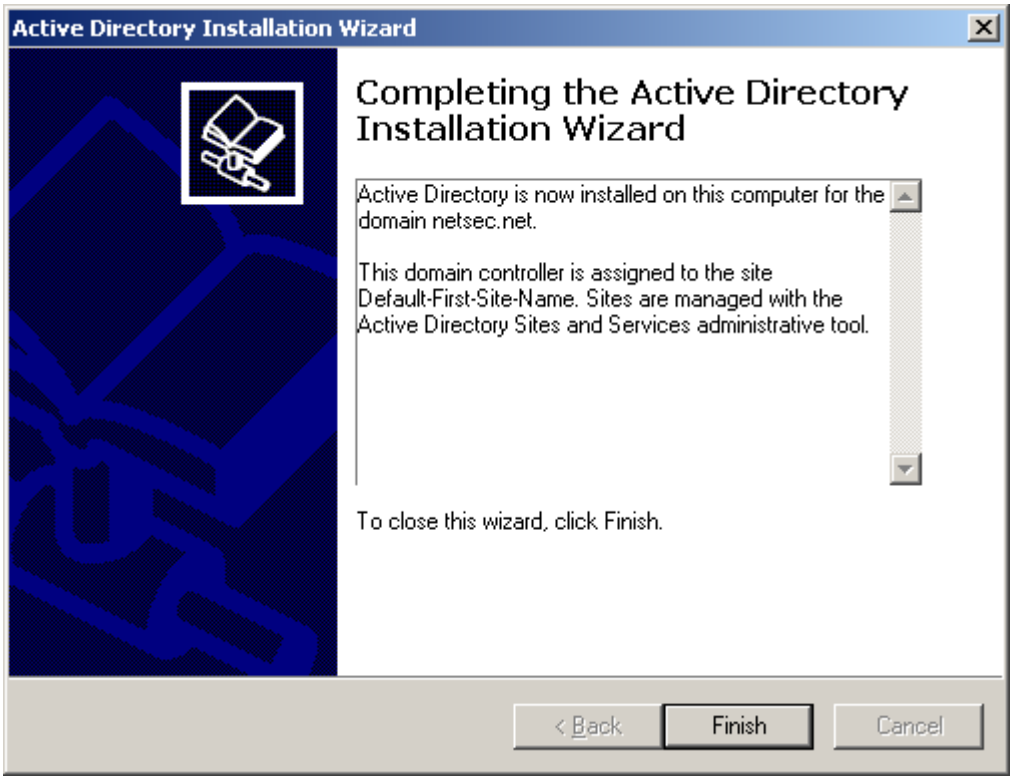
Click **Next.**

15. The ADS wizard now configures the computer with ADS.



Please wait. This operation can take a few minutes.

16. Once ADS had been installed, the completion dialog box appears.



Click **Finish.**

17. You are prompted to restart the computer. RESTART THE COMPUTER NOW.
18. Logon as administrator.

Viewing the Domain

In the following exercise you will use "My Network Places" to view the domain. This is similar to the style you used with Windows NT 4. Then you will use Active Directory to view the same information.

Estimated time to complete: 5 minutes

1. Double-click "**My Network Places**"
2. Double-click "**Entire Network"**
3. Double-click "**Microsoft Windows Network"**
4. Your domain should be visible in the window.
5. Double-click your domain.

   What is the name of the domain controller?

   _____

6. Close the window.

Now you will repeat this exercise but use Active Directory instead.

7. Click Start, All Programs, Administrative Tools, Active Directory Users and Computers
8. Expand your domain.

   What are the entries listed?

   _____

   _____

   _____

   _____

9. Expand "**Domain Controllers**"

   What is the name of the domain controller listed?

   _____

10. Close the window.

Lab 5

Creating Organization Units and Users

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Prepared by :
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LAB 5: CREATING ORGANIZATIONAL UNITS AND USERS

In this section you will use active directory to view the default settings that apply to user accounts when they are created. These settings can be overridden for a particular user, a group of users, or all users.

Estimated time to complete: 20 minutes

1. Logon as administrator
2. Click Start, All Programs, Administrative Tools, Active Directory Users and Computers.
3. Double-click the **Builtin** object.

What are the default groups and their types?

Name	Type

4. Double-click the **Administrators** group.

   Who are member of the administrators group?

_____

_____

_____

_____

Click **Cancel** to close Administrators Properties.

5. Double-click the **Users** group. Click the Managed By Tab. Click Change.
6. Click **Advanced** button.
7. Click **Find Now** button. Note that a list of user accounts is displayed. This means you can assign administrative control to a group.
8. Click **Cancel.**
9. Click **Cancel.**
10. Click **Cancel.**
11. Click on the **netsec.net** icon to select it.
12. Right-click the **Users** object and select properties. As you can see, little information is displayed. Click **Cancel.**
13. Click **View**, and click **Advanced Features** on the menu bar.
14. Right-click the **Users** object and select properties. There are now two additional tabs, Object and Security.
15. Click the **Security** tab. Click **Administrators** in the "Group or User names:" list. Note how the permissions change in the test box below. As you can see, a lot of control can be applied.
    Click *Authenticated* Users in the list.
    What operations are Authenticated Users permitted to do on user objects?

    _____

16. Click **Cancel.**

Creating Organization Units.

In the following exercise, you will create some organizational units that will acts as containers for some users. These organizational units model the departments within a small organization.

Estimated time to complete: 5 minutes

17. Click on the netsec.net icon to select it.
18. On the menu bar, click *Action*, *New*, *Organizational Unit*.
19. Enter **NWS1** as the name for the new organizational unit. Click OK.
20. Repeat step 17 to 19 to create the organization units **NWS2** and **NWS3.**

Creating organization units lets you place users directly into units and assign permissions and rights based on these units. This leads to better administration and delegation control than if you place users directly into the user container.

When users move from one department to another, it is a simple matter to move the user to the corresponding organizational unit. In this way, the inherit all the new features and rights and on the new organizational units, ensuring they have full access to all the resources they are entitled to.

Creating Users within Organizational Units

For proper control, it is better to create users within an OU rather than the users container. In the following exercise you will create a number of users, modify their properties, and move them from one organizational to another.

Estimated time to complete: 30 minutes

1. Click the **NWS1** OU to highlight it.
2. Create new user accounts for **Fari** and **Mcja**.
   2.1 Right-click **NWS1** and select New, User from the menu
   2.2 Enter the following details for Fari

First Name	Last Name	Full Name	User logon name
Fari	Ali	Fari Ali	fari.ali

2.3 Click Next.

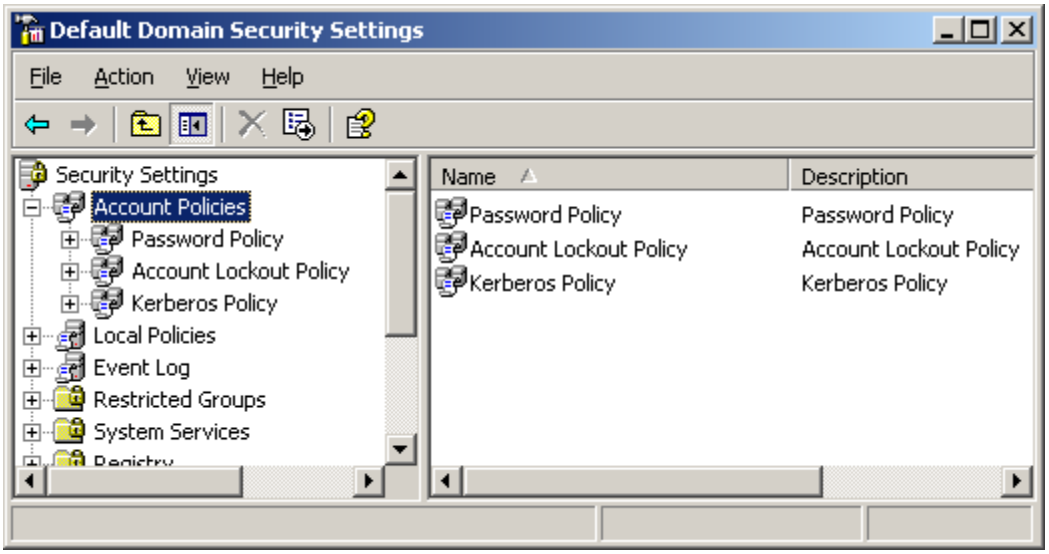
2.4 Enter the password as fari. Check the boxes “User cannot change password” and “Password never expires”, then click Next.

2.5 Click Finish to create the new user Zul.

2.5.1 The warning below will appear. This warning appear because your password does not meet the password policy requirements.

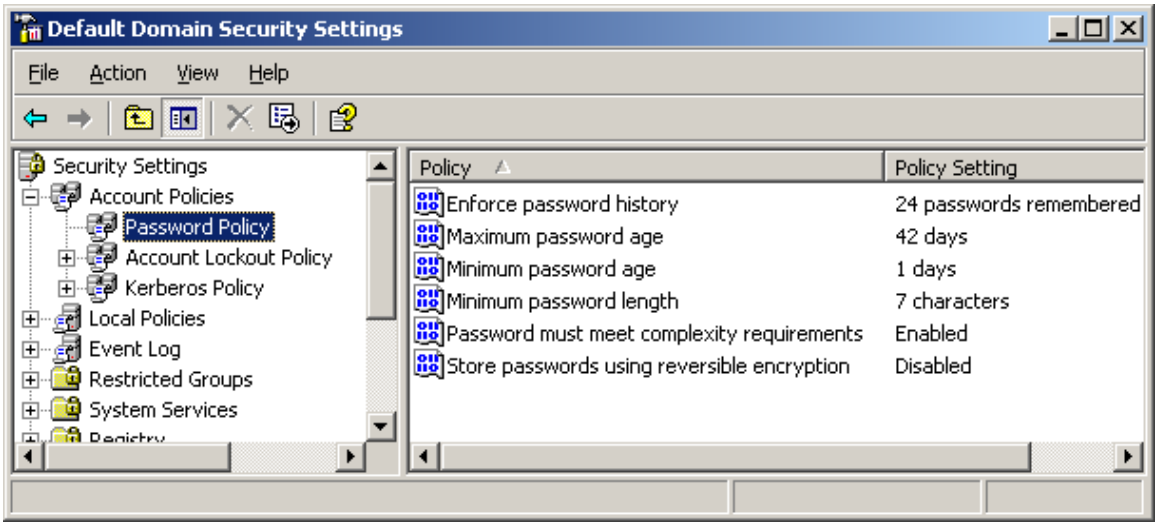


2.5.2 To disable this; click Start – All Programs – Administrative Tools – Domain Security Policy.



2.5.3 Expand Accounts Policies. Click Password Policy.

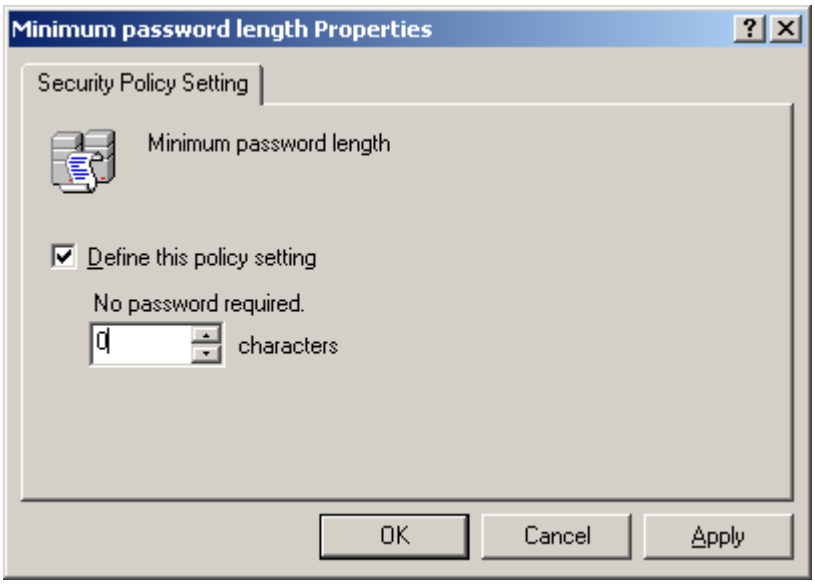
Right-click "**Password must meet complexity requirements"** and select **Properties.**



2.5.4 Select Disabled and click Apply. Click OK.



2.5.5 Now Right-click "**Minimum password length"** and select **Properties.**



Set No password required to **0** characters.

Click Apply.

Click OK.

2.5.6 Close all windows and **RESTART** your server.

After restarting server, login as Administrator and start create user Fari Ali again. There should be no problem anymore.

2.6 Now create the new user for Mcja using following properties.

First Name	Mcja
Last Name	Daud
Full Name	Mcja Daud
User logon Name	mcja.daud
Password	mcja
User cannot change password	
Password never expires	

3. Create the following user account in the **NWS2** OU.

First Name	Aj
Last Name	Nazuwan
Full Name	Aj Nazuwan

User logon Name	aj
Password	aj
User cannot change password	
Password never expires	

4. Create the following user account in the **NWS3** OU.

First Name	Azhar
Last Name	Amran
Full Name	Azhar Amran
User logon Name	azhar
Password	azhar
User cannot change password	
Password never expires	

5. Create the following accounts in the **NWS3** OU.

First Name	Fazlan
Last Name	Hatta
Full Name	Fazlan Hatta
User logon Name	fazlan
Password	fazlan
User cannot change password	
Password never expires	

First Name	Hafizi
Last Name	Md Yunos
Full Name	Hafizi Md Yunos
User logon Name	hafizi
Password	hafizi
User must change password at next logon	
Account is disabled	

5.1 Note the red X that appears on the icon for the User Hafizi Md Yunos, indicating this account has been disabled.

6. It is easy to delete, rename or move a user from an organization unit. In the above exercise the user Din Mahdin was inadvertently placed in the wrong OU. **Right-click** the user Hafizi and select **move** from the list. Click **NWS1** as the destination OU and click OK.

7. Expand the **NWS1** OU to confirm that user Hafizi is now a member.

You have now created a number of users within the organizational units created earlier. At this stage, you cannot see the benefits of doing this. However, the later exercises will start to illustrate why this has been done, by allocating resources to organizational units.

Thus, a user will get access to a resource based on their OU membership properties. If a user moves from one organizational unit to another, they will inherit all the resources associated with the new OU.

Default User Properties

In this exercise we will look at default user properties such as logon times and how often they need to change their password.

Active Directory allows organizations to store significantly more information than in previous versions of Windows. For example, you can store telephone and office information in the Active Directory with the user information.

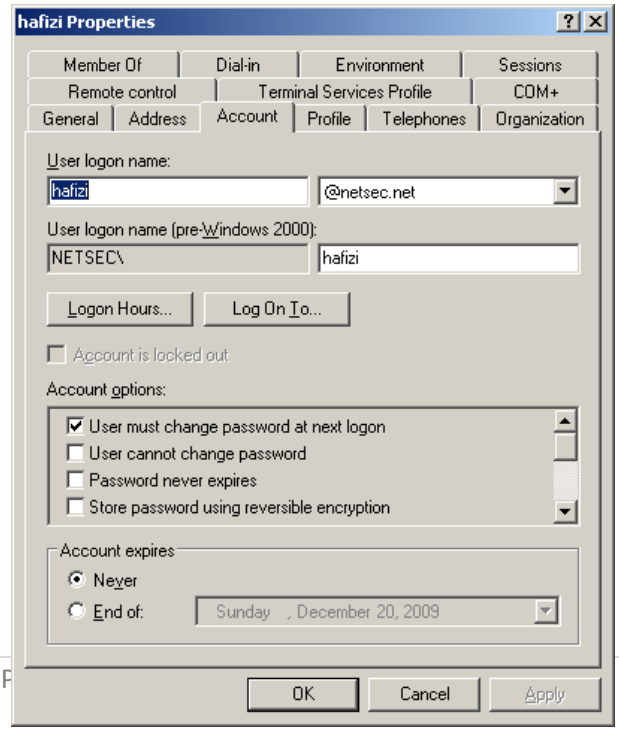
Estimated time to complete: 10 minutes

1. Double-click the user Hafizi in the **NWS1** OU.
2. Enter the following details.

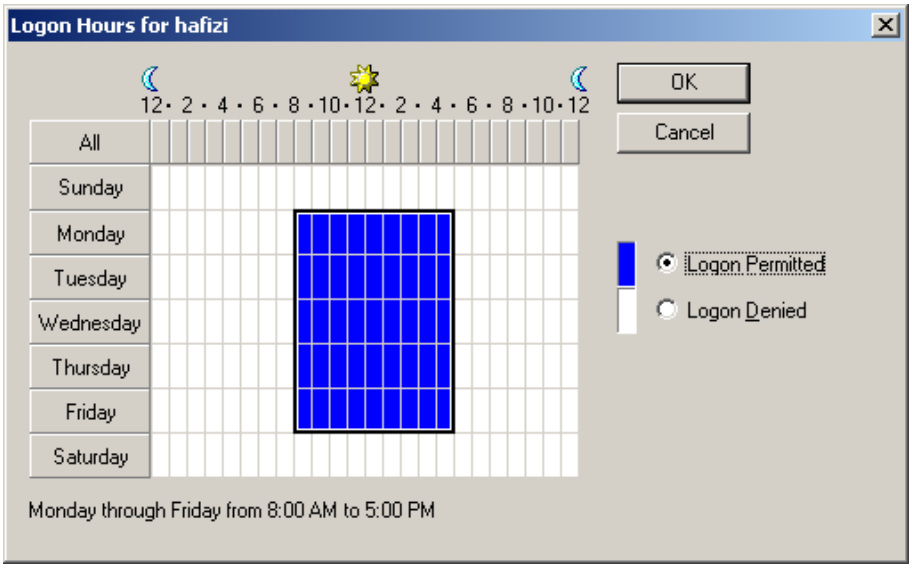
Office	Network Security
Telephone Number	019-5465285
Email	hafizi@netsec.net
Title (Organization)	Mr.
Department	Industrial Electronics
Company	GMi

2.1 Click OK to apply the changes.

3. Double-click the user Hafizi in the **NWS1** OU.
   3.1 Restrict the logon hours (under Account Tab) to Monday-Friday, 8am-5pm



Click the Logon Hours… button.



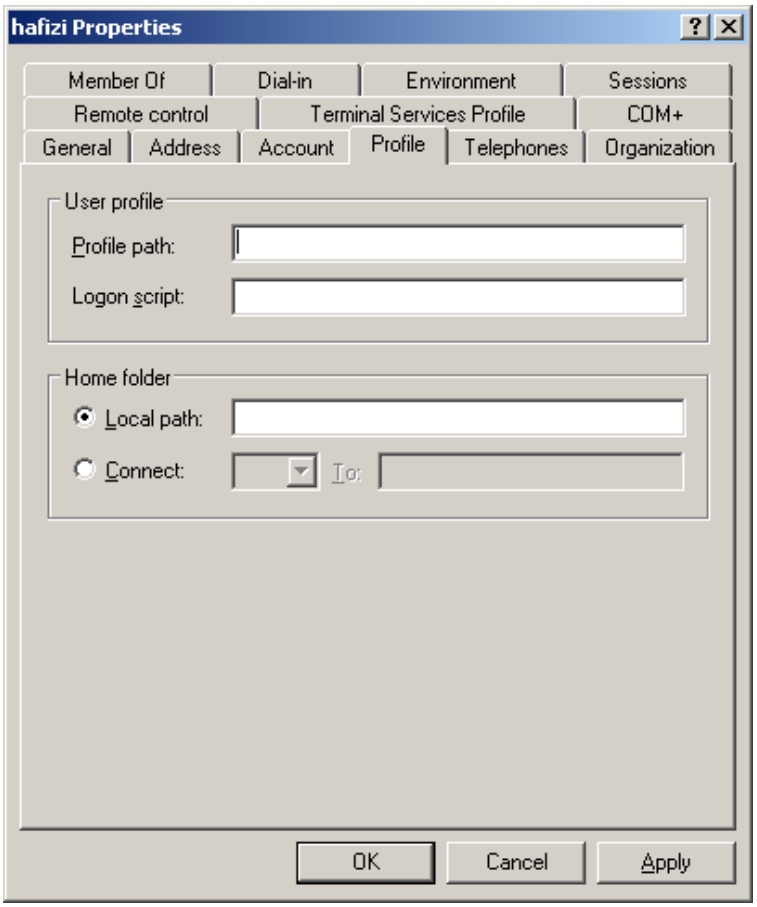
Select all areas and click Logon Denied.

Select the areas 8 to 5 and Monday to Friday.

Click Logon Permitted.

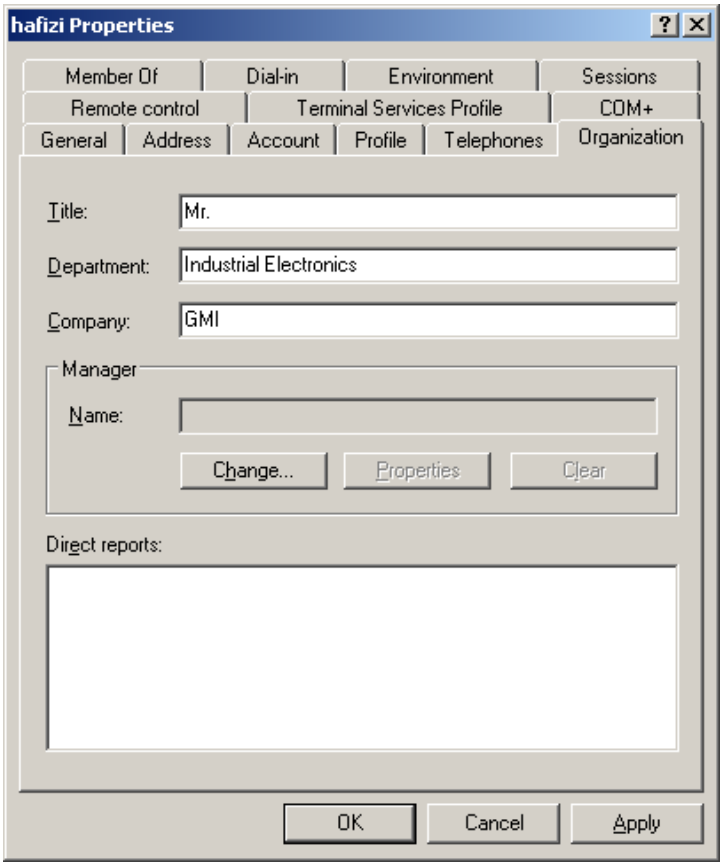
Click OK.

3.2 Click on the **Profile** tab.



It is here that you would specify a home directory for the user, so that when they log on to the network, a drive letter would be mapped to their folder on a network server where they can store their files.

3.3 Click on the **Organization** tab.



Click the Change button.

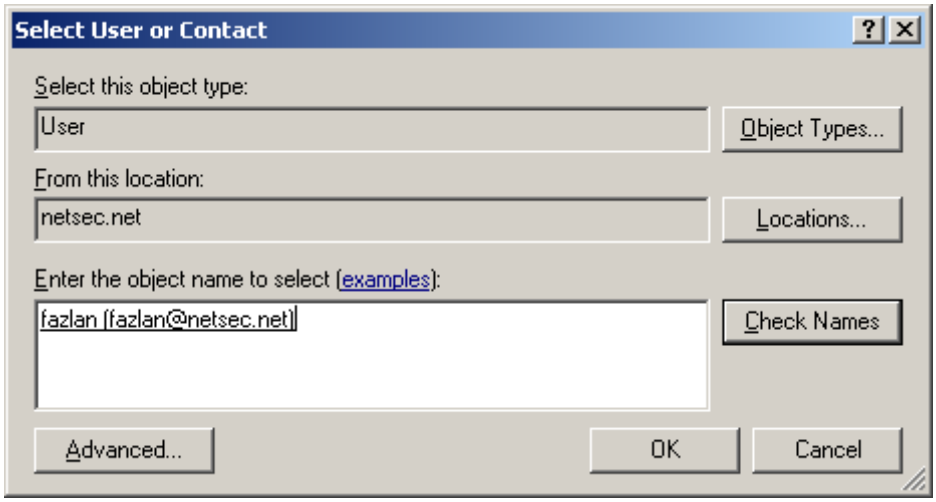
Type Fazlan in the "Enter the object name to select (examples):

Click Check Names button.

Select Fazlan from the list of accounts.

Click OK.

Fazlan will now be listed as a manager of this account.



3.4 If you wanted to assign dial-up access, which tab would you select?

_____

3.5 Click OK to close the Hafizi property box.

4. Log off the Server
5. Logon as mcja.daud
   5.1 Could you logon to the Windows Server 2003? YES or NO
   5.2 If there was a message displayed, what was it?

_____

_____

5.3 If you could no logon to the server, why do you think this is so?

_____

_____

5.4 Could you logon as any of the other user accounts you created? YES or NO
5.5 Which accounts (if any) could you logon as?

_____

_____

_____

In the above exercise you assigned some organizational information to a user. You also explored some of the properties that can be applied. In addition, you assigned a manager to add user, which will be used in future exercises.

Lab 6

Configuring Client Computer

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Prepared By,

Nurul Amirah bt Jamaluddin

LAB 6: CONFIGURE A CLIENT COMPUTER

In this section you will configure Windows XP Professional on the other computer that will be part of your network. This computer will act as a client computer that users of your network can use to access shared resources such as files, software and printers.

Estimated time to complete: 10 minutes.

Make sure that the Windows Server 2003 previously installed is running. **Please refer to the following table for client configuration.**

Name of This Computer	Client***SN***
Name of Organization	GMI
Role of This Computer	Client Workstation
Name of Installer	Administrator
Domain Name	Same domain name as you did for the Server
TCP/IP Address	192.168.10.***SN***
TCP/IP Subnet Mask	255.255.255.0
TCP/IP Gateway	192.168.10.***ServerNmber***

Note: ***SN*** = Station Number

Use the same domain name as you did for Server. The computer name is available from your instructor.

Network Setting (Windows XP)

1. Right click "**My Network Places**"
2. Select "**Properties**"
3. Right click "**Local Area Connection**"
4. Select "**Properties**"
5. Double click "**Internet Protocol (TCP/IP)**"
6.

- Use the following IP address :

IP address : 192.168.10.***SN*** (client station number)

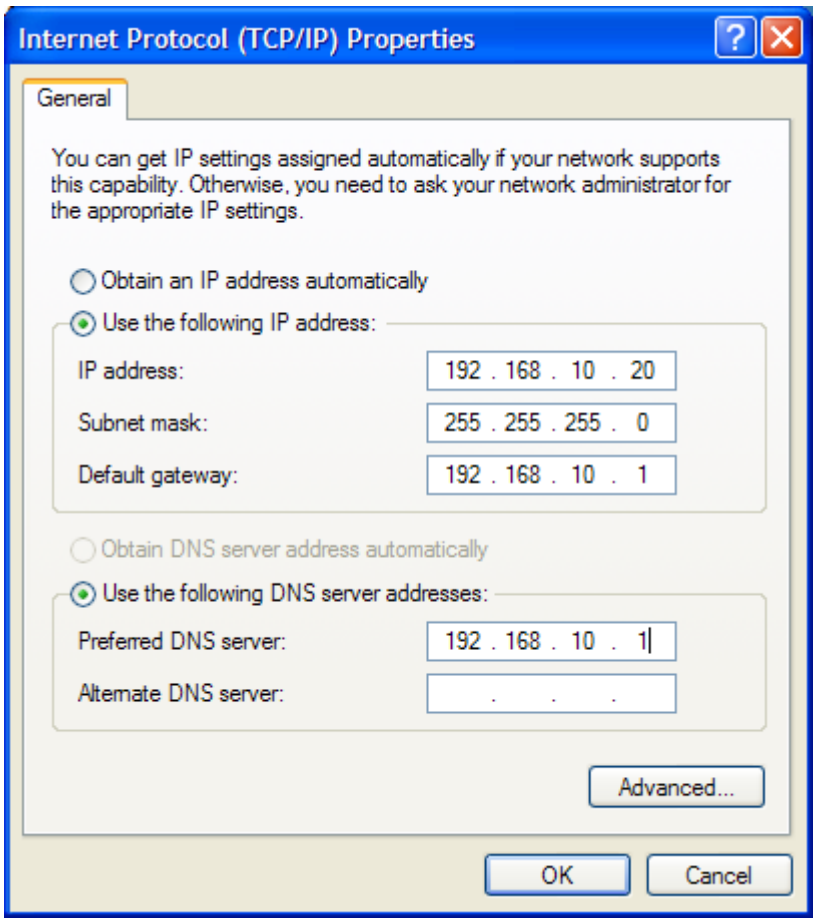
Subnet mask : 255.255.255.0

Default gateway : 192.168.10.***ServerNumber*** (server IP address)

- Use the following DNS server address :

Preferred DNS server : 192 . 168 . 10 . ***ServerNumber*** (1st server IP address)

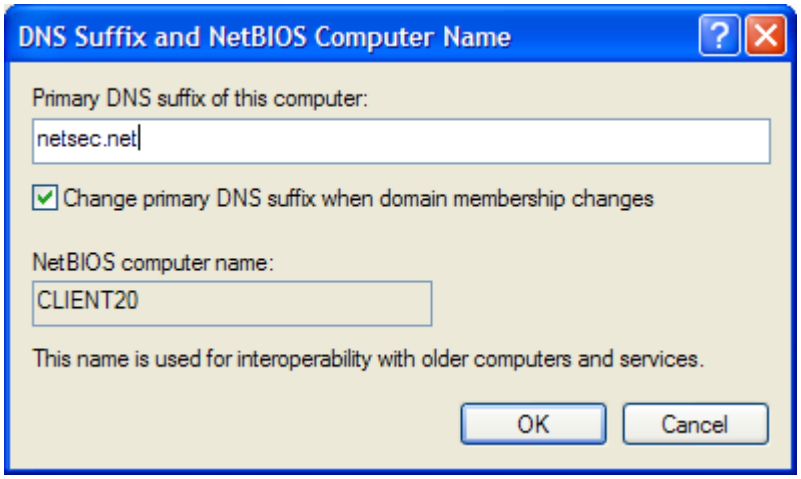
Alternate DNS server : ___ . ___ . ___ . ___ (2nd server IP address)



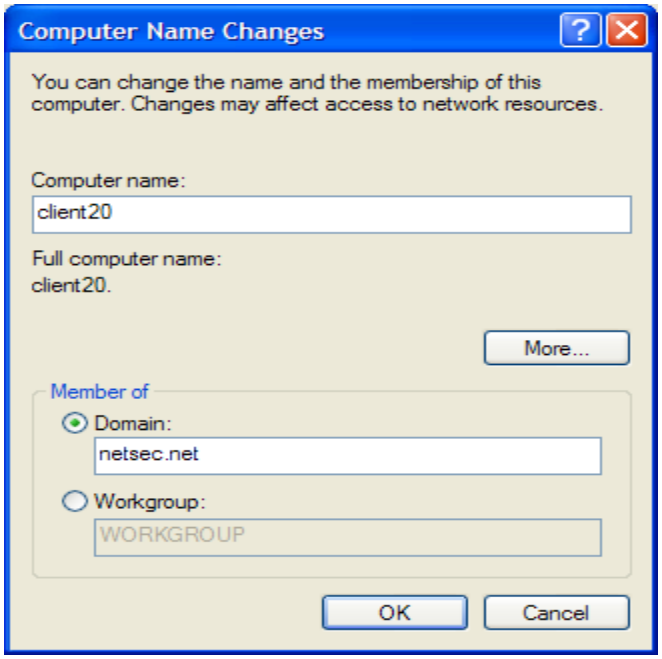
7. Click “Ok”
8. Click “Ok”

Joining Domain

9. Right-click **My Computer** and select **Properties**.
10. Click the **Computer Name** tab, then click **Change**.
11. Click More
12. Specify **domain.net** as the Primary DNS Suffix for This Computer.

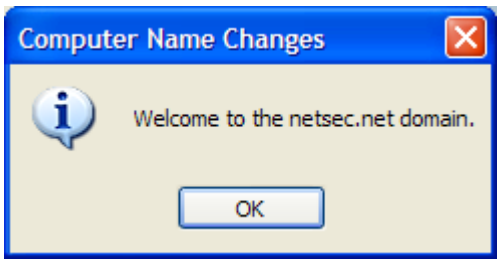


13. Click "**Ok**"
14. Select "**Member of ……Domain**" and enter the name of your Domain.

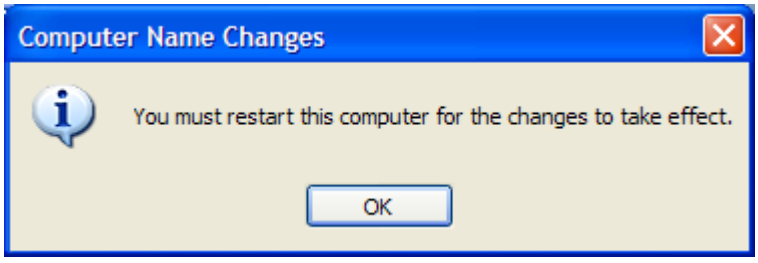


15. Click **OK**
16. Now Domain Server will prompt you for Username and Password. Enter any Username and password you have created before.

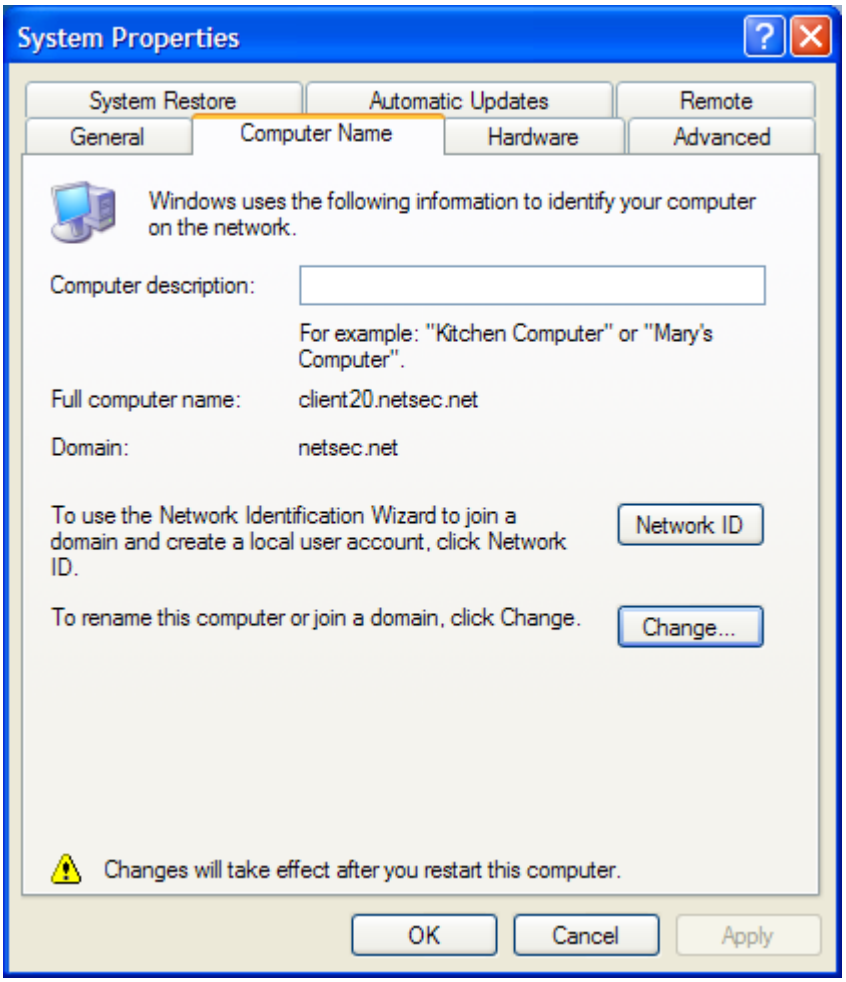
17. If you get this welcome message : Windows : “Computer Name Changes” – Welcome to the
    ……..domain”, it means you are successfully joining a domain.



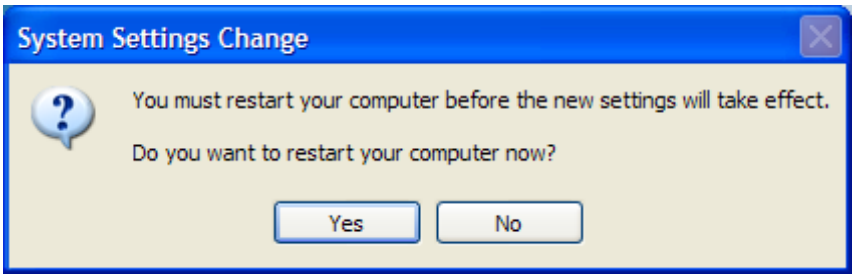
18. Since joining a domain is a major change is the security configuration of your system, you will be reminded that you have to restart your system. Click **OK**



19. You will be back in the “System Properties”, where you are now listed as being part of a domain.



20. Click **OK** to close the remaining dialog boxes, then click YES to restart the computer.



Lab 7

Viewing Computers In Active Directory

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LAB 7: VIEWING COMPUTERS IN ACTIVE DIRECTORY

In this section you will use Active Directory Users and Computers to view information for computers and server.

When a client workstation is installed using Windows XP Professional Windows 2000 Professional, it has its own accounts database and rights. When that client computer joins a domain or Windows 2003 Server network, this means that the domain wide accounts are available for use at the workstation. When a user logs on using the client computer, any policies are applied to the client computer.

Client workstation running Windows XP Professional have their own local accounts database. This means it is possible for an administrator on the workstation to create a **local workstation account,** which is not the same as the domain account, and allow users to logon to the local computer rather than the domain.

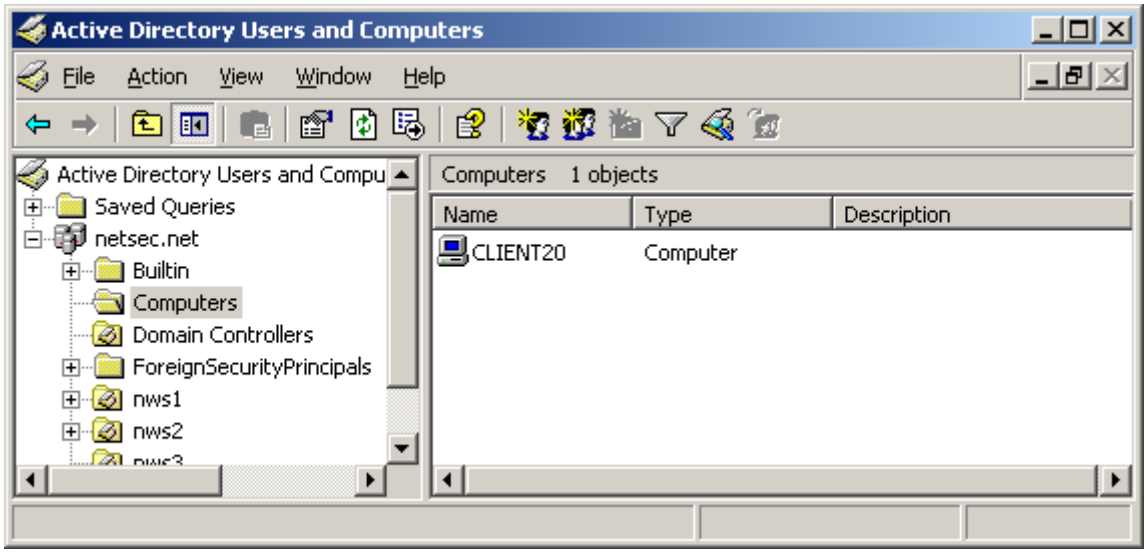
Currently, you should have Windows Server 2003 and Windows XP Professional client workstation running.
Logon on as administrator to the Windows Server 2003.

VIEWING COMPUTERS AND SERVER IN ACTIVE DIRECTORY

In this exercise, you will use Active Directory Users and Computers to view the workstation and servers in the domain.

Estimated time to complete: 15 minutes

1. Log on to the Windows 2003 Server as administrator.
2. Run Active Directory for Users and Computers.
3. Expand the domain icon.
4. Click on the Computer item from the list.



5. What is the name of the computer listed?

---

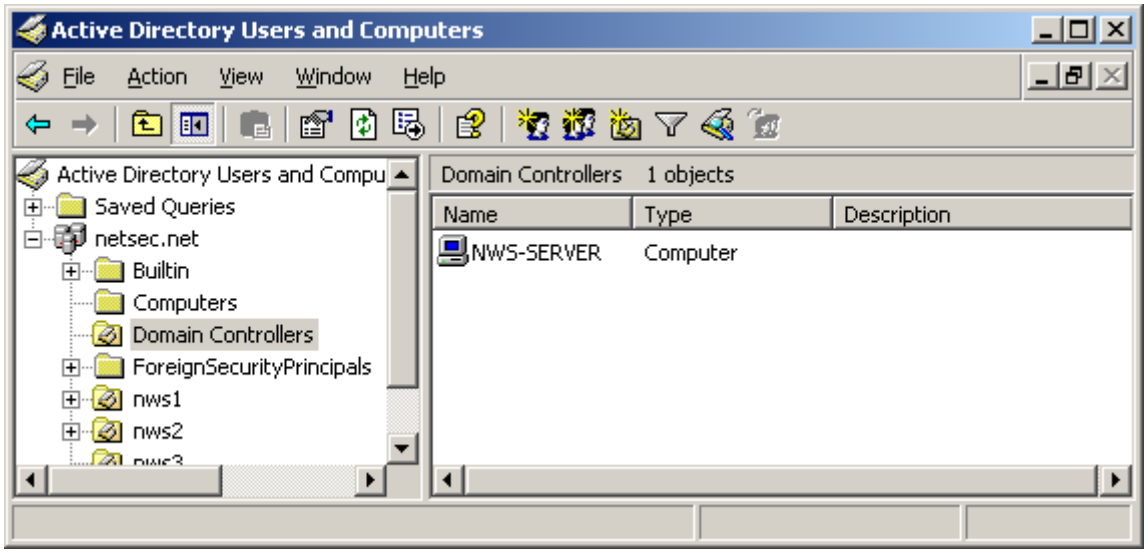
6. Double-click on the computer to display its properties.
7. What is the role of the computer?

---

8. Click on the **Operating System** tab.
9. What is the name of Operating System installed on this computer?

---

10. Close the properties box.
11. Click on **Domain Controller** folder in Active Directory.



12. What is the name of the domain controller listed?

---

13. Double-click on domain controller to display is properties.
14. Click on the **Operating System** tab.
15. Are there any service packs loaded on this computer?

---

16. Close the properties box and close all remaining dialog box.

In this exercise you viewed properties of workstations and servers in your network using Active Directory.

Using the Local Workstation Account

In this exercise you will log on the Windows XP Professional workstation using a local administrator account.

Estimated time to complete: 5 minutes.

1. Press **CTRL+ART+DEL** to display the logon dialog box.
2. Enter the user credentials as "**administrator**"**,** password as **"student".** Click **OK.**
3. Log off the client computer. Click Start – Shutdown, and select "**Logoff Administrator**" from the list. Click OK.

Using Domain wide accounts at the client computer

In this exercise you will log on to the client computer using a don=main account.

Estimated time to complete: 30 minutes

1. Press **CTRL+ART+DEL** to display the logon box.
2. Enter "**Fari.Ali**" as the username, "Fari" as the password, and click OK.
3. What happened? Could you log on?

4. If you could not log on, what do you suspect is the problem?

5. Click OK to dismiss the dialog box.
6. Look at the log on box. There is an extra field displayed, called "**Logon to...**" What does this field specify?

7. What other options are available in this field?

8. Select "Logon to DOMAINx". Enter the same user credentials as previously. Click OK.
9. What happened? Could you log on?

10. What do you think is difference between the two logon attempts?

11. Log off the client computer. Leave it running the Windows XP Professional (do not shut the computer down yet).
12. If you are currently logged in to the Windows 20-3 Server, log off.
13. Attempt to log on to the Windows 2003 Server as **Fari.ali**.

14. What happened? Could you log on?

15. If you could not log on to the Windows 2003 Server, why reasons might there be?

16. Log on to the Windows 2003 Server as administrator.
17. Run Active Directory for Users and Computers.
18. Click on the **Nws1** Organization Unit.
19. Double-click on the user **Fari.Ali** to display the properties box.
20. Click the **Members OF** tab.

    Click Add.
    Click Advanced button.
    Click Find Now button.
    Double-click **Server Operators** from the list.
    Click OK.

21. Click OK.
22. Log off the computer.
23. Attempt to log on the Windows 2003 Server as **Fari.Ali**
24. What happened? Could you log on?

25. What do you think is the difference between the two logon attempts?

Summary

Servers do not allow normal users to logon locally. Servers run the network and provide resources, which users connect to remotely across a network. Servers are not designed to have users physically sitting at theirs keyboards trying to log on and run programs. Users actually logon to a client computers in the network and access resources using a network connection.

Client computers running Windows XP Professional have their own accounts database.

Lab 8

Delegating Management Of Users

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LAB 8: DELEGATING MANAGEMENT OF USERS

In this exercise you will create new local groups and look at assigning managers to users and organizational units.

Estimated time to complete: 30 minutes

Delegating Control

In this portion of the exercise you will make Fari.Ali a manager of the Nws1 organizational unit. Once he is a manager, he will be able to modify user accounts within the Nws1 OU.

1. Log on as administrator to the Window Server 2003

2. Run Active Directory for Users and Computers

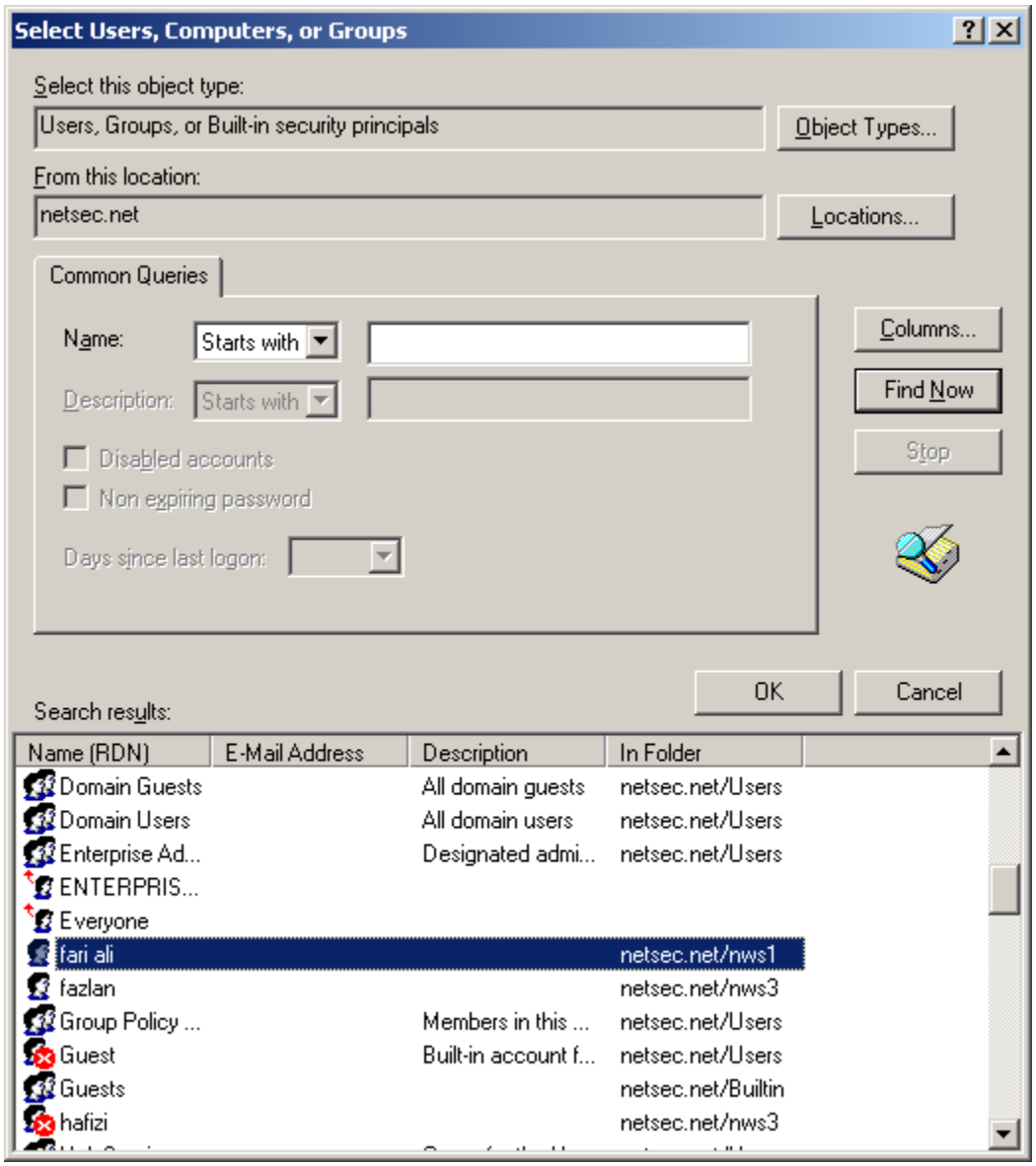
3. Expand domain.net

4. Right mouse-click the Nws1 OU and select Delegate Control

5. This starts the Delegation of Control Wizard.



6. Click the **Add** button, **Advanced**…..button, **Find Now** button to display a list of groups and users.

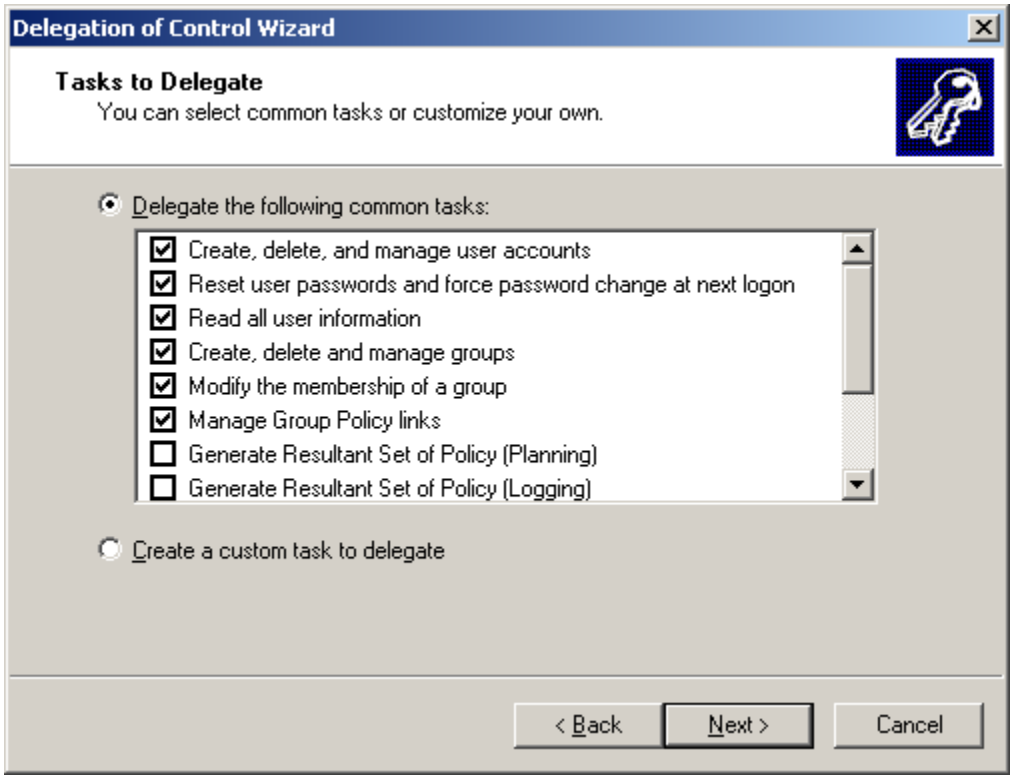


Highlight the user account **Fari.Ali** and click ok.

Click ok.

7. Click Next.

8. Delegate the following tasks as shown below.



Enable the delegated tasks as illustrated. Click Next

9. Click **Finish**

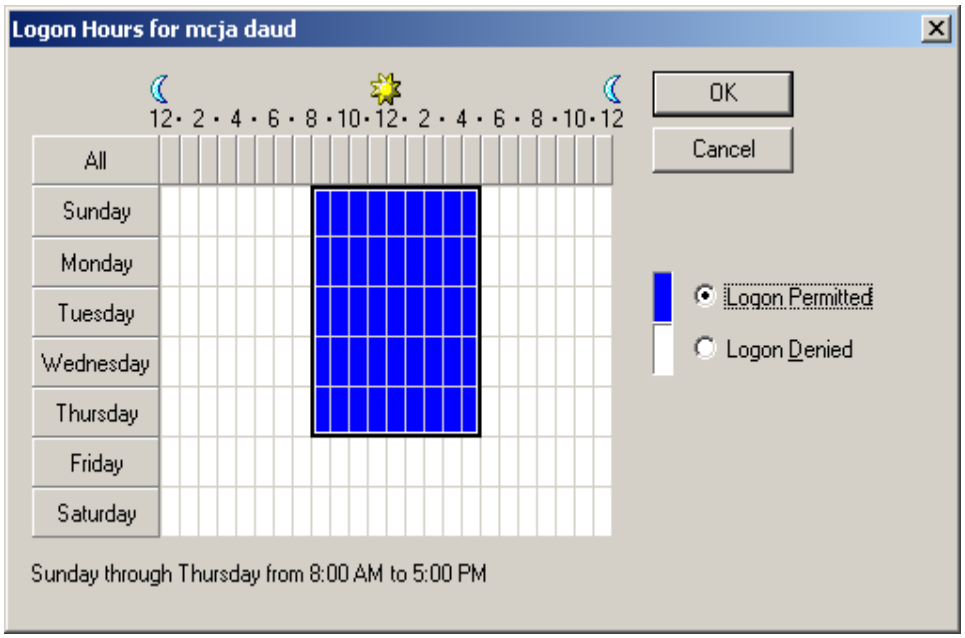
Managing Users

In this portion of the exercise you will log on as **Fari.Ali** and attempt to manage users.

10. Log on the Windows Server 2003 as **Fari.Ali.**

11. Run Active Directory for users and computers.

12. Click on the **NWS1** OU.



Double-click the user **Mcja daud**.

Click the account tab.

Click the **Logon Hours** button and change Mcja's logon hours to 8am-5pm Sunday to Thursday.

Click ok.

Click ok.

13. Were you able to successfully change **Mcja Daud** logon hours? YES NO

14. Click the Nws2 OU. Double-click the user account **Azhar Amran** to display the properties of this user.

15. Attempt to change the logon hours of this user. What message was displayed?

_____

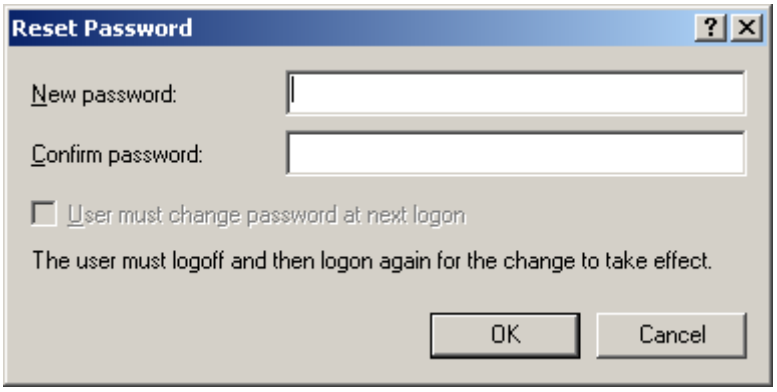
_____

16. Why do you think you are not able to modify this account?

_____

_____

17. Click the Nws1 OU. Right-click the user account **Mcja Daud** and select Reset password from the list.



This displays a reset password box that will allow the password to be change.

Click cancel.

18. Close all remaining windows.

19. Log off the user Fari.Ali

In the above exercise you delegated control of an Organizational Unit to a users. You then modified account details of users belonging to that OU as the designated manager of the OU.

Exploring Group Scopes and Types

In the following exercise you will create a number of groups. These groups will be used to demonstrate group scope. From the notes, group scope determines who can be a member and where that group can be used in the enterprise.

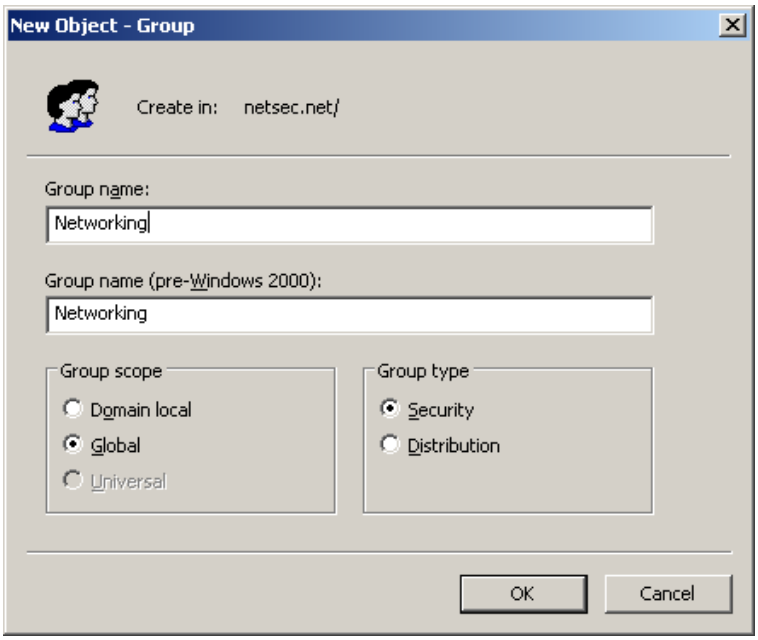
Group Type	**Scope**
Local	User accounts, Global groups and Universal groups from any domain in the forest, as well as local groups from the same domain.
Global	User accounts and global groups from the same domain.
Universal	User accounts, global groups and Universal groups from any domain in the forest.

The recommended strategy for using groups in Window 2003 is to use both global and domain local groups. Place users into global groups and then place the global groups into domain local groups and assign permissions to the domain local groups.

Global groups have access to account in the local domain. Where the enterprise consists of more than one domain, local groups allow the use of accounts across all the domains .Where the enterprise has combined a number of domains into a forest, Universal groups provide access to any accounts in the forest.

Estimated time to complete: 15 minutes

1. Log on as Administrator
2. Run Active Directory for Users and Computer
3. Right-click the domain icon and select **New-Group** from the list.

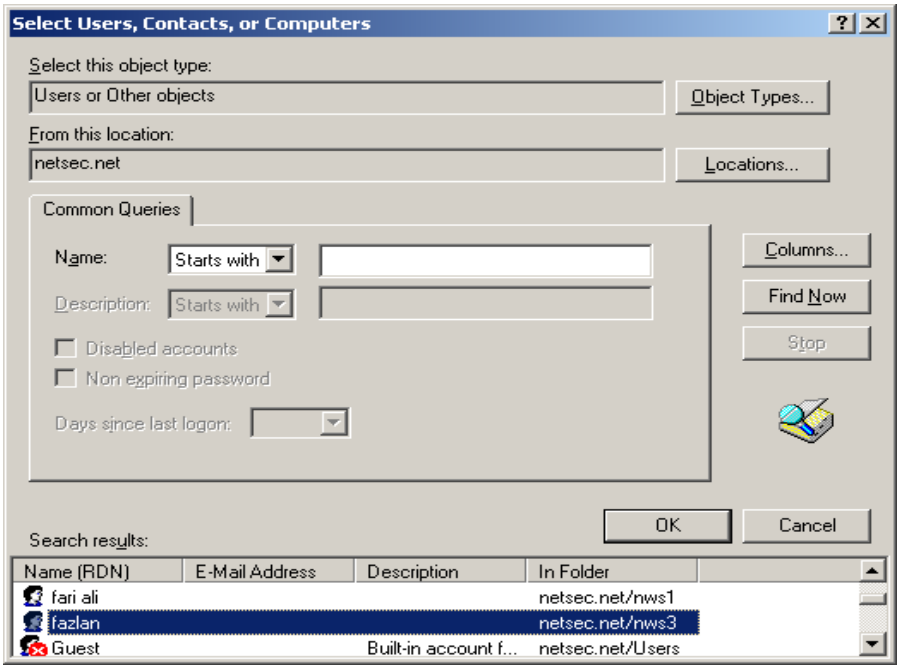


4. Create the **global** group called "**Networking**".

Verify Group scope set to **global**.

Verify the group types is set to **security**. Click OK.

5. Add **Fazlan** as a member of **Tech Support** (double-click Networking, click the **member** tab, click the **Add** button, click **Advanced**……button and then click **Find Now** button).



Highlight the user account **Fazlan** and click OK.

Click OK.

Click OK.

6. Create a new domain local group called “Intranet Users”
7. Access the group **Intranet Users** properties.
8. Add the group “**Intranet Users**” as a **member of Tech Support**. What happened?

9. Why do you think this is so?

10. Access the group **Tech Support** properties.
11. Add the group **Tech Support** as a member of **Intranet Users**. What happened?

12. Why do you think this is so?

13. Log off as administrator.

Summary

Delegating control of users using the delegation control wizard is simple. When control of users and groups is delegated, administrators can be relieved of simple

Administrative tasks such as resetting passwords and modification of users accounts.

Windows Server 2003 running in native mode supports the use of different group types. Global groups have access to user accounts and other global groups in the same domain. Local group allow you to access accounts outside the current domain, and universal groups provide access across organizations (forests)

Lab 9

Creating

And

Applying Group Policies

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Wan Muhammad Irsyad b. Wan Hashim

LAB 9. CREATING AND APPLYING GROUP POLICIES

In this exercise you will create a new group policy and apply it to users within an organizational unit.

Estimated time to complete: 30 minutes

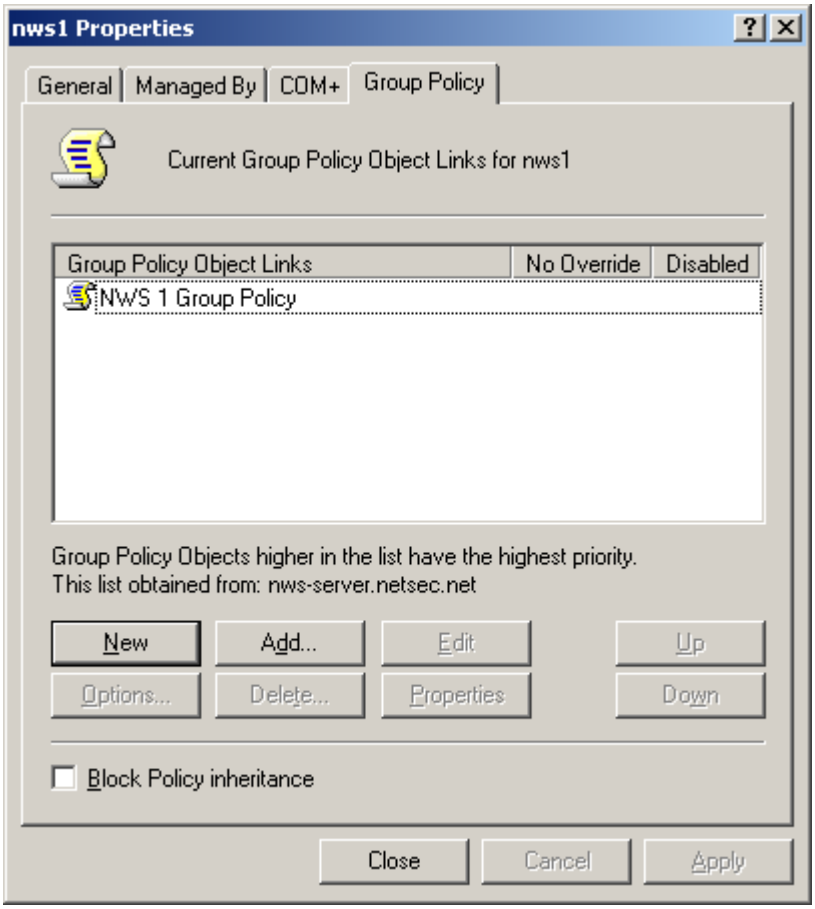
Group Policies

Group policies are settings or configuration that can be applied to users, groups, organizational units and domains. An administrator can create a group policy that configures the computer or user settings, such as menu and desktop settings, folder locations and default password.

Windows NT 4 and Windows 98 introduced system policies. Windows 2000 and 2003 extends these further using group policies.

Creating a group policy

1. Log on as Administrator to the Windows Server 2003.
2. Run Active Directory for Users and Computers.
3. Expand the domain and right-click on the **NWS 1** OU. Select ***Properties***.
4. Click the **Group Policy Tab**.



In this exercise, you will create a new group policy for the NWS 1 OU.

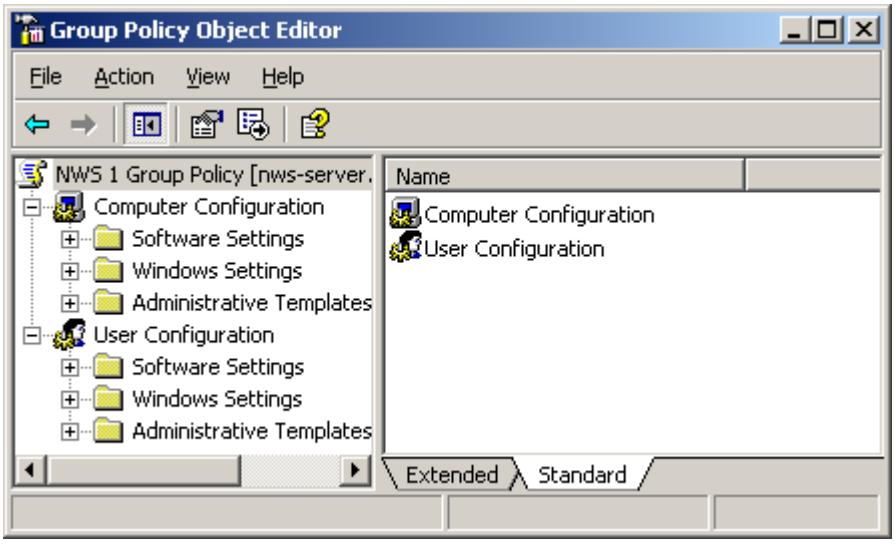
This new policy will apply to all members of the NWS 1 OU though in another exercise that follows, you will override this.

Click the button **New**.

Rename the policy as **NWS 1 OU Group Policy**.

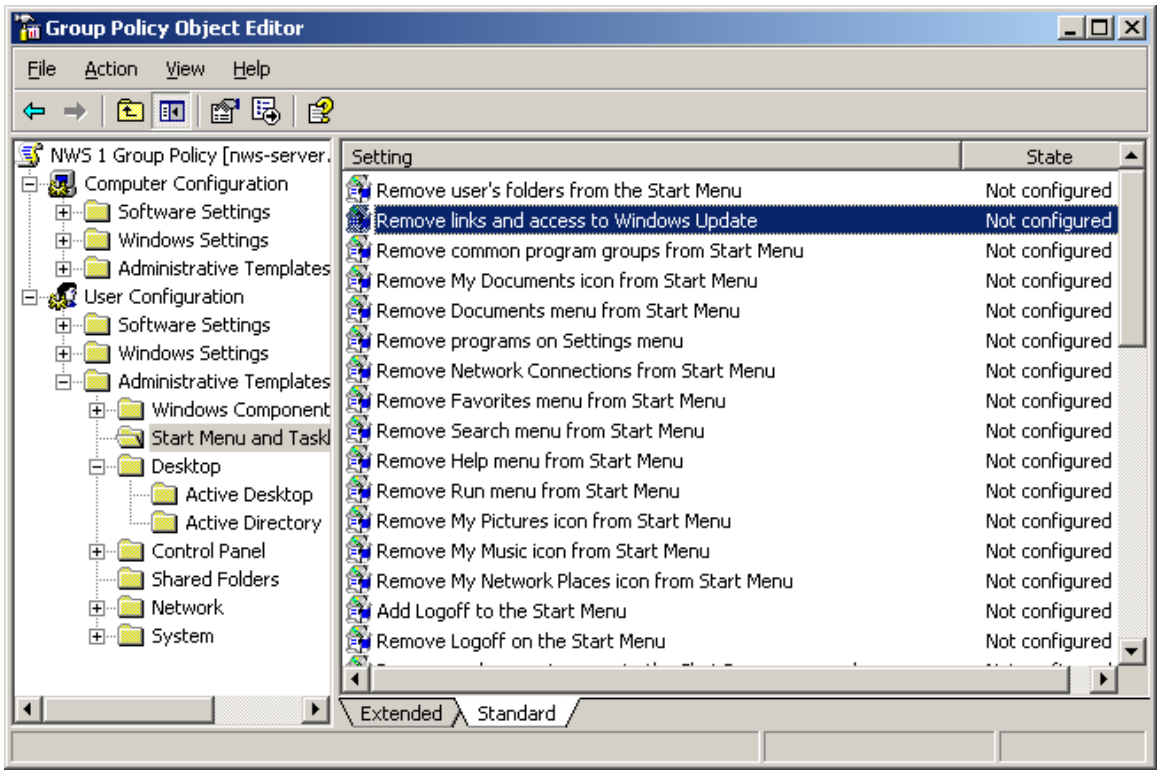
5. Select **NWS 1 OU Group Policy** and click the **Edit** button.

6. The group policy window opens.



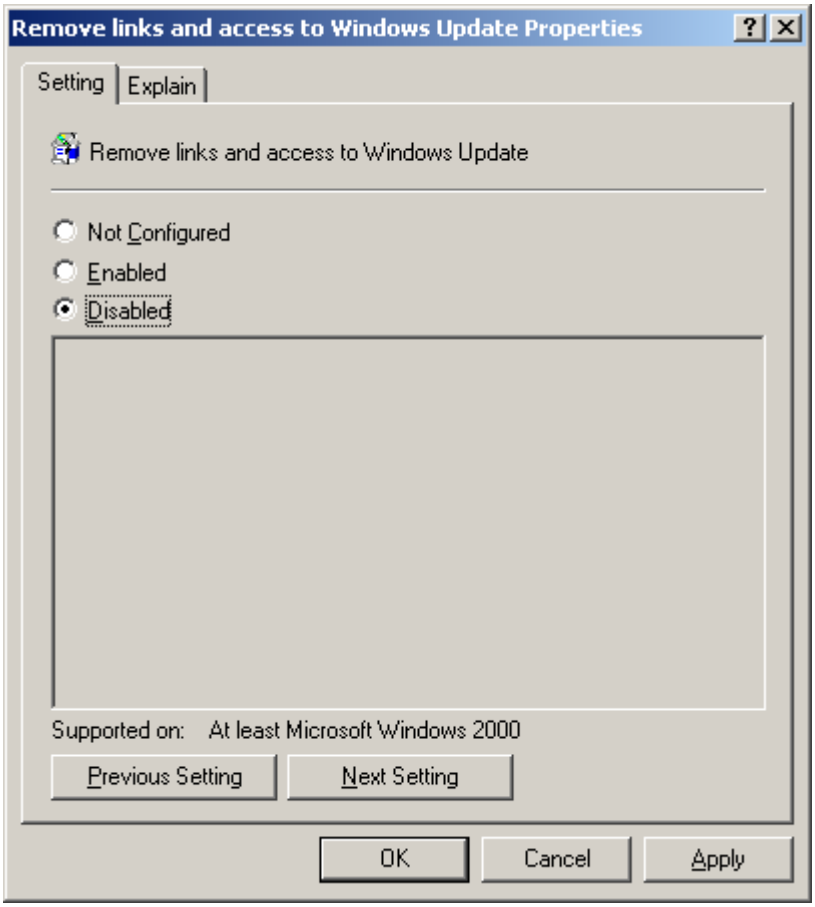
The group policy editor allows you to specify user and computer settings. In the following steps, you will change some of these settings.

7. Expand the **User Configuration**: and then expand the **Administrative Templates** folder. Expand the **Start Menu & Taskbar** folder.



A large list of selections is available. Double click the option **Remove links and access to Windows Update.**

8. The following dialog box appears.



Click the **Disabled** button to disable this setting.

Click **OK**.

The setting now displays as Disabled in the Group Policy Editor window.

9. Configure the following settings.

Remove Run menu from Start Menu – **Enabled**

Remove Search menu from **Start Menu** – **Enabled**

Desktop\Active Desktop\Enable Active Desktop – **Enabled**

Active Desktop Wallpaper – **Enabled**

Wallpaper Name **C:\WINDOWS\Web\Wallpaper\Ripple.jpg**

(This uses a wallpaper from the Windows XP Pro install on Drive C)

10. Close the group policy editor.

11. Close the NWS 1 OU properties window.

12. Click **Refresh** button and close Active Directory users and computers.

Testing the Group Policy

The group policy has been applied to members of the NWS 1 Organizational Unit. There are two members, fari ali and mcja daud. You will now test the policy to see if it works.

13. **Log off** the Server as administrator.

14. Log on to the Server as **fari.ali**.

15. You may get a dialog box asking you to configure Internet Explorer. If you do, configure it for a local LAN existing account.

16. Is the wallpaper displayed? YES NO

17. Do you have the RUN and SEARCH command on the Start Menu? YES NO

18. Now verify that the settings are applied to the client computer also. Log on to the Windows XP Professional computer as **mcja.daud**

19. Were the wallpaper, run and search options configured on the client computer?
    YES NO
20. Log off the client computer. Log off the Server.
21. Log on the client computer as **aj**
22. Were the group policy settings applied? YES NO
23. If not, why do you think this is so?

    _____

    _____

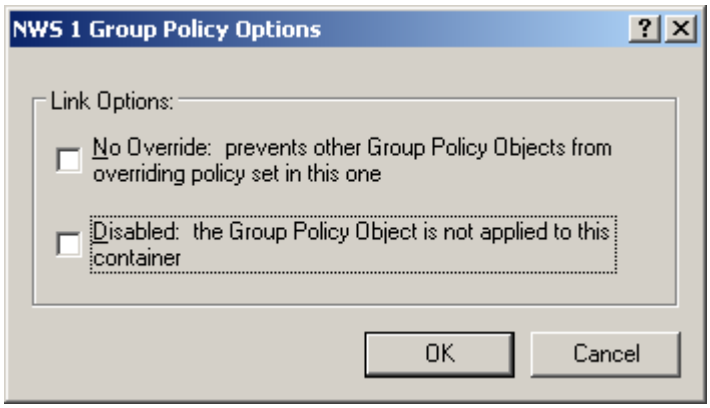
    _____

24. Log off the client computer.

Disabling the Group Policy

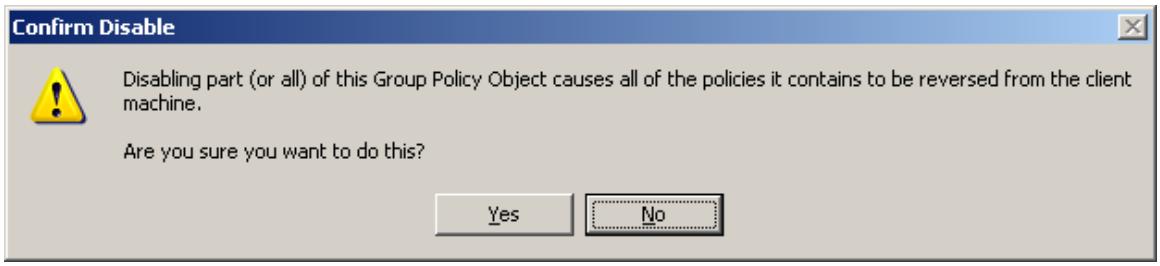
25. Log on the Server as Administrator. Run Active Directory for Users and Computers.

26. Access the **NWS 1** OU properties and select **Group Policy**. Click the **Options** button.



You are now going to disable the policy. This is a better option than removing the policy, as if you decide to re-implement the policy at a later date, it will still be there.

Check the Disabled box to disable the policy.



A confirmation box appears. Click **Yes**.

Click **OK** to close the dialog box.

27. Close all dialog box remaining and **Log off** the server as Administrator.

28. Log on to the server as **fari.ali** to test if the group policy is no longer applied.

29. Were the policies now disabled? YES NO

30. Log off the server.

Summary

In this exercise you created a group policy and applied it to an organizational unit. Only a fraction of the available settings were explored. Applying a group policy is a way of controlling security and configuring groups of users with common settings. This can help reduce the cost of ownership and the level of administrator support by restricting what users can do or change their computers.

Lab 10

Creating

And

Sharing Resources

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LAB 10. CREATING AND SHARING RESOURCES

One important aspect of a Windows Domain is the ability to share applications, files, printers and other resources on the network. Resources created on Windows Server computers are available to all users in the domain, and it is a simple administration task to allocate permission to users.

Estimated time to complete: 45 minutes

Preliminary Setup

Log on to the server as **Administrator**. Using Active Directory for Users and Computers, add **aj**, **mcja.daud**, and **azhar** to the **Intranet Users** group.

Creating and sharing a resource using Windows Explorer

In this exercise, you will use Explorer to create a folder and verify the NTFS file permissions. The folder will then be shared and permissions assigned. You will then access this shared resource from the client computer.

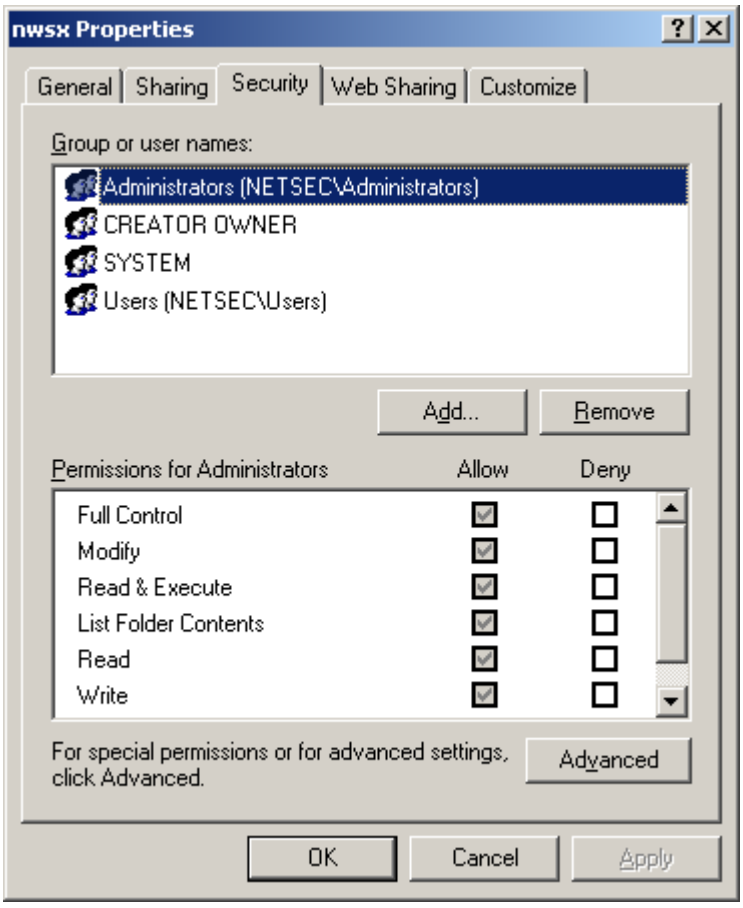
1. Log on to the server as Administrator.

2. Right-click **My Computer** and select **Explore.**

3. Access Drive E:
   (Make sure your drive E are NTFS formatted. If not, you have to convert or format it to NTFS.)

4. Create a folder named tempx(where x represents your Station Number).

5. Right-click the folder nwsx and select **Properties.**

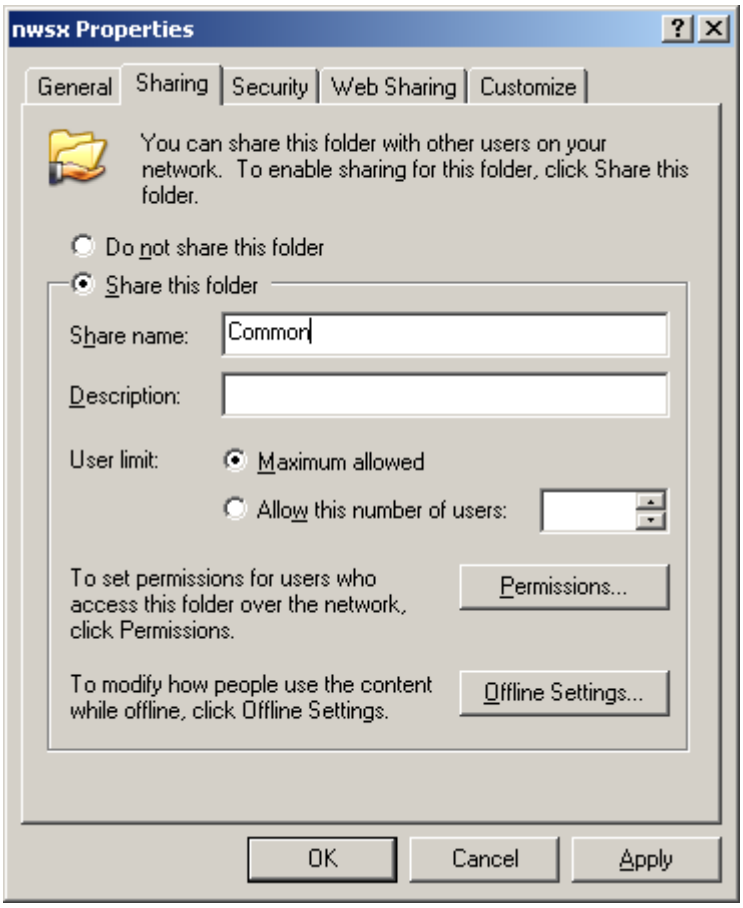


Click the **Security** tab. A list of security permissions is displayed. Note that the group Administrators is given Full Control access at the folder level.

When users access a folder across the network, both the share and NTFS permission lists define the user permissions.

Click **OK**.

6. Right-click the folder nwsx and select Sharing and Security.

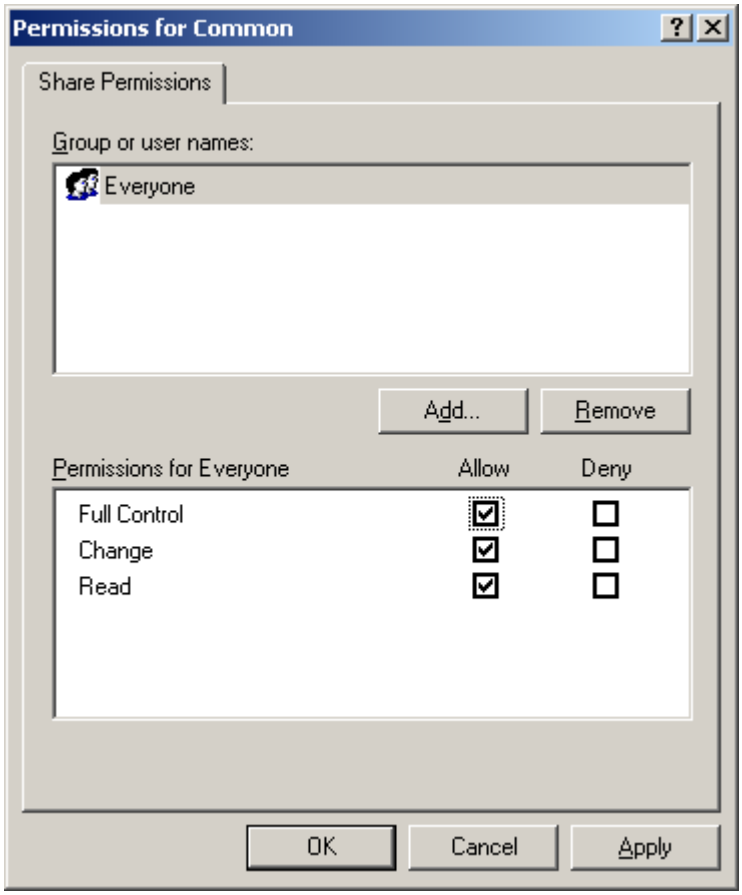


Enable the **Share this folder** option.

Specify the share name as **Common**.

Click the **Permissions** button.

7. Now you will restrict permissions at the share level. Remember that user permissions to a network resource are made up of the share permissions and the NTFS permissions.

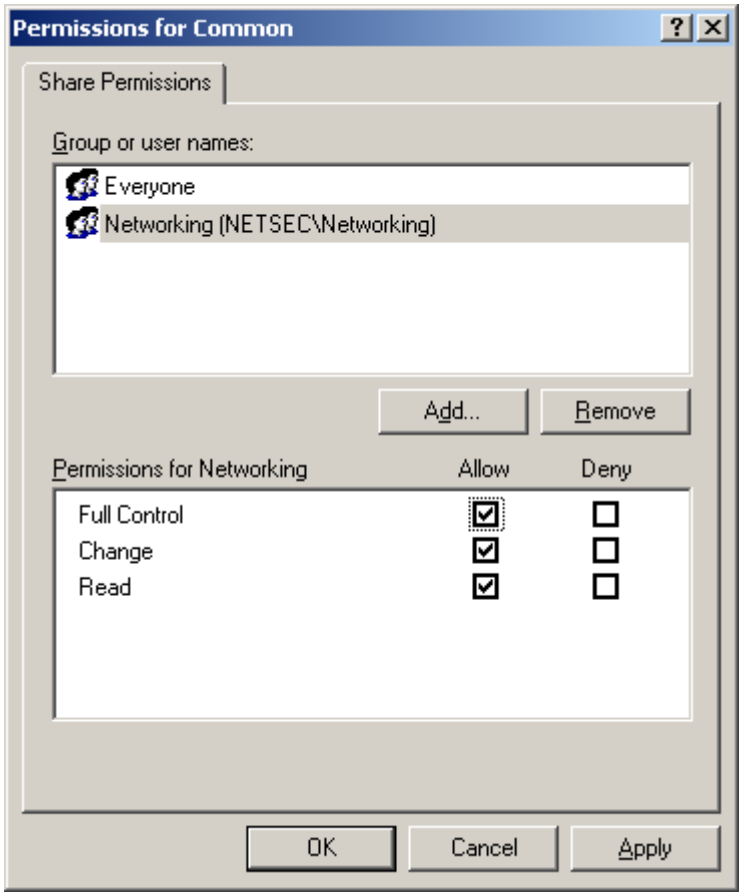


Remove the group Everyone. (Click Remove button).

Click on the **Add** button and add the **Networking** group with permissions of **Full Control**.

Click on the **Add** button and add the **Intranet Users** group with permission of **Read**.

8. The share permission should look like



Once you have set the permissions as described, click OK to close this dialog box.

Click OK to close the sharing dialog box for folder nwsx.

9. In the Explorer window you will note a little hand icon on the folder e:\tempx, which indicates the folder is now shared.

10. Log off the server.

11. Log on to the client computer as **mcja.daud**.

12. Using My Network Places, browse the entire network, select the server, and view the available resources. You should see the **Common** resource listed.

13. Double-click the resource so that you are connected to it. A new window will open up and display the contents of the folder (it will be empty as there are no files in the folder).

14. Attempt to create a new text file (right mouse click in the window and select **New, Text document**).

15. Could you create the file? YES NO

16. If No, why do you think this is so?

_____

_____

_____

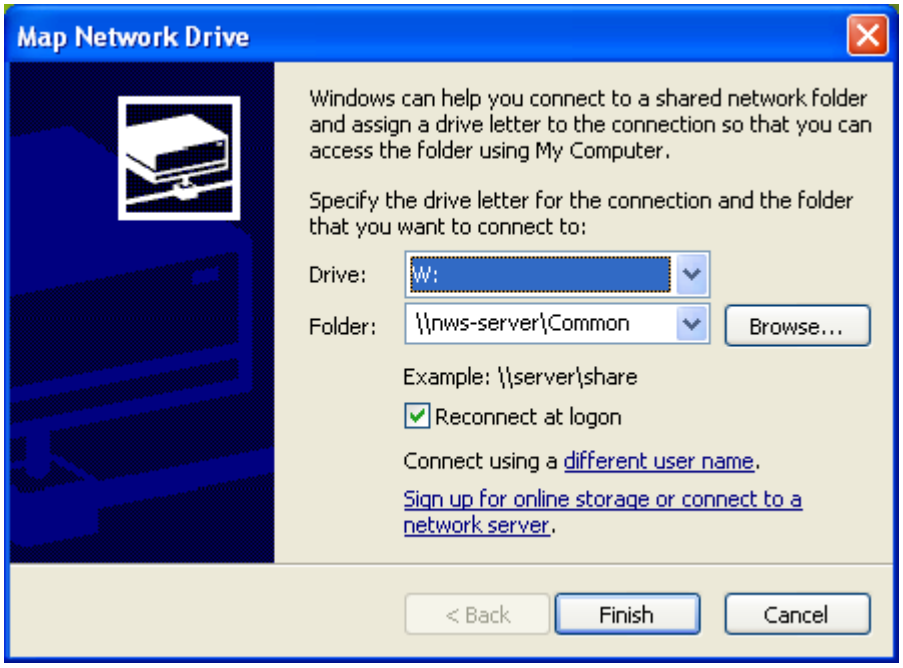
17. Log off the client computer.

18. Log on the client computer as **fazlan**.

19. Instead of using My Network Places, we will map a drive letter to the resource. This is an alternative way of accessing the resource, but requires that you know the location of the resource (using My Network Places allows you to view the available resources so you don't really need to know the location).

20. Right-click My Computer and select Map Network Drive.

21. Enter the location of the network resource in the dialog box.



You must specify the name of the server and the share name.

In this instance, it is \\nws-server\common.

Click Finish.

22. Attempt to create a new text file (right mouse click in the window and select New, text document).

23. Could you create the file? YES NO

24. If No, why do you think this is so?

_____

_____

_____

25. Log off the client computer.

Publishing a shared resource in Active Directory

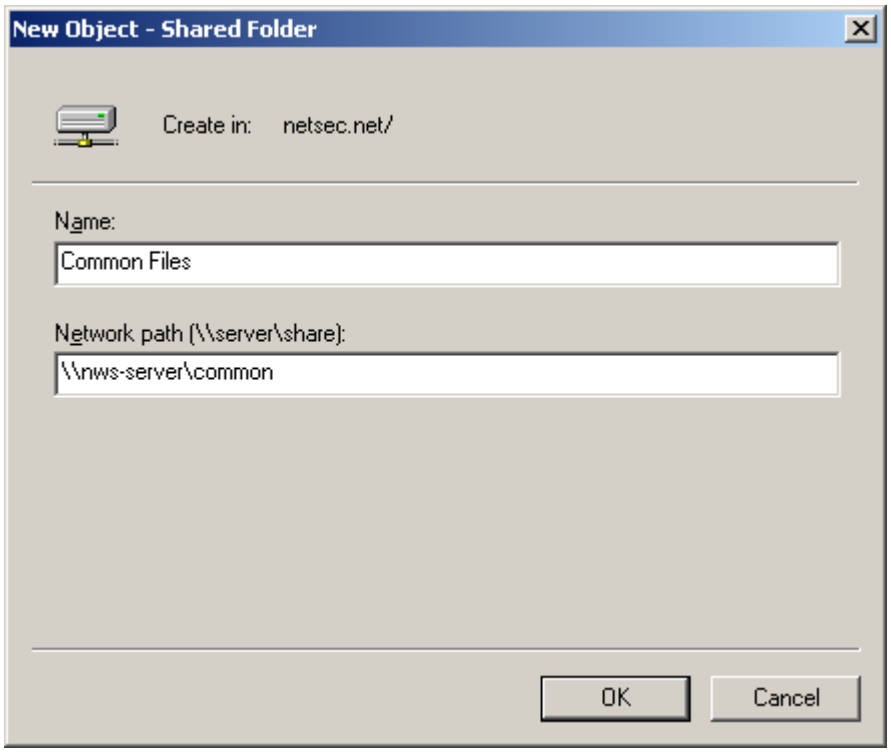
One of the problems of publishing shares in the way you have just done (which is the way they were done in NT 4 or 98) is that you have to browse the network or know which server the resource is located on in order to find it. This can be time-consuming and frustrating for users.

Resources can be published in Active Directory, making them easy to find. In the next step you will publish the resource into Active Directory.

1. Log on to the server as Administrator.

2. Run Active Directory for Users and Computers.

3. Right-click the domain and select **New**; **Shared Folder**.



Enter the name as **Common Files** and the Network path as your server name and share (in this example it is \\nws-server\Common).

Click OK.

4. The new shared folder appears in the right window pane of Active Directory.



5. Close Active Directory for Users and Computers windows.

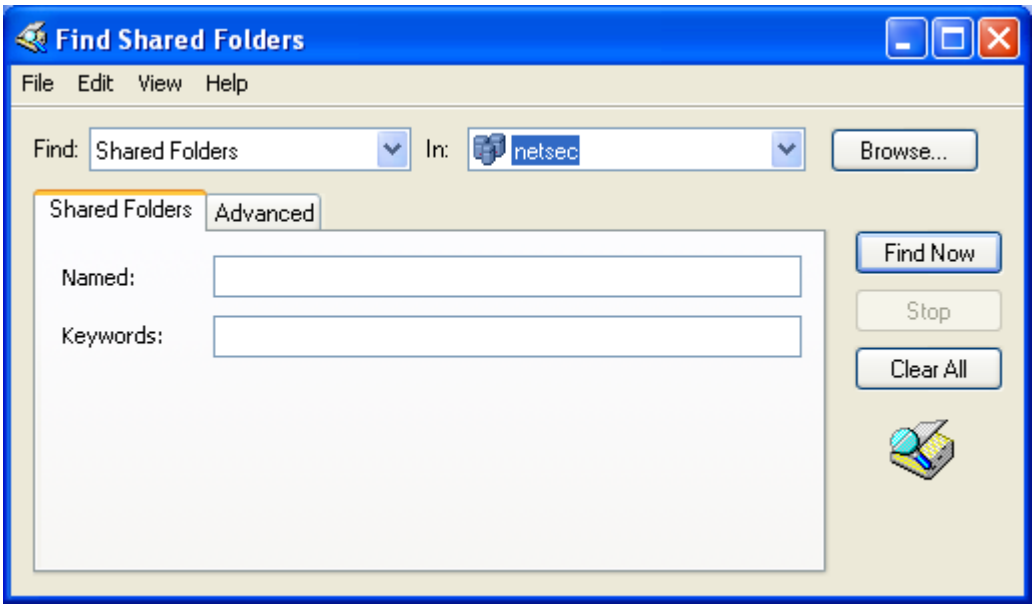
Locating a shared resource in Active Directory

Now that the shared folder is published in Active Directory, it is easy for users to locate and connect to the resource.

1. Log on to the client computer as **mcja.daud**.

2. Double click "**My Network Places**".

3. Click "**Search Active Directory**".

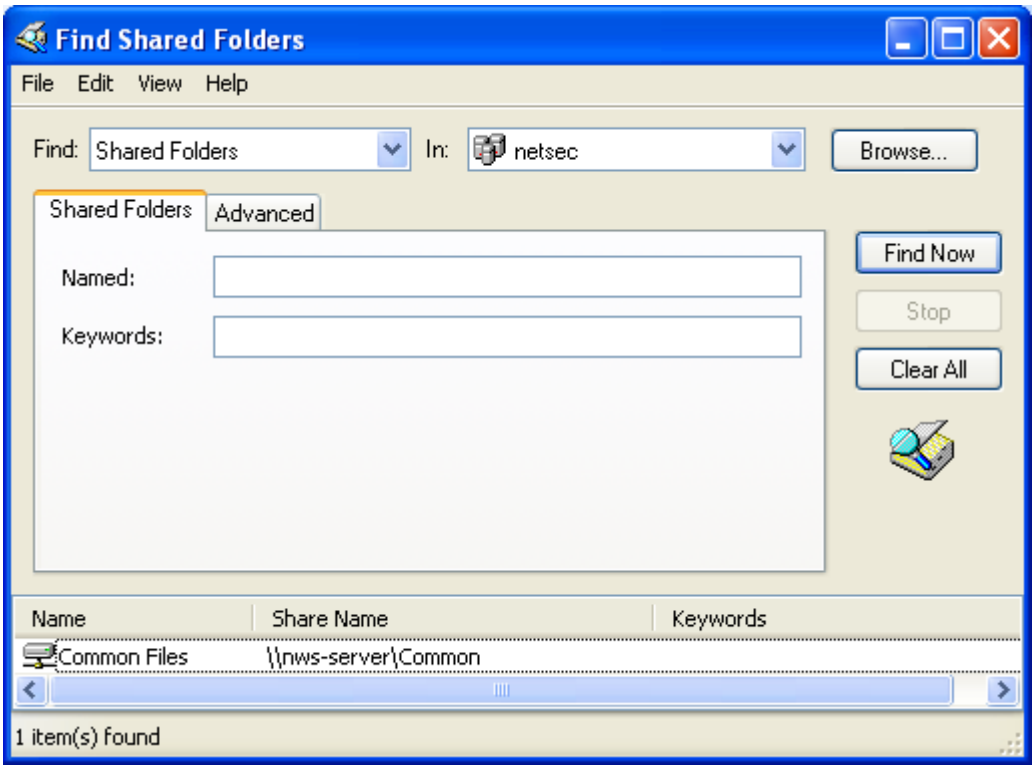


In the Find drop box, select **Shared folders**.

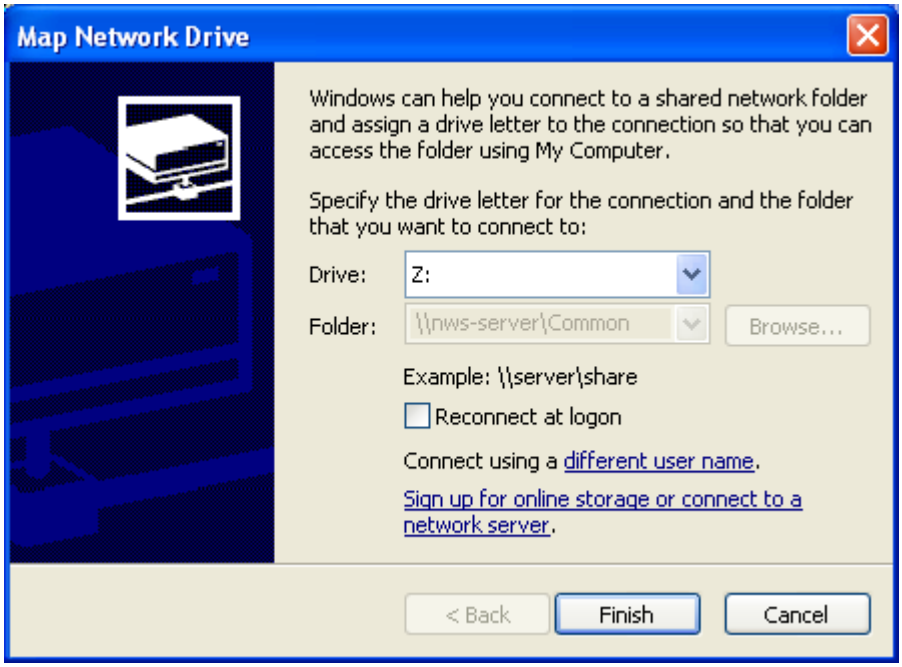
and in the In drop box, select **domain**.

Click **Find Now**.

4. A list of shared folders available is displayed.



Right-click the Common Files shared folder from the list and select Map Network Drive.



Note how the location for the server shared folder is filled in automatically.

Disable the checkbox Reconnect at logon.

Click **Finish**.

5. You are now connected to the shared resource.

6. Close all remaining windows.

7. Open “My Computer”. There are now one additional drive appears at the bottom. (Network Drive).

8. Log off the client.

Summary

Permissions are assigned at the SHARE and at the File system level. By default, Windows Server 2003 places every user created into the group EVERYONE, and, when creating a new directory or share, automatically assigns rights to that resource so the group EVERYONE can access it.

If you want to secure any resources by restricting access, you should ensure that the appropriate permissions have been set at both the share and file system level.

Publishing shared folders in Active Directory simplifies the task of locating resources.

Lab 11

Logon Scripts

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German-Malaysian Institute

Prepared By,

Wan Muhammad Irsyad b. Wan Hashim

LAB 11. SCRIPTS

In this exercise you will create logon and logoff scripts and apply these to users in an organizational unit. You will specify a network home directory for users and arrange for this directory to be mapped when the users logs on. Finally, you will specify disk space restrictions for specific users.

Estimated time to complete: 45 minutes

Logon Scripts

A logon script is a sequence of commands that executes when a user logs onto the network.

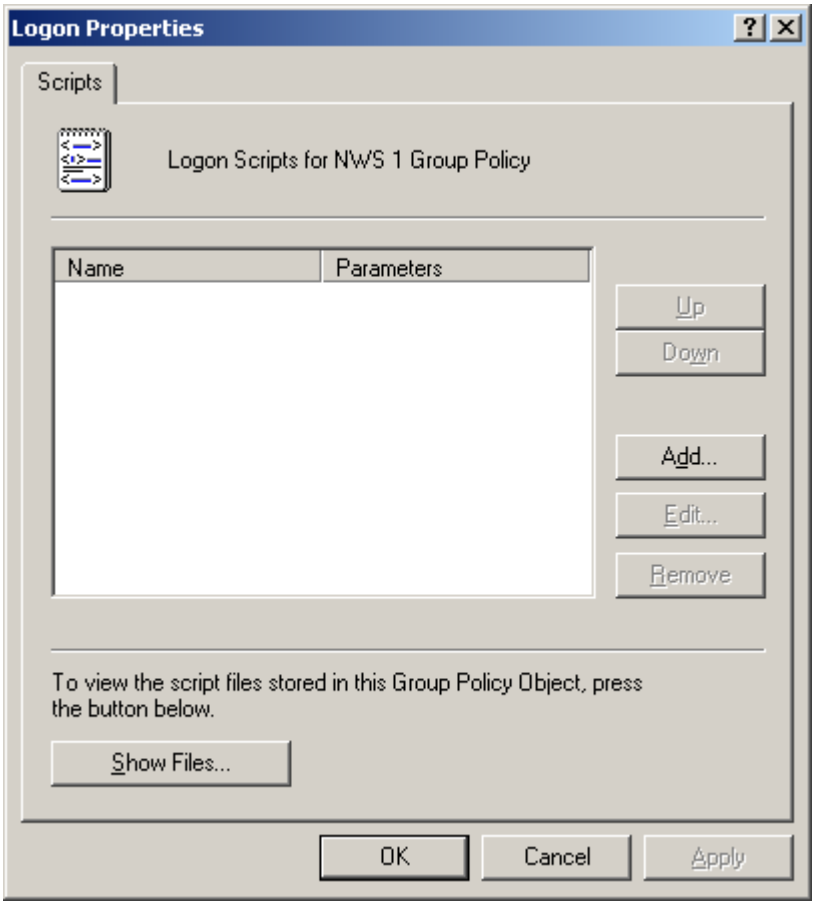
1. Log on to the Server as Administrator.

2. Run Active Directory for Users and Computers.

3. Right-click the **NWS 1** Organizational Unit and select **Properties**.

4. Click the **Group Policy tab** and then click the **Edit** button to edit the **NWS 1** OU Group Policy.

5. Expand User Configuration, Windows Settings, Scripts(Logon/Logoff) and double-click Logon.



In the Logon Properties window, click **Show Files**.

6. Right-click inside the new window and select New; Text document. Rename the text document as **NWS 1.cmd**.

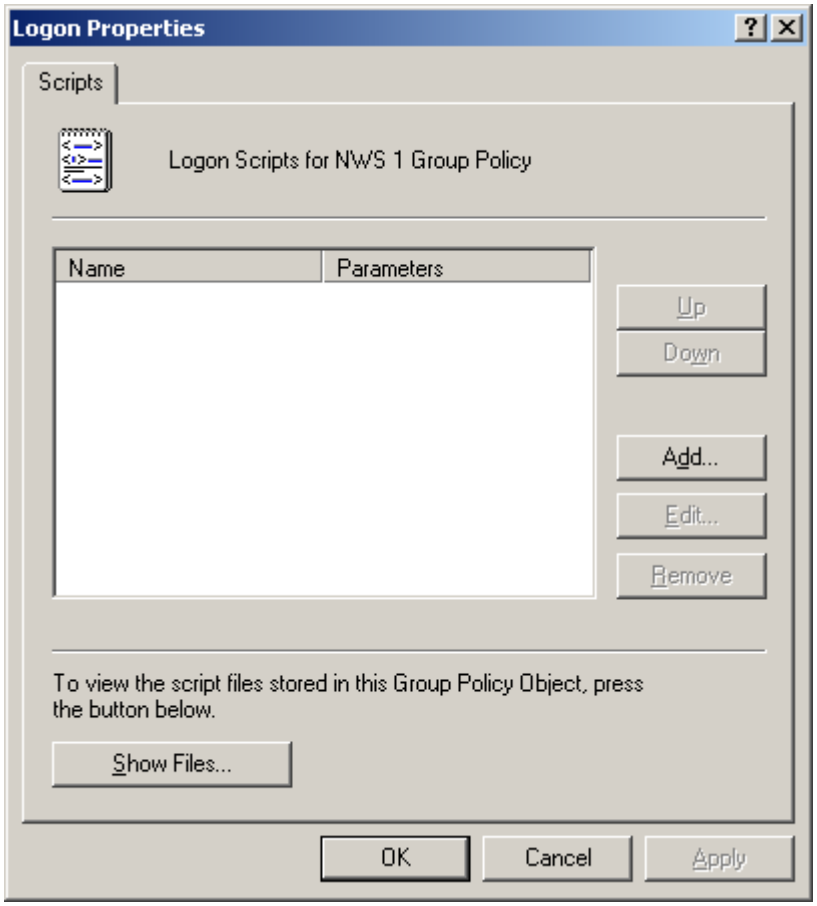
7. Right-click NWS 1.cmd and select Edit.

8. Click “Open”. This will load the Notepad editor. Type the following text into the file.

```
echo off
cls
echo this is a log on script for the NWS 1 OU
echo Welcome %USERNAME%, member of the NWS 1 OU
Pause
```

9. Save the file. Close Notepad.

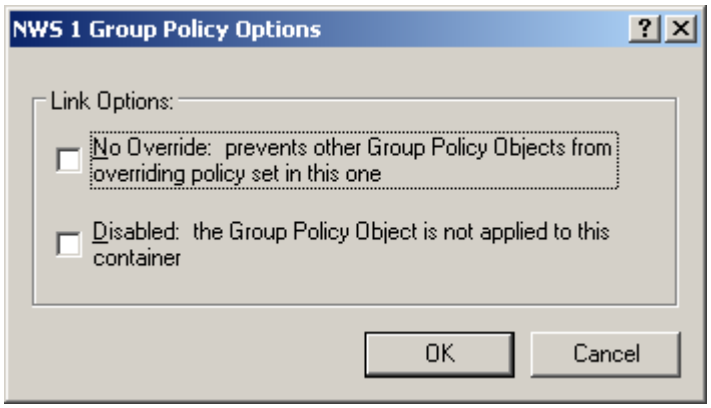
10. Click the Add button of the Logon Properties window. Enter **NWS 1.cmd** as the script name and click **OK**.



11. Click **OK** to close the Logon Properties window.

12. Click Options button.

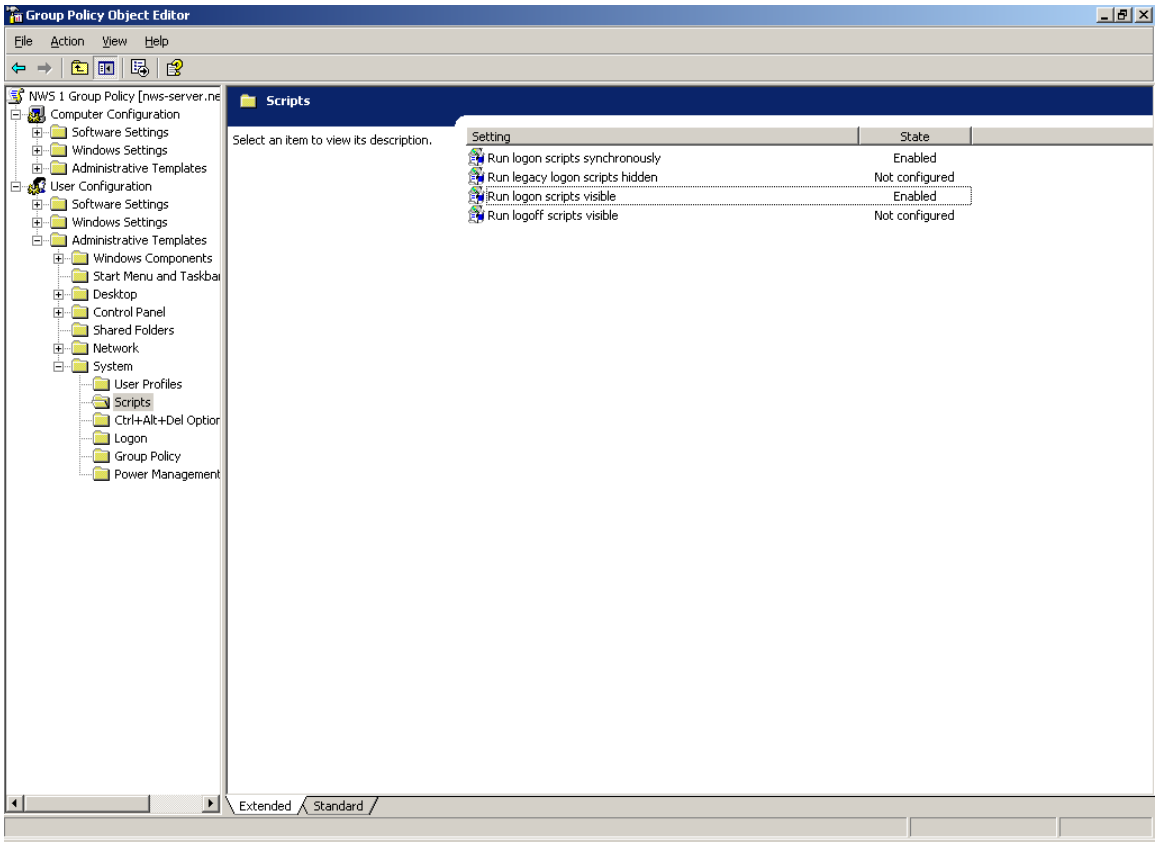
13. Uncheck all options in the NWS 1 OU Group Policy Options.



14. Click **OK** to close the NWS 1 Properties window.

15. In "**Group Policy Object Editor"**, expand User Configuration, Administrative Templates, System, Scripts. Double-click **Run Logon Scripts Visible**. **Enable** the option.

16. In the same folder, enable **Run logon scripts synchronously**.



17. Close the Group Policy Object Editor.

18. Click **Refresh** button and **Close** Active Directory for Users and Computers.

Testing the logon script

19. Log on to the client computer as mcja.daud.
20. Did the logon script execute? YES NO
21. If yes, what was displayed?

_____

_____

_____

22. If the script did not execute, why do you think this was so?

_____

_____

_____

23. Log off the client computer.
24. Log on the client computer as **fazlan**
25. Did the logon script execute? YES NO
26. If the script did not execute, why do you think this was so?

_____

_____

_____

27. Log off the client computer.

Summary

Scripts allow for both user and computer environments to be configured. The four scripts available are startup, shutdown, logon and logoff.

Lab 12

Home Directories

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Prepared by :
N MUHAMMAD ASNAWI B. MOHD ABD. GHANI

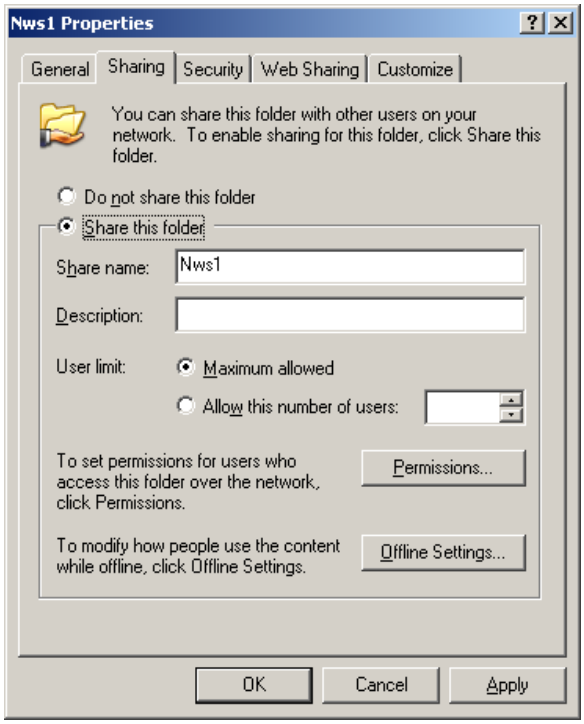
LAB 12. HOME DIRECTORIES

Home Directories

In this section, you will create a shared folder on the server that will be used for user home directories. You will map a home directory for a specific user, so that when they log on to the network, they will have a drive mapped to their home directory on the server.

1. Log on to the server as Adminstrator.
2. Run Active Directory for Users and Computers.
3. Expand the **NWS1** Organizational Unit.
4. Right-click the user **Fary** and select properties.
5. Run Windows Explorer (Start;All Programs; Accessories;Windows Explorer).
6. Create a new folder on drive E: called Fary

   (replace x with your group ID, e.g., e:\users1)
7. Right-click the fary folder and select Sharing and Security.



Specify the share name as Users.

Click **OK.**

8. In Windows Explorer, expand the new Fary folder.

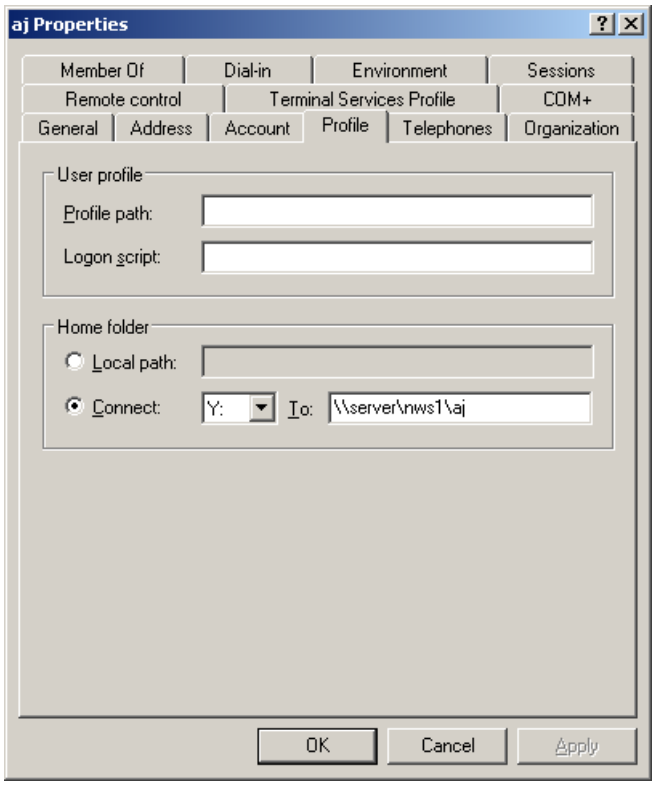
9. What are the sub-folders under Fary?

_____

_____

_____

10. Switch back to the Propeties window for Fary.



Click the Profile tab.

connect drive Y: to \\serverSN\user\fary

(specify the name of your server instead of serverSN as in this example)
Click OK.

11. Click Refresh button and Active Directory for Users and Computers.

12. Look in the Windows Explorer window.
    What are the directories listed under D:Fary?

13. Access the folder Fary and display the security permission (right-click the folder and choose Propeties, then click the Security tab).

14. Who has Full Control rights to this sub-folder?

15. Close Windows Explorer.

16. Log on to the client computer as Fary.

17. What, if any, is drive Y: mapped to (double-click My computer to list the available drives)?

    _____

    _____

    _____

18. Log off the client computer.

Summary

Home directories allow users to store their files on the network. This is especially suited to roaming users.

LAB 13

DISK QUOTAS

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German - Malaysian Institute.

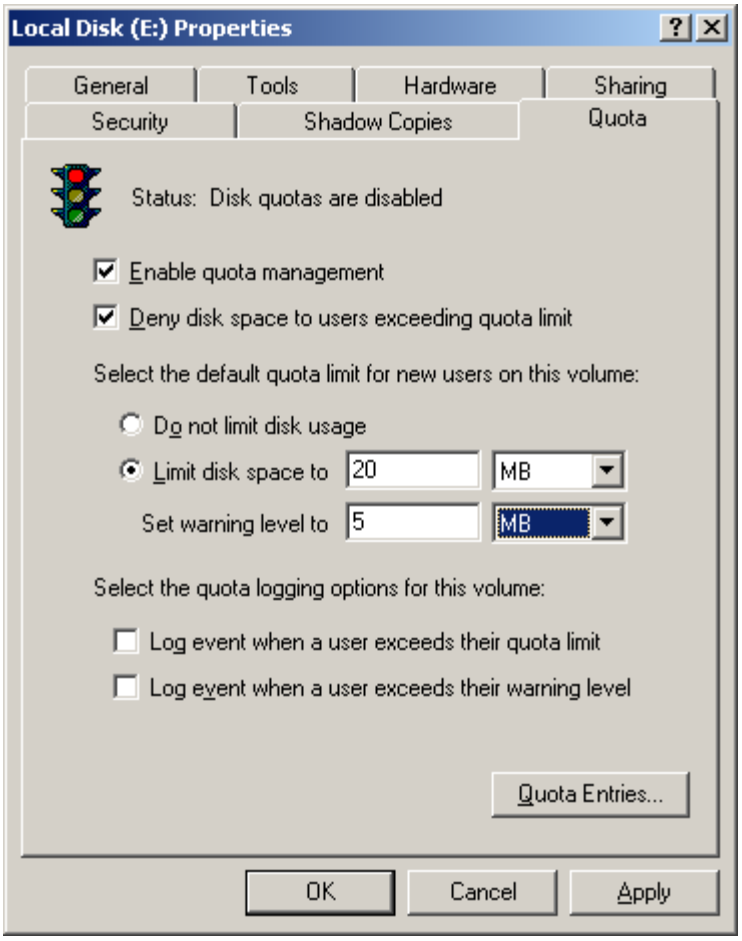
Prepared by :
N MUHAMMAD ASNAWI B. MOHD ABD. GHANI

LAB 13. DISK QUOTAS

Disk Quotas

In this exercise you will apply disk space restrictions to users.

1. Log on the servers as administrator.
2. Run Windows Explorer. Right-click on the Drive E: icon and select **Propeties**.
3. lick the **Quota** tab**.**



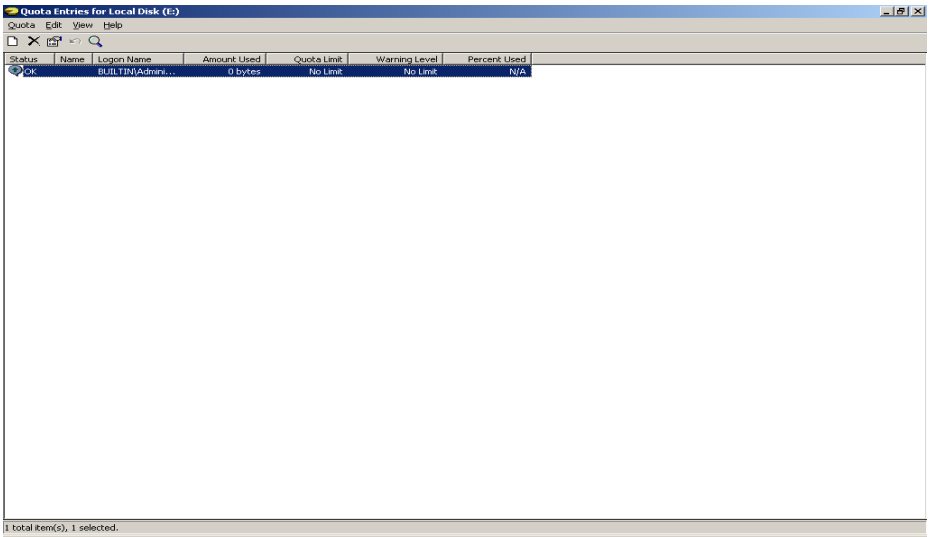
Enable the check box **Enable quota management**.

Enable the check box Deny **disk space to users exceeding quota limit.**

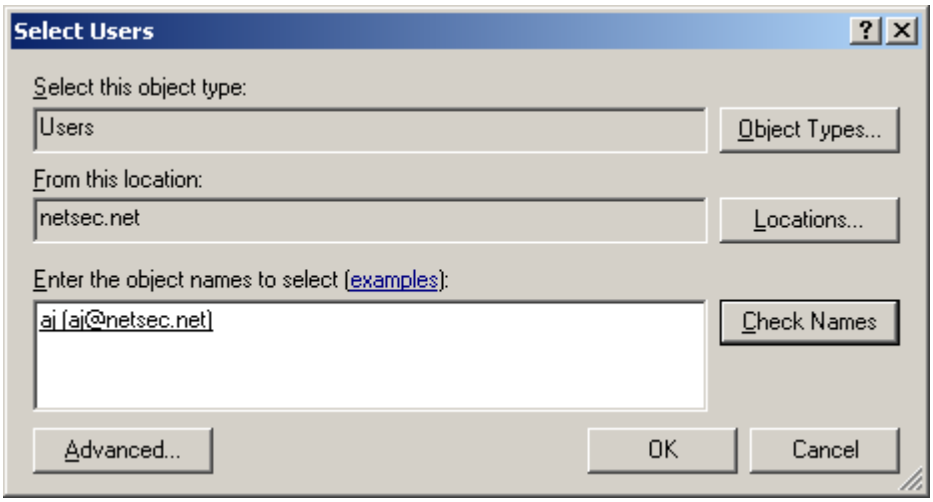
Set the **Set warning level** option to **5MB.**

Click the **Quota Entries** button.

4. A list of quota entries will be displayed (this may take a few minutes to complete).



5. On the Menu Bar, click Quota, New Quota Entry.
6. Key-in fary and click “Check Names” button.

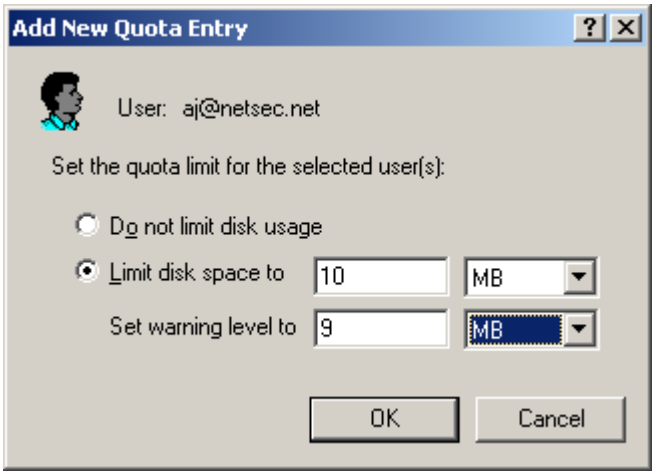


7. Click **OK**.

8. Set the following parameters.

   Enable the option **Limit Disk Space to** and set the value to **10MB.**

   Set the values for **Set warning level to** option to **9MB.**



   Click **OK**.

9. Close the Quota Entries window.

10. Click OK to close the Local Disk E: Propeties window.

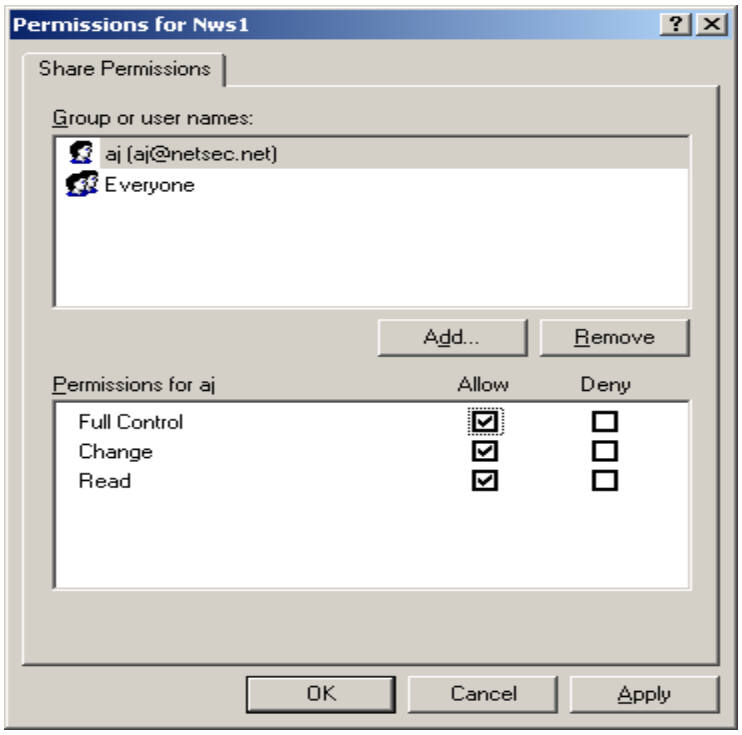
11. Click OK to enable the quota system now.

12. Right click folder **fary**(folder you created in the earlier exercise), select "sharing and Security".

13. Click "Permission" button.

14. Click "Add" button.

15. Key-in **fary** in the box and click "Check Names" button.
16. Click **OK**.
17. Set permission for Fary **Full Control**.



18. Click OK.
19. Click OK.
20. Log on the client computer as fary
21. Access drive Y:where fary home directory is located.
22. Use Windows explorer to access the C:\\Windows\Web\Wallpaper sub-folder.
23. Click on the file **Bliss.bmp** and copy it (it is around 1407 KB in file).
24. Use windows Explorer to access the Y:
25. Paste the file **Bliss.bmp** into Fary home directory on drive Y:
26. Copy and paste another file into Fary home directory on drive Y:
27. Continue pasting copies of the file until the disk quota warning appears.
28. Remove the all file from Fary home directory on drive Y:
29. Log off the client computer.
30. Log off the server.

Summary

Disk quota allow administrator to restrict disk space to users so that disk space can be effectively managed.

LAB 14

Managing Software Applications

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Prepared by :
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LAB 14. MANAGING SOFTWARE

In this exercise you will deploy software to a Windows 2003 client computer. You will deploy WinZip, a file compression program that does not have an associated MSI file. This means you will need to create a ZAP file in order to publish the application.

In the second part of this exercise you will publish this software to members of the **NWS2 OU**, then test the deployment of the software.

Estimated time to complete : 45 minutes

Establish a Software Distribution Point

To support this exercise, you will need a shared folder on the network that contains the software applications that will be deployed.

1. Log on to the server as Administrator.
2. Run windows Explorer.
3. Create a new sub-folder on drive E: named SoftDistx (where x presents your group ID).
4. Share this folder as Esoftware. Ensure that the NWS2 OU and Administrator has read access to the share.

Copy Software Application files to the Software Distribution Point

The next step is to copy some software applications to the distribution share.

5. Download file **WinRar 3.7** from NOZ server at http://download.nos.net/compression.htm and save to the software distribution share point (or download it from the Internet from www.winrar.com)
6. Download fie **sample.rar** from NOZ server at http://download.nos.net/compression.htm to the software distribution share point (or create a rsr file that has a readme.txt file in the archive).

Create A ZAP file for the application

To deploy the WinRar application. deploy the WinRar application, you will need to create a ZAP file, as no MSI file is available.

7. Create New text document inside E:\SoftDiskx, and rename the text document as winrar.zap.
8. Double click winrar.zap. This will load the Notepad Editor. Type the following text into the file.

[Application]
FriendlyName = Win Rar
SetupCommand = \\server1\esoftware\wrar370.exe
DisplayVersion = 3.7
Publisher = Win Rar
URL = http://www.winrar.com
[EXT]
RAR=

Publish the Software Application to Users of the Production OU

In this step, you will edit the group policy for the NWS2 OU and specify a new software installation for users.

9. Run Active Directory for Users and Computers.
10. Right-click the **NWS2** OU and select Properties.
11. Click the Group Policy tab, create new group policy (click **New** button) name it as **NWS2 Policy**.
12. Highlight the group policy and click the **Edit** button.
13. Expand the User Configuration tree, Software Settings, right-click Software Installation and select New.
14. Click Package.
15. Enter the path to the shared folder containing the ZAP file (\\server1\esoftware\winrar.zap)

16. In the Files of Type box, click ZAW Down-level application package (*.zap)
17. Click the winrar.zap file to select it.
18. Click Open.
19. Select published.
20. Click OK.
21. Click Refresh and close the Group Policy editor. Close the NWS2 properties window.
22. Click Refresh and close Active Directory for Users and Computers.

Test software deployment

In this step, you will log on to the client computer and test to see if the software can be deployed. In order for the software to install however, the user need sufficient rights on the local computer.

21. Log on to the client computer as Administrator.

Select Start; Control Panel.

Click Perfomance and Maintanance.

Click Administrative tools and run Computer Management.

Expand System Tools; Local Users and Groups; Groups.

Double-click Power Users and click the Add button.

Type fary in the box and click Check Names button.

Enter username fary and her password.

Click OK.

Click OK.

Exit Computer Management.

Log Off the client computer.

22. Log on to the client computer as fary.
23. Copy the file from \\server1\esoftware\sample.rar to the folder e:\temx (where x is the number of your group ID) on the local computer.

24. Double-click the file sample.rar. What happened?

25. Log off the client computer.

Installing Application with MSI support

In this exercise you will deploy FrontPage 2003

1. Log on to server as Administrator.
2. Access the E:\SoftDistx folder
3. Create a subfolder called FrontPage.
4. Insert the FrontPage 2003 CD and copy all files and folders in the FrontPage 2003 CD to the E:\SoftDistx\FrontPage folder.
5. Run Active Directory Users and computers.
6. Right-click the NWS1 OU and select **Properties**.
7. Click on the **Group Policy** tab, select the group policy file and click **Edit**.
8. Expand Users Configuration; Software Settings. Right-click on Software Installation and select **New**, **Package**.
9. Access the frontPage folder and select **FP11.msi**. Click **Open**.
10. Select Advanced. Click OK.
11. Click thee Deployment tab. Select Assigned. Click OK.
12. Click Refresh and close the group policy editor.
13. Click OK to close the NWS1 properties windows.
14. Click Refresh and close Active Directory for Users and Computers.

Now you will test the deployment of FrontPage 2003 by logging onto the client computer as a member of the NWS1 OU.

15. Log on to the client computer as Administrator.
16. Add to the Power Users group.
17. Log off the client computer.
18. Log on to the client computer as
19. Click Start, All Programs, Microsoft Office, Microsoft Office FrontPage 2003. Note how FrontPage appears on the start menu.

20. The installation process will begin. When requested, enter the CD key and click OK.

21. After running FrontPage 2003, log off the client computer.
22. Log on to the client computer as
23. Is FrontPage 2003 available on the Start menu? YES NO
24. If your answer is NO, why do you think the software applictaion it is not available.

---

25. Log off the client computer.
26. Log off the server.

Summary

In this exercise you deployed a software application to a group of users. The application was not supported by Windows Installer so required you to create a ZAP file.

The software application and Zap file were placed on a network share. This software was then associated with a group policy for the Organizational Unit. The software deployment was then tested when a user of the OU logged onto a client computer.

In installing software on the client computer, the installer needed the required permissions. In this exercise, the users were made members of the Power Users group to enable the installation of the software. In actual use, members would be set up with the required permissions, rather than perhaps being made a member of this group on the local computer.

Managing the software distribution can simply the administrator of the network and ensure that users only get the applications that have been assigned to them.

Lab 15

Viewing Events

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German - Malaysian Institute.

Prepared by :
Mohammad Nadzir bin Nazuwan

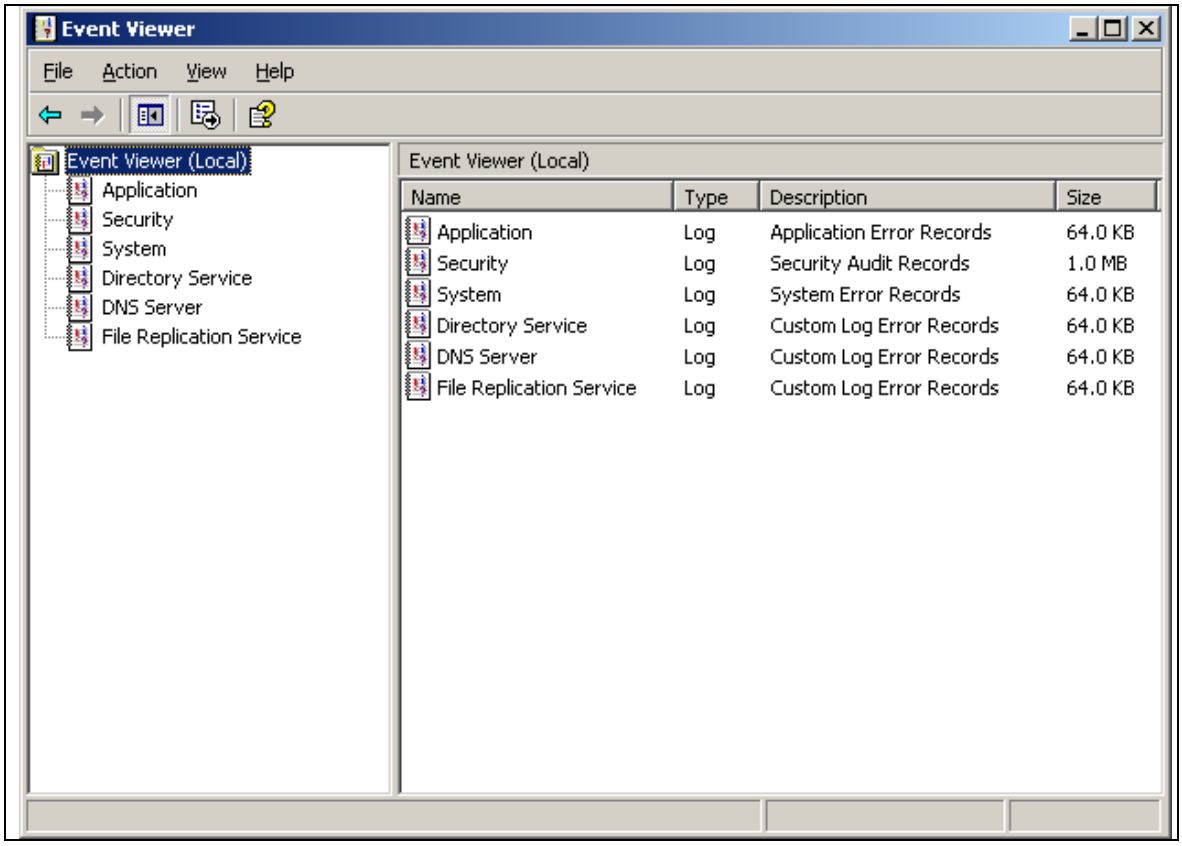
LAB 15: VIEWING EVENTS

In this exercise you will look at events generated on the server. This is important because when there is a problem, often the cause is logged by the system. The event logs are a good source to look for problems in configuration or access.

Estimated time to complete: 15 minutes

Running Event Viewer

1. Log on to the server as Administrator.
2. Run Event Viewer by accessing Start; All Programs; Administrative tools; Event Viewer
3. The Event Viewer window displays the current event logs. There are a number of logs available.



Viewing the Different Log Files

To view events, you need to select a specific log file.

4. Click the System Log to select it.

5. Note the large number of events that are listed in the right window.

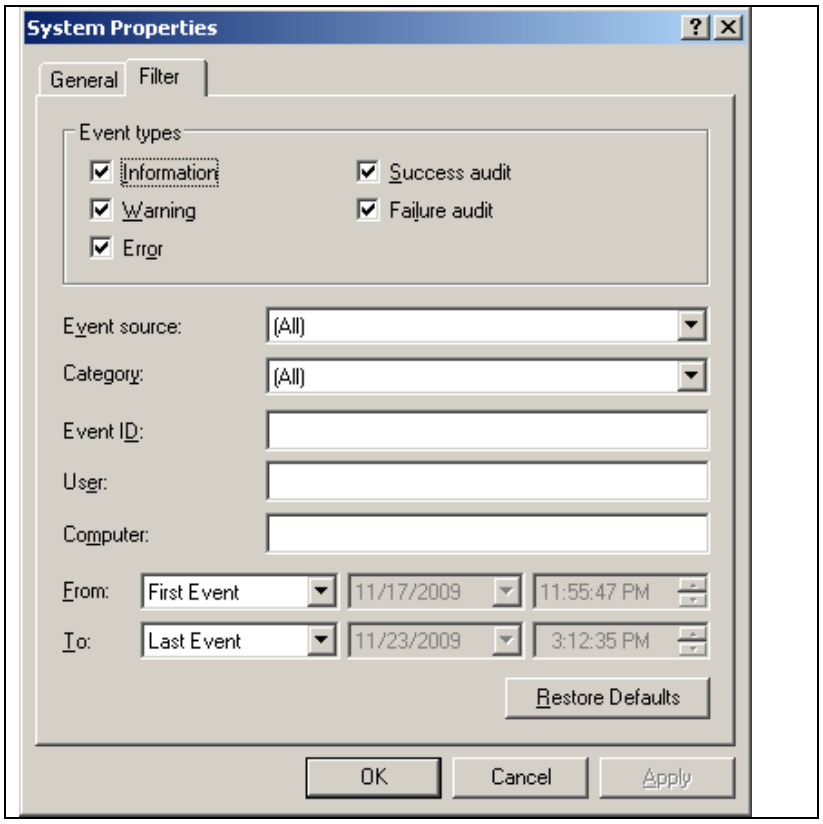
6. All events have a source and category. Note these two columns in the window.

7. It is handy to sometimes restrict the events being viewed to just those events that are of interest.

Filtering Events

In this exercise you will use the filtering function to display only those events of interest. Often the event log has hundreds of events listed, so you need the ability to look for only those events that are relevant to what you are trying to resolve.

8. Click **View** then **Filter** from the menu.

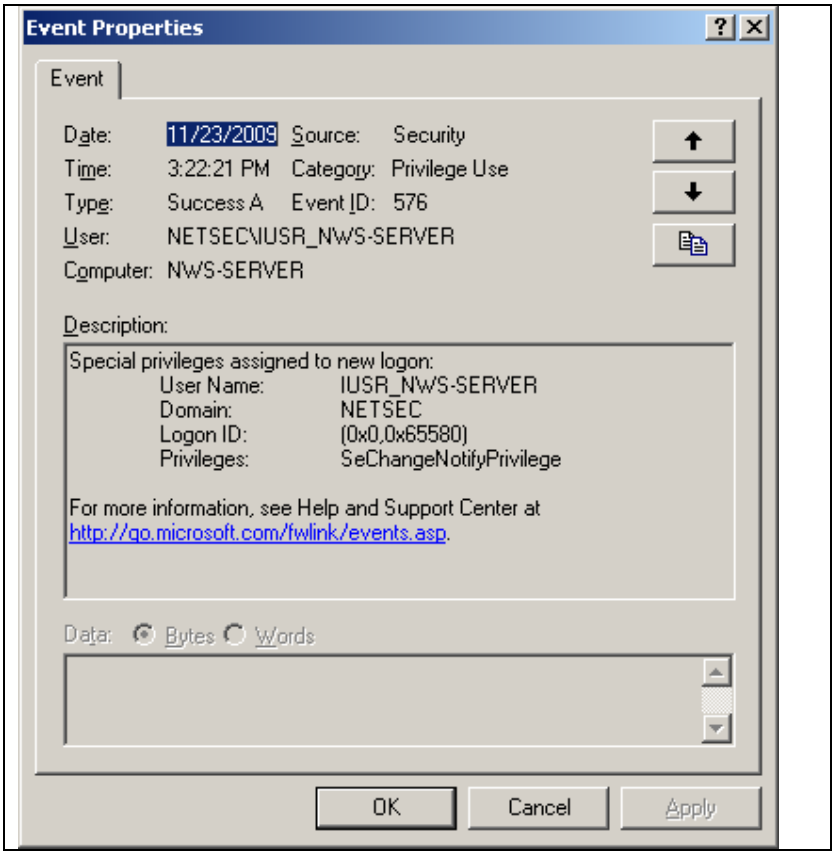


9. In Event Source, select **Security.**

10. Set the Category to **Privilege Use.**

11. Click **OK.**

12. Note that only the Security events are now listed. Double-click the event at the top of the list.



13. The dialog box gives an indication of the event (including the event ID, which is helpful when exploring TechNet as to possible problems).

14. Click OK.

15. Close event viewer.

16. Log off the server.

Summary

Windows 2003 logs activity to event logs. These events can be viewed with Event Viewer. Typical events are printing, security, auditing, logon and logoff, as well as other events generated by application software or other services such as DNS. Events are helpful in determining problems with configuration or security.

Lab 16

Auditing

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TTO, Network Security

German-Malaysian Institute

Prepared By,

Wan Muhammad Irsyad b. Wan Hashim

LAB 16. AUDITING

In this exercise, you shall look at enabling auditing on selected resources, so that their usage and access can be monitored. You will use event viewer to view the logged accesses. Often, if you find that you cannot resolve problems in user access, enabling auditing and viewing the audit logs with event viewer can help you to determine the cause of the problem.

Estimated time to complete: 30 minutes.

Enabling Auditing for the Server using Group Policies

You will define a Group Policy for Domain Controllers that specifies Auditing.

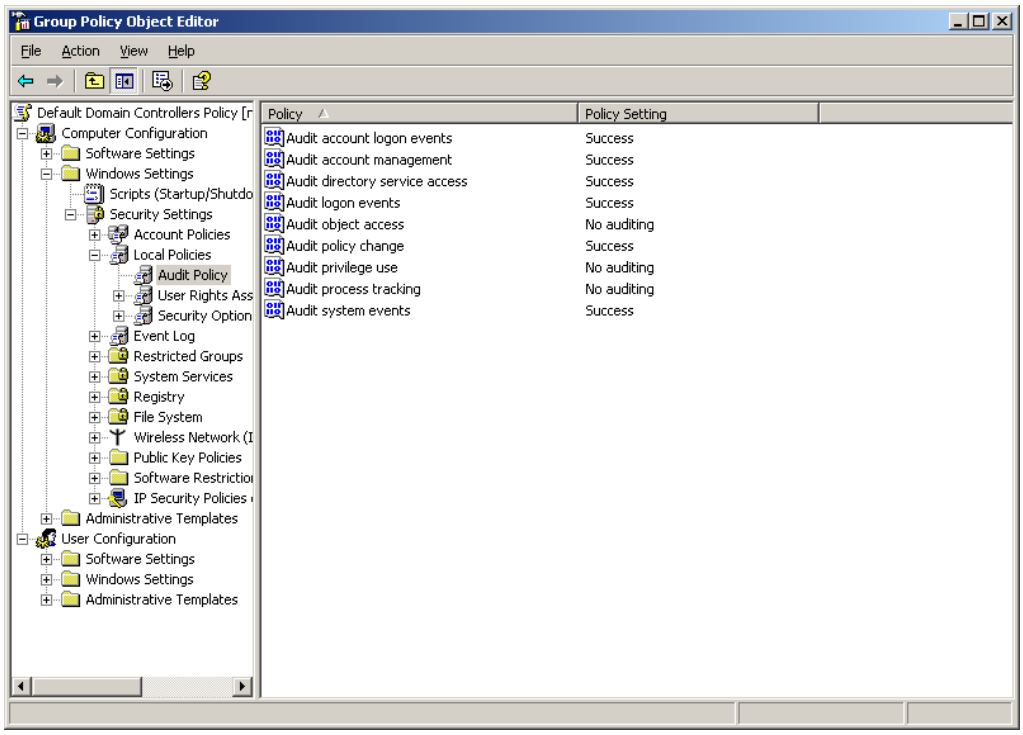
1. Log on to the server as Administrator.

2. Run Active Directory for Users and Computers. Expand Domainx.com

3. Right-click Domain Controllers and select Properties. Select Group Policy and click Edit button.

4. Expand Computer Configuration, Windows Setting, Security Settings, Local Policies, Audit Policy.

5. Enable the following events



Audit account logon events, success.

Audit logon events, success and failure.

Audit object access, success and failure.

6. Close the group policy editor.

7. Click **OK** to close the Domain Controller Properties window.

8. Close Active Directory for Users and Computers.

Set the Local Security Policy on the Server to Enable Auditing

9. Click Start, All Programs, Administrative Tools, Domain Security Policy.

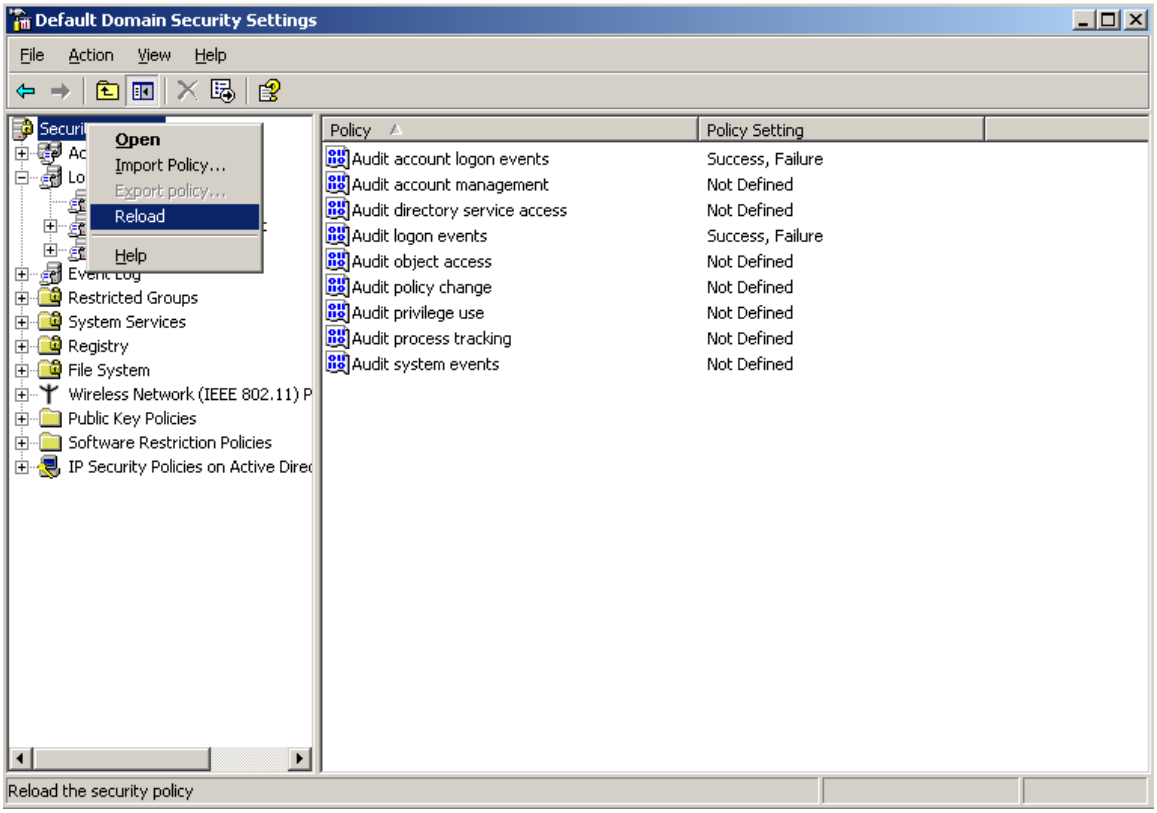
10. Expand Local Policies, Audit Policy.

11. Enable the following events

    Audit Account Logon events – Success and Failure

    Audit Logon Events – Success and Failure

12. Right-click Security Settings at the top of the tree. Select **Reload**.



13. Close the Local Security Settings.

Set Auditing at the file object level

14. Run Windows Explorer and expand drive E:

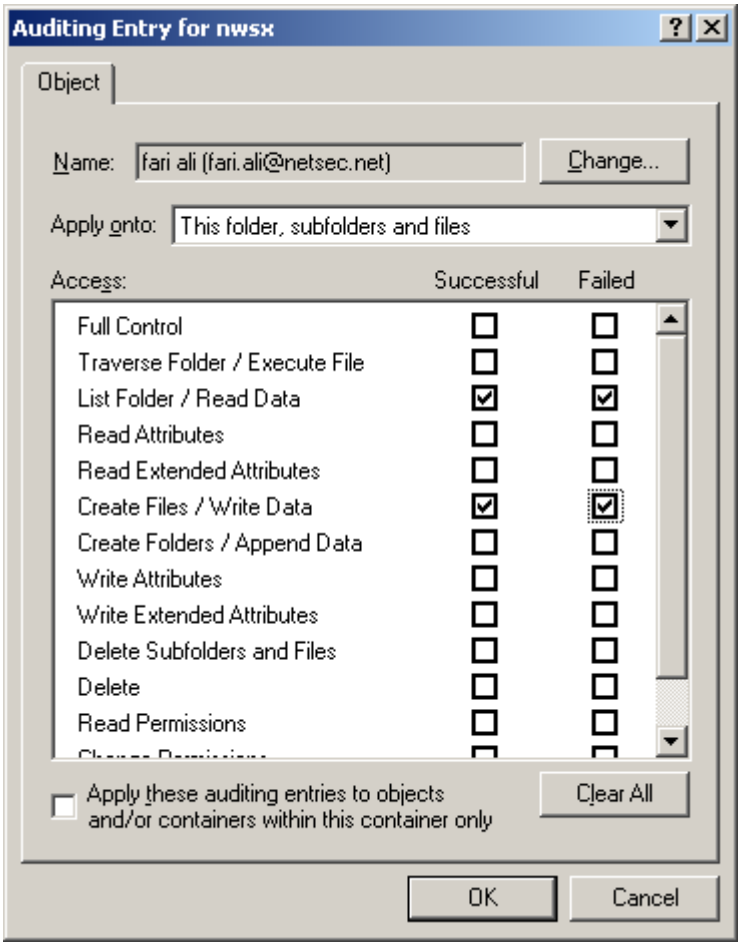
15. Right-click E:\nwsx and select **Properties**. Click the Security tab.

16. Click the **Advanced** button.

17. Click the **Auditing** tab.

18. Click the **Add** button, type **fari.ali** in the box, and click **Check Names** button. Click OK to add his account.

19. Enable the following options.



List Folder/Read Data [success and failure]

Create Files/Write Data[success and failure]

20. Click **OK**.

21. Click **OK**.

22. Click **OK**.

23. Log off the server.

Access the resource to generate the audit event

Now it is time to test the auditing. What you did in the previous steps was setup a group policy for domain controllers. You then enabled auditing on the server using Local Security Policy. Next, you enabled auditing on the files and sub-folder E:\tempx. In the next step you will log on and access this resource, thus generating an audit event.

24. Log on to the server computer as fari.ali.

25. Access the folder E:\nwsx.

26. Create a new text file called Readme.txt and write a short sentence. Save the file. Close notepad.

27. Log off the server.

View the audit events

In the last step, you accessed the resource and this would have generated an audit event. These events are stored in the security log and are viewed with event viewer.

28. Log on to the server as Administrator.

29. Run Event Viewer and access the Security Log.

30. Filter by Security, then Object Access.

31. Look for the first event of type 560.

32. What was the name of the object that was opened for event type 560?

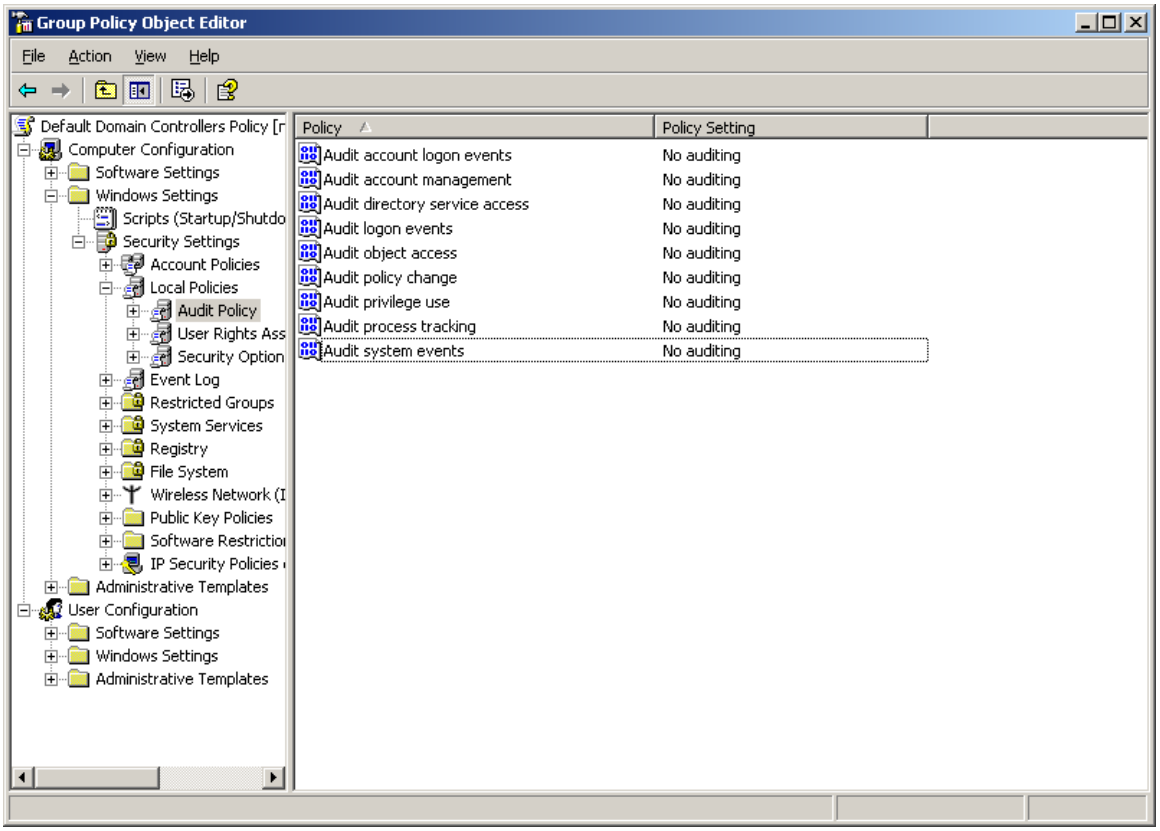
    _____

33. Close Event Viewer.

Disable Auditing

Auditing places a performance penalty overhead on the computer. In this step, you will disable auditing.

34. Access the group policy for the Domain Controllers and change the auditing to No Auditing.



35. Run Windows Explorer, right-click the folder E:\tempx and select Properties.

36. Click the **Security** tab.

37. Click the **Advanced** button.

38. Click the **Auditing** tab.

39. Click **Remove**.

40. Click **OK**.

41. Click **OK**.

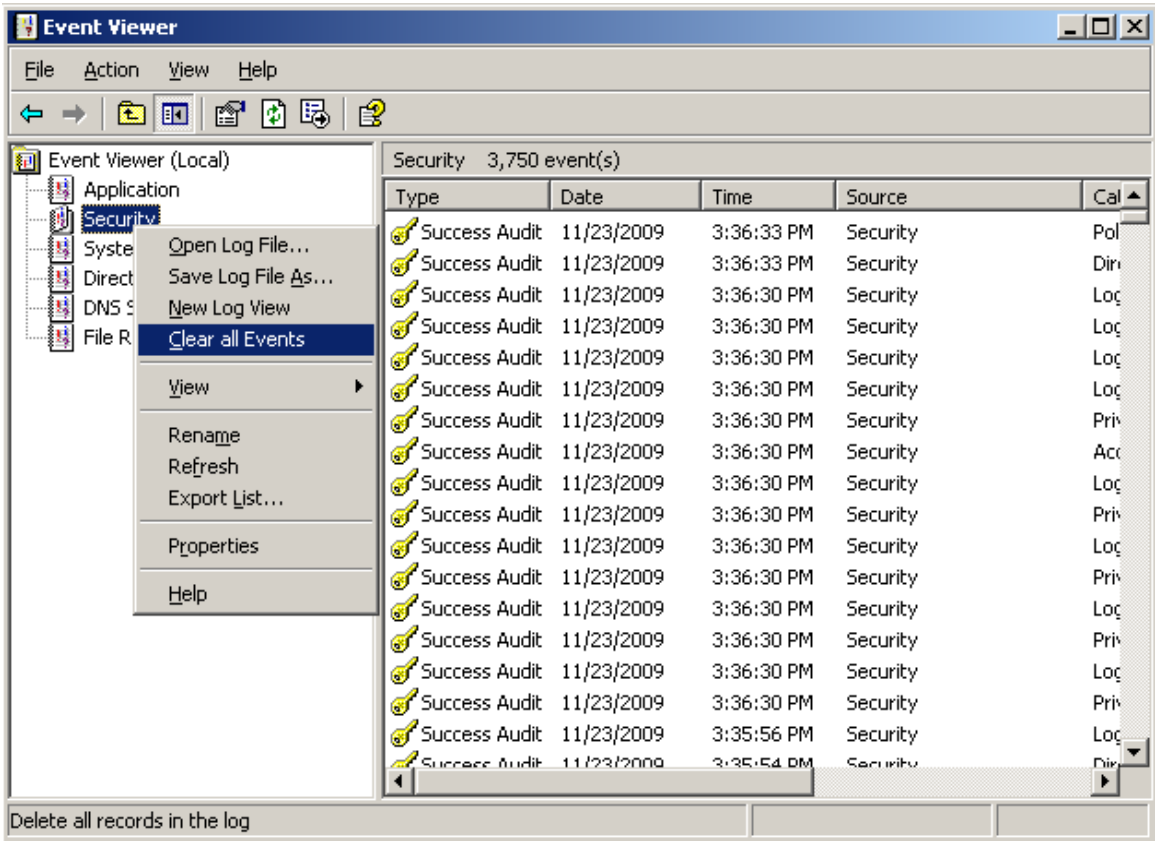
42. Close all application open window.

Clear the Security Log Event

In this step you will clear all the events in the Security Log.

43. Run Event Viewer and right-click the **Security** Log.

44. Select **Clear All Events**.


Event Viewer
File Action View Help
Event Viewer (Local)
Application
Security
Open Log File...
Save Log File As...
New Log View
Clear all Events
View
Rename
Refresh
Export List...
Properties
Help
Security 3,750 event(s)
Type Date Time Source
Success Audit 11/23/2009 3:36:33 PM Security
Success Audit 11/23/2009 3:36:30 PM Security
Success Audit 11/23/2009 3:35:56 PM Security
Delete all records in the log

45. Click **NO** so that the events are not saved.

46. Close Event Viewer.

Summary

Both directories and files can be audited. When auditing is enabled: events are written to an event log, which can be viewed in Event Viewer.

Lab 17

Managing Printing

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Prepared by :
MUHAMMAD HAFIZI BIN MD YUNOS

Lab 17 : MANAGING PRINTING

In the exercise, you shall look at creating a local printer on the server and access that printer remotely from the client computer. This exercise used an HP LaserJet 1015 printer attached to the network.

Estimated time to complete : 45 minutes

Installing the printer

1. Log on to the server as administrator
2. Click start; Printers and faxes.
3. Double-click the **Add Printer** icon to run the Add Printer Wizard.
4. Click **Next**.
5. Select Local Printer. Disable the "Automatically detect and install my Plug and Play printer" checkbox. Click Next.
6. Select **Create a new port**.
7. Select Standard **TCP/IP Port** in the "Type of port." Drop down menu. Click Next.
8. Click Next.
9. Type printer IP address (192.168.10.23).
10. Click next.
11. Click finish.
12. In the list of Manufacturers, select **HP**.
13. In the list of printers, select **HP Laser Jet**. Click **Next**.
14. Enter **HP LaserJet 1015** as the printer name. Click **Next**.
15. Enter **HPLAser** as the shared printer name. Click **Next**.
16. Enter "**Stkm**" for the location field. Click **Next**.
17. Select **NO** so that a test page is not printed. Click **Next**.
18. Click **Finish**.
19. Windows server 2003 now copies the necessary drivers and installs the printer.

Viewing the installed Printer Setting

20. Open Printer and Faxes.
21. Right- Click the installed printer and select **Printing Preferences**.
22. Click the **Advanced** button.
23. Set the paper siza to that specified by our instructor.
24. Click **OK** to close the dialog box.
25. Click **OK** to close the printer preferences windows.

Assign a print manager for the printer

In this section, you will assign a user to manage the printer. This printer is delete jobs and performs other administrator task.

26. Right-click the installed printer and select **SHARING**. You will see that windows server 2003 has already shared the printer.
27. Click the **Security** Tab.
28. The current security setting for the printer is similar to you will note that everyone (all users) has print access, whilst Administrators have all rights. Print operators also have all rights.
29. Click the **add** button. Click **advanced** button. Click **find now** button. Select from the list and click **ok**. Click **ok**.
30. Give ocah.blue full right to this printer. Ocah should have full right to the printer. This effectively makes her a manager for this printer.
31. After setting the right as indicated, click **OK**.
32. Close the printer windows.

Locating Printers using Active Directory

In this section you will use active directory to locate printers.

1. Run Active Directory for User and computer.
2. From the menu, click Action, Find.
    a. In the find list, choose **Printers.**
    b. In the Location Field, enter **GMI**.
    c. Click **Find Now**.
3. What information did the search result display.

    Name _____
    Location _____
    Model _____
    Server Name _____

4. Close Active directory For User And Computer.

Accessing the printer from the client computer

In this section you will log on to the client computer and set up access to the shared printer on the server.

5. Log on to the client computer as ocah.blue
6. Click Start ; Printer And Faxes.
7. Click the add printer Icon to run the Add printer wizard. Click **Next**.
8. Select **Network Printer**. Click **Next**.
    a. Select **Find Printer in the directory**.
    b. Click Next.
9. This option makes finding a printer easier as you do not need to know the name of the server on which the printer is located.
    a. Enter **Stkm** in the location field.
    b. Click **Find Now**.
    c. Select the printer in the search result at the bottom of the windows.
    d. Click **OK**.
10. Windows XP professional is now ready to connect to the specified printer.
    a. Click finish.

Setting Client document defaults

11. Open Printer and Faxes.
12. Right-click the printer and select **Printing Preferences**.
13. Click the **Advanced** button.
14. Set the paper size to A4.
15. Click **OK**.
16. Click **OK**.

Printing a file

In this section you will print a page to the printer

17. Right-click the printer and select **Properties.**
18. Click the button **Print test Page**.
19. Click **OK**.

20. Click **OK**.

Managing the printer

In this section you will manage the printer by deleting all print jobs, and then pausing the printer.

21. Make the printer **ERROR** (takeout printer toner).
22. Open notepad. Type a few words and print.
23. Right-click the printer and select **Cancel all Documents**.
24. Click **Yes**.
25. Log off the client computer.
26. Log on the client computer as mad.akmal.
27. Connect to the printer in the same ways you did previously when logged on as ocah.
28. Right-click the printer and select cancel **all document**.
29. Click **Yes**.
30. What was the message displayed?

    _____

    _____

    _____

31. Why do you think this is so?

    _____

    _____

    _____

32. Log off the client computer.
33. Log on the client computer as ocah.blue.
34. Right-click the printer select **pause Printer**.
35. Log off the client computer.

SUMMARY

In this exercise you established a network printer and connected to it using print manager responsible for the printer was established and you tested management function. You also learnt to locate a printer using the directory, and to set document defaults such as page size for the printer.

Lab 18

Other Administrative Tools

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Muhammad Rafi'uddin Bin Rozman

LAB 18. OTHER ADMINISTRATIVE TOOLS

In this exercise you will look at other administrative tools.

Backup

Automated System Disk

Disk Management – Chkdsk and Defrag

Safe Mode

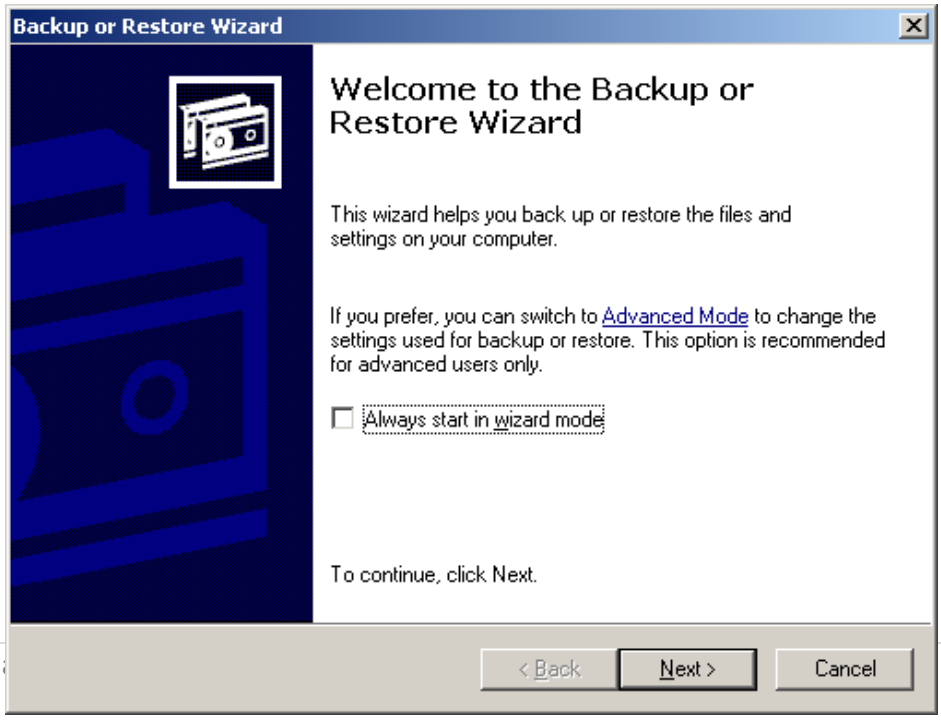
Directory Service Repair mode

Estimated time to complete : 40 minutes

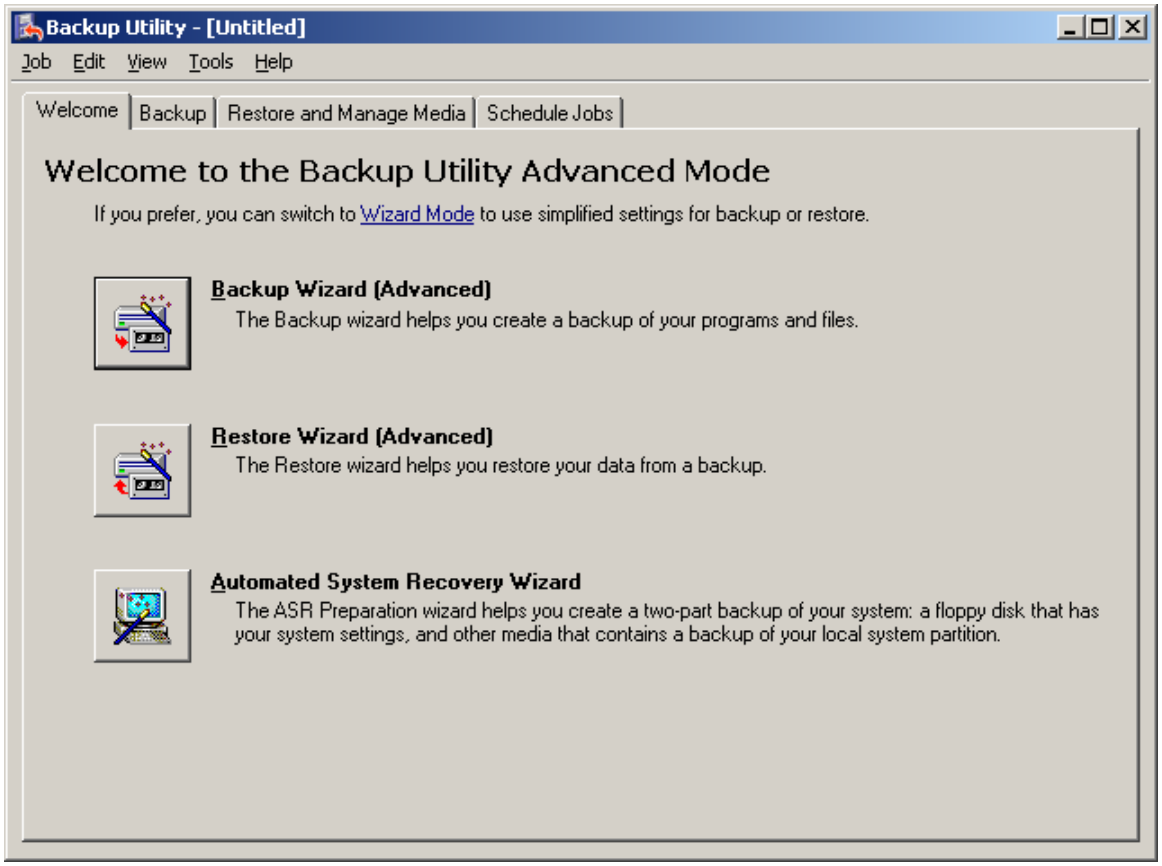
BACKUP

In this exercise you will use the Backup utility provided with Windows Server 2003 to perform a selective backup and restore of files.

1. Log on the server as Administrator.
2. Run Backup by selecting Start, All Program, Accessories; System Tools, Backup.
3. Uncheck **Always start in wizard mode** box in the Welcome Wizard dialog box.
4. Click Cancel.

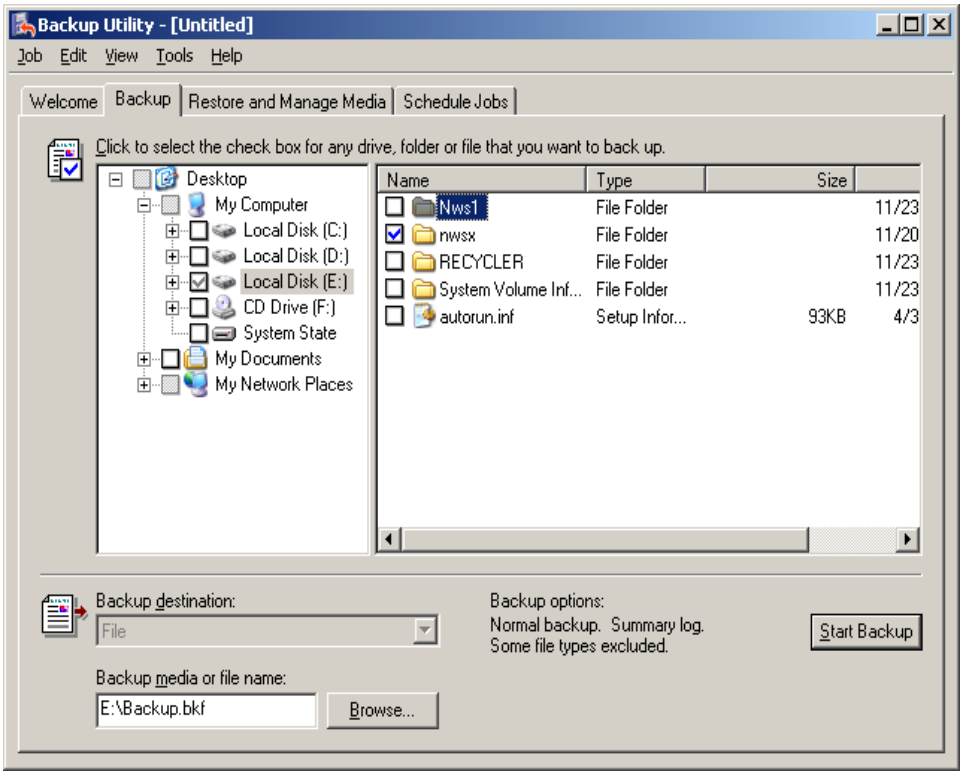


5. Run Backup again.



Backup displays the initial screen, offering the choice of Wizards, and creating an Automated System Recovery Disk.

6. Click the **Backup** tab.
7. Backup displays a list of folders and drives.

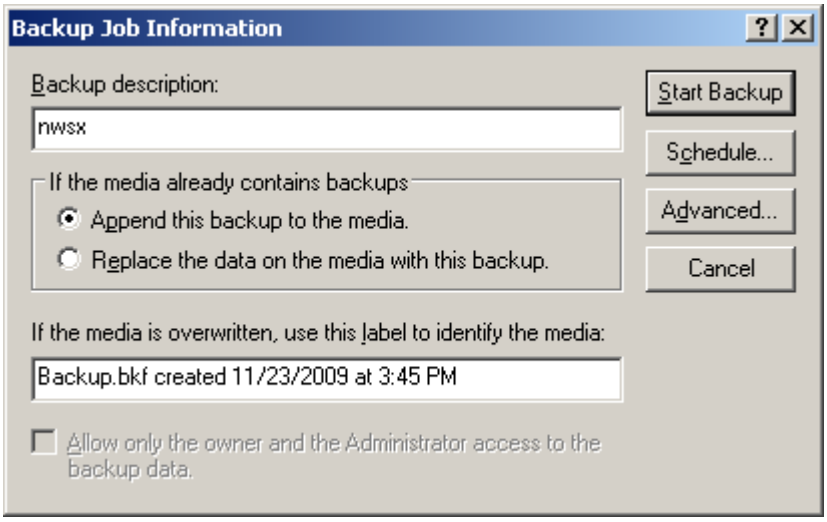


Select the folder **E:\tempx** for backup.

Specify the backup media as **E:\backup.bkf**

Click Start **Backup**.

8. Backup displays a Job Information dialog box.

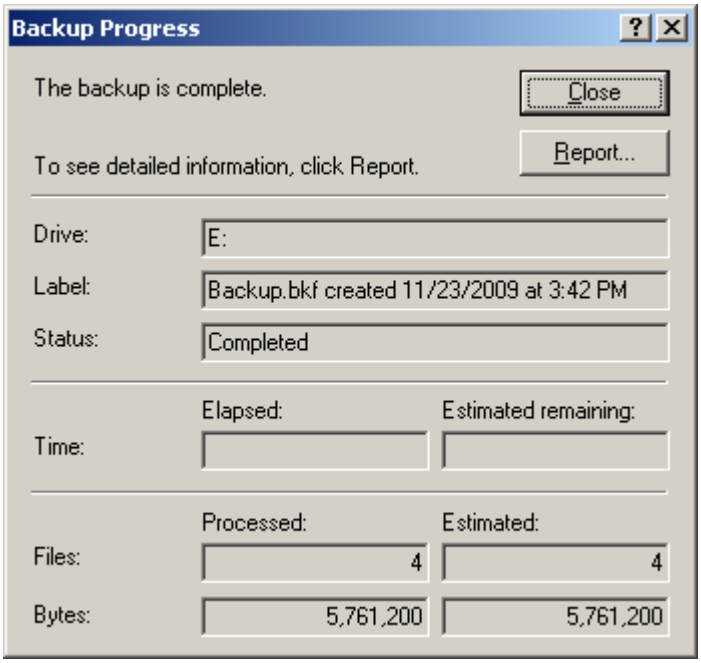


Enter the backup description as **Temp Folder**.

Click **Start Backup**.

9. Backup will display a progress box.

10. After all the files have been archived Backup displays a completion summary.

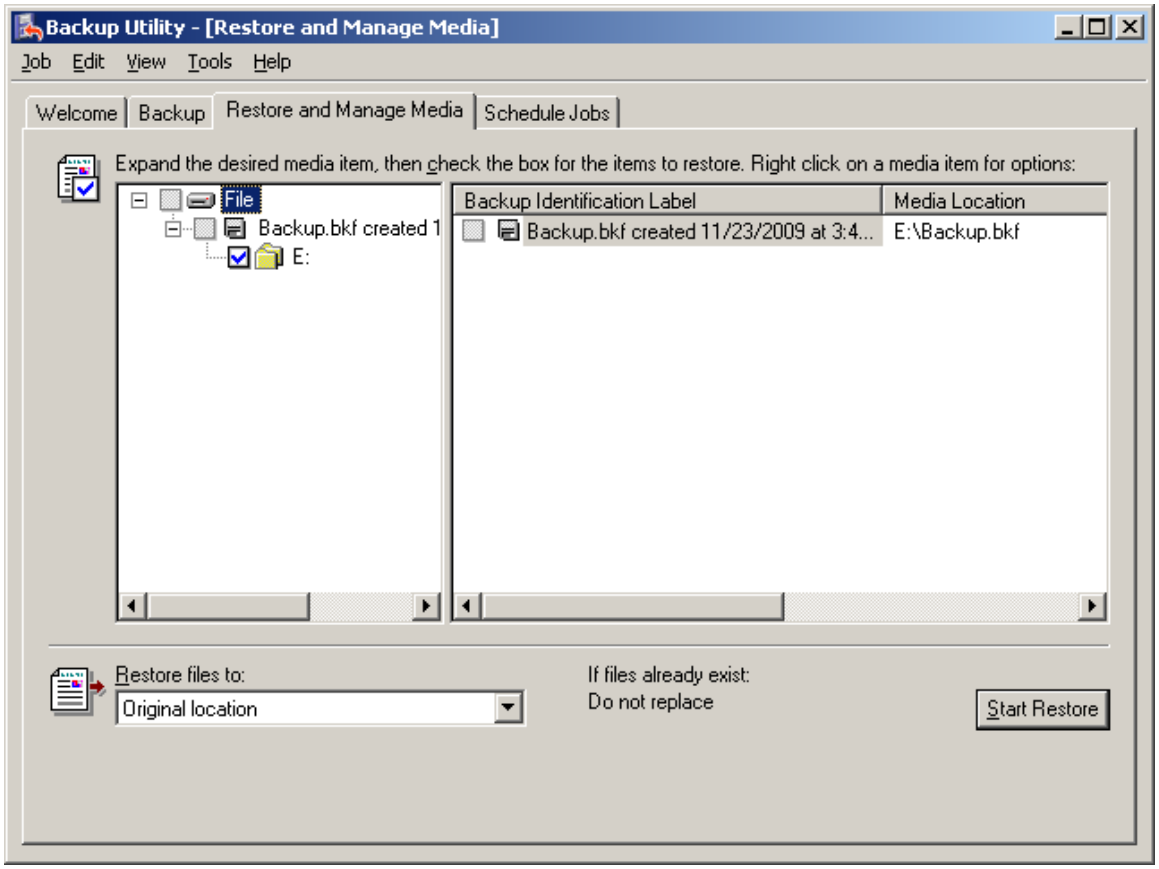


Click **Close**.

11. Open windows Explorer and confirm that the file E:\backup.bkf exists.
12. Delete the folder **E:\tempx**.

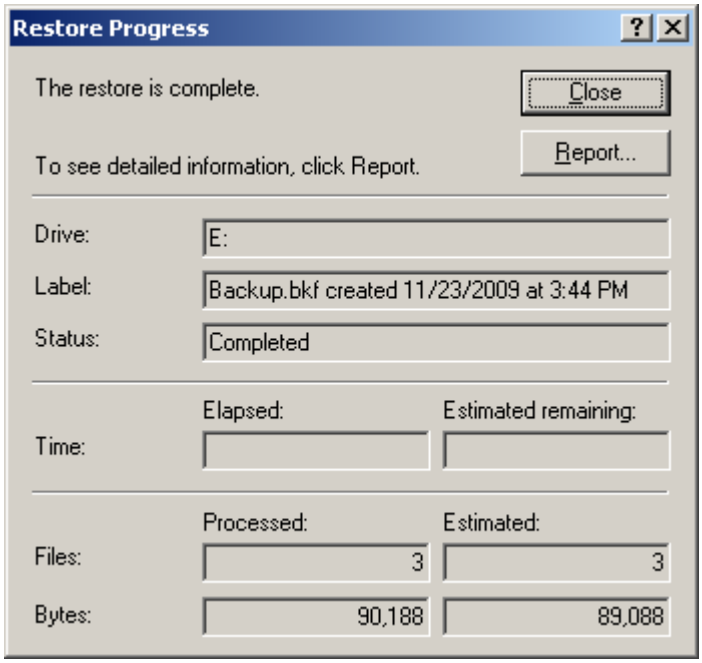
Restoring Files

13. Run Backup.
14. Click **Restore and Manage Media** tab.
15. Expand File; **Backup.bkf** and select E:



16. Click **Start Restore** button.
17. Click OK.
18. Restore will display a progress box.

19. After all the files have been restore, Restore display a completion summary.



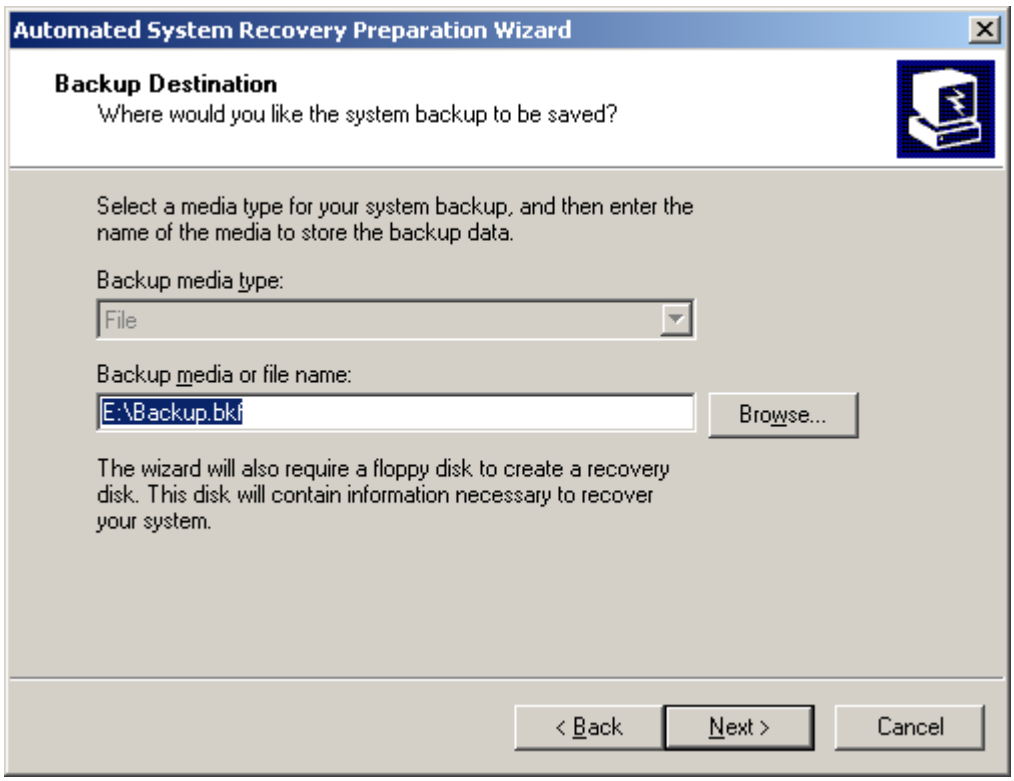
20. Click Close.
21. After the restore is complete, verify that the files exist using Windows Explorer.
22. Delete the file **E:\backup.bkf**

Leave Backup running and proceed to the next exercise.

Create an Automated System Recovery Disk

In this exercise you will create an emergency repair disk. Backup should already be running from the previous exercise. You will required a blank formatted diskette for the exercise.

1. Insert the diskette into Drive A.
2. Click Create Emergency Repair Disk from the Tools menu of Backup. Click Next.
3. Backup Destination dialog box appear. You can set different backup destination or just leave the default name.



4. Click Next.
5. Click Finish.
6. Backup will display a progress box.

7. Make sure have insert blank diskette in drive A and click OK.
8. Now Backup will create ASR diskette.
9. After ASR diskette has been created, remove the diskette and click OK.
10. Click Close
11. Exit Backup Utility.

An automated system recovery disk is useful in restarting the computer in the event of failure. A new diskette should be created after installation of service packs, upgrades or device driver changes. Label the diskette and store it in a secure location.

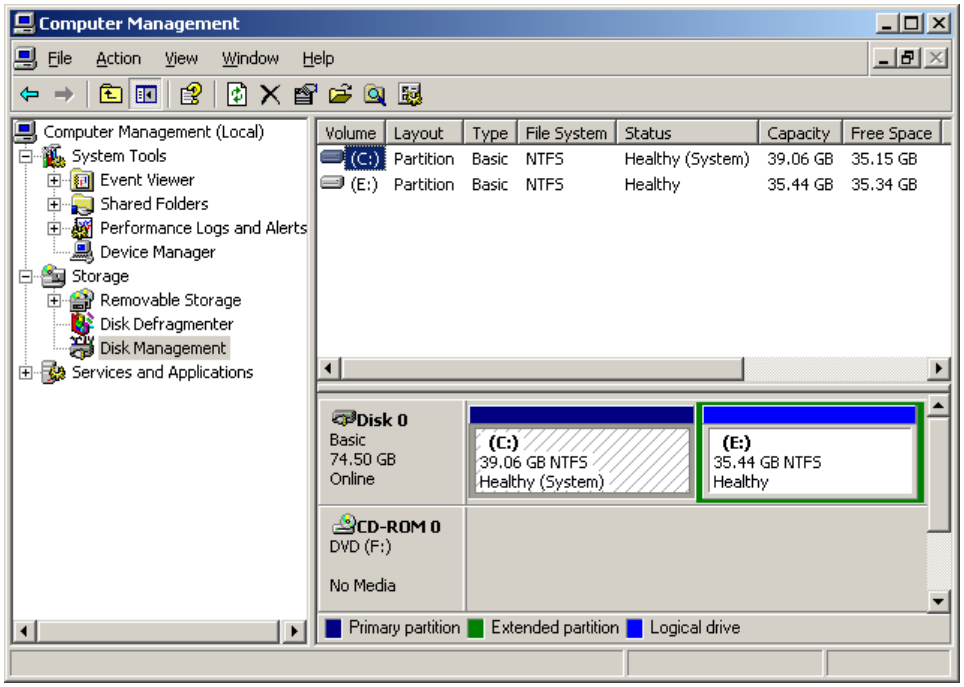
Computer Management

This is an administrative tool that allow you view the physical drives, file-systems, partitions, and logical drives on the computer. This tool can also be used to check the file-systems and defragment then.

Checking the File-system

In this section you will use Computer Management to check the file-system. If files are currently in use, Windows 2003 is unable to check the state of the file-system, and will flag the file-system for checking on the next reboot.

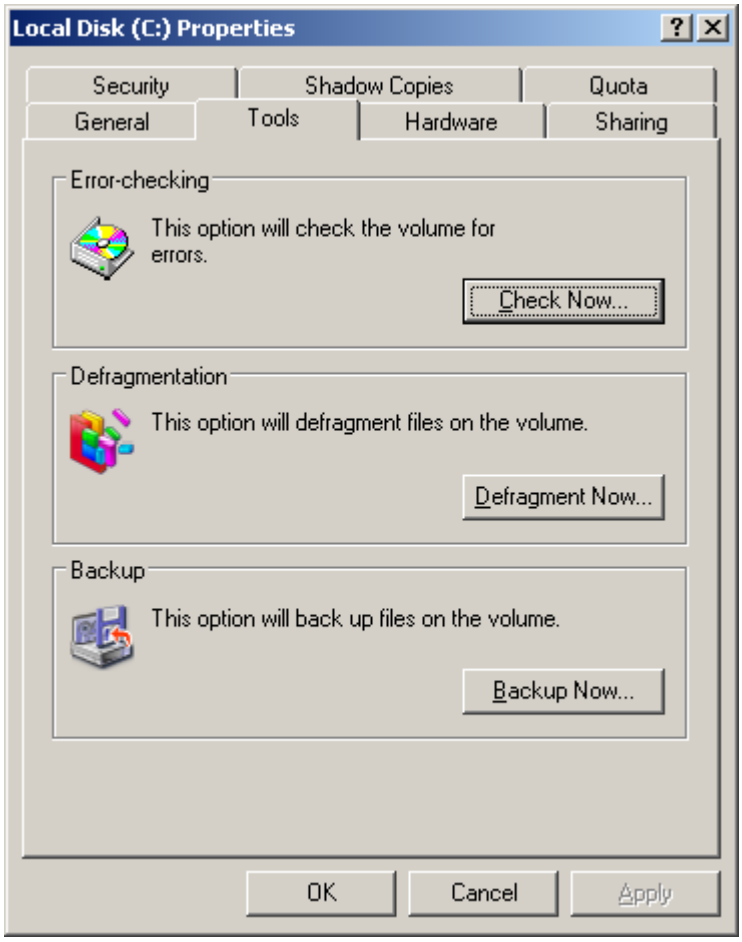
1. Click Start; All Program; Administrative Tools; Computer Management.
2. The main window will display a number of folders and options.



Expand Storage, Disk Management.

Right click drive C: and select Properties.

3. From the Properties window, click the Tools tab.

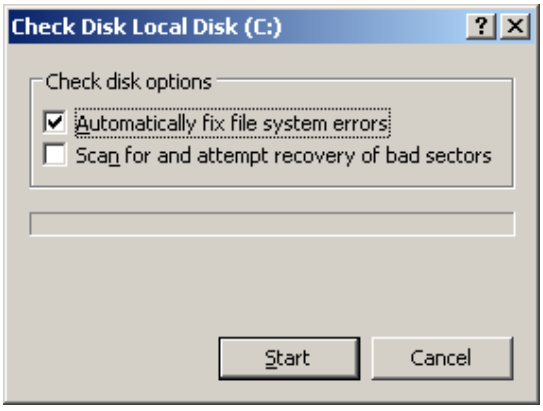


This tab display options for you to check the file-system, backup files, or defragment the drive.

Click the **Check Now** button to check the drive for errors.

Check the option **Automatically fix file system errors**.

Click **Start**.



4. If drive C: is not in use, check disk will now scan the drive for errors. If the drive is in use, you will be presented with the option to schedule the disk check when the computer is restarted. Click Yes, then restart the computer to perform the file-system check.

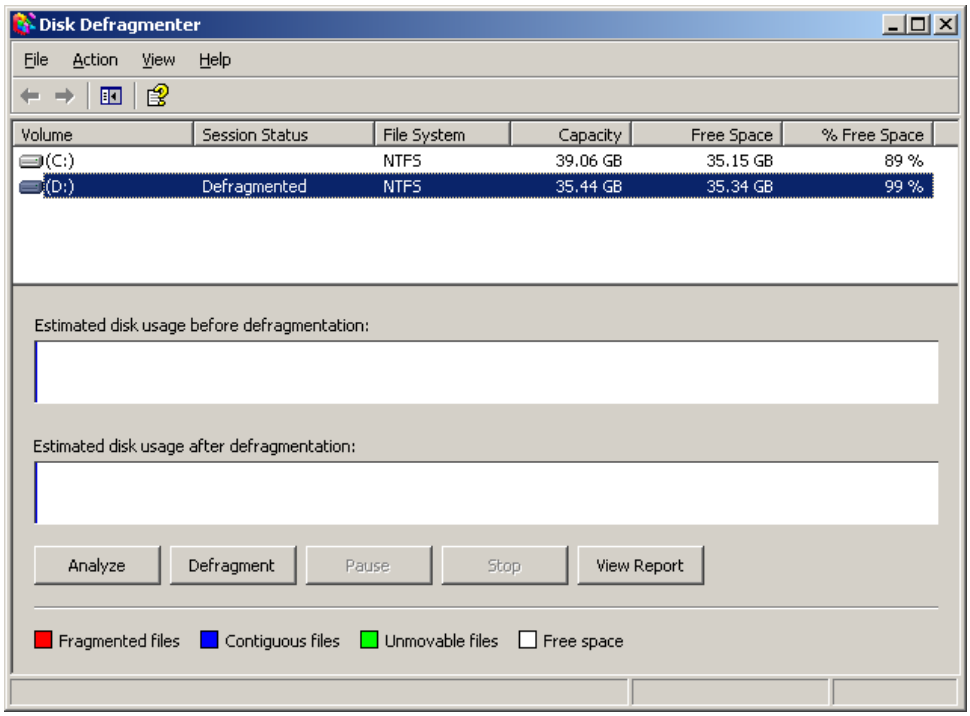
5. Use the same procedure to scan drive D:
6. Restart the computer by selecting Start; Shutdown; Restart. You will be able to observe the process of checking the file-system occurring once the computer restarts. Once this process have finished, the computer will restart and load Windows Server 2003.

The file-system should be checked on a regular basis for integrity by running chkdsk. Unfortunately, this process often requires restarting the server.

Defragmenting the file-system

Over a period of time, portions of files can become scattered over the surface of the disk and this makes accessing files slower. The process of defragmenting a disk involves moving the portions of each file back together so they are all next to each other. In this section you will defragment the current drive.

1. Log on to the server as Administrator and run Computer Management.
2. Expand Storage; Disk Management, right click drive C: and select Properties.
3. Click the **Tools** tab.
4. Click **Defragment Now**
5. Select drive **D**: and click **Defragment**.



This drive will now be defragmented by moving fragmented files so they are contiguous, making them quicker to load.

6. After the drive has been defragmented, close Disk Defragmenter and close Computer Management.

Defragmenting the file-system should occur on a regular basis to ensure files can be accessed and loaded quickly. Files in use cannot be defragmented, so administrator should schedule this to occur during periods of inactivity. A heavily fragmented file-system is often the cause of poor performance.

Safe Mode

In this section you will restart the computer in Safe Mode. This a special mode only available when the computer is restarted and you press F8 before the computer starts loading Windows 2003.

Safe mode provides a means of recovering from loading device drivers that do not work properly. For instance, an administrator might install a new graphics card, and rather than let Windows 2003 install the appropriate drivers, may select an alternative driver. This can result in a system that results in an unreadable screen display. To recover from such a possibility, Windows 2003 provides Safe Mode.

Starting the computer in safe mode

1. You should be currently logged on to the server as Administrator.
2. Select Restart from Start; Shutdown; Restart.
3. When the computer restarts, press **F8** key when it displays the boot sequence at the bottom of the screen.
4. Highlight the option **Safe Mode** and press **Enter**.
5. As this computer is dual booting, select **Domainx.net** and press **Enter**.
6. After the computer has started in safe mode, shut the computer down, and when it restarts, press **F8**.

Active Directory Service Repair Mode

In this section you will boot the computer using a startup option by pressing F8 at startup. This will allow you to enter a mode where you can repair the Active Directory, or back-up and restore Active Directory. The active directory is stored in the file ntds.dit in the folder NTDS.

As changes occur to Active Directory over time, the database file becomes fragmented. An administrator should perform a backup of the Active Directory database file.

7. Select the option **Directory Services Restore Mode** and press **Enter**.
8. As this computer is dual booting, select **Domainx.net** and press **Enter**.
9. Wait till the computer has successfully started in Safe Mode.
10. Log on to the server as Administrator with Active Directory password you set earlier exercise.
11. Open a command prompt window.
12. Access the folder d:\w2000x\ntds
13. Copy the file **ntds.dit** to a new file named **ntds1.dit**
14. Delete the file **ntds.dit**
15. Rename the file **ntds1.dit** as **ntds.dit**
16. Close the command prompt window.
17. Restart the server.

In this exercise you learnt how to make a backup copy of the active directory database by copying it to another file.

Lab 19

Installing And Configuring DHCP Server

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Prepared by :
Muhammad Rafi'uddin Bin Rozman

Lab 19. INSTALLING AND CONFIGURING DHCP SERVER

"Dynamic Host Configuration Protocol (DHCP) is an IP standard designed to reduce the complexity of administering IP address configurations.-"Microsoft's definition. A DHCP server would be set up with the appropriate settings for a given network. Such setting would include a set of fundamental parameters such as the gateway, DNS, subnet masks, and a range of IP addresses. Using DHCP on a network means administrators don't need to configure these setting individually for each client on the network. The DHCP would automatically distribute them to the clients itself.

In this exercise you will set DHCP server and deploy DHCP to a Windows 2003 client computer. You will configure DHCP service and limit it to 3 host.

Estimated time to complete: 45 minutes

Preliminary Setup

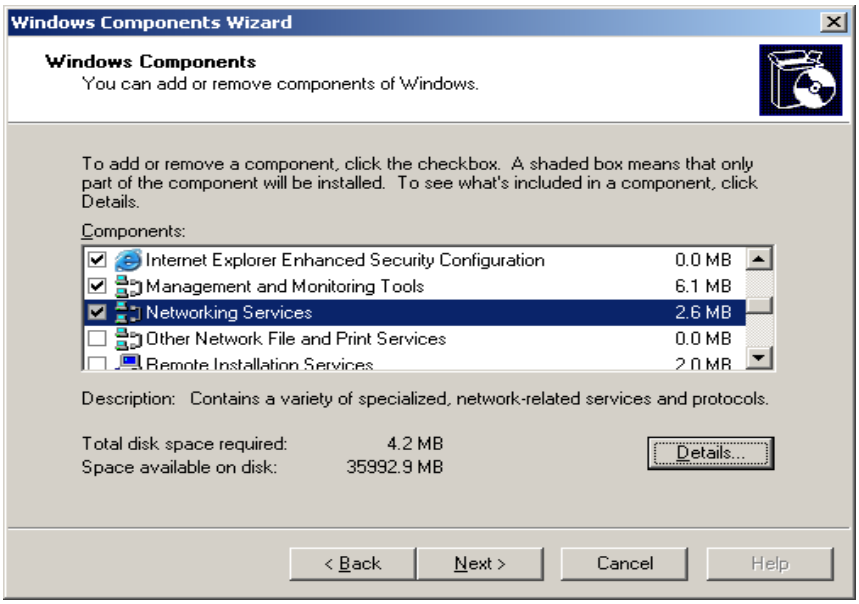
To support this exercise, you will need to change your network cable from straight cable to **cross cable** and hook-up cross cable to your server and your client.

Installing DHCP

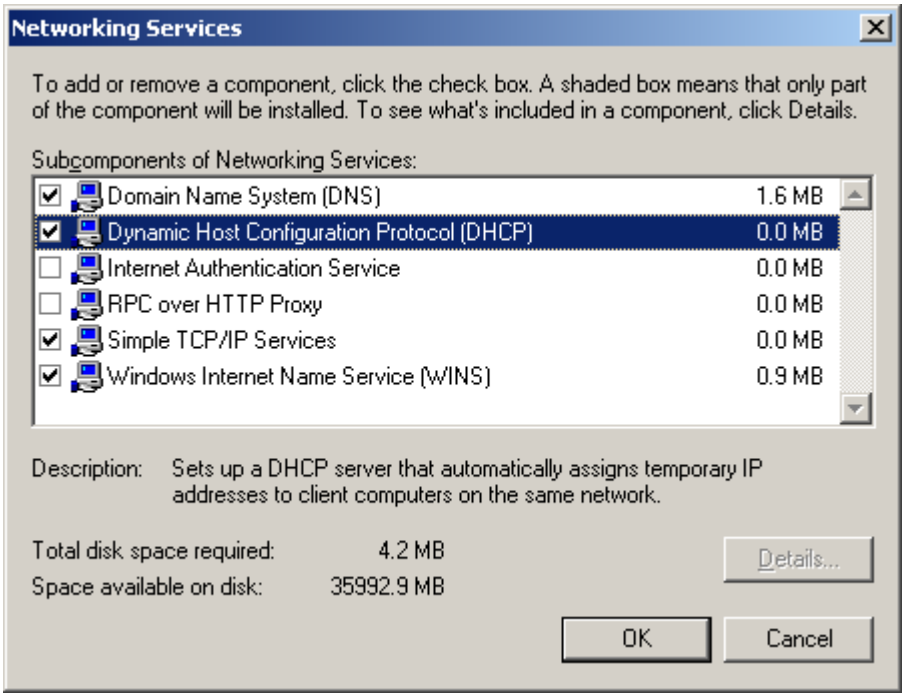
This will serve as a step-by-step guide on how to setup a DHCP server.

1. Log on to the server as **Administrator.**
2. Click Start; Control Panel; Add or Remove Programs.
3. Click **Add/Remove Windows Components**.

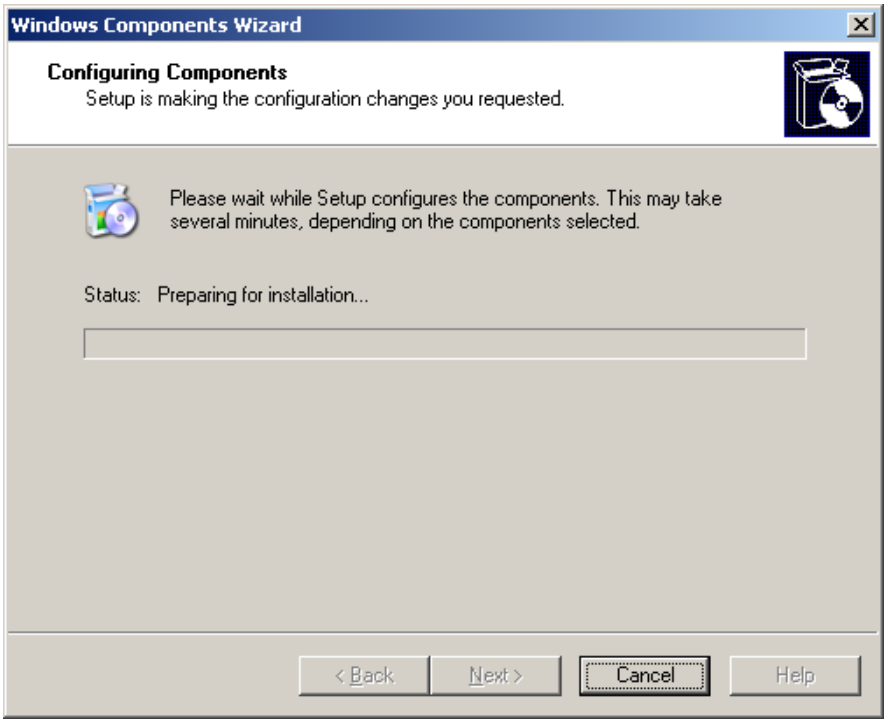
4. Highlight **Networking Services** and click **Details** button.



5. Mark **Dynamic Host Configuration Protocol (DHCP)** box.



6. Click OK.
7. Click Next; windows will start installing DHCP.



8. After complete installing DHCP, click **Finish.**



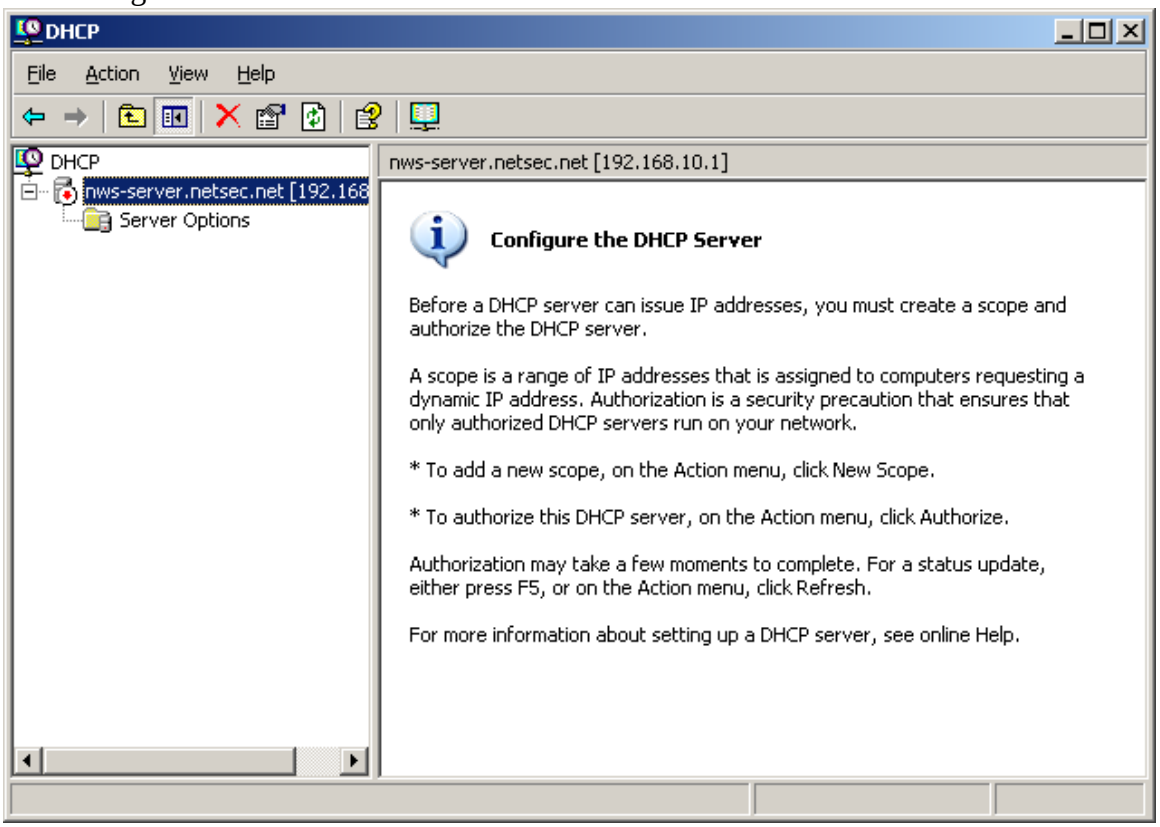
9. Close Add or Remove Programs.

Creating a Range of Address : DHCP "Scopes"

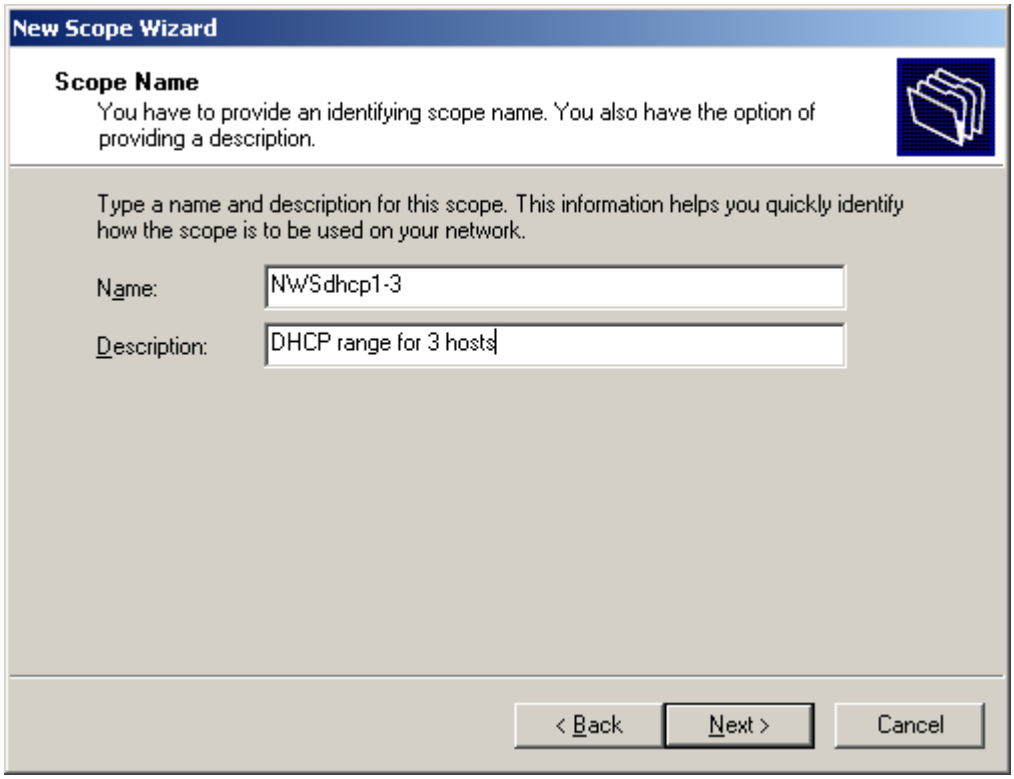
To specify range of IP address.

Creating The Scope

1. Log on to the server as **Administrator.**



2. Click Start; All Programs; Administrative Tools; **DHCP**.
3. Select your **Domain.net** icon.
4. On the menu bar, click **Action; New Scope**. This will start New Scope Wizard.
5. Click Next.
6. Fill in "**Name**:" as **dhcp1-3** and "**Description**:" as **DHCP range for 3 host**.



7. Click Next.

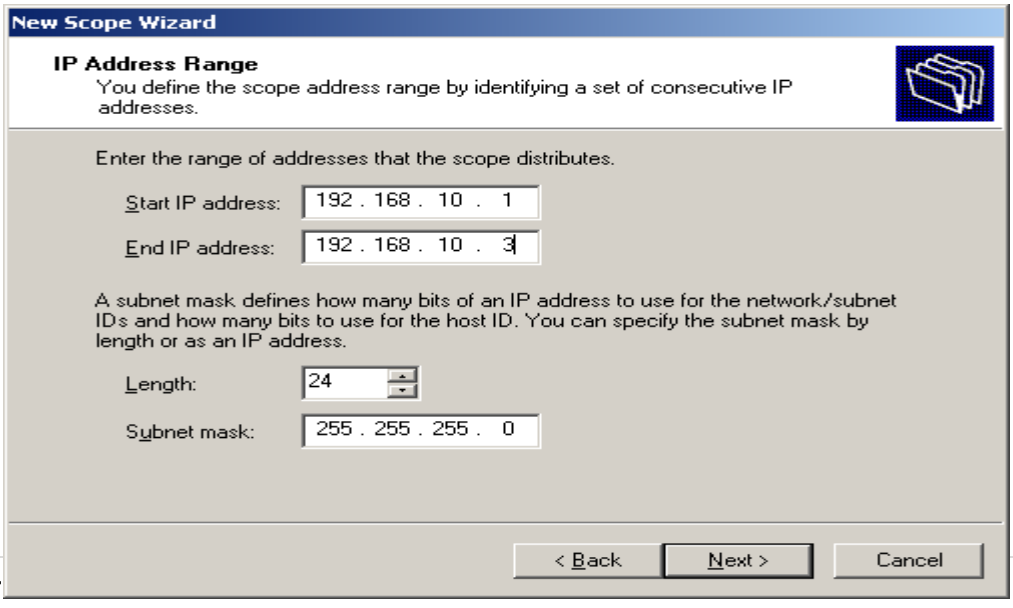
Specifying IP Address Range

Now you will configure DHCP service and limit it to 3 host.

8. Set “Start IP Address:” and “End IP Address:” as following:-

Start IP Address : 192.168.10.**Server Number**

End IP Address : 192.168.10.**Server Number + 2**



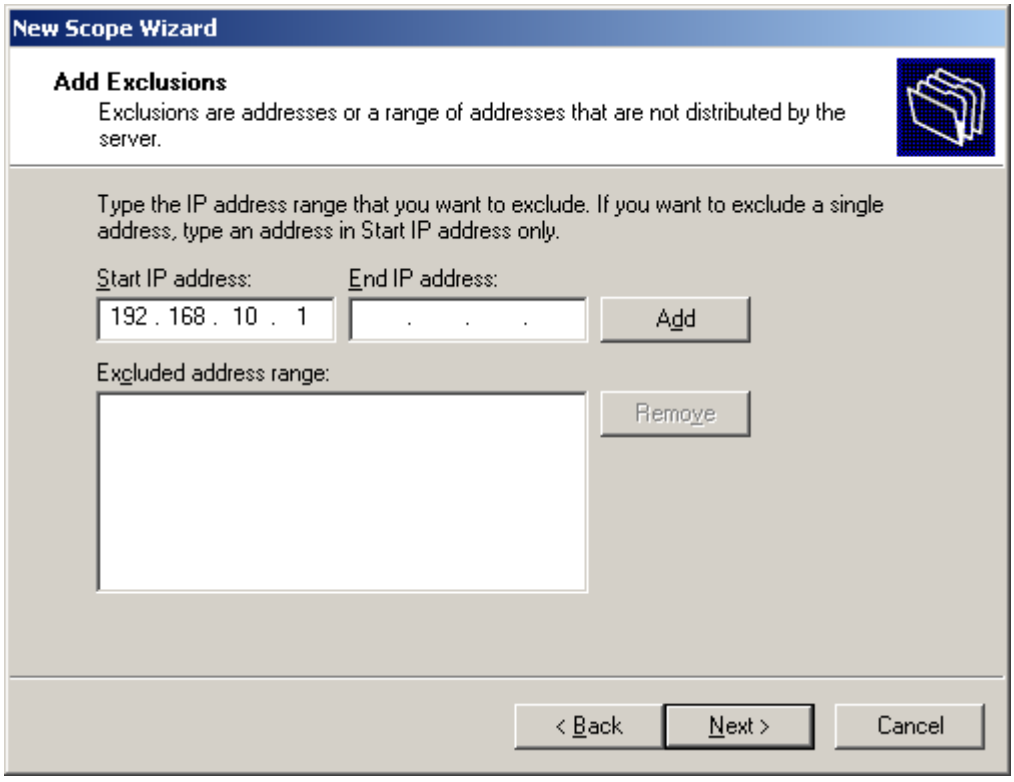
9. Set “Length:” and “Subnet mask:”

   Length : **24** (This number automatically set by windows)

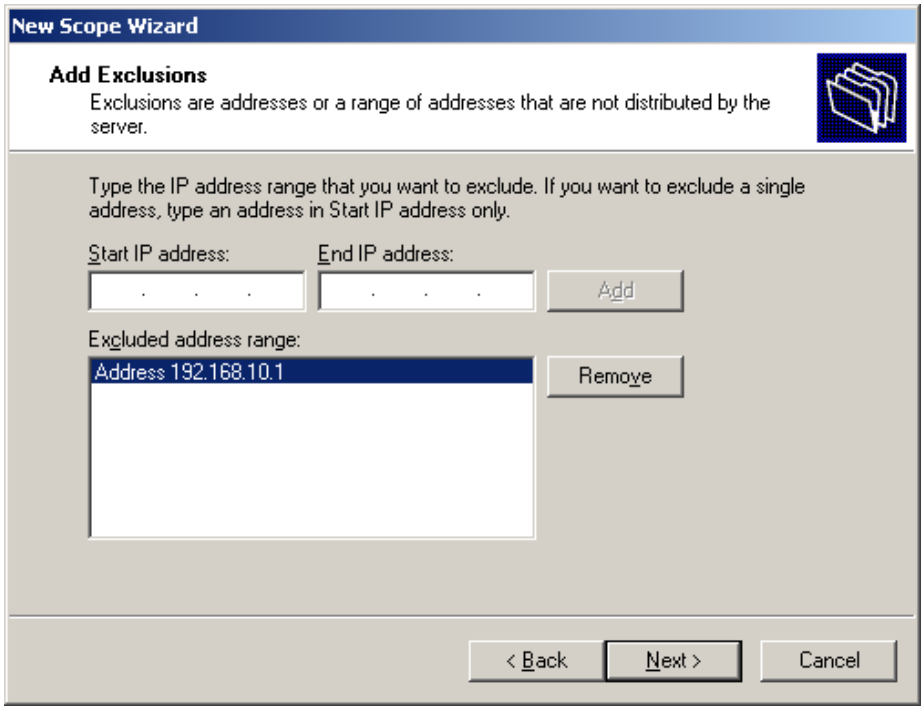
   Subnet mask : **255.255.255.0** (Set your subnet mask here)

10. Click Next.

11. Enter your **server IP** address at “**Start IP Address:**” to exclude it IP from distributed by DHCP server.



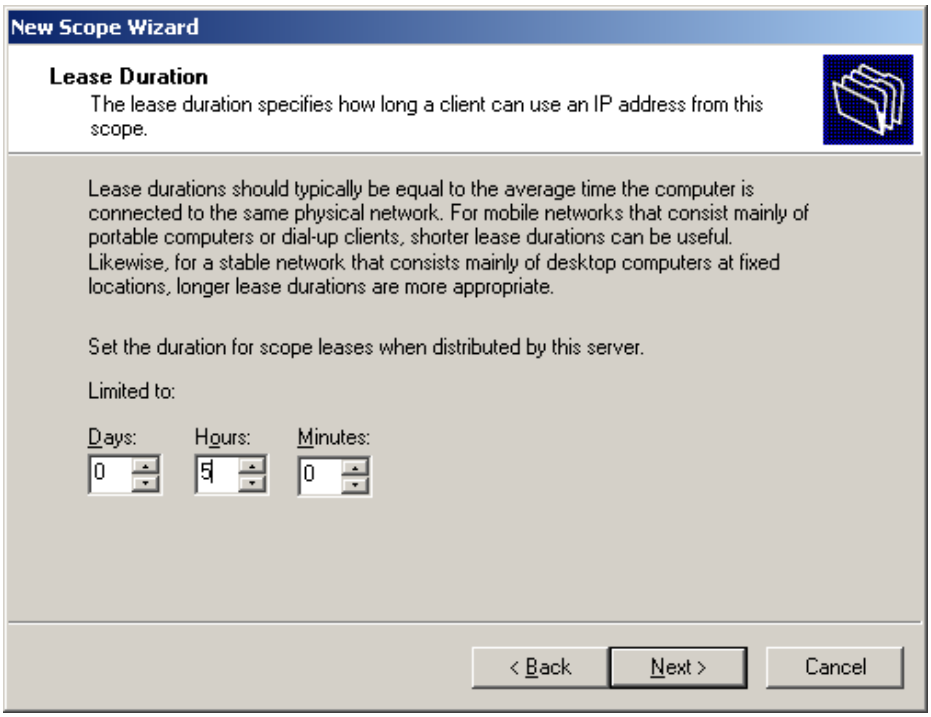
12. Click **Add** button.
13. Click Next.



Setting Lease Durations

The lease duration specifies how long a client can use an IP address from this scope.

14. Set lease Duration to **5 hours**.



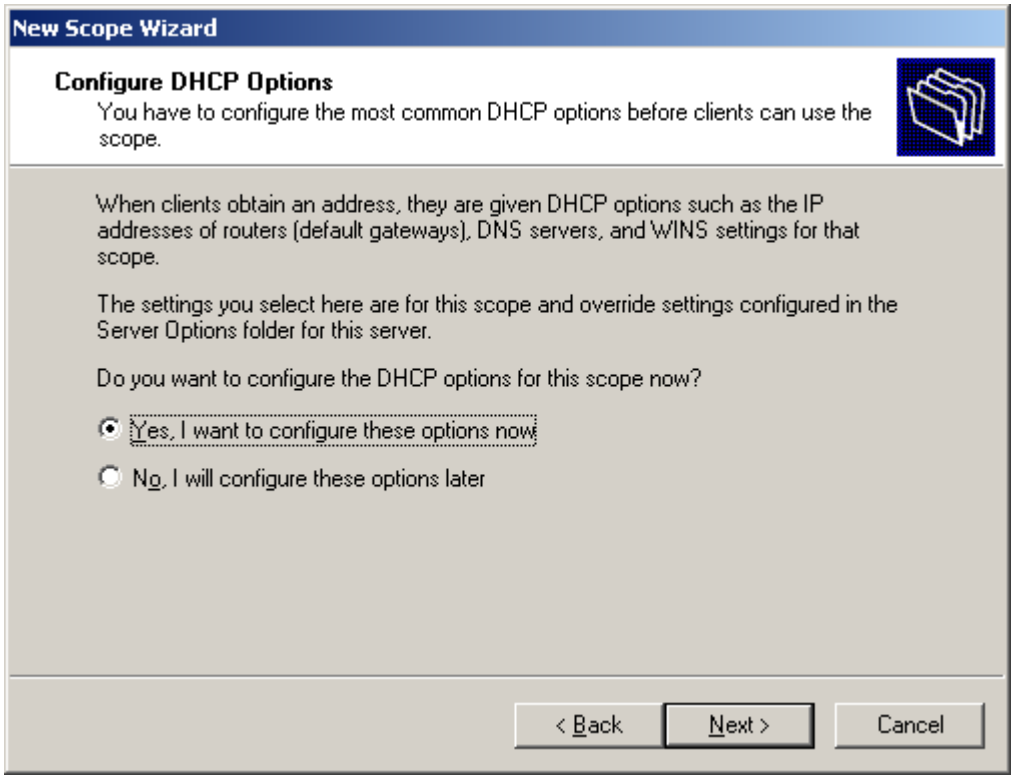
15. Click Next.

Setting Client Options

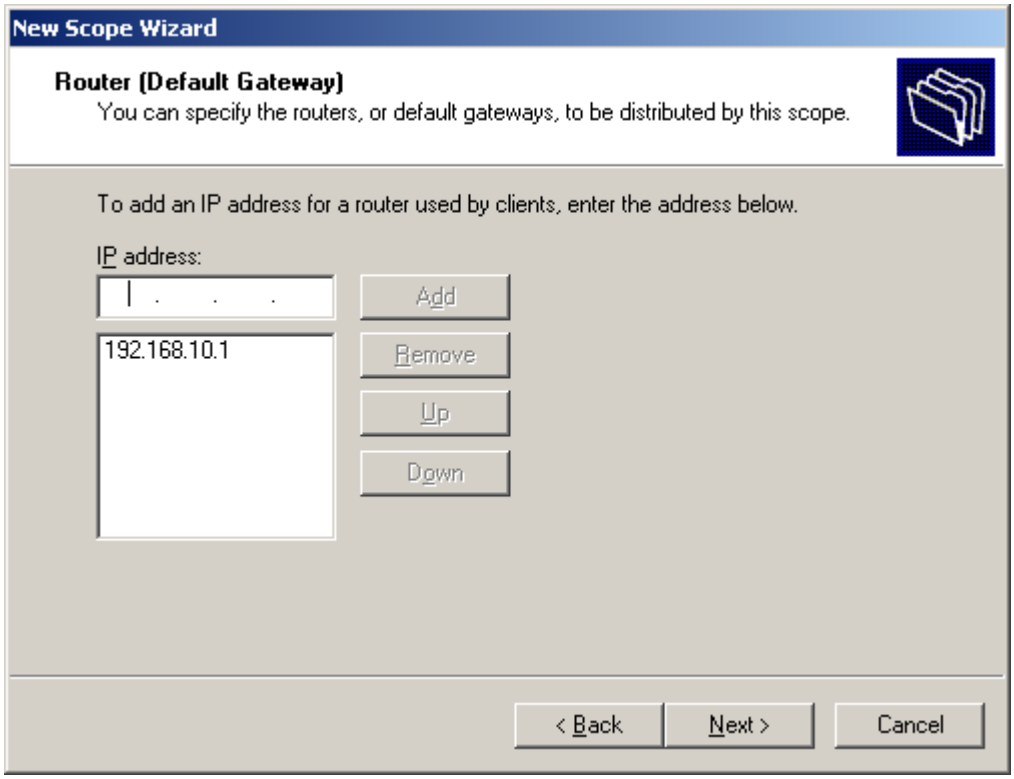
DHCP can provide default values for a whole host of TCP/IP parameters, including these basic items:-

- Default Gateway
- Domain Name
- DNS Server
- WINS Server

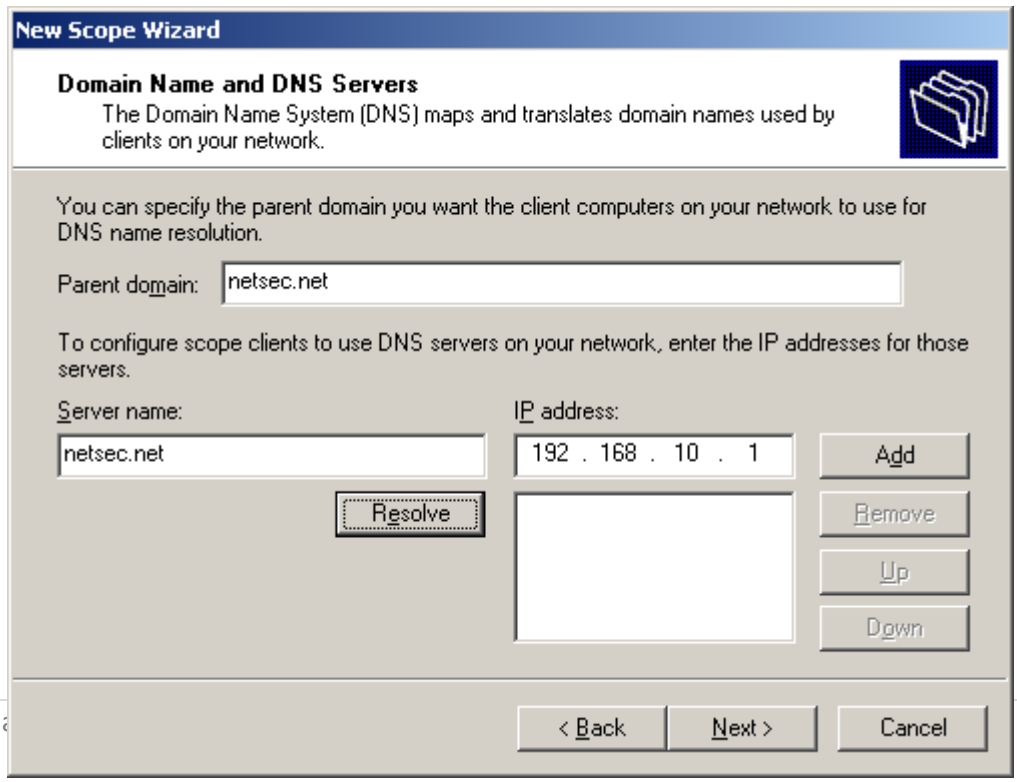
16. Select “Yes, I want to configure these options now”



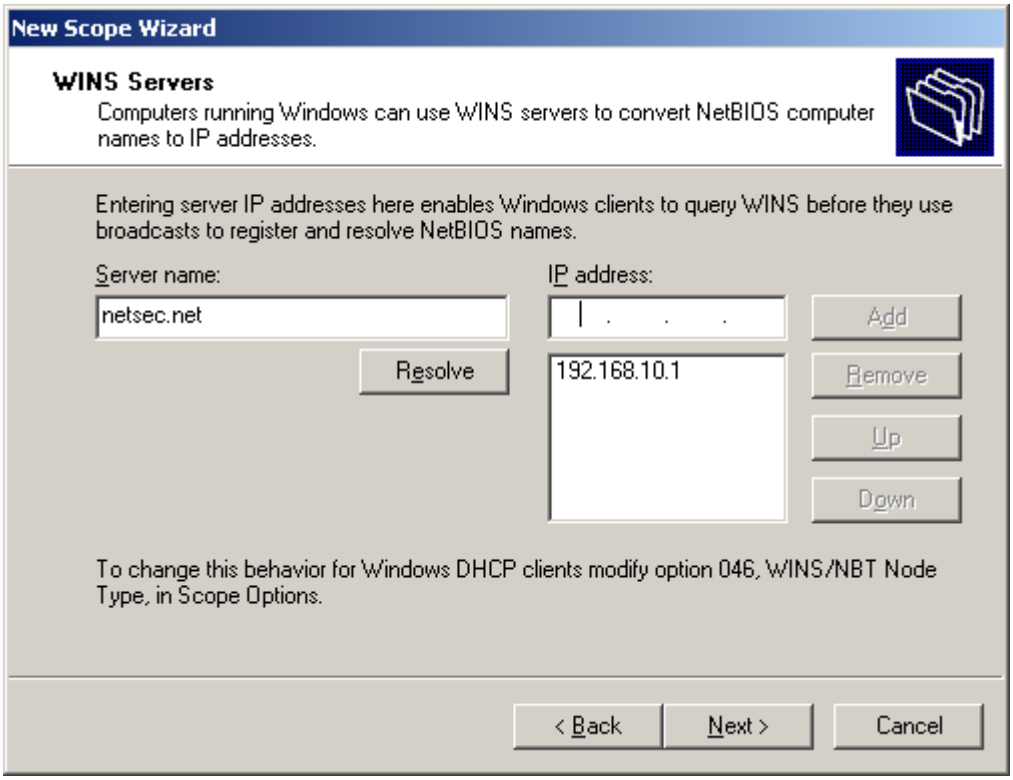
17. Click Next.
18. Set default gateway to your **server number**.
19. Click Add button.



20. Click Next.
21. Set Parent Domain same as your domain.net (e.g netsec.net)
22. Set DNS Server name and IP address (set this if you want client to use DNS server on your network)



23. Click Add button.
24. Click Next
25. Set WINS server name and IP address.

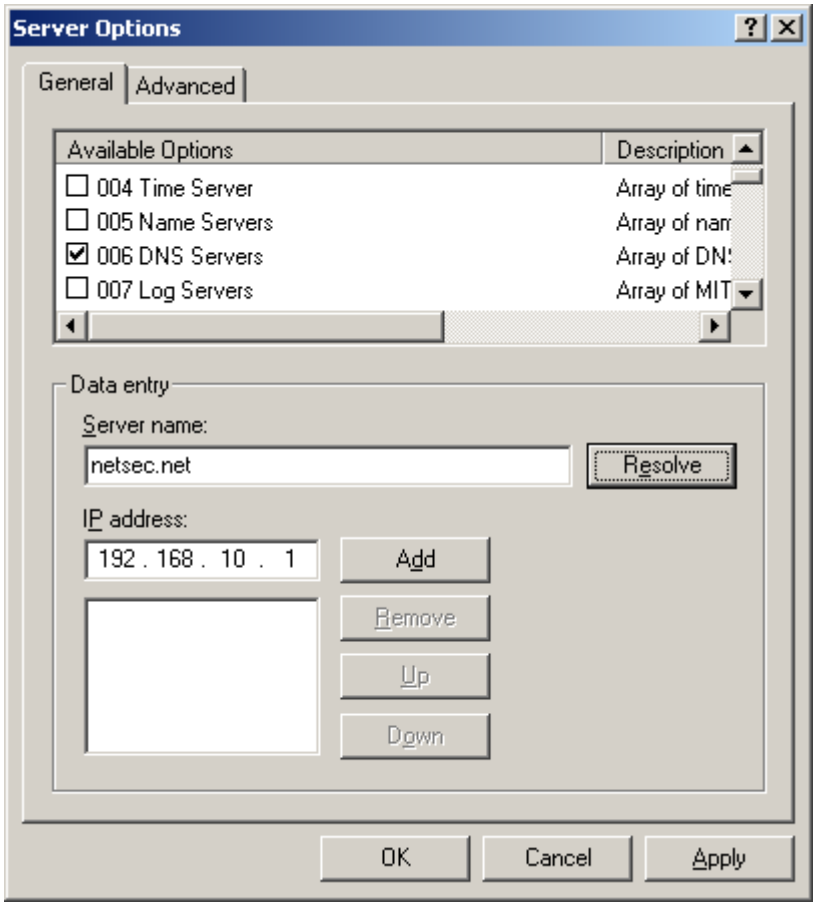


26. Click Add button.
27. Click Next.
28. Select “yes, I want to activate this scope now” to activate the scope.
29. Click Next.
30. Click Finish.

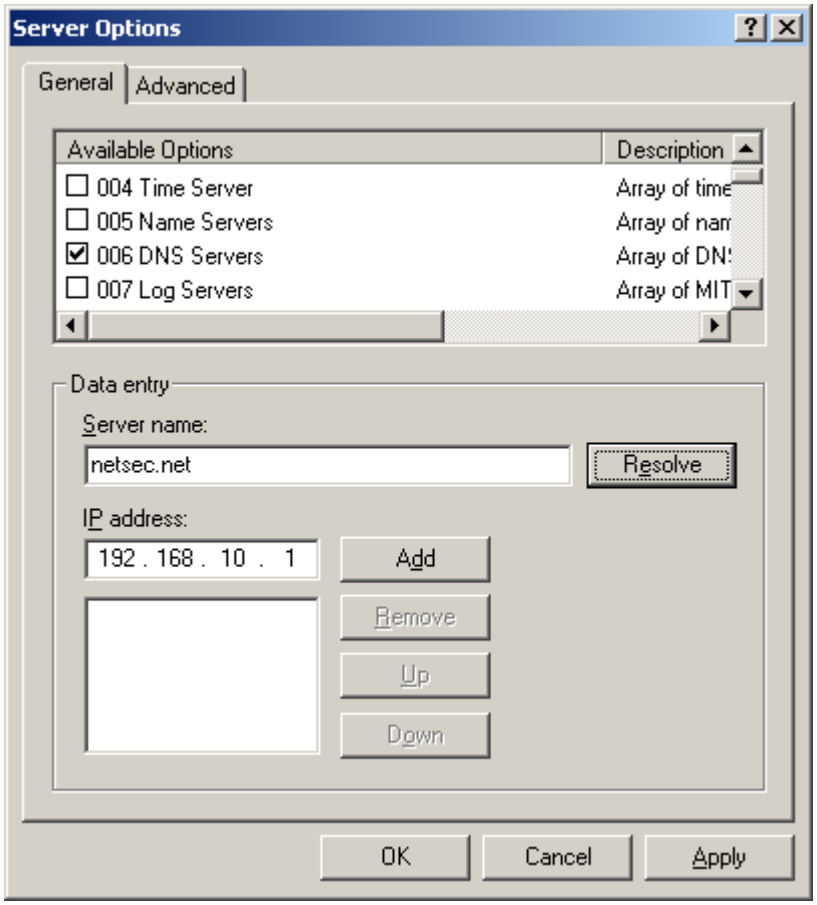
Setting Options For All Scopes

Server options are useful when you're putting more than one scope on a server.

1. Log on to the server as **Administrator.**
2. Open DHCP (Click Start; All Programs; Administrative Tools; **DHCP**)
3. Expand your **Domain.net** icon.
4. Right click **Server Options** folder.
5. Choose **Configure Options..**
6. Select **DNS Servers**



7. Type your DNS server name in the **Server name** : and click **Resolve** button.
8. Click **Add** button to add your DNS server IP address.



9. Click OK.

Authorizing DHCP Servers (For Active Directory Users)

10. Click your **server** icon.
11. On the Menu Bar, select **Action; Authorize**.
12. Click **Refresh** button (or press F5 to refresh).

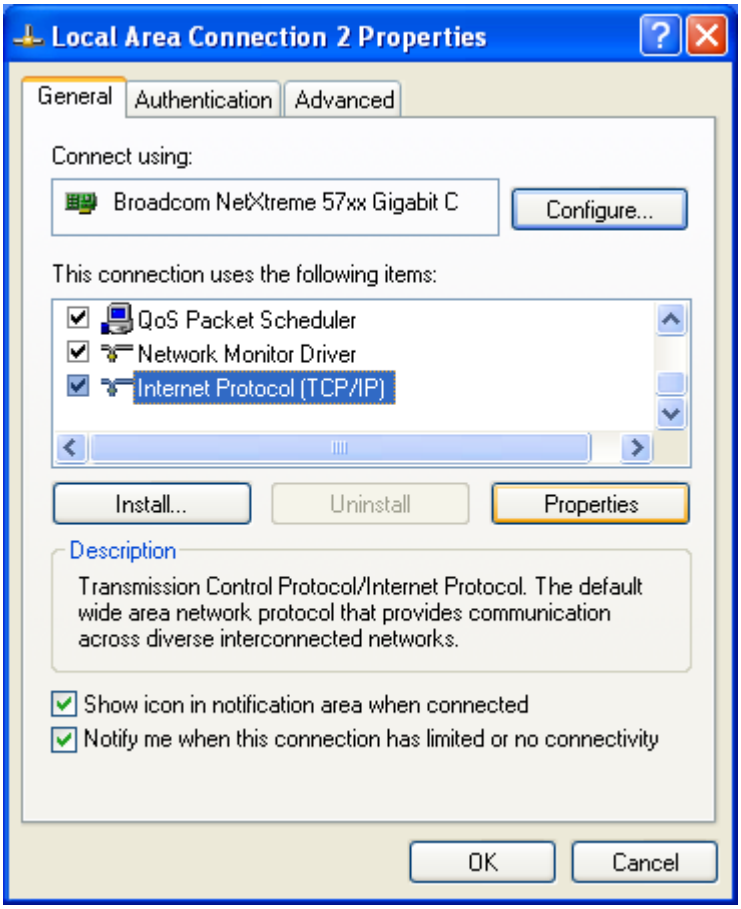
You'll notice the arrow beside server icon turn from red to green. This means your DHCP server is now active.

13. Close DHCP snap-in.

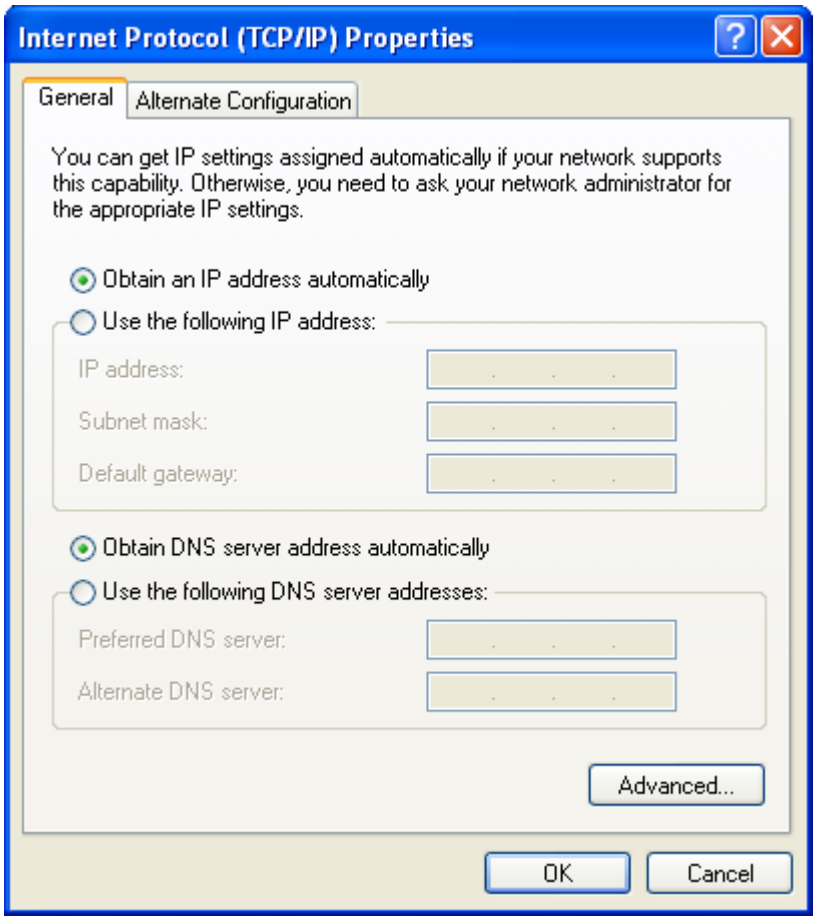
Testing DHCP Server

In this exercise you will test your DHCP server functionality.

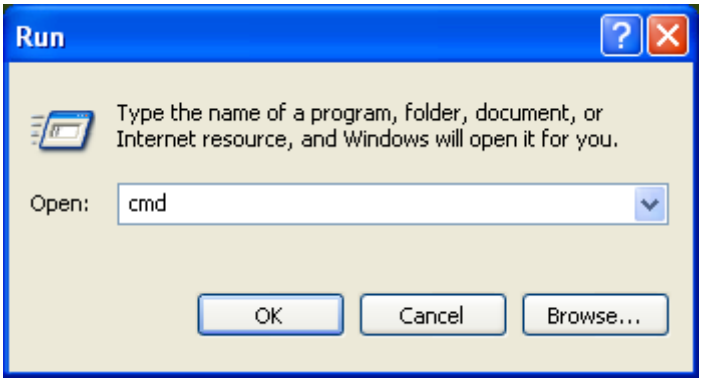
1. Log on to the client computer as **Administrator** with **abc123!@#** as password [select Log on to : CLIENT (this computer)].
2. Open **Network Connections** (Click Start; All Programs; Accessories; Communications; Network Connection)
3. Right click **Local Area Connection** and select **Properties**.
4. Select **Internet Protocol (TCP/IP)** and click **Properties**.



5. **In Internet Protocol (TCP/IP) Properties**, select **Obtain an IP Address automatically** and select **Obtain DNS server address automatically**.

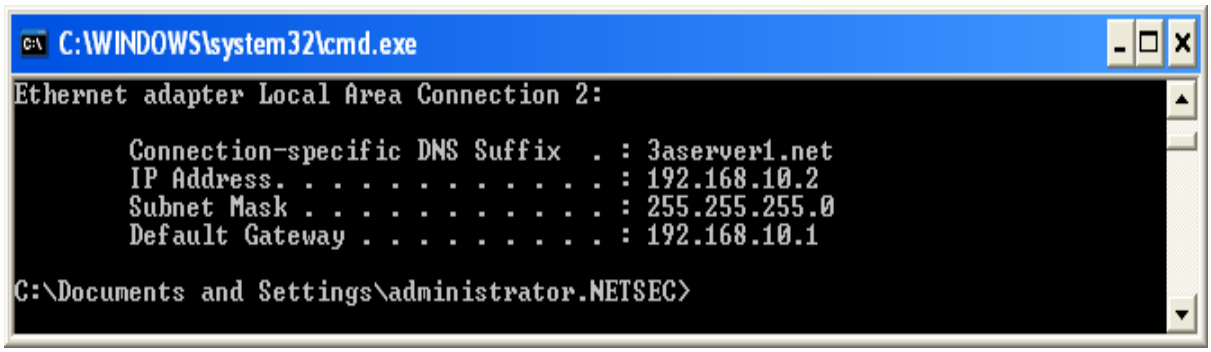


6. Click OK.
7. Click Close.
8. Open **Command Prompt** (click Start; Run; type **cmd** and press Enter)



9. Type **ipconfig** and press Enter.

This will display the IP address, subnet mask and default gateway for your Ethernet adapter.



10. What are the IP address listed?

    _____

11. What are the Subnet Mask listed?

    _____

12. What are the Default Gateway listed?

    _____

    Now your client computer are set to obtain an IP address automatically from DHCP server. So you can see the IP address have change accordingly to the IP range you have set in the DHCP server setting earlier.

13. Type **ipconfig/?** and press Enter.
14. Type **ipconfig / all** and press Enter.
15. What are the different between **"ipconfig"** and **"ipconfig / all"** command?

    _____

    _____

16. Now, type ipconfig /release and press Enter. What happen?

    _____

    _____

17. Type ipconfig /renew and press Enter.

18. What happen? What is this command stand for?

    _____

    _____

19. Log off client computer.

Summary

In this exercises, you setting up a DHCP server. The DHCP server provides you with an easy way of assigning IP addresses to workstations on your network. You were shown how to install and configure a DHCP Server and how to avoid overlapping scopes.

Lab 20

Installing And Configuring Web Server

Mr Razali bin Abu Bakar
TTO, Network Security
German - Malaysian Institute.

Prepared by :
Muhammad Rafi'uddin Bin Rozman

LAB 20. INSTALLING AND CONFIGURING WEB SERVER

In this exercise, you will install and configure your server to run as Web Server. This exercises also describes the basics of managing a Web site's insfrastructure, from setting a site home directory and default Web Page, to redirecting requests and dynamically altering Web pages.

Internet Information Services (IIS) 6.0 overview

Used with the Microsoft Windows Server 2003 family, Internet Information Services (IIS) provides integrated, reliable, scalable, secure, and manageable Web server capabilities over an internet, the Internet, or through an extranet. You can use IIS 6.0 to create a strong communications platform of dynamic network applications. Organizations of all sizes use IIS to host and manage Web pages on the Internet or an intranet, to host and manage File Transfer Protocol (FTP) sites,and to route news or mail using the Network News Transfer Protocol (NNTP) and the Simple Mail Transfer Protocol (SMTP). IIS 6.0 capitalizes on the latest Web standards like ASP.NET, Extensible Markup Languange (XML), and Simple Object Access Protocol (SOAP) for the development, implementation, and management of Web applications. IIS 6.0 includes new features designed to help organizations, IT professionals, and Web administrators achieve their goals of performance, reliability, scalability, and security for potentially thousands of Web sites either on a single IIS server or on multiple servers.

Installing IIS and Web Server

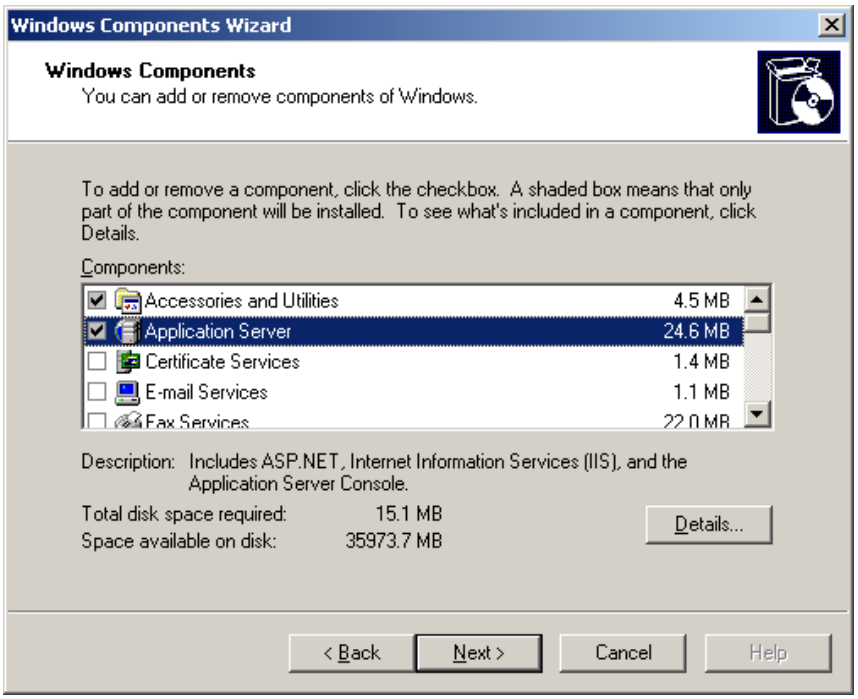
When you install IIS initially, the service is installed in a highly secure mode. Because, IIS only servers static content by default, you must enable features such as ASP, ASP.NET, Common Gateway Interface (CGI), Internet Server Application Programming interface (ISAPI), and Web Distributed Authoring and Versioning (WebDAV), if you need them.

During installation, IIS installs optional components such as common files and IIS Manager. You can choose not to install the optional components. However, if you do not install specific components, you can decrease IIS functionality or disable IIS services. If you are unfamiliar with the optional components and how they affect IIS, install IIS with the default setting.

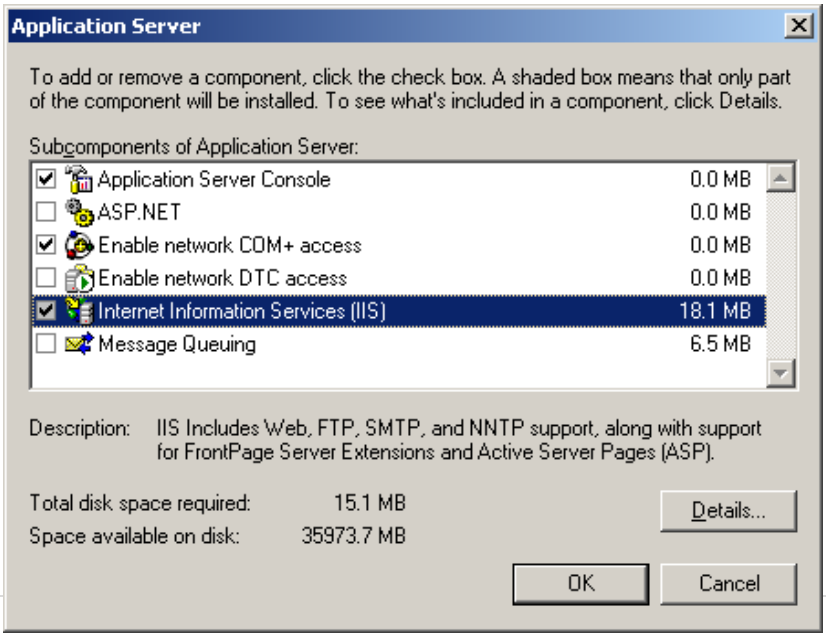
To install Internet Information Services (IIS) 6.0

1. Log on to the server as **Administrator**.
2. Open the Windows Components Wizard.

   (Start; Control Panel; Add or Remove Programs)
3. Click **Add/Remove Windows Components**.
4. Under **Components**: highlight **Application Server**, and then click **Details**.

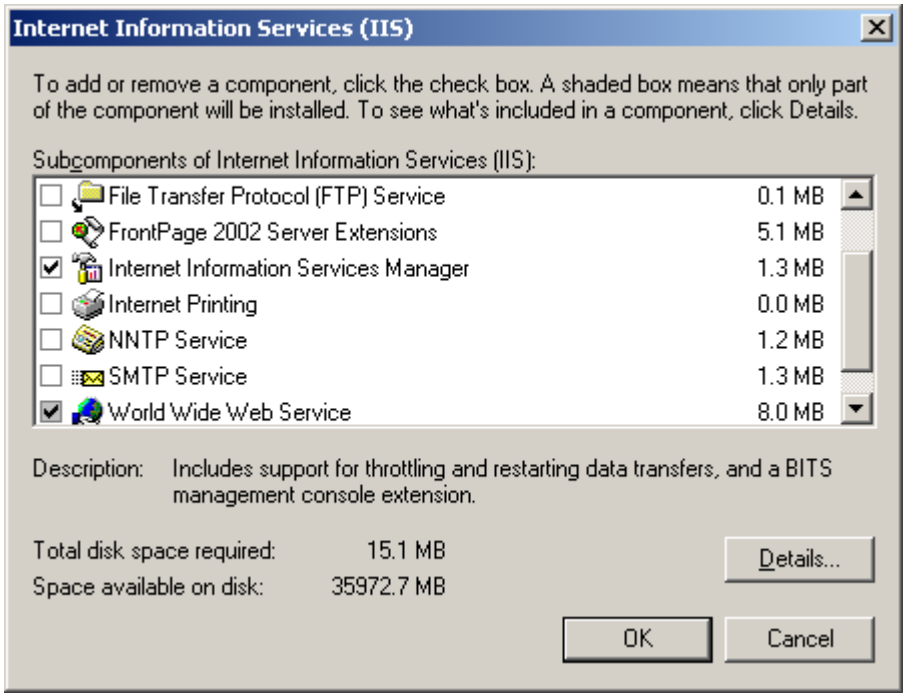


5. In the **Application Server** dialog box, under **Subcomponents of Application Server**, select **Internet Information Services (IIS)**, and then click **Details**.



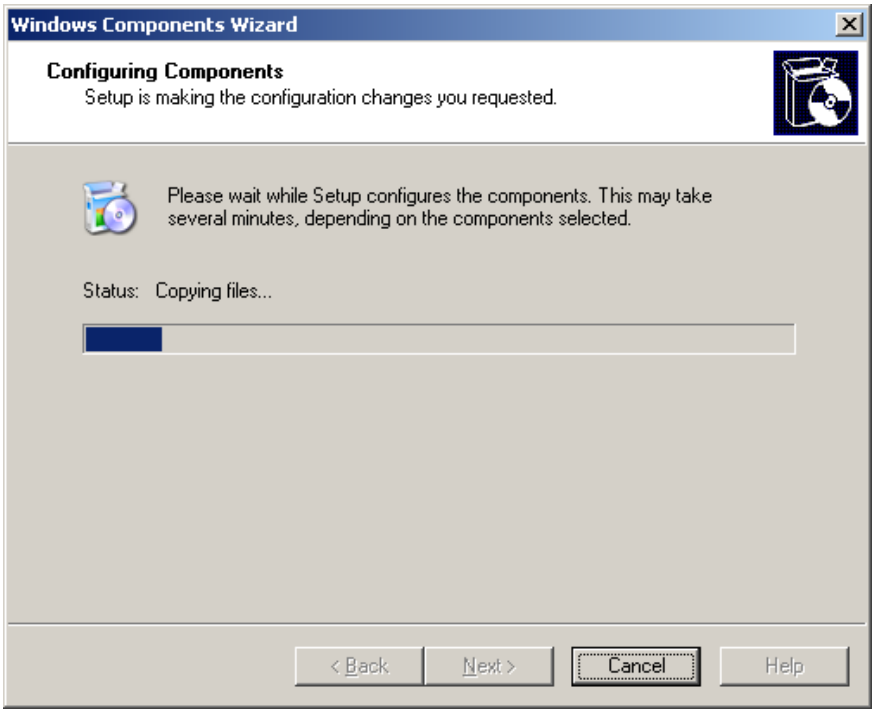
To Install Web Server

6. In the **Internet Information Services (IIS)** dialog box, under **Subcomponents of Internet Information Services (IIS)**, select item as the following:
    - Common Files
    - Internet Information Services Manager
    - World Wide Web Service



7. Click **OK** until you return to the **Windows Components Wizard.**

8. Click **Next**, and windows will start installing IIS.



9. After installation completed, click **Finish**.



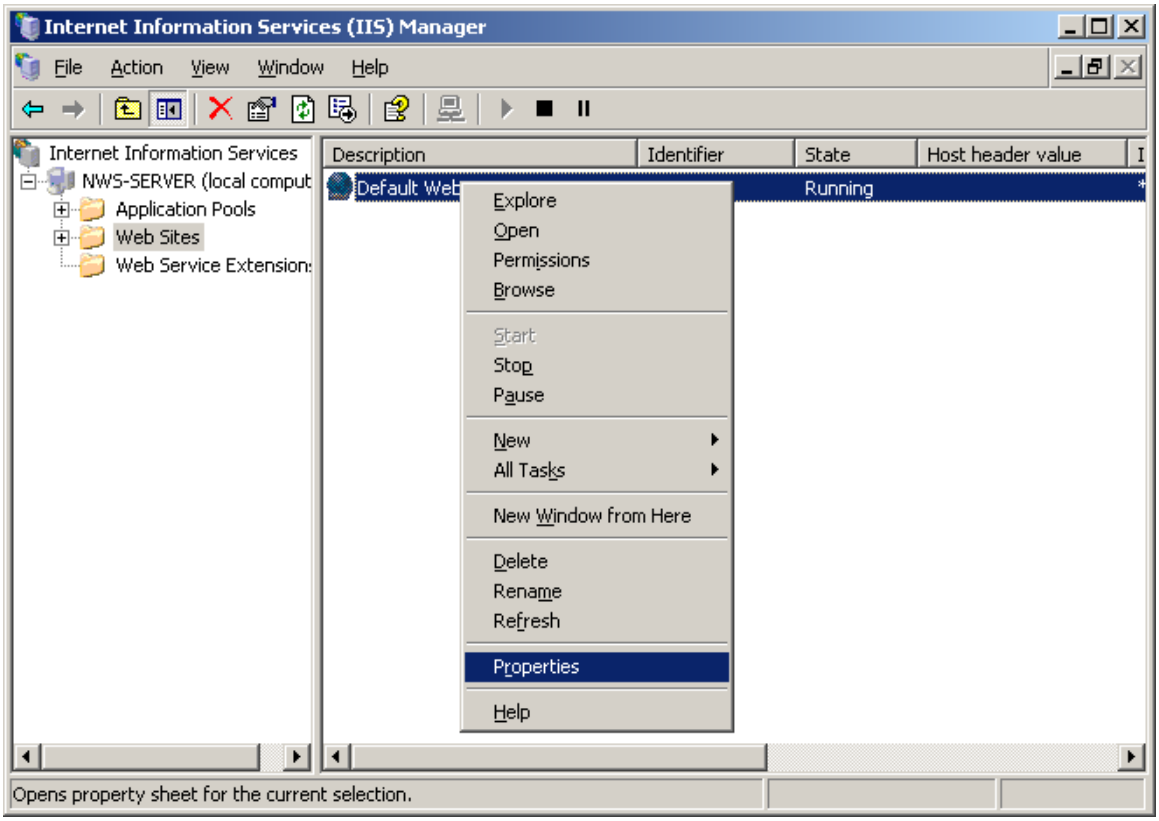
Configuring Web Server

IIS creates a default Web site configuration on your hard disk at the time of installation. You can use the \Inetpub\Wwwroot diretory to publish your Web content or create any directory or virtual directory you choose.

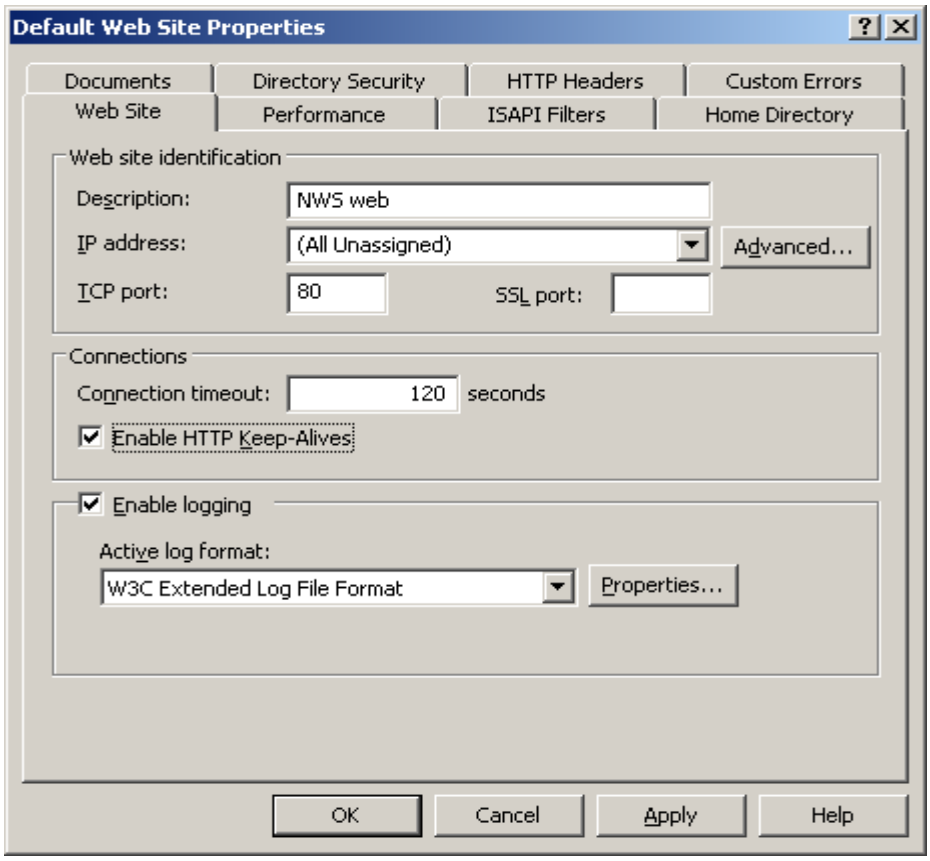
Creating a Web site using IIS Manager does not create content, but merely creates a directory structure and configuration files from which to publish the content.

To use the default Web site

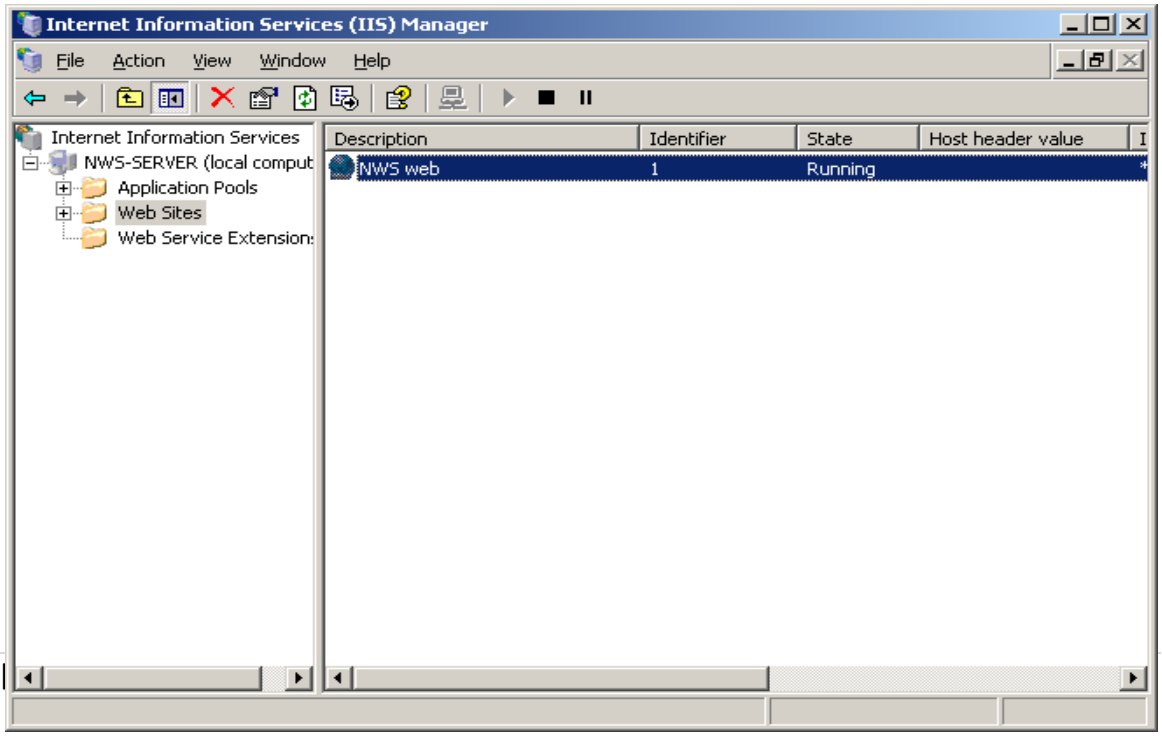
1. Log on to the server as **Administrator**.
2. Open **IIS Manager**.
   (Start; Administrative Tools; Internet information Services (IIS) Manager)
3. In IIS Manager, expand the local computer, expand the **Web Sites** folder, right-click **Default Web Site**, and click **Properties**.



4. On the **Web Site** tab, under **Web site description**, type the name of your Web site in the **Description** box. (e.g Server 1 Web Site)



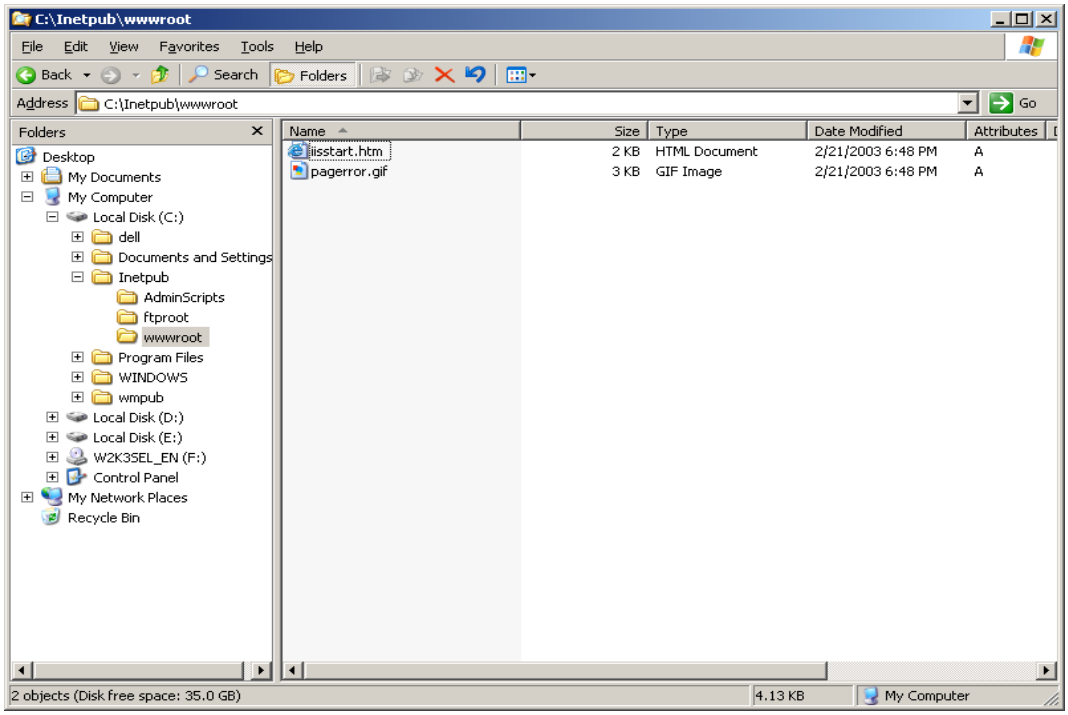
5. Click **OK**. The name of the new site appears in IIS Manager.



6. Click **Refresh** button and close the IIS Manager.

To create a simple Web page

7. Open Windows Explorer. Go to *D:\Inetpub\Wwwroot* directory.



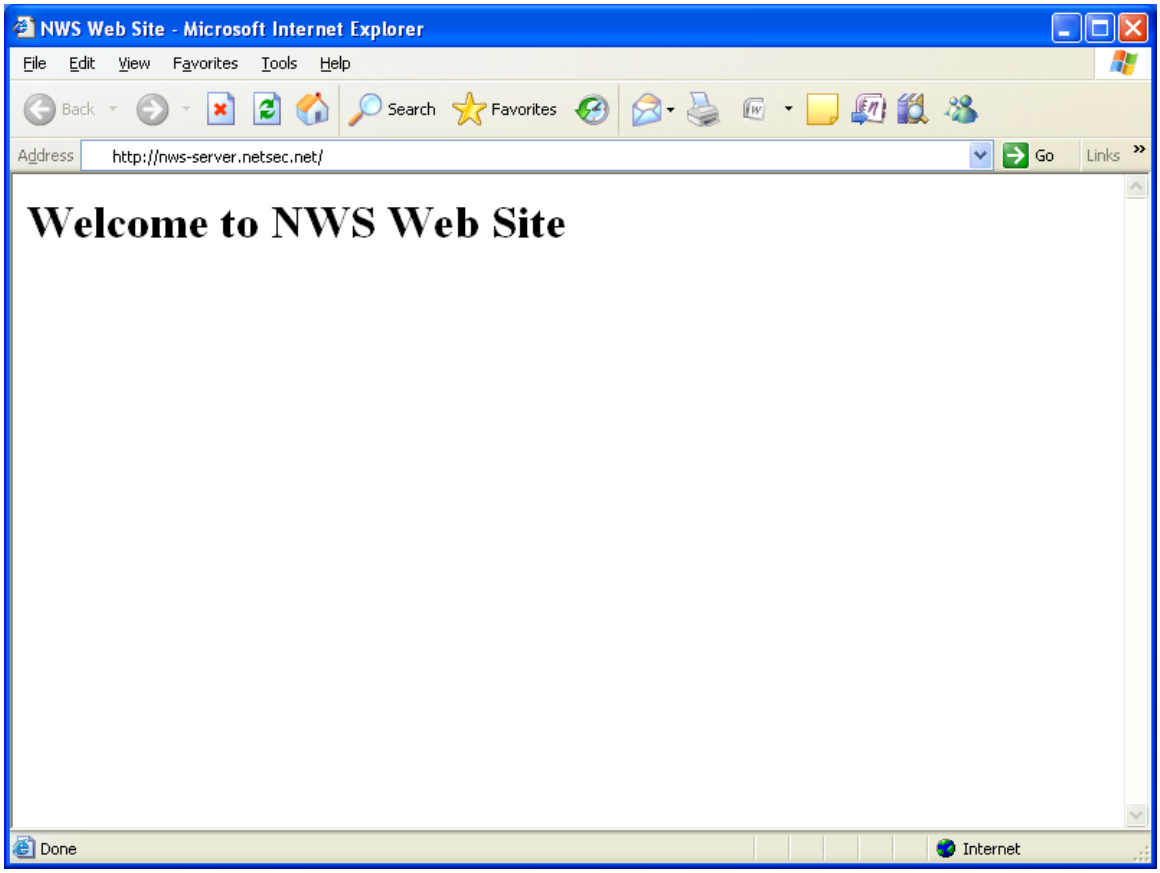
8. Create **New Text document** inside D:\Inetpub\wwwroot, and rename the text document as **index.htm**.
9. Right click **index.htm** and select **Edit**. This will load the Notepad Editor. Type the following text into the file:

```
<html>
<head>
<title>Web Server<\title>
</head>
<body>
<p><h1>Welcome To My Web Server</h1></p>
</body>
</html>
```

10. Save and close this file.

Test The Web Server

11. Log on to the **client computer** as **Administrator**.
12. Open **Internet Explorer**. (Start; All Programs; Internet Explorer).
13. On the Address box, key-in **http://serverSN.yourdomain.net** (e.g http://nws-server.netsec.net) and click **GO** button.



14. Your webpage will appear in the browser.

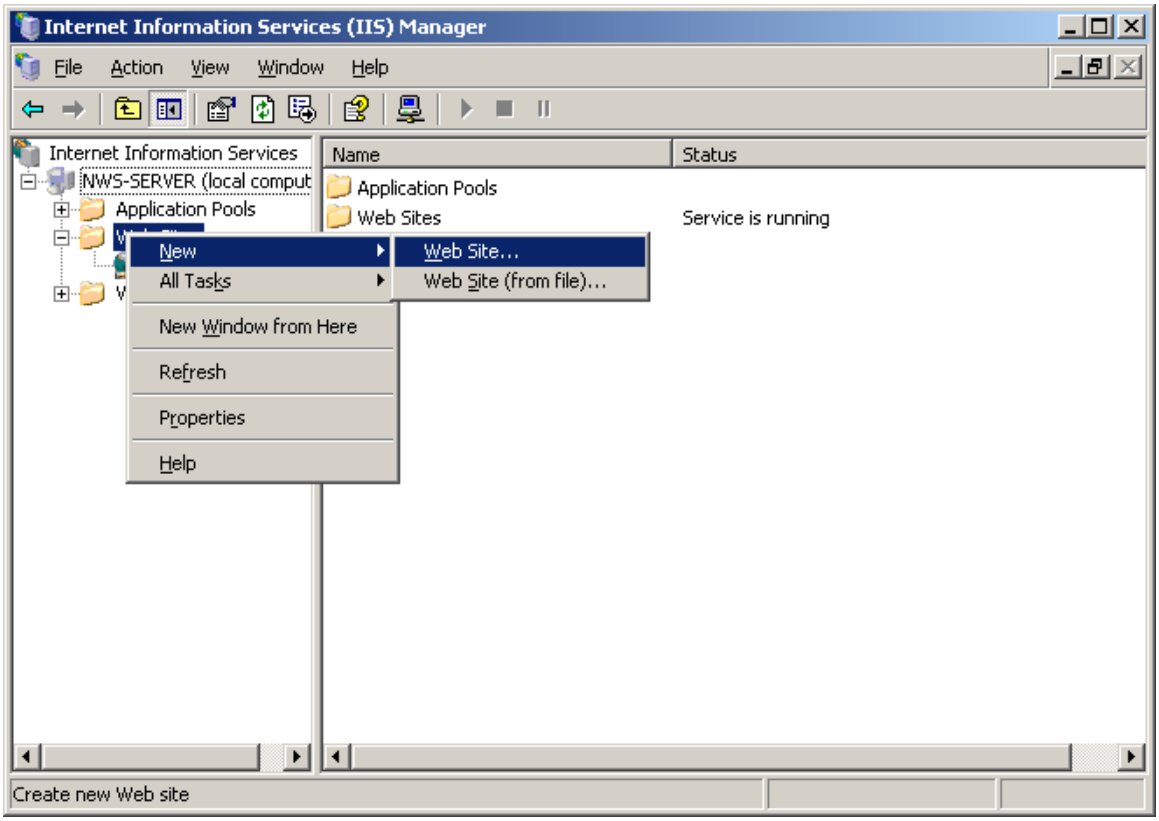
To create a new Web site

1. Log on to the server as **Administrator**.
2. Open **Windows Explorer**.
   (Start; All Programs; Accessories; windows Explorer)
3. Create new sub-folder on drive E: named **WebServer**.
4. Create **New Text Document** inside E:\WebServer, and rename the text document as **default.htm**.
5. Right click **default.htm** and select **Edit**. This will load the Notepad Editor. Type the following text into the file:

   <html>
   <head>
   <title>New Web Site</title>
   </head>
   <body>
   ** <h1 align="center">Welcome To My New Web Site</h1>**
   **<h3 align="right">Hosted by MyWeb Server</h3> **
   </body>
   </html>

   5.1 Save and close this file.

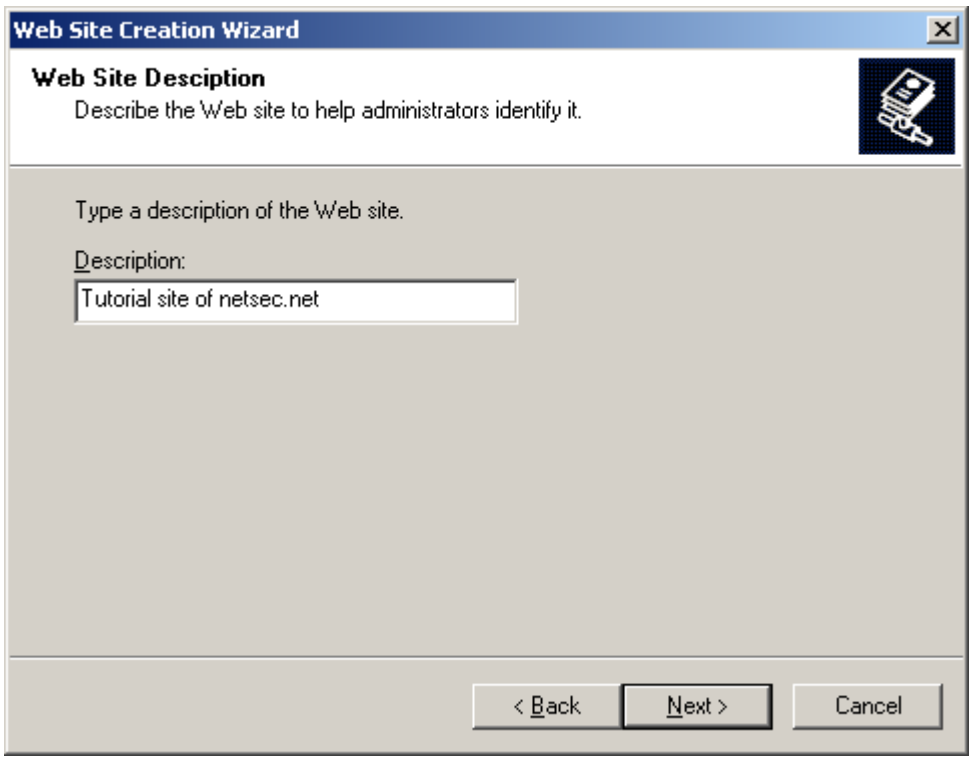
6. Open IIS Manager. **(Start; Administrative Tools; Internet Information Services (IIS) Manager)**
7. In IIS Manager, expand the local computer, right click the **Web Sites** folder, point to **New**, and then click **Web Site**.



8. The **Web Site Creation Wizard** appears. Click **Next**.

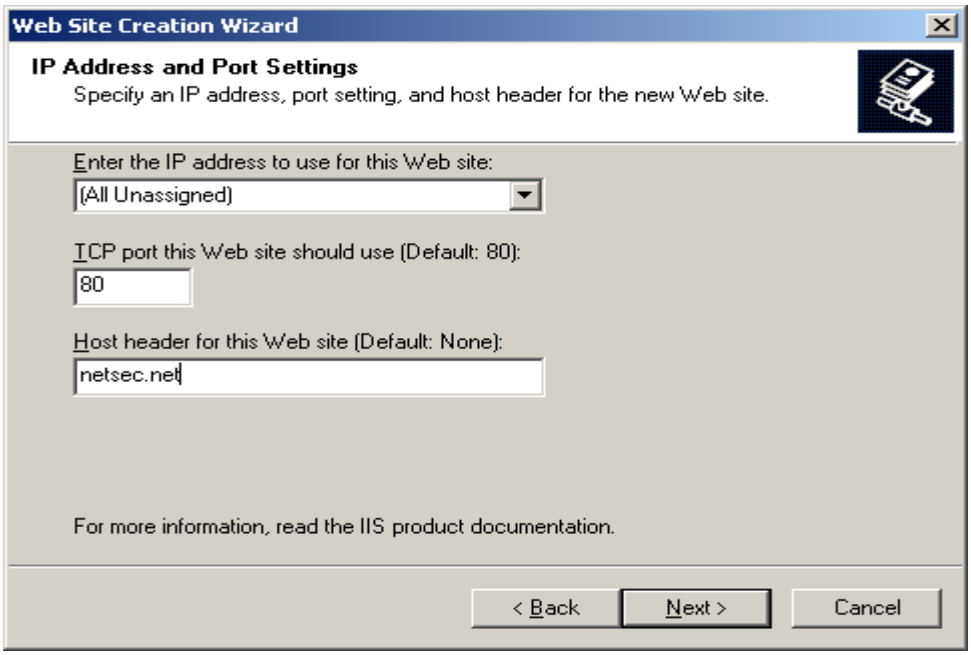


9. In the **Description** box, type the name of your site (e.g. Tutorial Site of netsec.net),

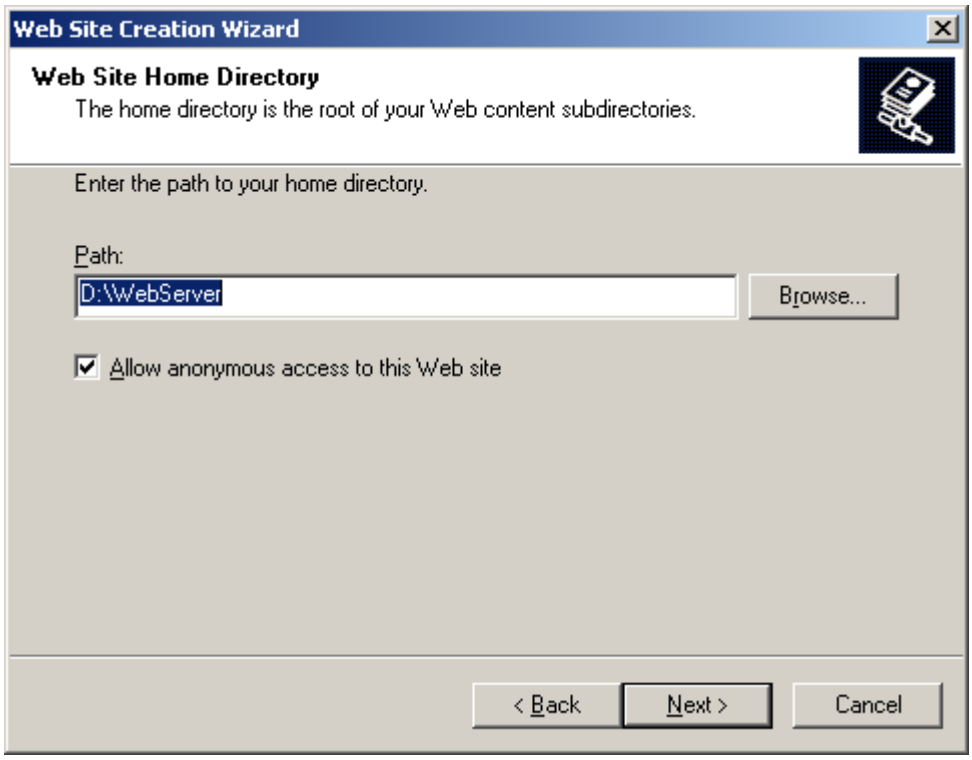


and then click **Next**.

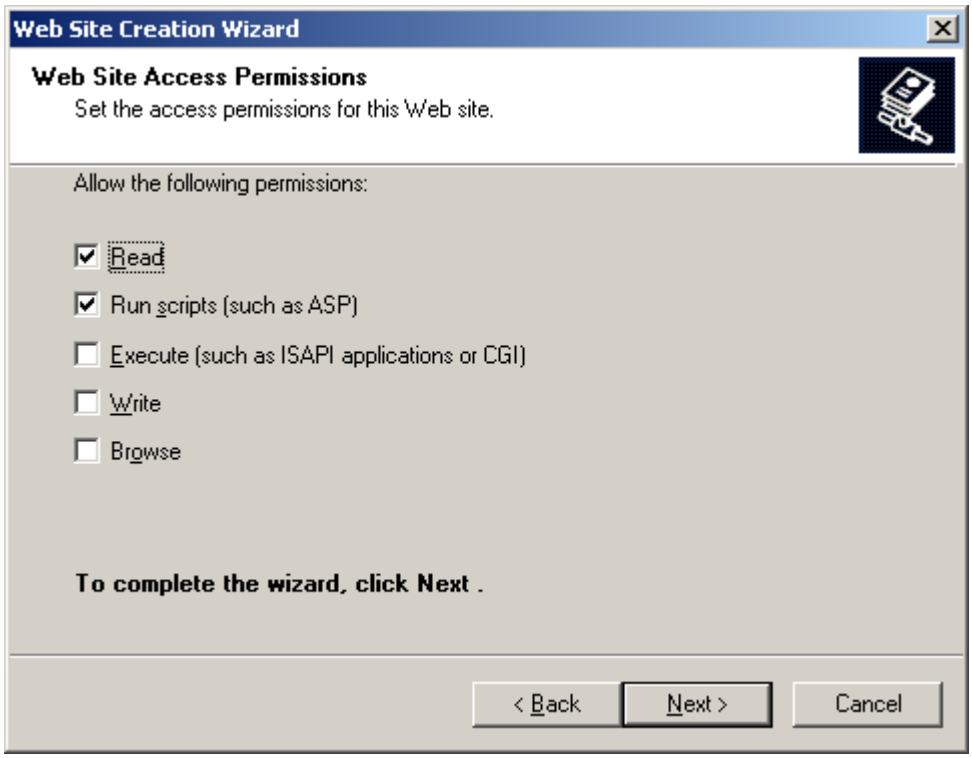
10. Type or click the IP address (the default is All Unassigned), TCP port, and host header (for example, www.netsec.net) for your site, and then click **Next**.



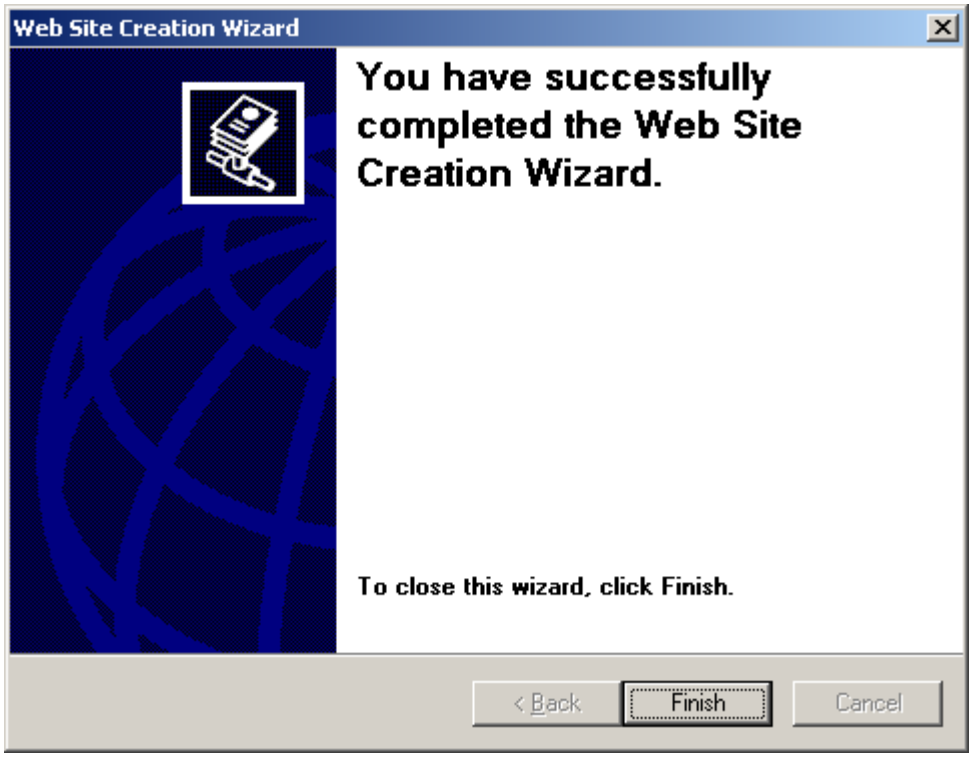
11. In the **Path** box, type or browse to the directory that contains the site content (**E:\WebServer**), and then click **Next**.



12. Select the check boxes **Read** and **Run Script** for the Web site access permissions you want to assign to your users, and then click **Next**.



13. Click **Finish**.

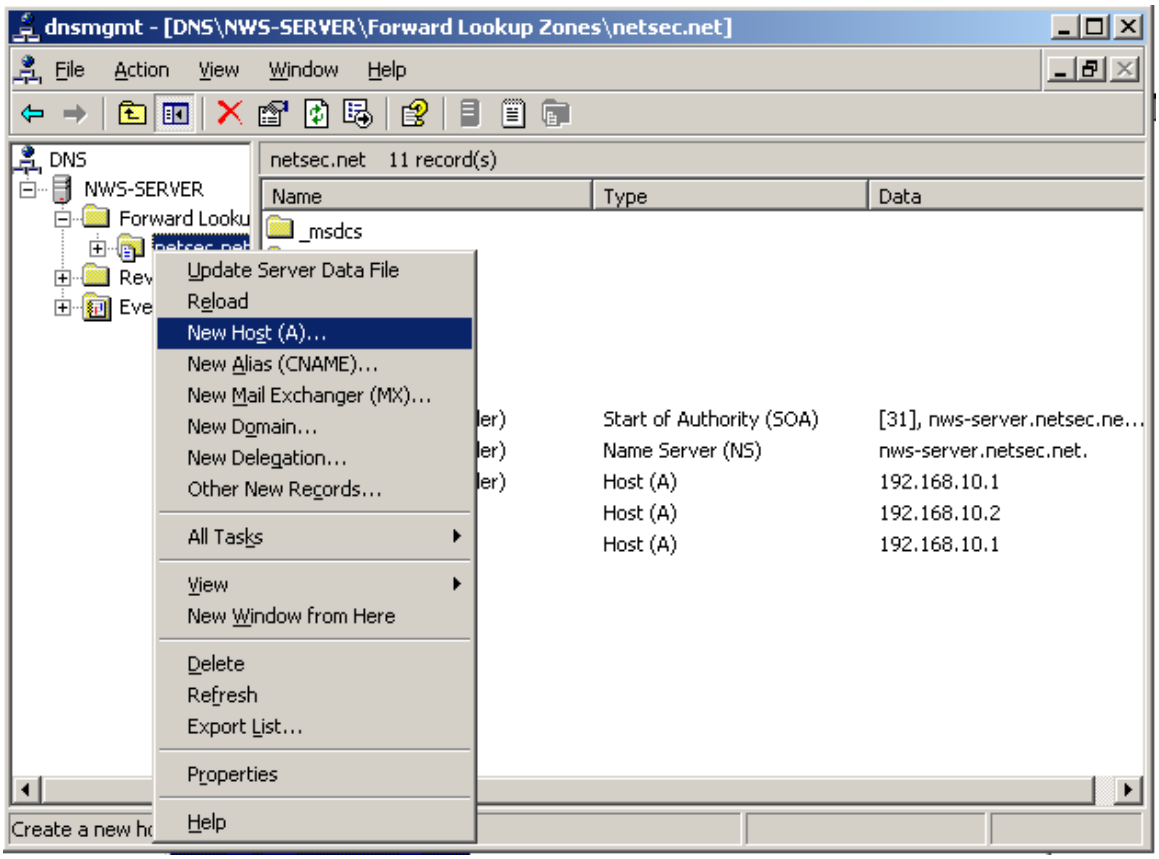


14. Click **Refresh** button and close the IIS Manager.

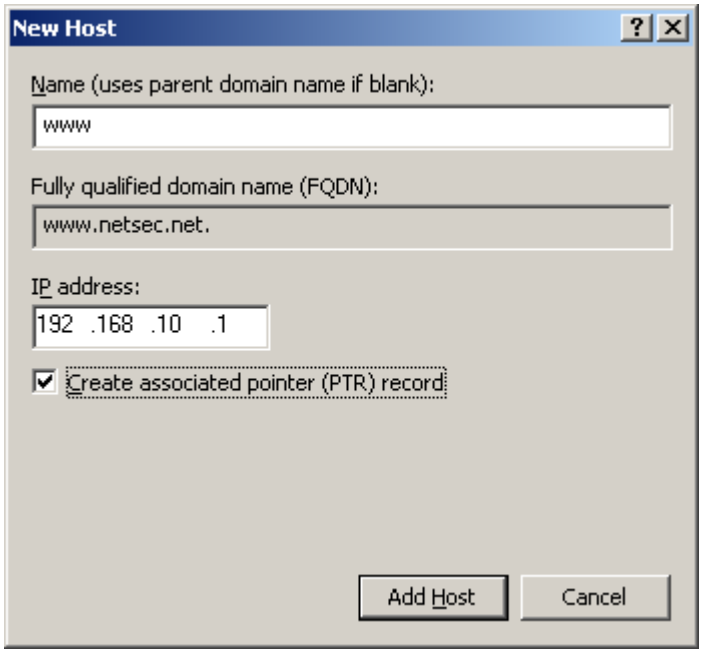
Note: To change these and other setting later, right-click the Web site, and click **Properties**.

Configure DNS Service for Host Header

15. Open the **DNS** manager. (Start; All Programs; Administrative Tools; DNS)
16. In the DNS manager window, double-click the computer icon and expand the **Forward Lookup Zones** field.
17. Right-click your **domain.net** and select **New Host (A)…**



18. In the Name box type **www**, and in the IP address box type your server IP. Make sure you check **Create associated pointer (PTR) record**.



19. Click **Add Host** button.
20. Click OK.
21. Click **Done** button.
22. Click **Refresh** button and close the DNS manager.
23. Log off the server computer.

Test The New Web Site on Web Server

24. Log on to the **client computer** as **Administrator**.
25. Open Internet Explorer. (**Start; All Program; Internet Explorer)**
26. On the Address box, type http://www.yourdomain.net (e.g.http://www.netsec.net) and click GO button.

27. Your webpage will appear in the browser.
28. Log off the client computer.

Summary

Whether your site is on an intranet or the Internet, the principles of providing content are the same. You place your Web files in directories on your server so that users can establish an HTTP connection and view your files with a Web browser. But beyond simply storing files on your server, you must manage how your site is deployed, and more importantly, how your site evolves. Today, an engaging Web site is seldom a static collection of pages. Most successful Web administrators are kept busy accommodating ever changing Web content.

Each Web site must have a home directory. The default Web site home directory is *LocalDrive*:\Inetpub\Wwwroot. You can change a Web site home directory using IIS Manager.

Lab 21

Installing and configuring FTP Server

Mr. Razali bin Abu Bakar
TTO, Network Security
German Malaysian Institute

Prepared by,
Siti Nurul Farhana bt Mohd Rasid
Nur Faridatul Asma bt Ali

LAB 21: INSTALLING AND CONFIGURING FTP SERVER

Installing and configuring FTP Server

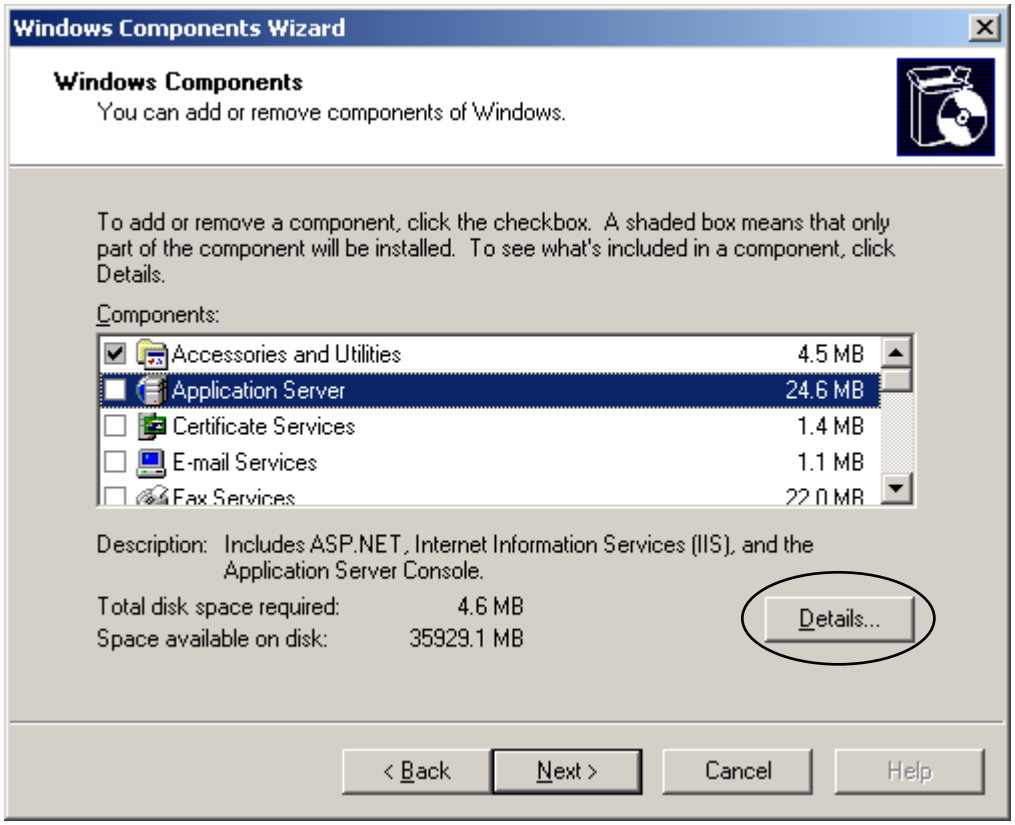
In this exercise, you will install and configure your server to run as FTP server. This exercise also describes installation of the FTP service, and changing default FTP setting globally and for specific FTP sites.

IIS includes the File Transfer Protocol (FTP) service for publishing and managing files. This version of IIS secures and commercializes FTP services for their customers.

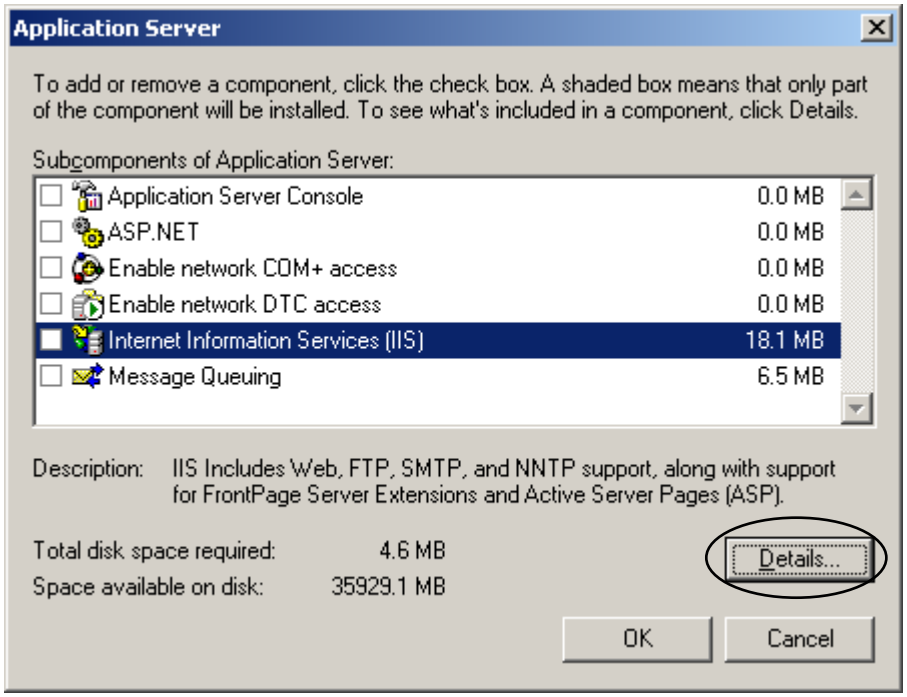
The FTP service is no installed by default. To set up an FTP site, you must first install the FTP services through the Control Panel. Installing the FTP services create a default FTP site, which you can then customize to you needs using IIS manager.

To install FTP Server

1. Log on to the server **Administrator**.
2. Open the **windows Components Wizard**.
   (Start; Control Panel; Add or Remove Program)
3. Click **Add/Remove Windows Components**.
4. Under **Components**: highlight **Application Server**, and then click **Details**.

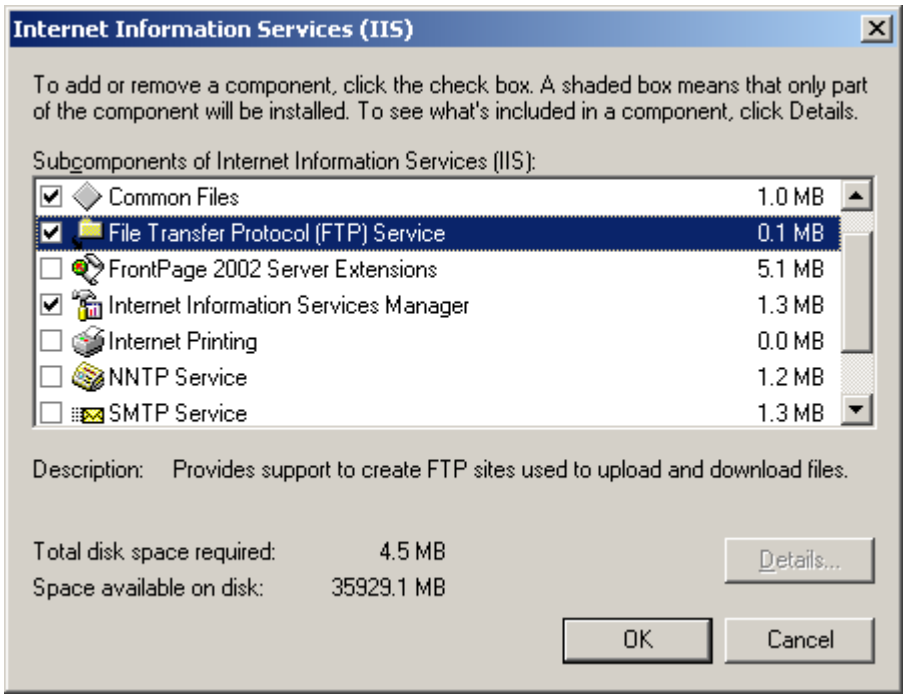


5. In the **Application Server** dialog box, under **Subcomponents of Application Server**, select **Internet Information Services (IIS)**, and click **Details.**



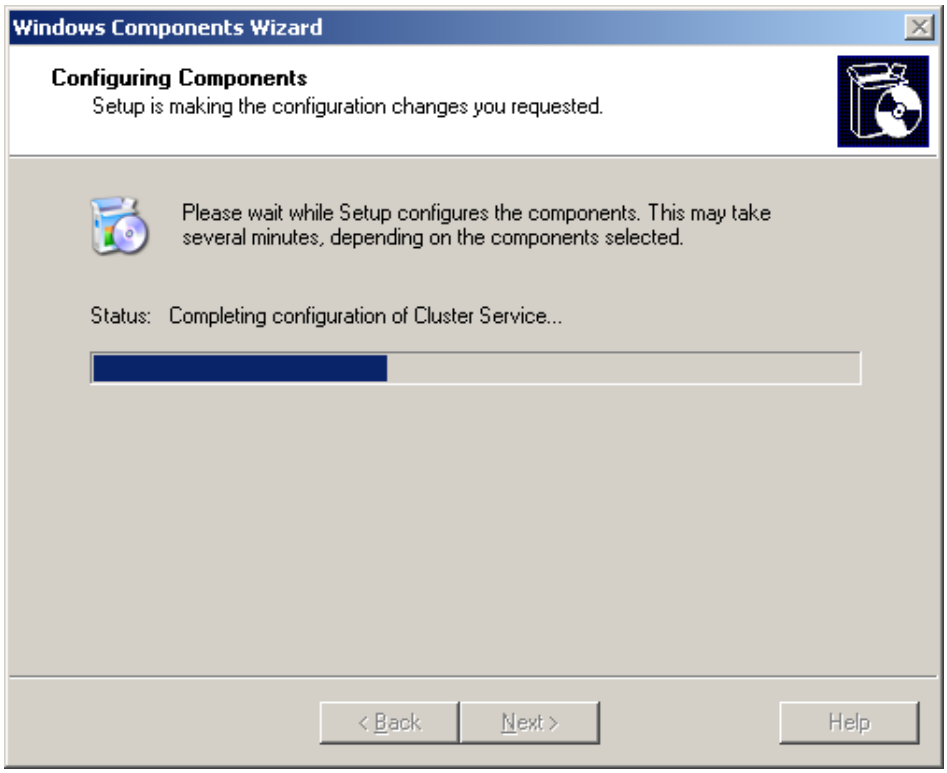
6. In the **Internet Information Services (IIS)** dialog box, under **Subcomponents of Internet Information Services (IIS)**, select item as the following.

- Common Files.
- Files Transfer Protocol (FTP) Services.
- Internet information Services Manager.



7. Click OK, until you return to the **Windows Component Wizard**.

8. Click Next, and windows will start installing IIS and FTP server.



9. After installation completed, click **Finish**.



Configuring FTP Server

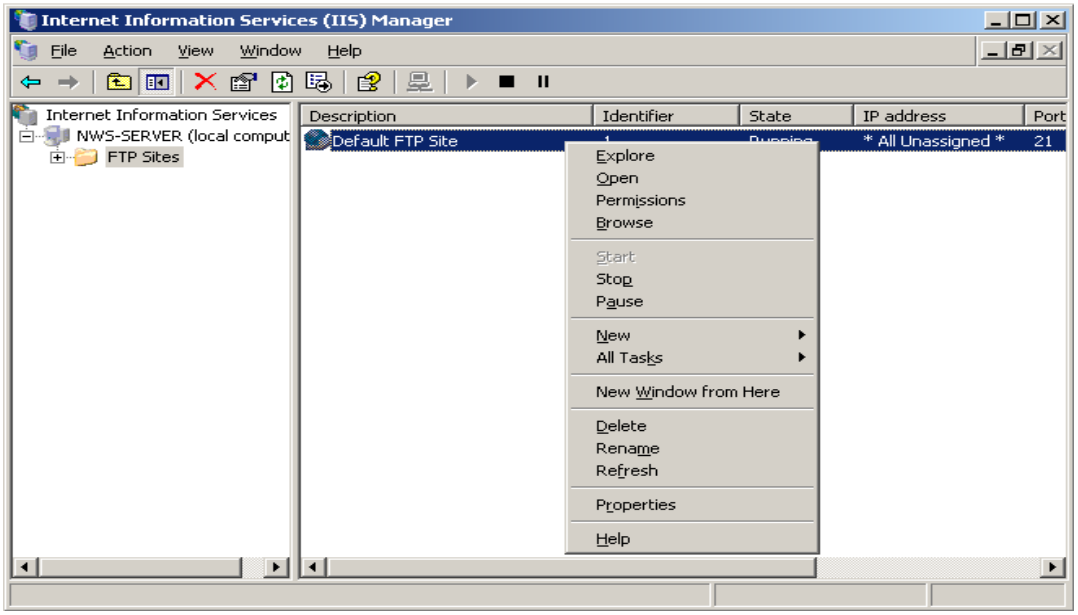
IIS creates a default FTP site configuring on your hard disk at the time of installation. You can use the \netpub\ftproot directory to store your Ftp files, or create any directory or virtual directory you choose.

Setting up the FTP services for the first time involves first setting global FTP settings, then settings for the default FTP site, and finally adding the content to the FTP site. IIS uses an inheritance model, which means that settings on higher levels are automatically inherited by lower levels. Settings at lower levels to can be edited individually override inherited settings from the next level up.

If you change a setting at a lower level, then later change a setting at a higher level that conflicts with the lower-level setting, you will be prompted to choose whether you want to change the lower-level setting to match the new higher-level setting.

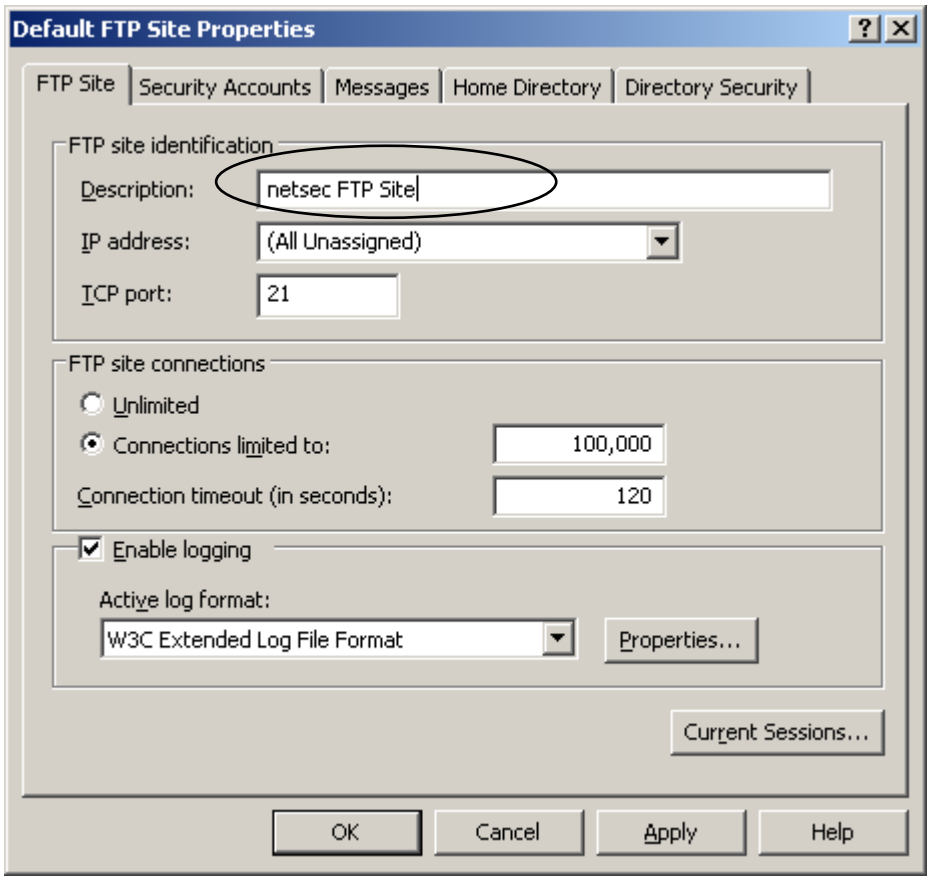
To change settings for the default FTP site.

1. Log on to the server as Administrator.
2. Open **IIS Manager**.
   (Start; Administrative Tools; Internet Information Services (IIS) Manager)
3. In IIS Manager, expand the local computer, expand the **FTP Sites** folder, right – click **Default FTP Site,** and click **Properties.**

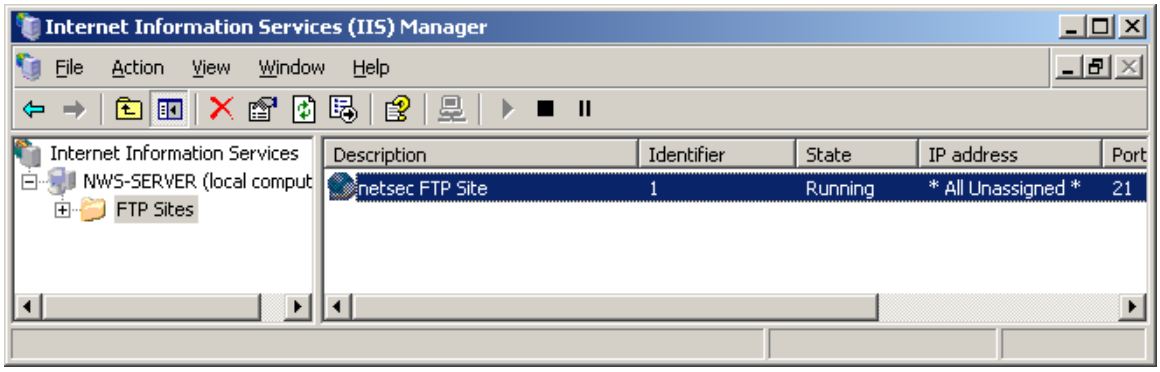


4. On the **FTP Site** tab, under **FTP Site description**, type the name of your FTP Site in the **Description** box.

(e.g : netsec FTP Site)



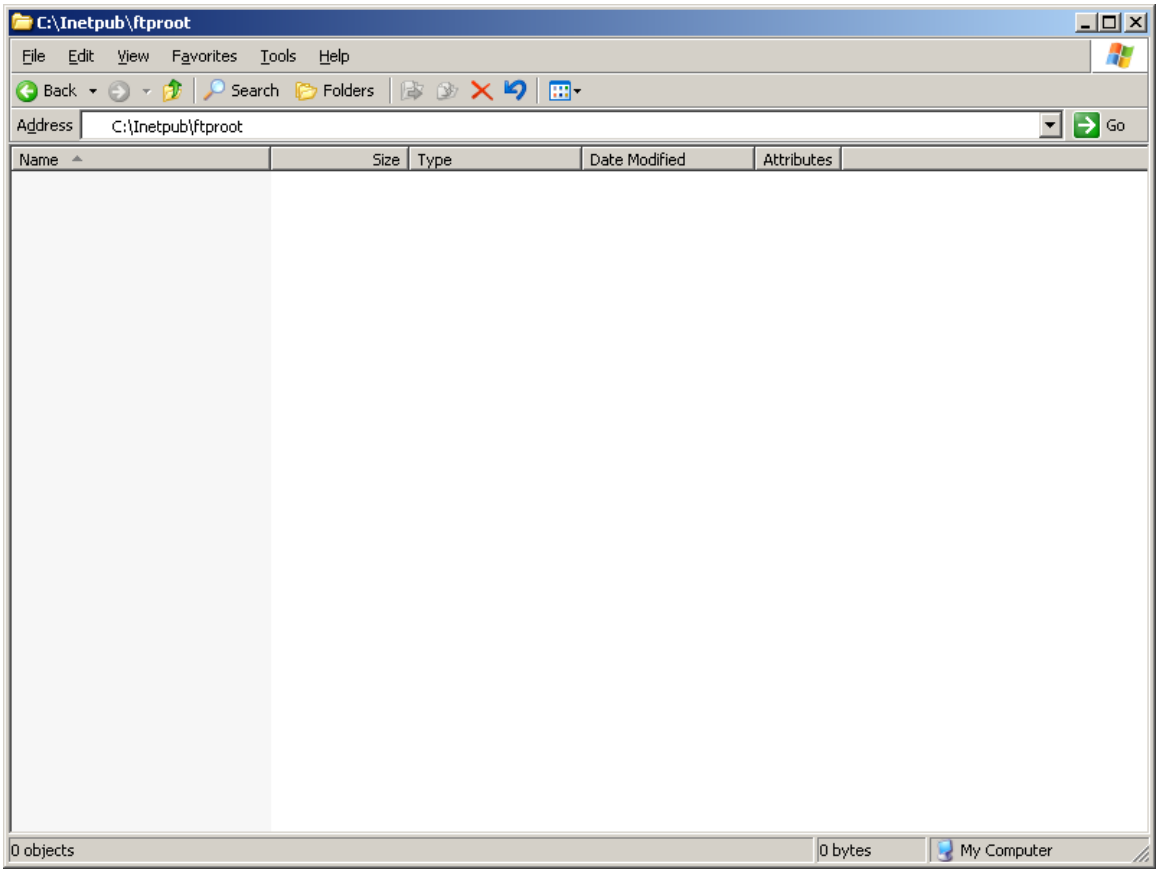
5. Click **OK**. The name of the new site appears in IIS Manager.



6. Click **Refresh** button and close IIS Manager.

To create Text Document in Default FTP folder.

7. Open **Windows Explorer**. Go to ***D:\netpub\ftprott*** directory.



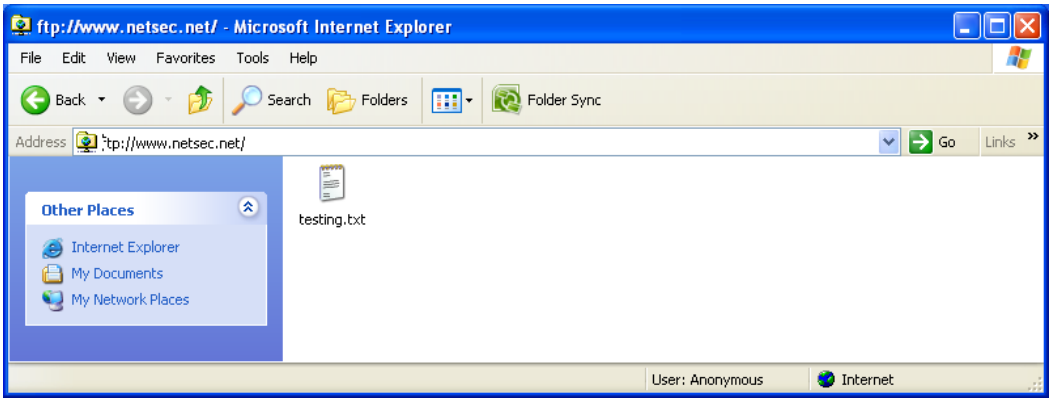
8. Create **New text document** inside D:\netpub\ftprott, and rename the text document as ***testing.txt.***
9. Right-click testing.txt and select **Edit.** This will load the NotePad Editor. Type the following text into the file :

   This only test document to test FTP server.

10. Save and close this file.

Test the default FTP Site

11. Log on to the **client computer** as **Administrator**.
12. Open **Internet Explorer** . (Start ; All Program ; Internet Explorer)
13. On the **Address** box, key-in **ftp://www.yourdomain.net.**
    (e.g : ftp://www.netsec.net) and click **GO** button.



14. Your FTP site will appear in the browser.
15. Now try key-in **ftp://serverSN.yourdomain .net** (e.g : ftp://www.netsec.net) in the Address box and click **GO** button.
    What is the contents of this FTP Site?

    _____

    Were the contents of this FTP Site same with the previous FTP Site? YES / NO

    Why this happen?

    _____

16. Attempt to create the new folder (right click mouse in the windows and select New ; Folder)

    Could you create the folder? YES / NO

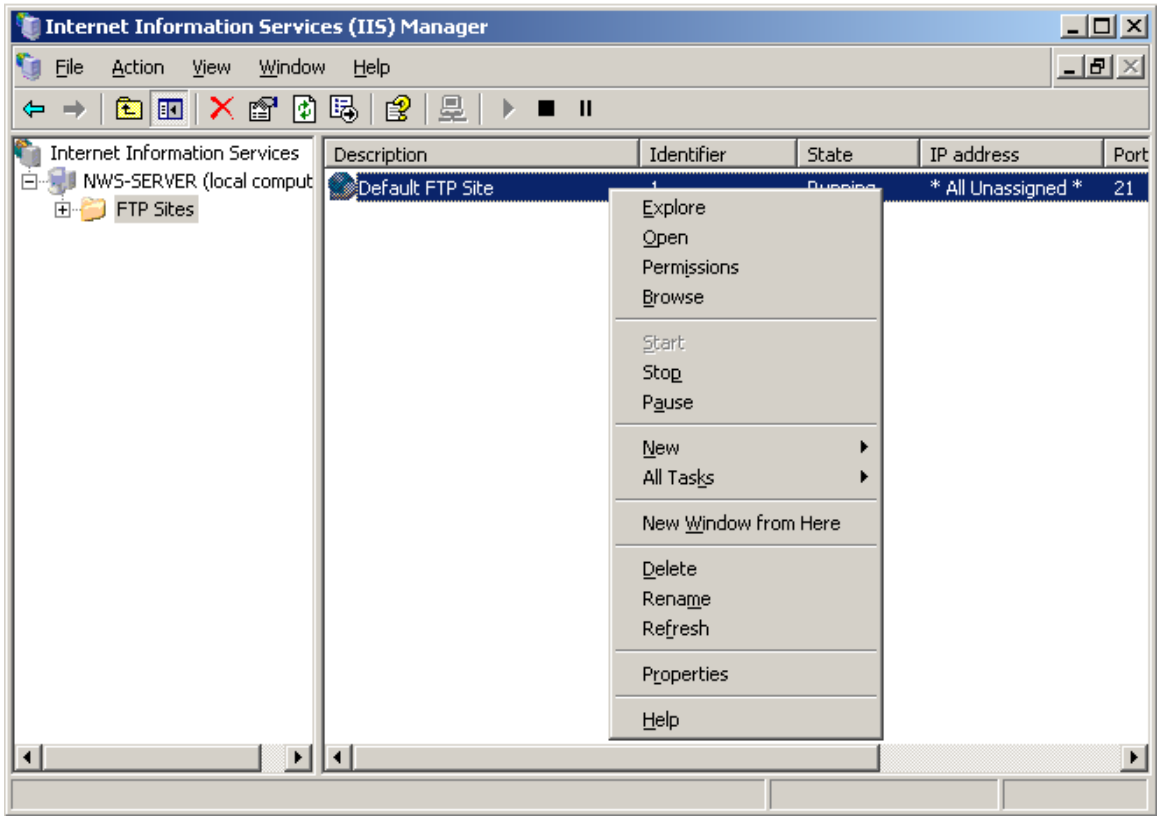
    If NO, why do you think this is so?

    _____

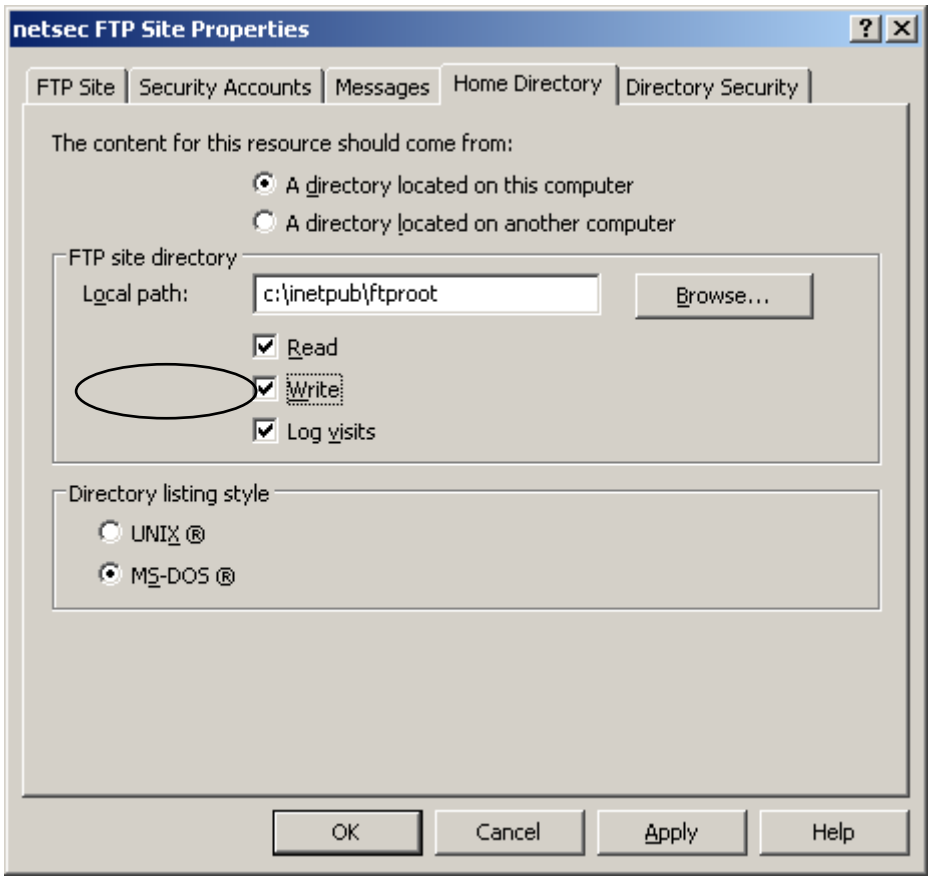
17. Log off the client computer.

To Allow User to Upload or Modify File and Directory Inside Default FTP Site.

1. Log on to the server as **Administrator**.
2. Open IIS Manager. (Start ; Administrative Tools ; Internet Information Services (IIS) Manager)
3. In **IIS Manager**, expand the local computer, expand the **FTP Sites** folder, right-click **Default FTP Site** and click **Properties**.



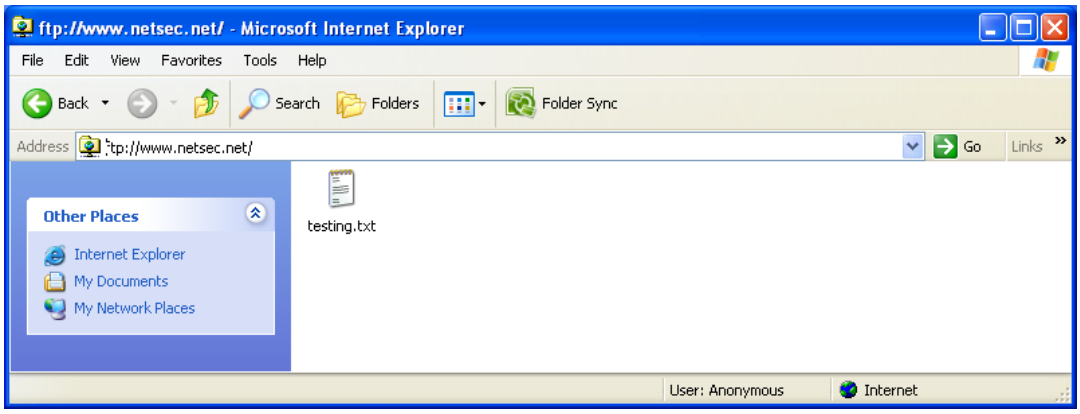
4. Click **Home Directory** tab. Under **FTP Site Directory**, tick the **Write** option.



5. Click **OK**.
6. Click **Refresh** button and close the IIS Manager

Test The Default FTP Site

7. Log on **the client computer** as **Administrator.**
8. Open **Internet Explorer**. (Start ; All Programs ; Internet Explorer)
9. On the **Address** box, key-in **ftp://www.yourdomain.net** (e.g. ftp://www.netsec.net) and click **GO** button.



10. Your FTP Site will appear in the browser.
11. Attempt to create a new folder (right-click mouse in the window and select New;Folder)

    Could you create the folder? YES / NO

12. Now try copy any files and paste it to this FTP Site.

    Could you paste any files? YES / NO

    Why do you think this is so?

    _____

13. Log off the client computer.

Creating New FTP Site Using Multiple IP Address

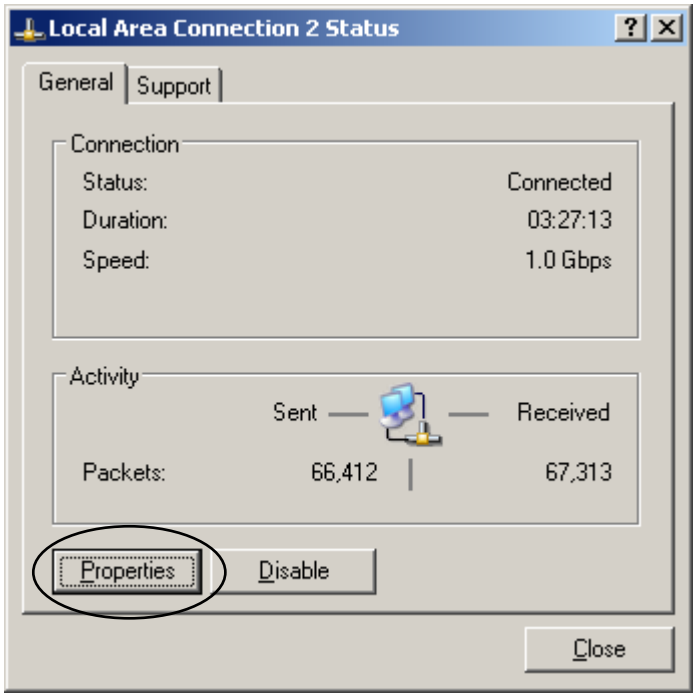
You can create multiple FTP Site using multiple IP addresses and multiple ports. While creating multiple sites with multiple IP Addresses is a common and recommended practice, it can be more complicated because, by default, clients call port 21 when using the FTP protocol. Therefore, if you create multiple FTP using multiple ports, you need to inform users of the new port number so their FTP clients can locate and connect to the port.

If you create a new site using the same port as an existing site with the same IP Address, the new site will not start. The general rule is that you can have multiple site using the same IP and Port, but only one site from this group can run at a time. If you try to start another site from this group, you receive an error message.

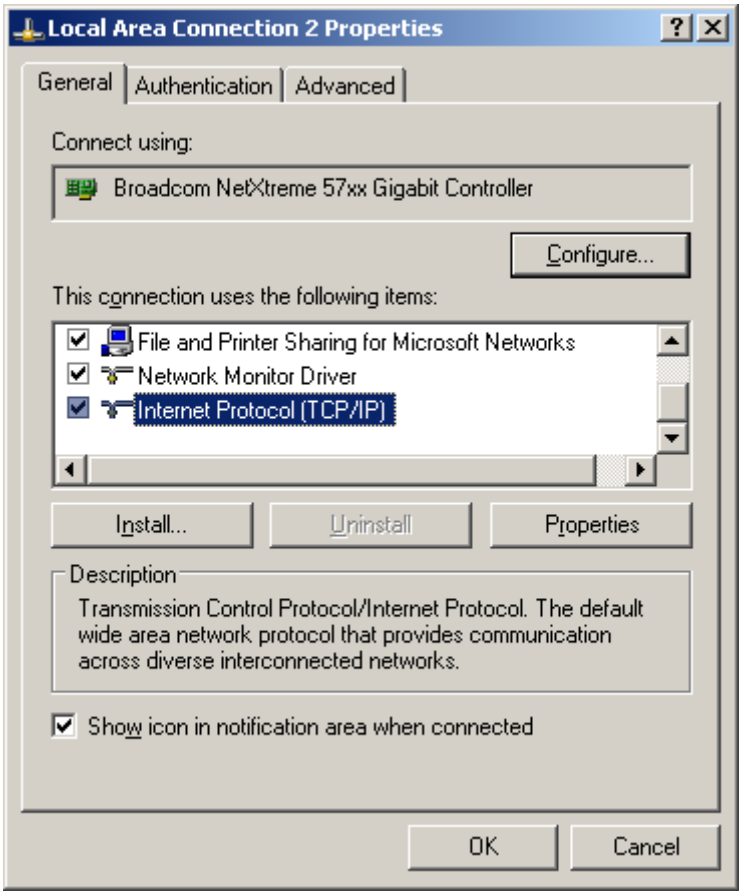
Before your start create multiple FTP site using multiple IP address, you need to make sure your server have set with multiple IP address. If not, you have to set your server to use multiple IP address.

Creating Multi IP Address in Single NIC

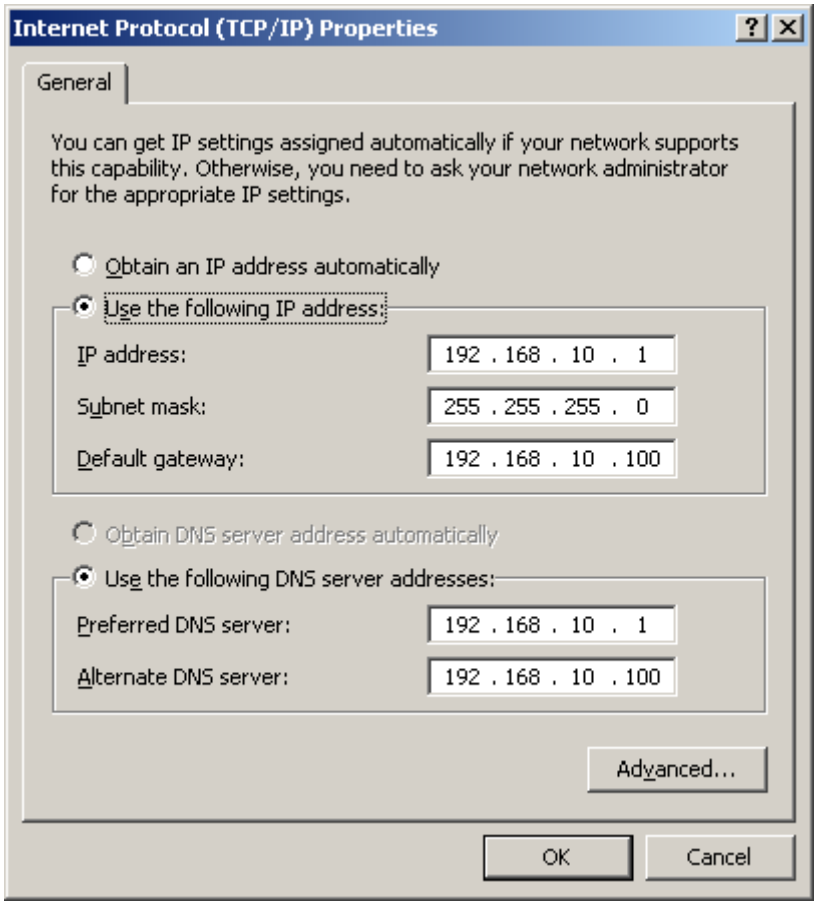
1. Log on to the server as **Administrator**.
2. Open **Local Area Connection**. (Start – Control Panel –Network Connection – Local Area Connection)
3. Click **Properties** button.



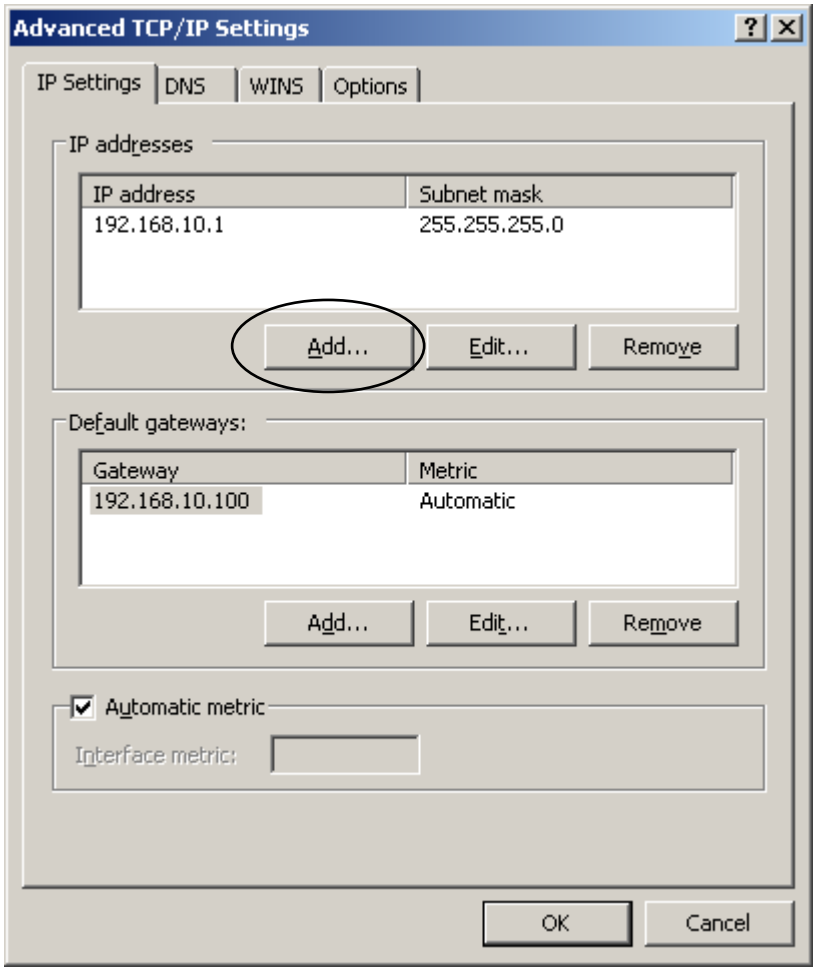
4. Select **Internet Protocol (TCP/IP)**



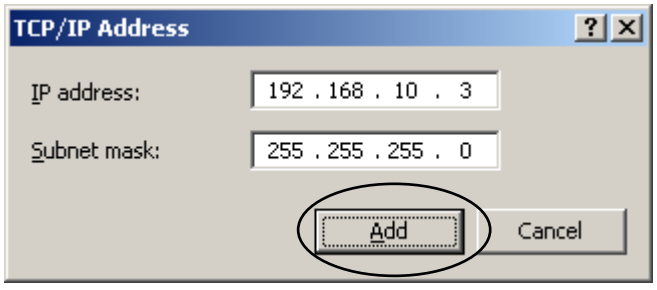
5. Click **Properties b**utton.



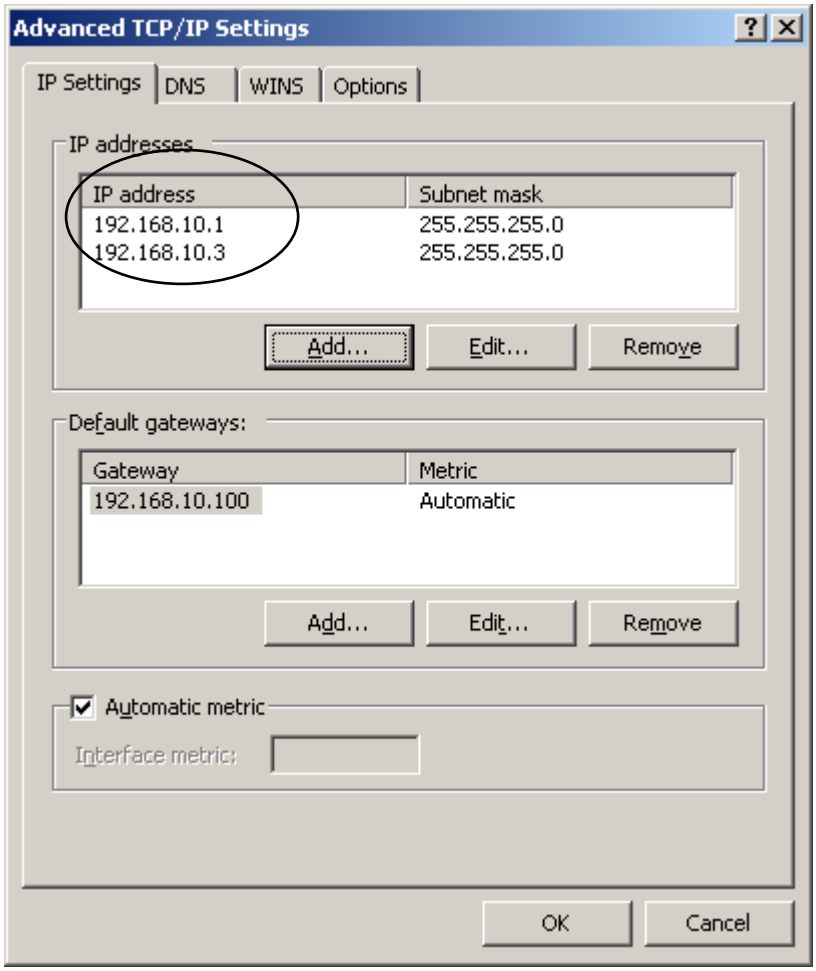
6. Click **Advanced** button.
7. Select the **IP setting** tab.
8. Under **IP address** field, click **Add** buton.



9. Enter your second IP address for your server. (e.g : 192.168.10.**ServerNumber**)



10. Enter your subnet mask number. (e.g : 255.255.255.0)
11. Click **Add** button.



As you can see, now your server has 2 IP address

12. Click OK.
13. Click Ok.
14. Click Close.
15. Click Close.

Creating a FTP Drop Site.

FTP Drop Site can be useful for businesses that need to allow anonymous users to upload large files for your company via FTP. By setting appropriate NTFS permissions on a drop folder on your FTP server, you can allow users to upload files without being able to view or even know about files uploaded by other users.

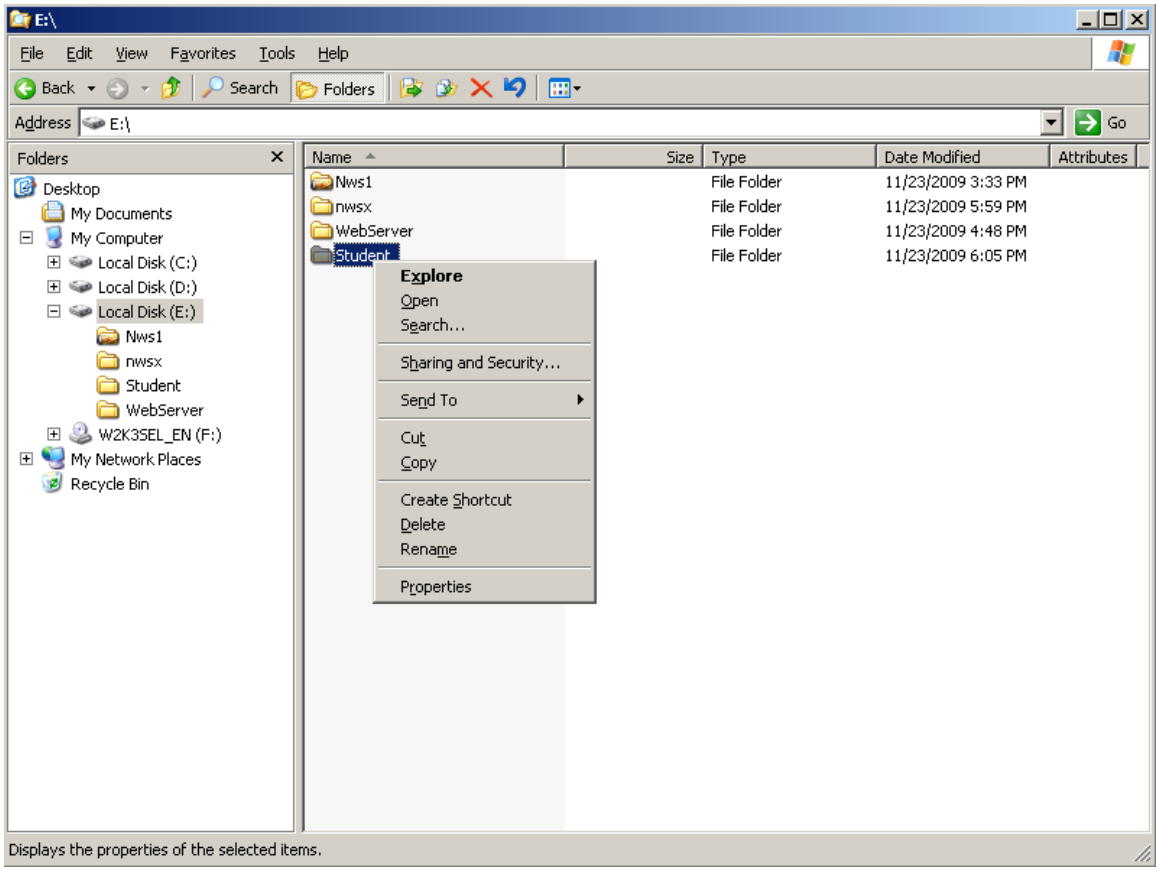
Before your start creating new FTP drop site, you need to create a new FTP root folder for this site.

Let create a new FTP site for student called **Student** that makes use of this new features.

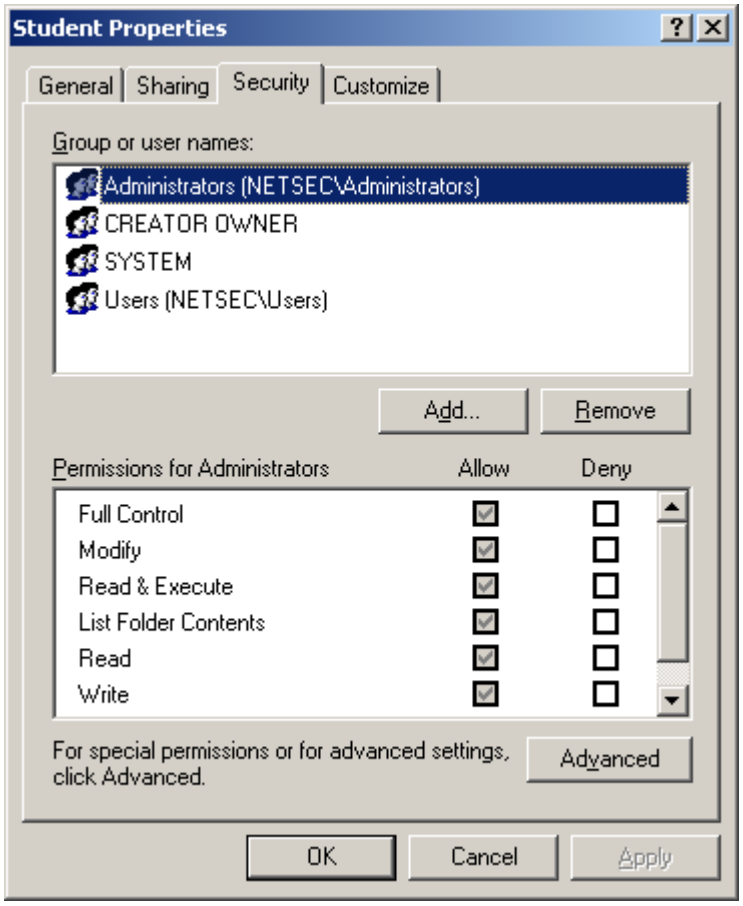
1. Log on to the server as **Administrator** .
2. Open **Windows Explorer**. (Start ; All Programs ; Accessories ; Windows Explorer)
3. Expand drive **E:**
4. Create new sub-folder on drive E: named **Student** as the root directory for the site and 192.168.10.3 for the site's IP address.

Preparing the Drop Box

5. Create new sub-folder called **Drop** under **E:\Student** which you will use as your FTP drop box.
6. View the default permission for your new **Student** folder. (Right-click folder **E:\Student**; Properties).

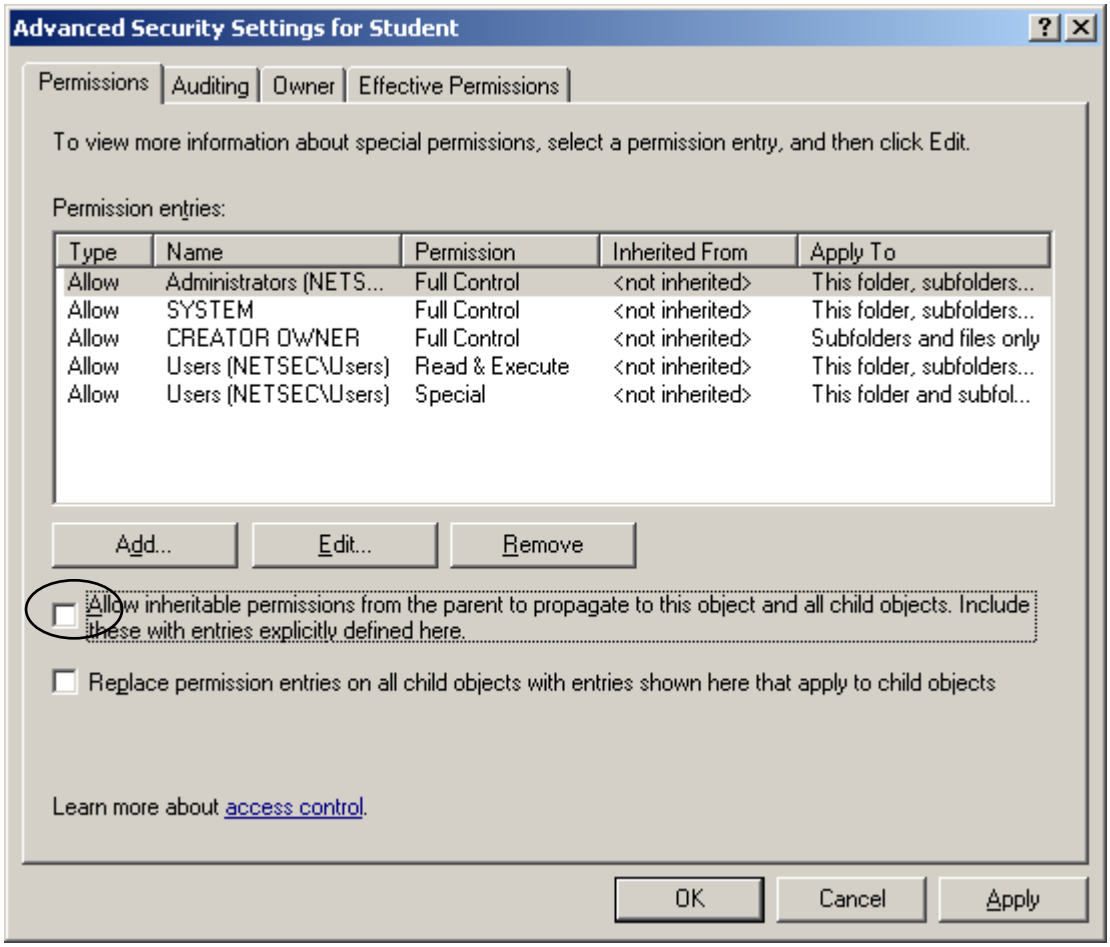


7. Click **security** tab. You should see your default folder security setting permissions for your new **Student** folder.



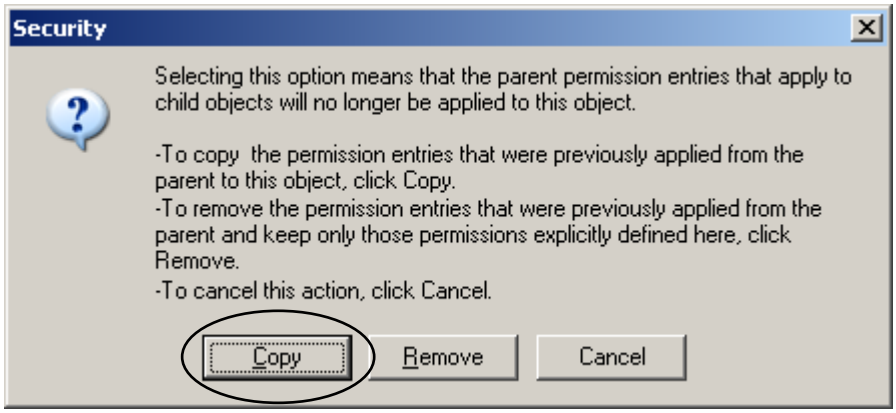
Let's change the permissions on the Student folder to make them more suitable for our purpose.

8. Click **Advanced** button to open the Advanced Security Setting dialog box.
9. Clear the check box labeled "**Allow inheritable permissions from the parent to propagate to this object……**"



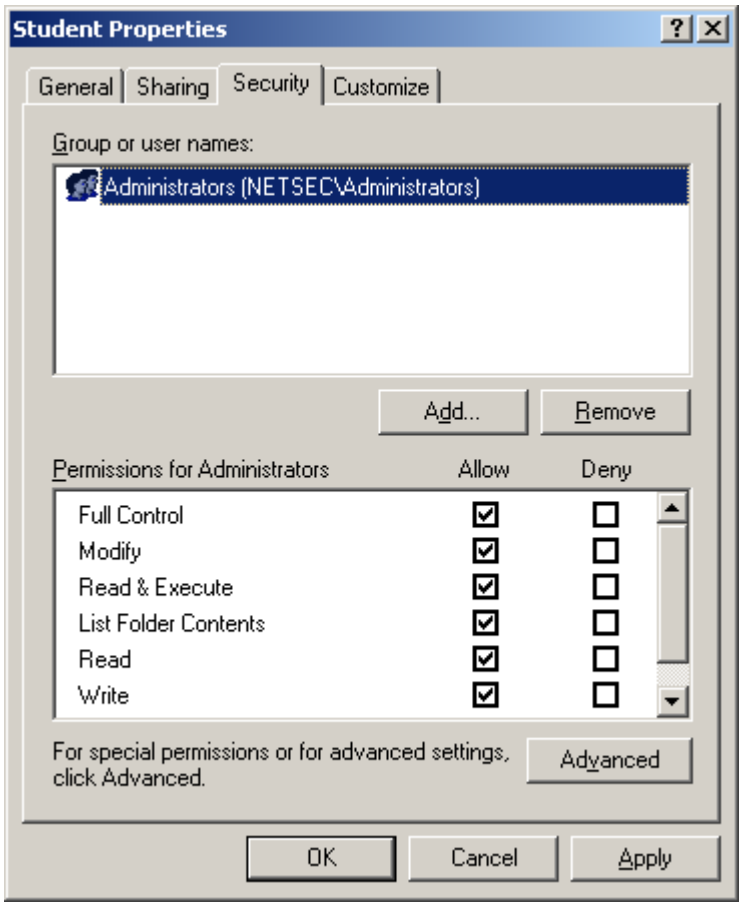
10. When a new dialog box asking you whether you want to copy or remove the existing inherited permissions on the folder, select **Copy**.

The result is that existing permissions on the Student folder are the same as default permissions but they are now explicit permissions rather than inherited.



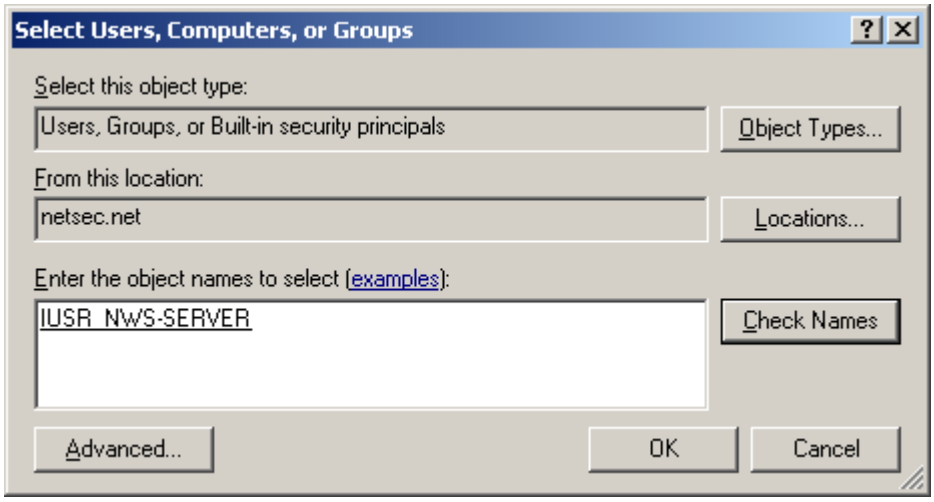
11. Click OK to close the Advanced Security Setting dialog box and return to the properties for the Student folder.
12. Remove the ACEs for the following three groups: CREATER OWNER, SYSTEM, and Users. At this point the only ACE on the Student folder is Administrators have Full Control.

To delete users, **highlight** users and click the **Remove** button.
(e.g : highlight **SYSTEM**; click **Remove** button).

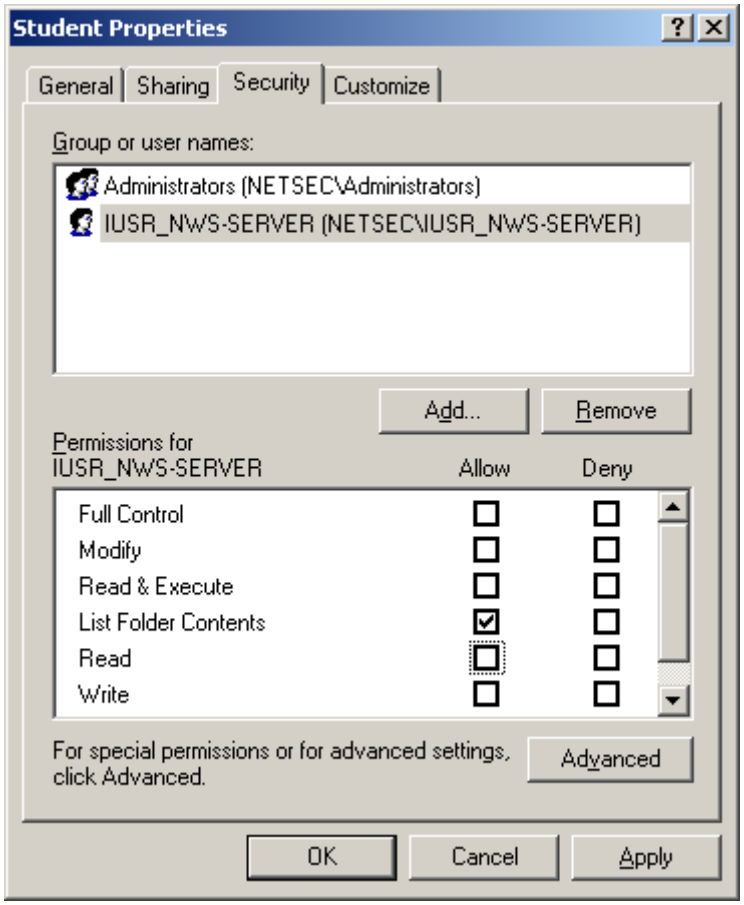


13. Now click **Add** button.
14. Add an ACE for the IIS anonymous user account, which by default is named IUSR_servername where servername is the name of your Windows Server 2003 machine.

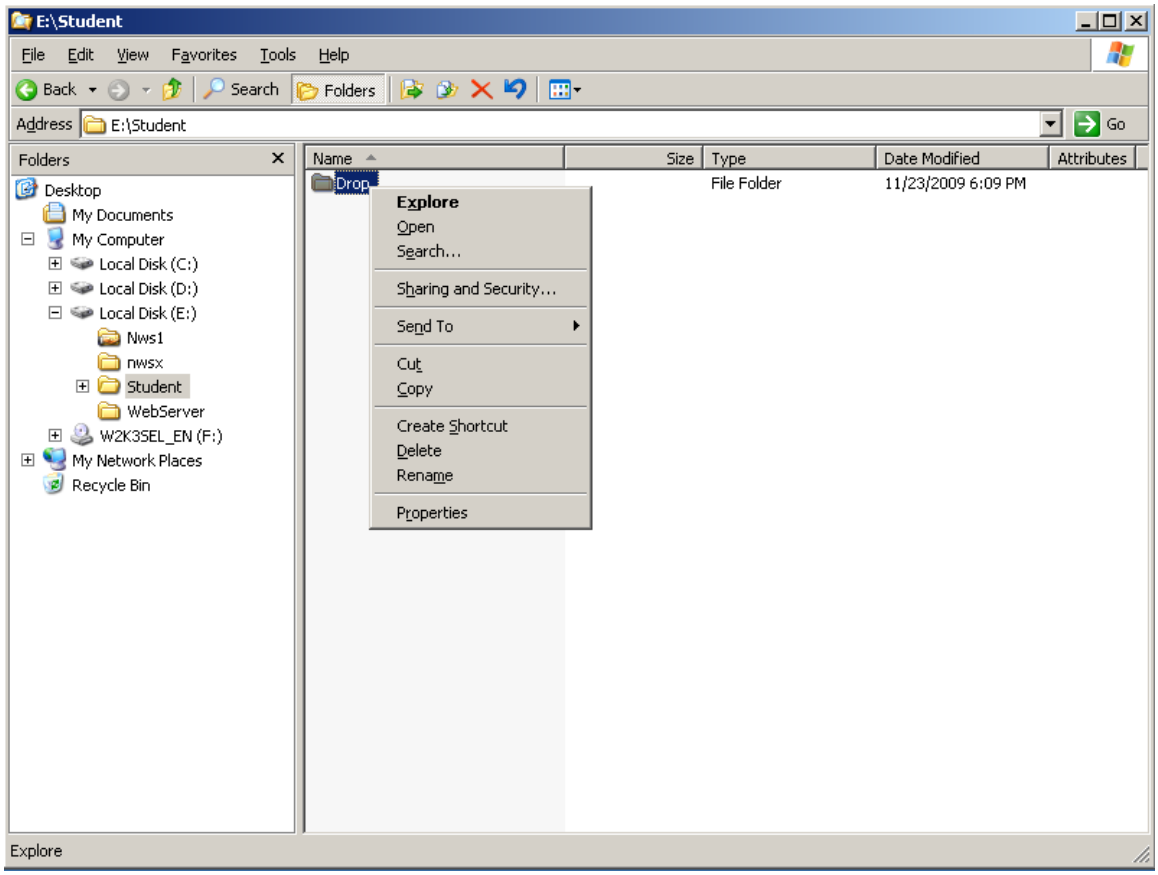
Key-in **IUSR_yourservername** (e.g : IUSR_netsec) and click **Check Names** button.



15. Click **OK**.
16. Configure the ACE for IUSR_yourservername so that the account has only **List Floder Contents** permissions allowed as shown below.

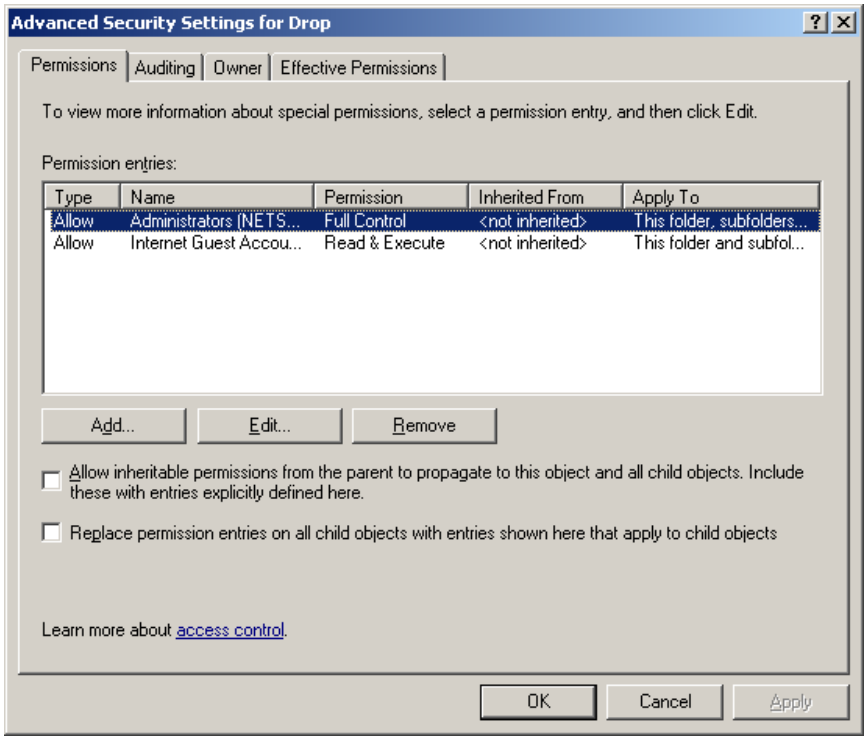


17. After finish configure , click **OK**.
18. What we have at this point is an Student folder that allow Administrators full access but anonymous users only the abililty to list the folder ‘s contents. These permissions by default are also inherited by the drop subfolder beneath Student folder, and those permissions need some modification too so, now open the **Properties** sheet for the **drop folder** (Right-click folder **E:\Student\Drop**; Properties).



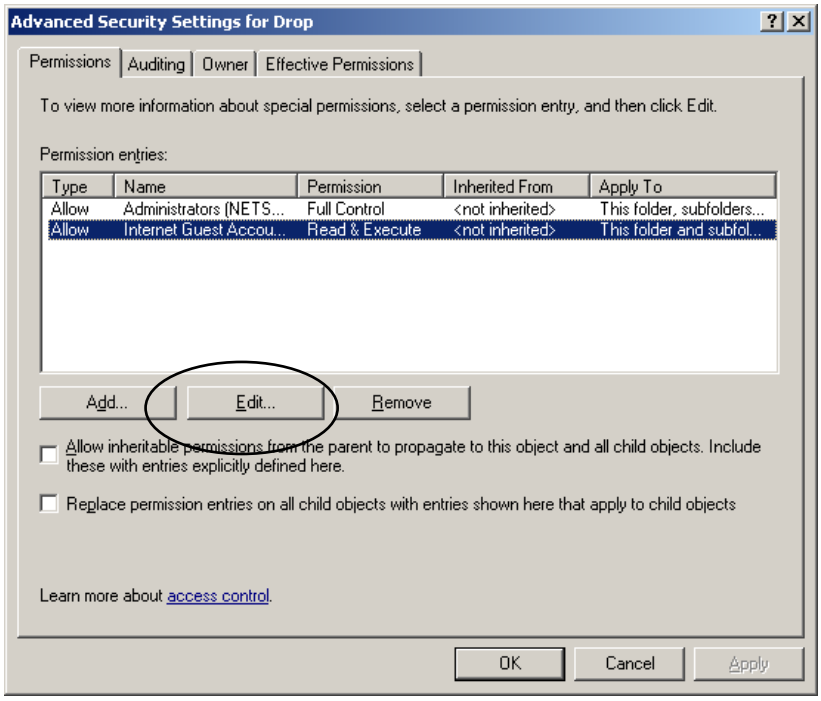
19. Click **Security** tab.
20. Click **Advanced** button.
21. Clear the “Allow **inheritable permissions..**” checkbox.
22. Click **Copy** , and click **OK** so we can manually modify permissions on the drop folder.

23. Click the **Advanced** button again to open the Advanced Security Setting for the drop folder, which should now look like Figure below :

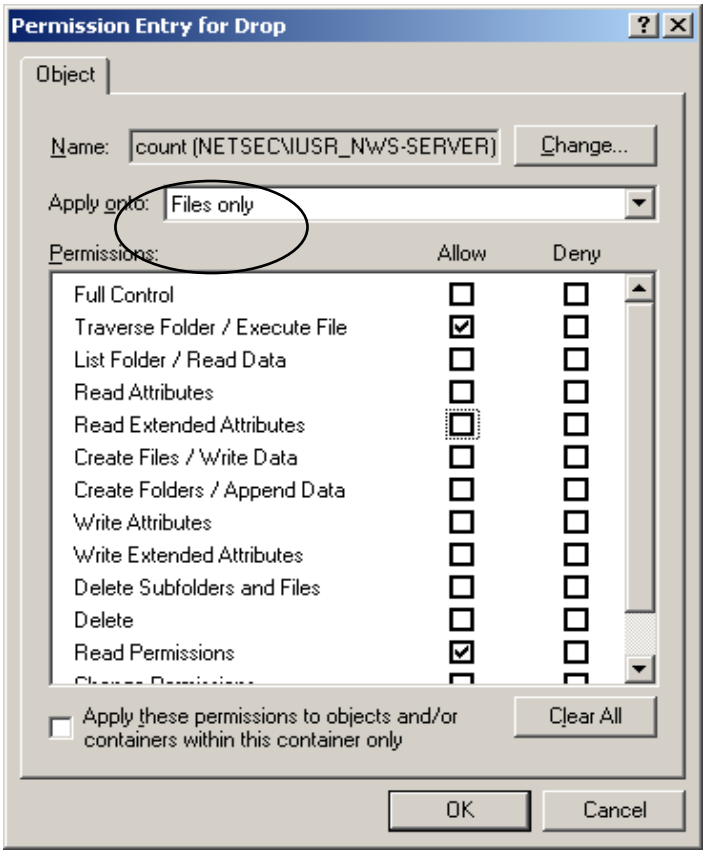


24. Note that although the standard permissions (ACE on the Properties page) for IUSR_yourservername was List Folder Contents, here these show up as Read &Execute. We need to tighten the permissions for IUSR_yourservername even further,

So select this identity as shown in Figure below and click the **Edit** button to display the special NTFS permissions for IUSR_yourservername.



25. When the permissions Entry dialog box appears, Changing the **Apply Onto** : dropdown list box to **Files Only.**

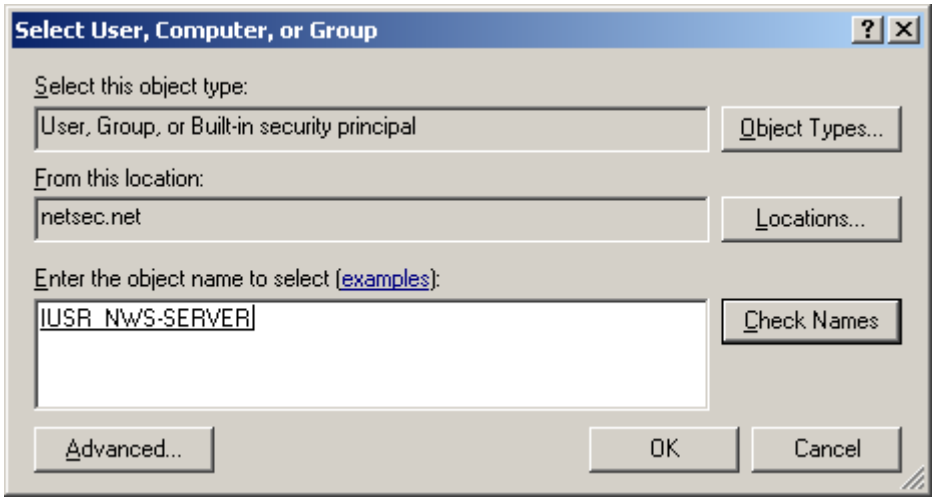


26. Remove checkmarks beside everything **except List Folder / Read Data** and **Read permissions**.
27. Click **OK** to close the permissions Entry box and modify the special permissions for IUSR_yourservername.
28. Now we need to create another ACE for IUSR_yourservername, but this time it will apply to the drop folder itself and not to files within that folder as the ACE we did previously.

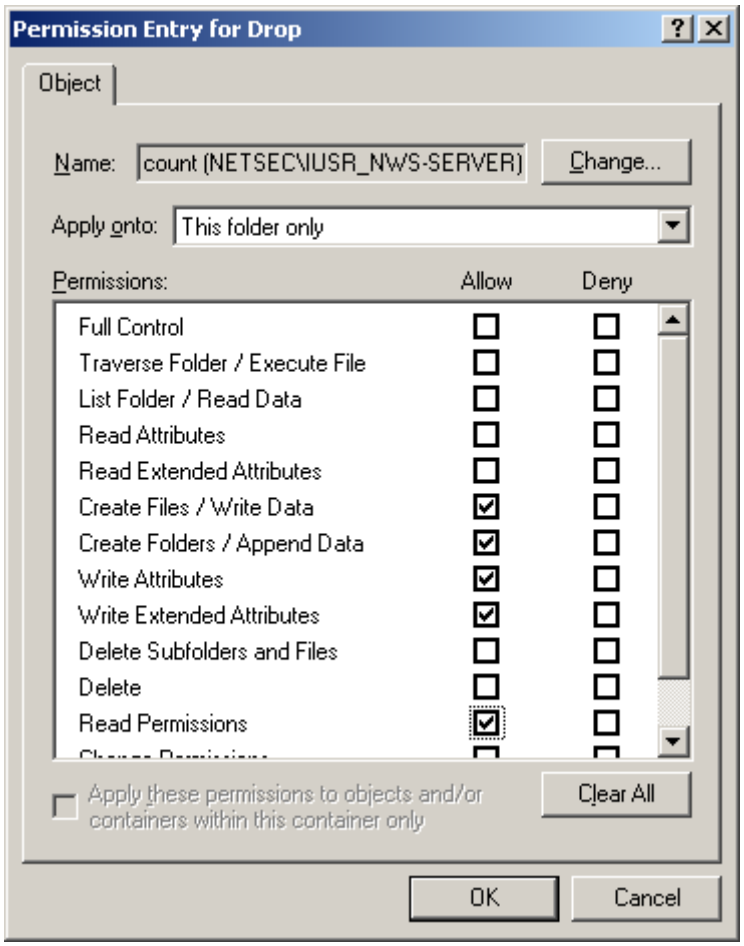
To create this new ACE, click the **Add** button, specify IUSR_yourservername as (e.g : IUSR_netsec) the account you want to apply permissions to.

Click **Check Names** button.

Click **OK**



29. Configure the permissions as shown in Figure below.



30. Click OK.
31. Click OK.
32. Click OK.
33. Close all remaining dialog boxes and you're done.

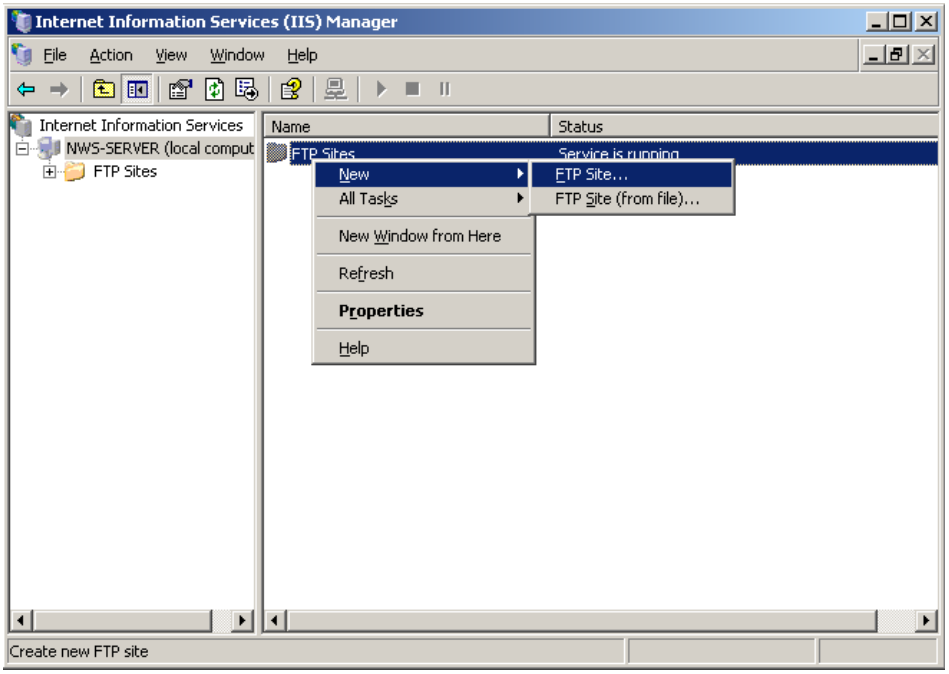
Creating New FTP Site Using Multiple IP Address

You can create multiple FTP sites using multiples IP addresses and multiple ports. While creating multiple sites with multiple IP addresses is a common and recommended practice, it can be more complicated because, by default, clients call port 21 when using the FTP protocol. Therefore, if you create multiple ports, you need to inform users of the new port number so their FTP clients can locate and connect to the port.

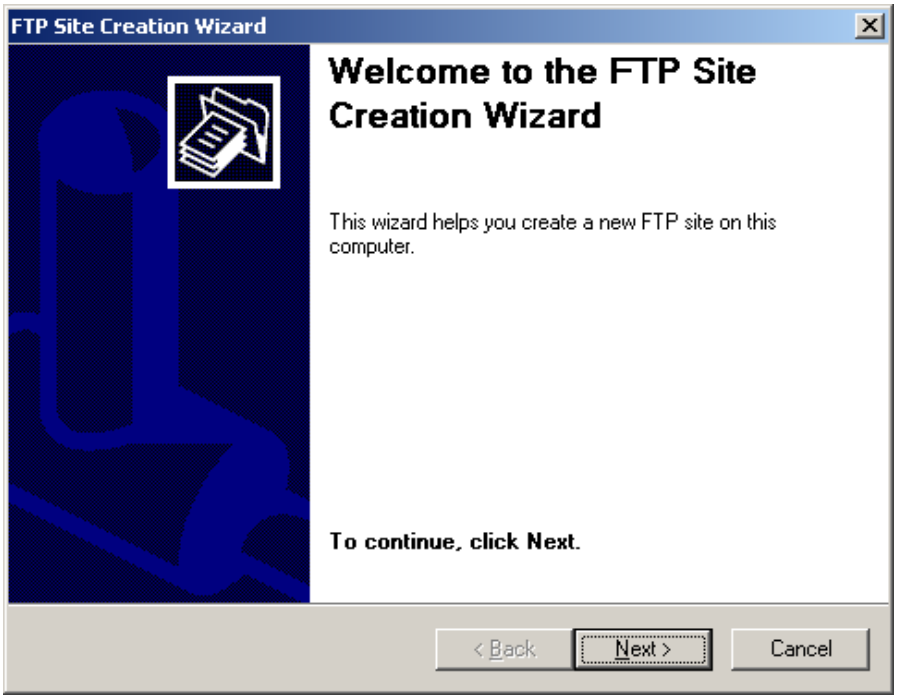
If you create a new site using the same port as an existing site with the same IP address, the new site will not start. The general rule is that you can have multiple sites using the same IP and port, but only one site from this group can run at a time. If you try to start another site from this group, you receive an error message.

Let create a new FTP site for student called **Student** that makes use of this new features.

1. Log on to the server as **Administrator**.
2. Open **IIS Manager**. (Start; Administrative Tools; Internet Information Services (IIS) Manager)
3. In IIS Managers, expand the local computer, right-click the **FTP Sites** folder, point to **New**, and click **FTP Site**.

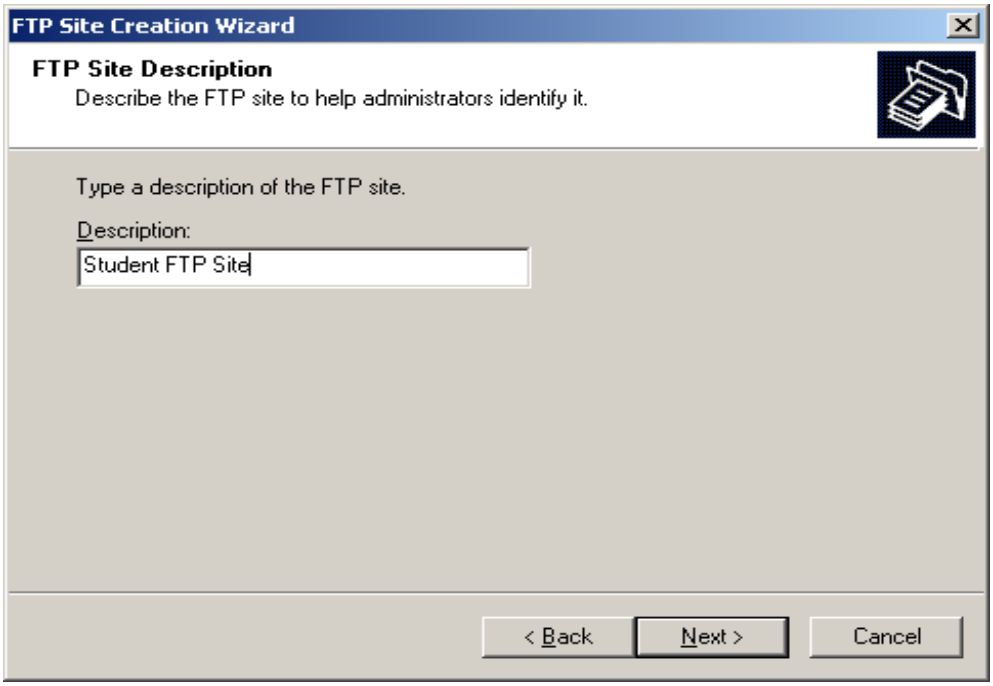


4. FTP site Creation Wizard appear. Click **Next**.

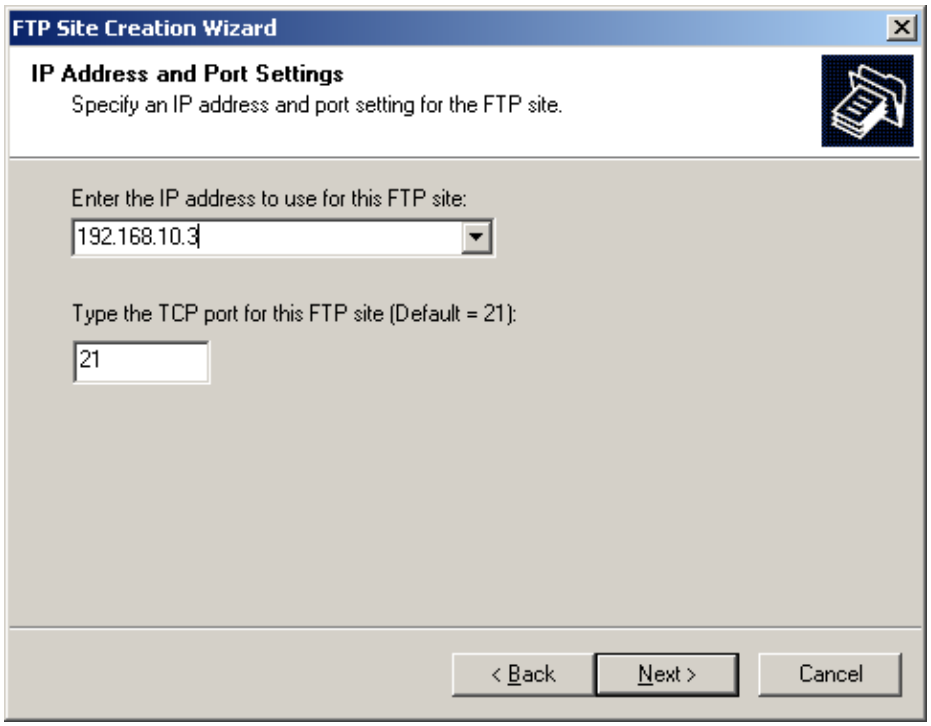


5. **FTP Site Description** dialog boxes appear. Key-in **Student FTP Site** in the **Description**: box

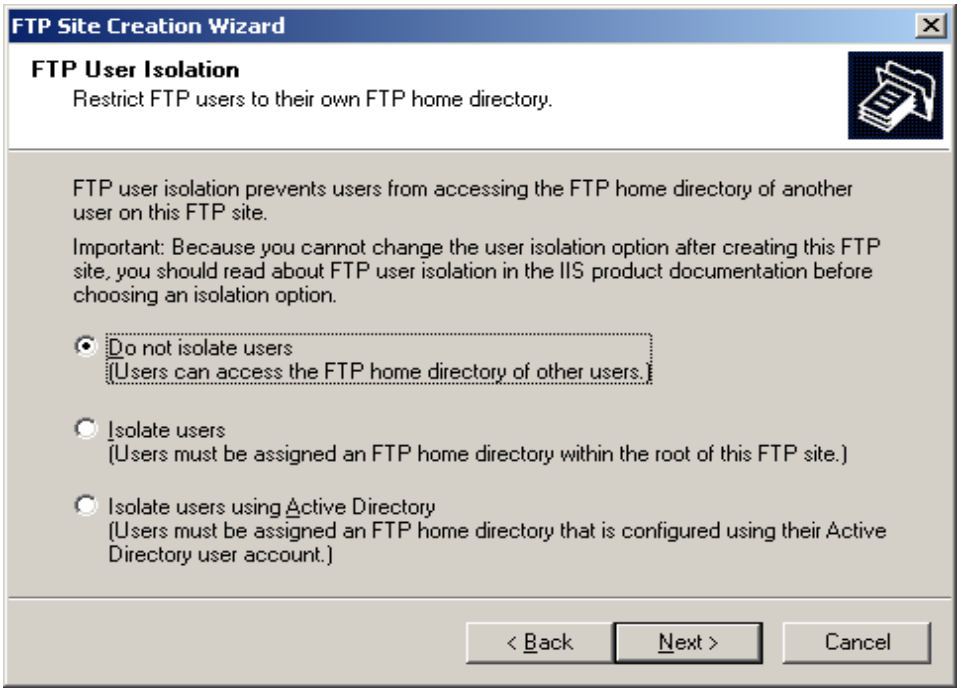
   Click **Next.**



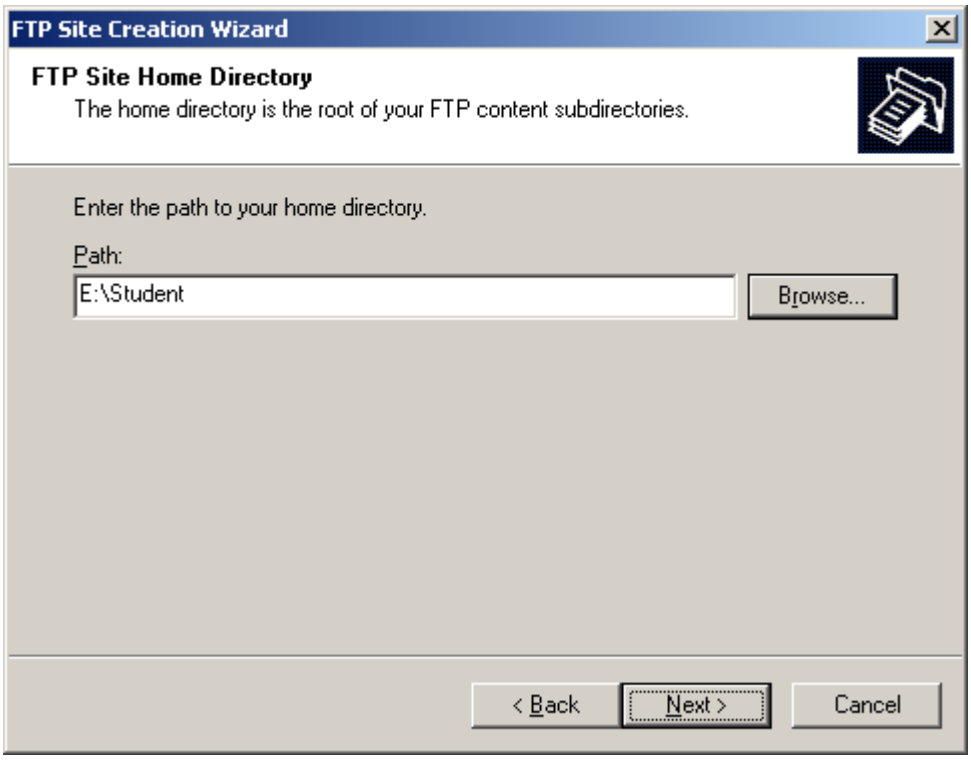
6. Now the wizard asking for **IP Address and Port Settings**, key-in your server second IP address (e.g. 192.168.10.3) and just leave the port default setting (Default = 21). Click **Next**.



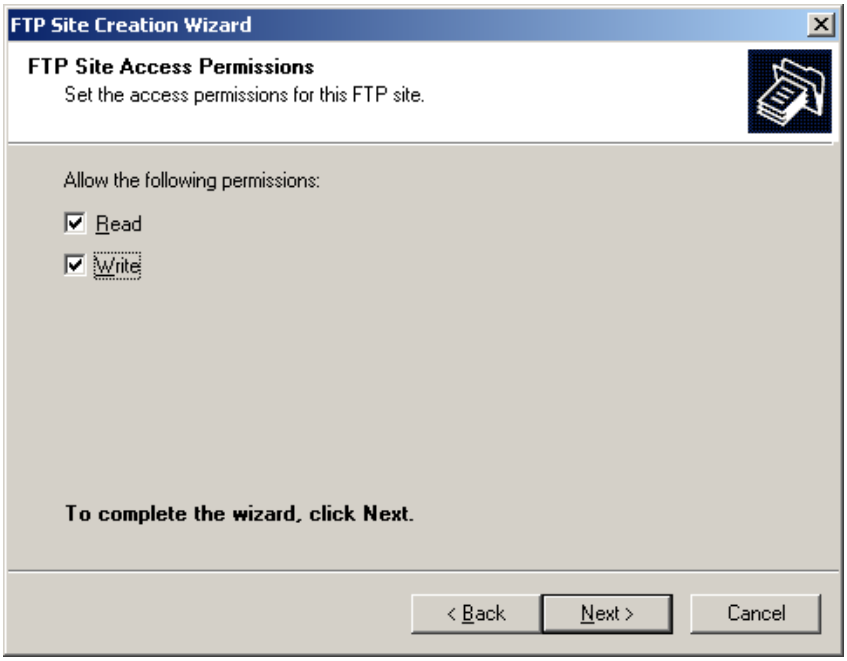
7. In the **FTP User Isolation** dialog box, click **Isolate users**, and click **Next**.



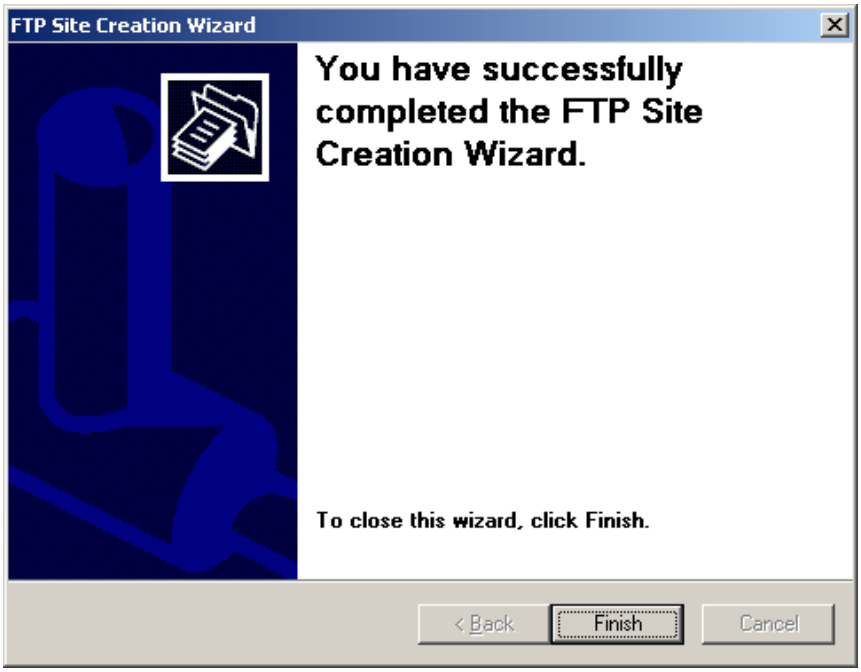
8. Set the FTP Site Home Directory. Under the **Path:** field key-in the FTP site home directory (e.g. **E:\Student**).



9. Click **Next.**
10. Set the FTP site access permission to Read and Write to allow user upload and modify the FTP site contents.



11. Click Next.
12. Click Finish.

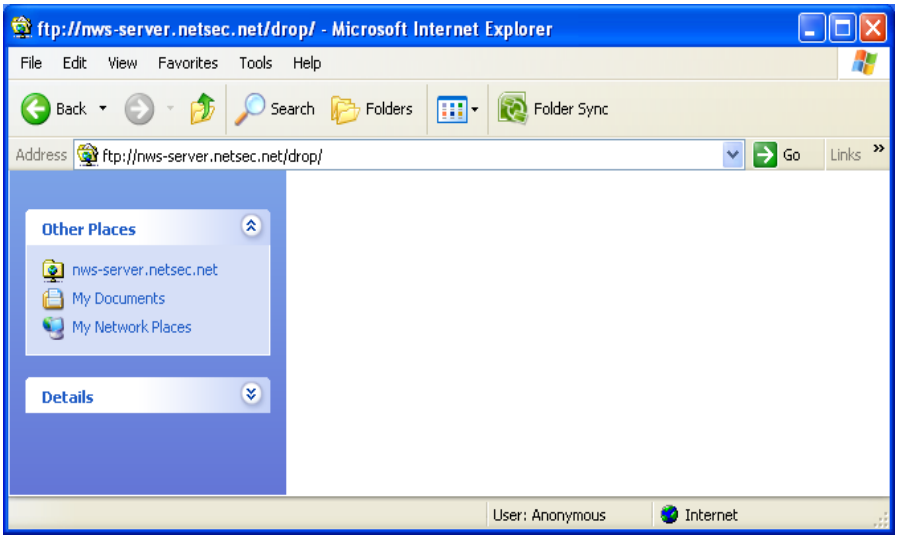


13. Log off the server computer.

Test The Drop Box

1. Log on to the **client computer** as **Administrator**.
2. Open Internet Explorer. **(Start; All Programs; Internet Explorer)**.
3. On the **Address** box, key-in **ftp://serverSN .yourdomain.net/drop** or **ftp://your Domain_second_IP/drop**

   (e.g. ftp://server 1.netsec.net/drop or ftp://1902.168.10.3/drop) and click **GO** button.



4. Your FTP site will appear in the browser.
5. Use Windows explorer to access the C:\Windows\Web\Wallpaper sub-folder.
6. Click on the file **Bliss.bmp** and copy it (it is around 1407 KB in file size).
7. Paste the file **Bliss.bmp** into the drop folder on the FTP server.
8. Now try copy any files and paste it to this FTP server.

   Could you paste any files? YES/NO
9. Press **F5** key.

   What happen?

   _____

   _____

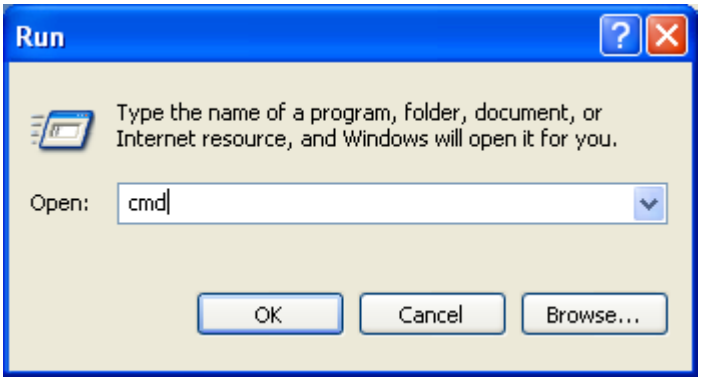
   Why do you think this is so?

   _____

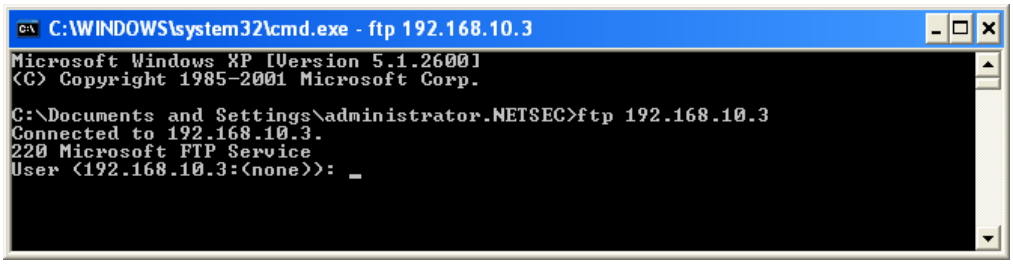
   _____

10. If you refresh the view in Internet Explorer by pressing F5, the displayed files disappear. In other words, once files are uploaded they can't be seen anymore or by anyone else who accesses the drop folder using an FTP client. To double-check this, you can connect to your drop folder using the Windows command-line ftp.exe client and try to view the contents of the drop folder after uploading some files to it (see Figure below). The result is that you are denied access when you try to list the contents of the drop folder.

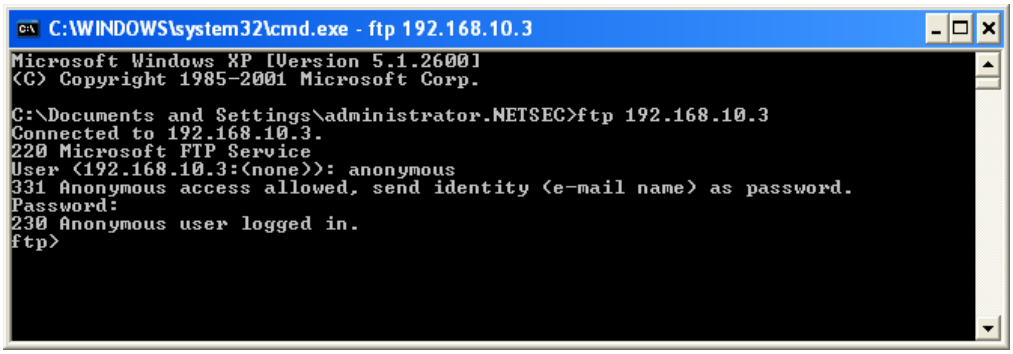
10.1 Open **Command Prompt** (click Start; Run; type **cmd** and press Enter)



10.2 At command prompt, key-in **ftp yourFTPserver_second_IP** and press **Enter**.



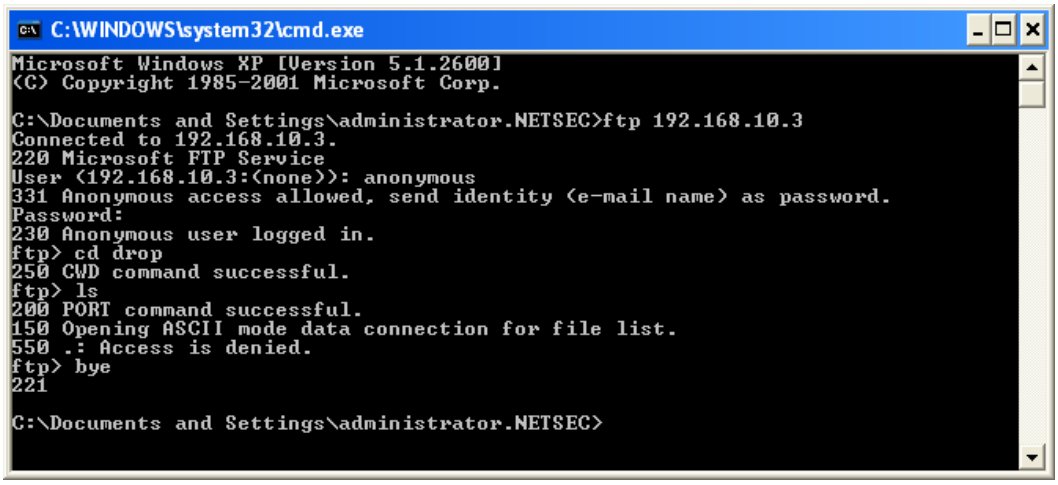
10.3 You'll be asking to enter the username. Key-in **anonymous** as username and just press **Enter** for password (NO Password).



10.4 Key-in **cd drop** and press **Enter** to go inside the **Drop** folder.

10.5 Key-in **Is** and press **Enter** to display contents of **Drop** folder.

As you can see, your access are denied.



10.6 Key-in **Bye** and press **Enter** to logout from FTP server.

11. Log off the client computer.

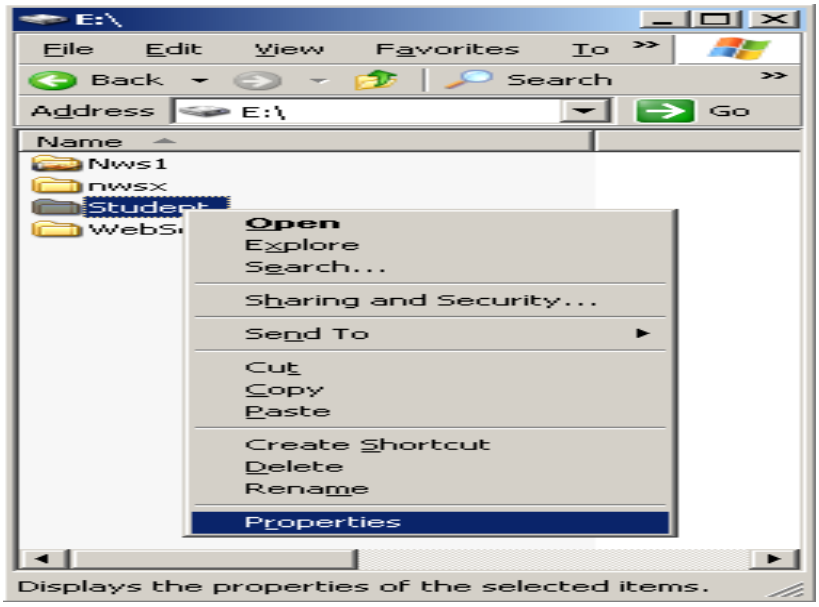
Creating a FTP Site for Specific User

FTP Site can be set to be login only by specific user. You can allow specific users to establish an FTP connection and transfer files with an FTP client or FTP-enabled Web site is deployed, and more importantly, how your site evolves. This section presents the basics of managing the infrastructure of an FTP site, from securing your site to hosting multiple sites.

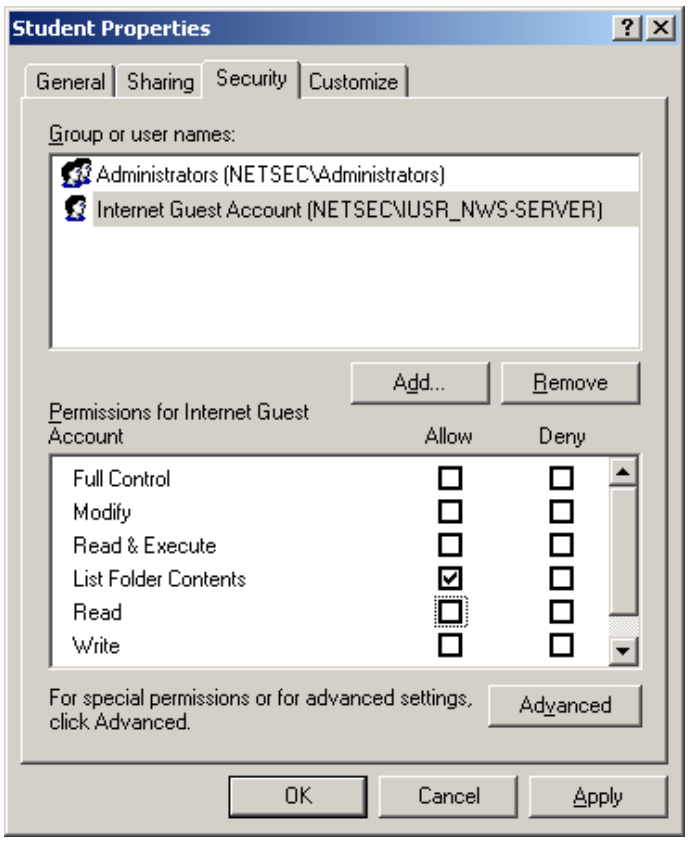
This exercise to help administrator, and particularly Internet hosting providers, efficiency secure and commercialize the FTP services for their customers.

Let's say we want to set **Azhar** as administrator for the **Student FTP Site** you created earlier.

1. Log on to the server as **Administrator.**
2. Open **Windows Explorer.**
   (Start ; All Program ; Accessories ; Windows Explorer)
3. Expand drive **E**:
4. View the default permissions of your **Student** folder. (Right-click folder **E:\Student** ; Properties)

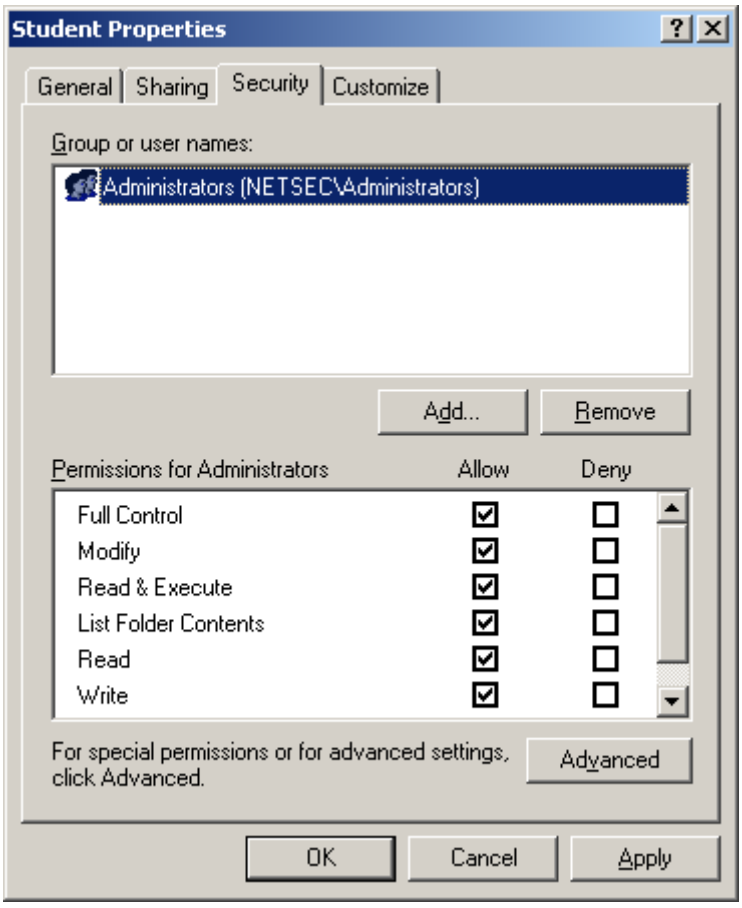


5. Click **Security** tab. You should see your default folder security setting permissions for your new **Student** folder.

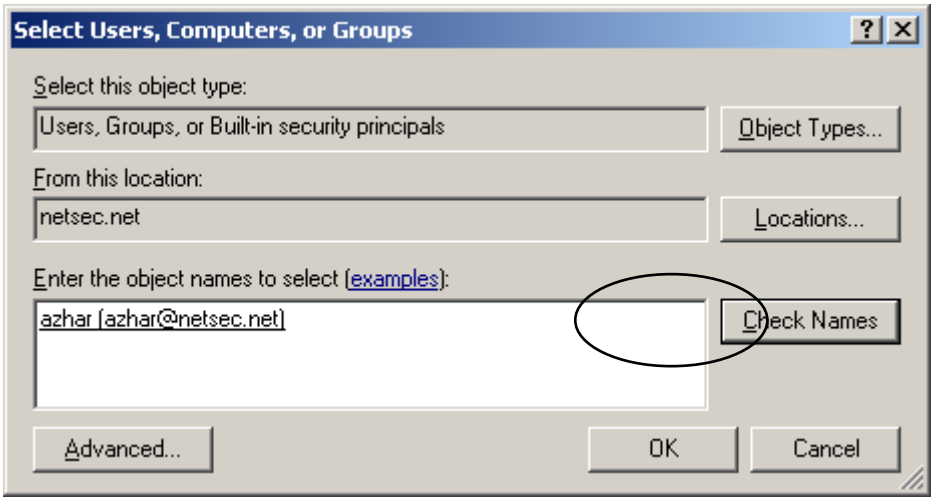


6. Delete the **IUSR_yourservername** (e.g : IUSR_netsec) user.

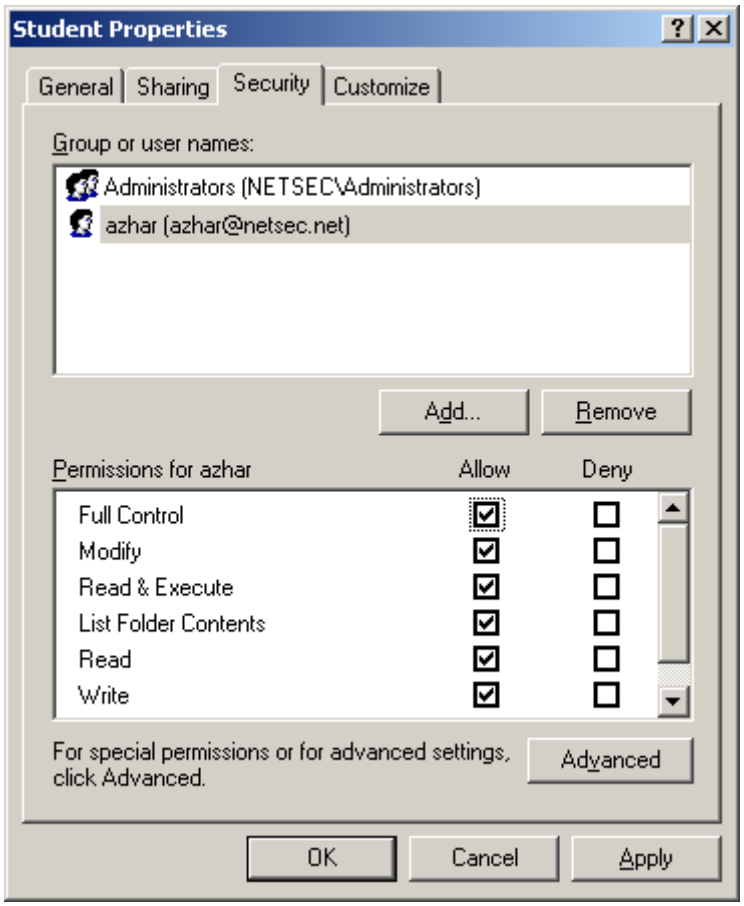
   To delete users, **highlight** the **IUSR_yourservername** (e.g : IUSR_netsec) and click **Remove** button.



7. Now click **Add** button.
8. Key-in **Azhar** to add Azhar and click **Check Names** button.



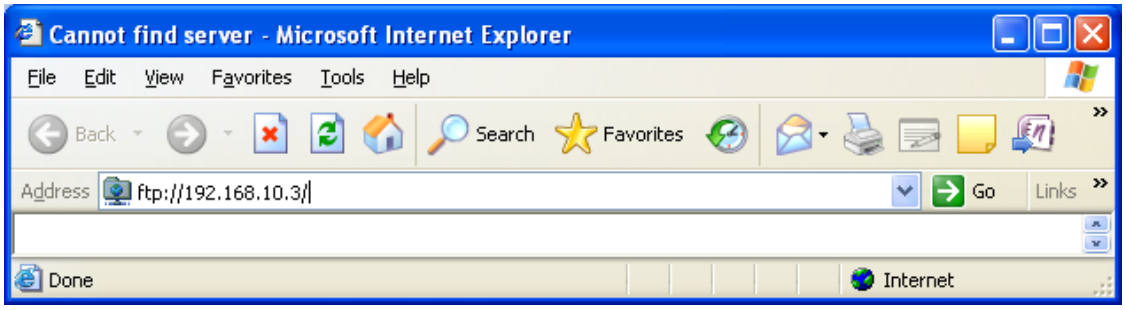
9. Click **OK**.
10. Give Azhar Amran **Full Control** of this FTP site because we want her to be as administrator for the Student FTP site.



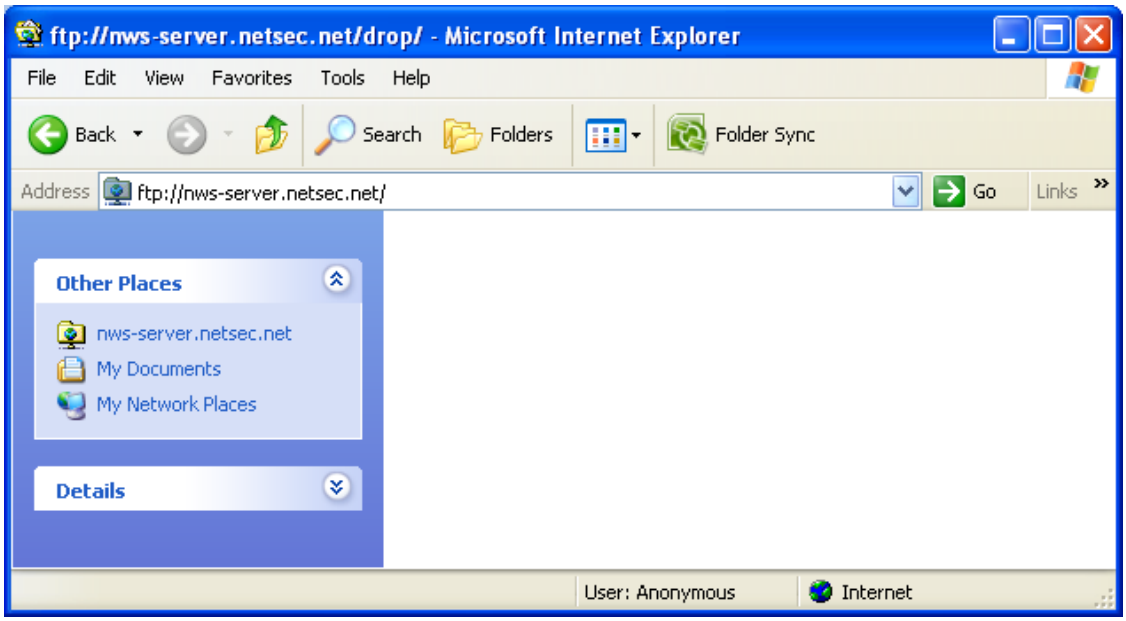
11. After finish configure, click **OK**.
12. What we have at this point is an Student folder that allow Administrators and user Azhar full access to the folder's contents. These permissions by default are also inherited by the Drop subfolder beneath Student folder.
13. Close all remaining windows.

Test FTP site for specific User

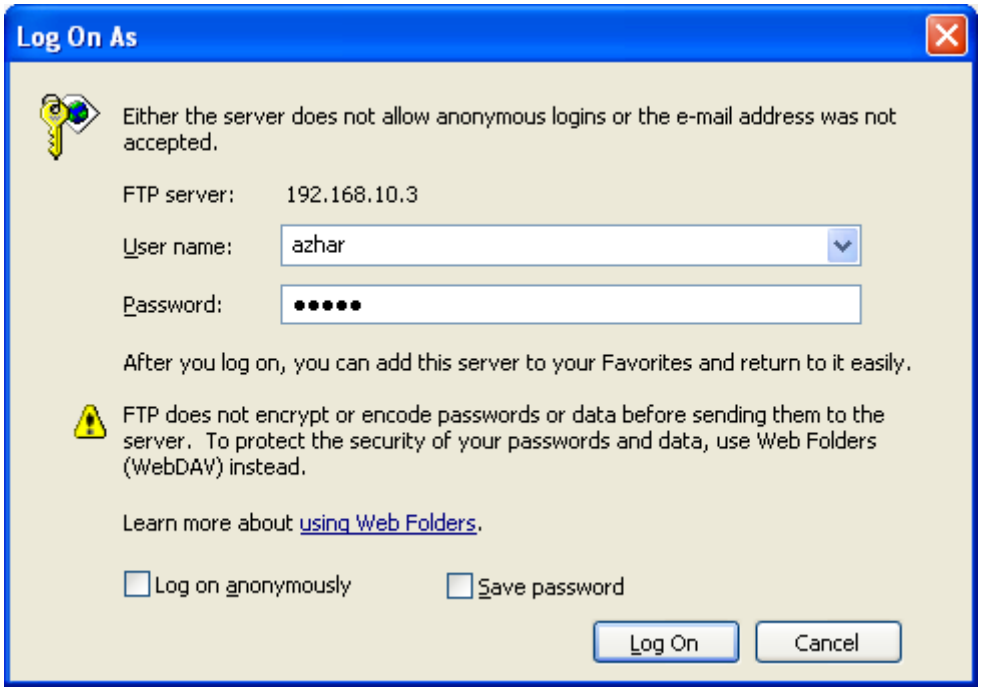
1. Log on to the **client computer** as **Administrator**.
2. Open Internet Explorer. **(Start; All Programs; Internet Explorer).**
3. On the **Address** box, key-in **ftp://yourDomain_second_IP** or **ftp://serverSN.yourdomain .net**
   (e.g. ftp://192.168.10.3 or ftp://server1.netsec.net) and click **GO** button.



Or



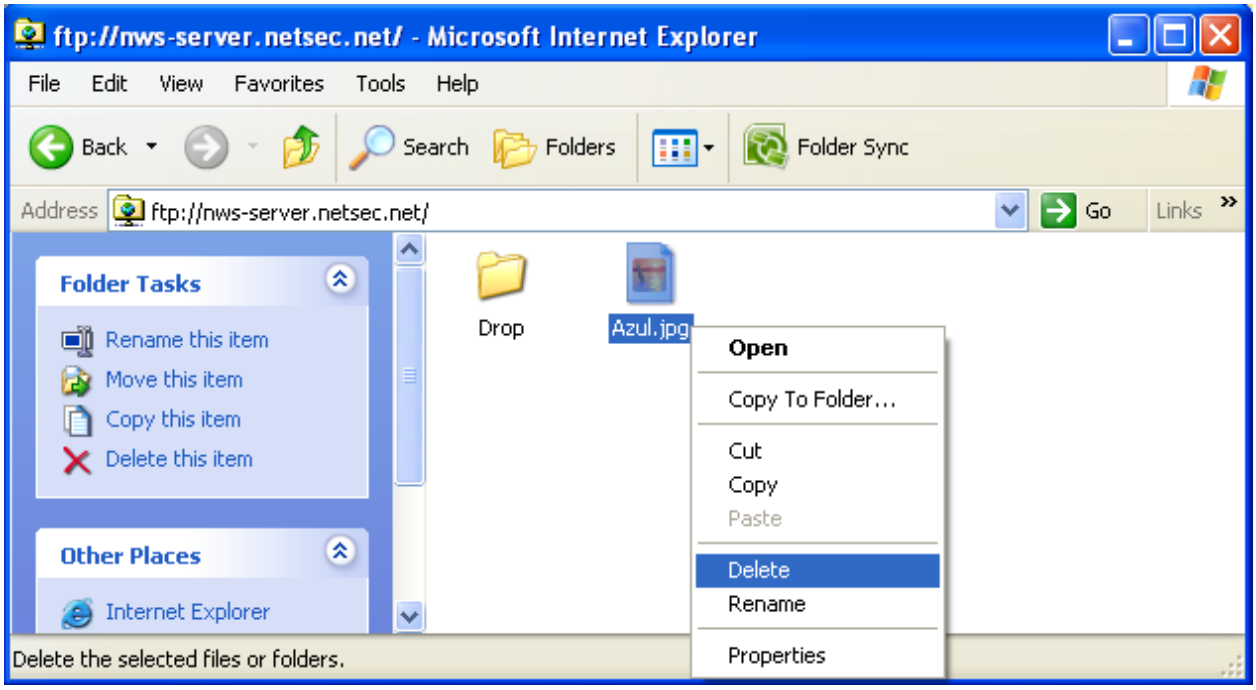
4. You will be asking for username and password. Key-in **ain.syahmi** as username and **ain** for password. Click **Log On** button.



5. Your Student FTP site will appear in the browser.
6. Use window explorer to access the **C:\Windows\Web\Wallpaper** sub-folder.
7. Click on the file **Azul.bmp** and drag and drop it into the **Student FTP site** window.

8. Now try copy any files and paste it into this FTP server.to
   Could you paste any files? YES/ NO

9. Try to delete the **Azul.bmp** file.



What happen? Could you delete the files? YES /NO.

Why do you think this is so?

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10. Log off the client computer.

Changing FTP Site Home Directories

Each FTP sites on a computer must have its own home directory. The default home directory for the default FTP site is LocalDrive :\netpub\FT proot.

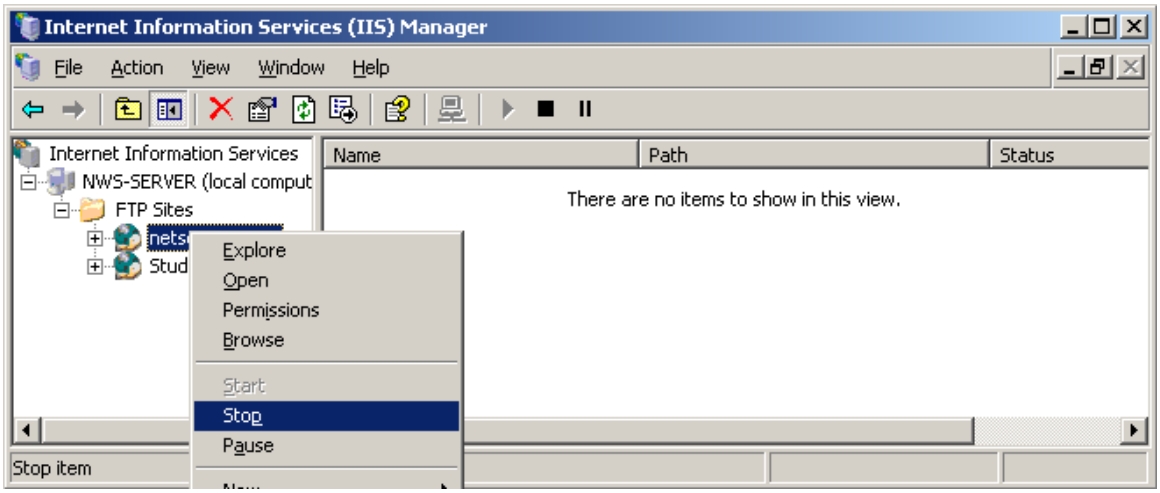
There are two ways to change the home directory of an FTP site :

- Use IIS Manager
- Edit the MetaBase.xml file directory.

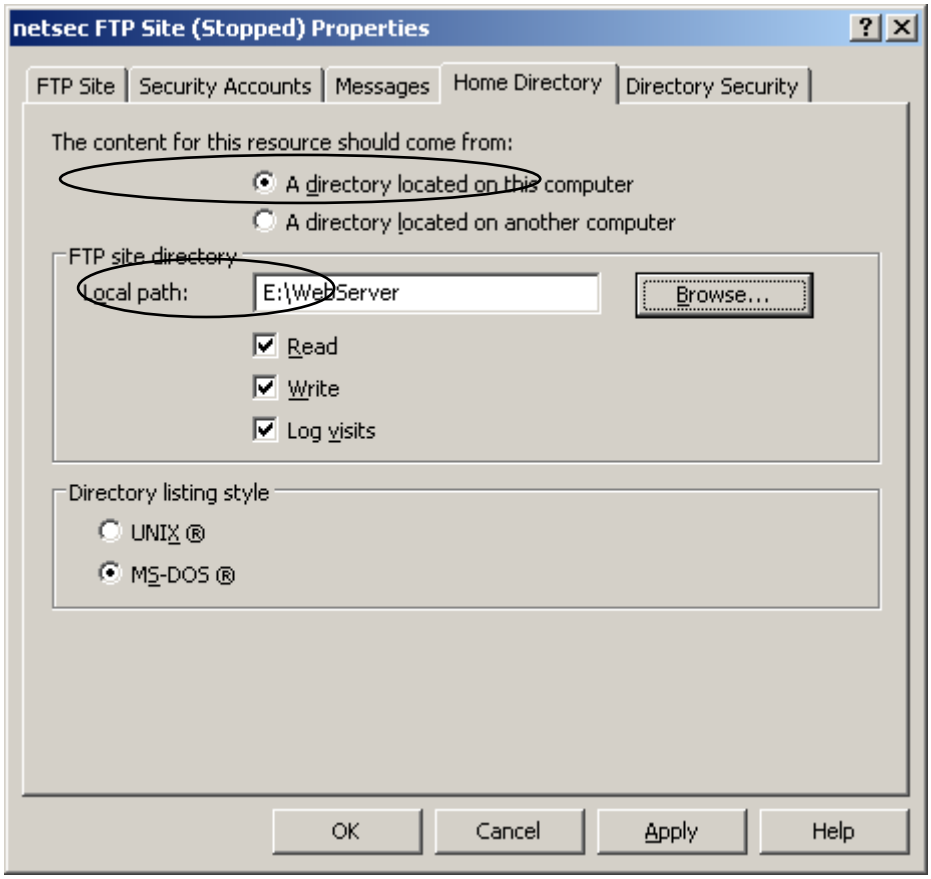
But in this exercise we only use IIS Manager.

To change the home directory of an FTP site using IIS Manager

1. Log on to the server as **Administrator.**
2. Open **IIS Manager**. (Start ; Administrator Tools ; internet Information Services (IIS) Manager).
3. in IIS Manager, expand the local computer, expand the **FTP Sites** folder, right-click the **Default FTP Sites** (e.g : Netsec FTP Site) and click Stop.



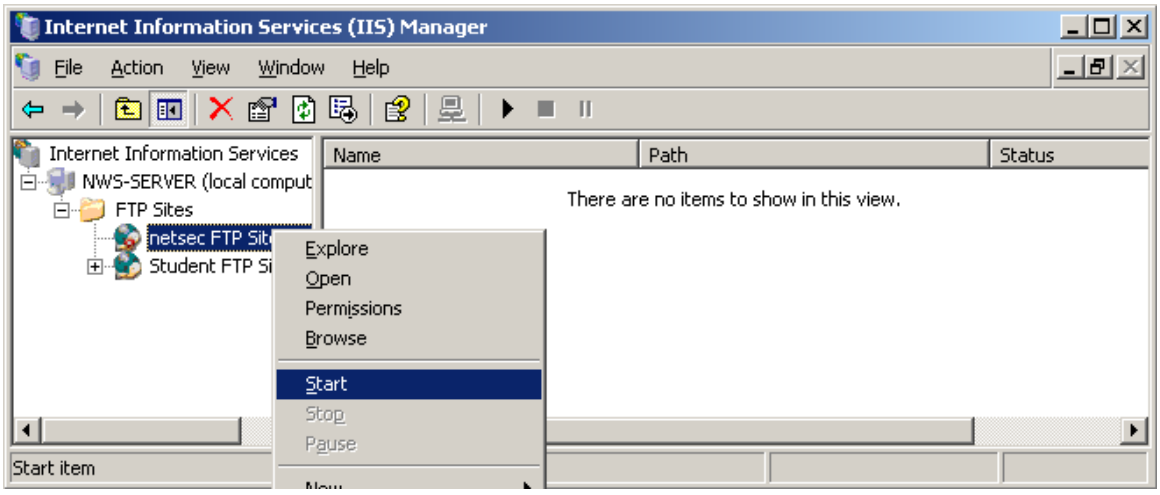
4. Right-click the **Default FTP Site** again, and click **Properties.**
5. Click the **Home Directory** tab.
6. Under **The content for this resources should come from,** click the location of your home directory under **Local path :** box (e.g: E:\WebServer) or press **Browser** button to find the location of your home directory.



Note :

If you select a directory on a network share, you might need to enter a user name and password to access the resource. IUSR_computername is the default account used if another account is not specified. If you use an account with administrative credentials on the server, clients can gain access to server operations. This seriously jeopardizes the security of your network. For more information on security see, Security Best Practices in Windows Help.

7. Click **Apply,** and then click **OK.**
8. Right-click the FTP site you've just changed, and click **Start**.



9. **Optional :** Move or copy the content for the FTP site from the original home directory to the new home directory.
10. Close all remaining window.

Summary

In this section you have learn how to:

- Changing FTP Site Home Directories: Describe the concept of a home directory and methods for changing the home directory of an FTP site.
- Naming FTP Sites: Describes assigning a descriptive name to an FTP site.
- Stopping and Starting FTP Site: Describes why you would need to stop and restart your FTP sites and how to perform these actions.
- Changing Default FTP Sites Setting : Describes how change default settings globally or on an individual site.
- Creating Multiple FTP Sites : Describes how to use IP addresses or port numbers to differentiate multiple FTP sites.
- Adding FTP Sites to Your Server : Describes the process Of adding a new FTP site to a server running IIS.
- Securing FTP Sites: Describes some of the misconceptions about FTP security and how to establish a secure FTP site.
- Isolating FTP Users: Describes the concept of FTP user isolation and which type of isolation to use to restrict users to their own directories.